

# **ACADEMIC REGULATIONS COURSE STRUCTURE AND DETAILED SYLLABUS**

**CHOICE BASED CREDIT SYSTEM**

**MLR18**

## **INFORMATION TECHNOLOGY**

for

### **Bachelor of Technology(B.Tech)**

B. Tech. - Regular Four Year Degree Program  
(For batches admitted from the academic year 2018 - 19)

&

B. Tech. - Lateral Entry Scheme  
(For batches admitted from the academic year 2019 - 20)



# **MLR Institute of Technology**

(Autonomous)

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## FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

**PRINCIPAL**

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**B. Tech. - Regular Four Year Degree Program**  
**(For batches admitted from the academic year 2018 - 19)**  
**&**  
**B. Tech. - Lateral Entry Scheme**  
**(For batches admitted from the academic year 2019 - 20)**

For pursuing four year under graduate Bachelor Degree Programme of study in Engineering (B.Tech) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2018-19 onwards. Any reference to "Institute" or "College" in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

**1. ADMISSION**

**1.1. Admission into first year of four year B. Tech. degree Program of study in Engineering:**

**1.1.1. Eligibility:**

A candidate seeking admission into the first year of four year B. Tech. degree Program should have:

(i) Passed either Intermediate Public Examination (I.P.E) conducted by the Board of Intermediate Education, Telangana, with Mathematics, Physics and Chemistry as optional subjects or any equivalent examination recognized by Board of Intermediate Education, Telangana or a Diploma in Engineering in the relevant branch conducted by the Board of Technical Education, Telangana or equivalent Diploma recognized by Board of Technical Education for admission as per guidelines defined by the Regulatory bodies of Telangana State Council for Higher Education (TSCHE) and AICTE.

(ii) Secured a rank in the EAMCET examination conducted by TSCHE for allotment of a seat by the Convener, EAMCET, for admission.

**1.1.2. Admission Procedure:**

Admissions are made into the first year of four year B. Tech. Degree Program as per the stipulations of the TSCHE.

(a) Category A seats are filled by the Convener, TSEAMCET.

(b) Category B seats are filled by the Management.

**1.2. Admission into the second year of four year B. Tech. degree Program in Engineering**

**1.2.1 Eligibility:**

A candidate seeking admission under lateral entry into the II year I Semester B. Tech. degree Program should have passed the qualifying exam (B.Sc. Mathematics or Diploma in concerned course) and based on the rank secured by the candidate at Engineering Common Entrance Test ECET (FDH) in accordance with the instructions received from the Convener, ECET and Government of Telangana.

**1.2.2 Admission Procedure:**

Admissions are made into the II year of four year B. Tech. degree Program through Convener, ECET (FDH) against the sanctioned strength in each Program of study as lateral entry students.

## **2. PROGRAMS OFFERED**

**MLR Institute of Technology**, an autonomous college affiliated to JNTUH, offers the following B. Tech. Programs of study leading to the award of B. Tech. degree under the autonomous scheme.

- 1) B.Tech. - Aeronautical Engineering
- 2) B.Tech. - Computer Science and Engineering
- 3) B.Tech. - Electronics and Communication Engineering
- 4) B.Tech - Electrical & Electronics Engineering
- 5) B.Tech. - Information Technology
- 6) B.Tech. - Mechanical Engineering

## **3. DURATION OF THE PROGRAMS**

### **3.1 Normal Duration**

- 3.1.1 B. Tech. degree program extends over a period of four academic years leading to the Degree of Bachelor of Technology (B.Tech.) of the Jawaharlal Nehru Technological University Hyderabad.
- 3.1.2 For students admitted under lateral entry scheme, B. Tech. degree program extends over a period of three academic years leading to the Degree of Bachelor of Technology (B. Tech.) of the Jawaharlal Nehru Technological University Hyderabad.

### **3.2 Maximum Duration**

- 3.2.1 The maximum period within which a student must complete a full-time academic program is 8 years for B. Tech. If a student fails to complete the academic program within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.2 For students admitted under lateral entry scheme in B. Tech. degree program, the maximum period within which a student must complete a full-time academic program is 6 years. If a student fails to complete the academic program within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.3 The period is reckoned from the academic year in which the student is admitted first time into the degree Program.

## **4. AWARD OF B.Tech. DEGREE**

A student will be declared eligible for the award of the B.Tech. degree if he/she fulfils the following academic regulations:

- 4.1 The candidate shall pursue a course of study for not less than four academic years and not more than eight years.
- 4.2 The candidate shall register for 160 credits and secure 160 credits.
- 4.3 The degree will be conferred and awarded by Jawaharlal Nehru Technological University Hyderabad on the recommendations of the Chairman, Academic Council.

## **5. PROGRAMME STRUCTURE**

- 5.1 UGC/AICTE specified Definitions/ Descriptions are adopted appropriately for various terms and abbreviations used in these Academic Regulations/ Norms, which are listed below.

**Semester Scheme:**

Each UGP is of 4 Academic Years (8 Semesters), each year divided into two semesters of 22 weeks ( $\geq 90$  working days), each Semester having - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)'. Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) as denoted by UGC, and Curriculum/Course Structure as suggested by AICTE are followed.

5.1.2 The B.Tech. Programmes of MLR Institute of Technology are of semester pattern, with 8 Semesters constituting 4 Academic Years, each Academic Year having TWO Semesters (First/Odd and Second/Even Semesters). Each Semester shall be of 22 Weeks duration (inclusive of Examinations), with a minimum of 90 Instructional Days per Semester.

5.1.3 Credit Courses:

a) All Courses are to be registered by a student in a Semester to earn Credits. Credits shall be assigned to each Subject/ Course in a L: T: P: C (Lecture Periods: Tutorial Periods: Practical Periods : Credits) Structure, based on the following general pattern ..

- One Credit - for One hour/Week/Semester for Theory/Lecture(L)/Tutorial(T) Courses; and
- One Credit - for Two hours/Week/Semester for Laboratory/Practical (P) Courses, Mini Project...
- Mandatory Courses will not carry any Credits.

5.1.4 **Course Classification:**

All Courses offered for the UGP are broadly classified as:

- Basic Science Courses (BSC) include Mathematics, Physics, Chemistry, Biology etc,
- **Engineering Science Courses (ESC)** courses include Materials, Workshop, Basics of Electrical/ Electronics/ Mechanical/Computer Science Engineering, Engineering Graphics, Instrumentation, Engineering Mechanics, Instrumentation etc.
- **Humanities and Social Science including Management Courses (HSMC)** courses include English, Communication skills, Management etc.
- Professional Core Courses(PCC) are core courses relevant to the chosen specialization/branch.
- **Professional Elective Courses (PEC)** are courses relevant to the chosen specialization/ branch offered as electives.
- **Open Elective Courses (OEC)** courses from other technical and/or emerging subject areas offered in the College by the Departments of Engineering, Science and Humanities.
- **Mandatory Course:** Course work on peripheral subjects in a programme, wherein familiarity considered mandatory. To be included as non-Credit, Mandatory Courses, with only a pass in each required to qualify for the award of degree from the concerned institution.
- **Project Work** and/or internship in industry or elsewhere, seminar.

5.1.5 **Course Nomenclature:**

The Curriculum Nomenclature or Course-Structure Grouping for the each of the UGP E&T (B.Tech. Degree Programmes), is as listed below (along with AICTE specified Range of Total Credits).

| <b>S. No.</b>                                    | <b>Broad Course Classification</b> | <b>Course Group/ Category</b>                              | <b>Course Description</b>                                                                                                                      | <b>Range of Credits</b> |
|--------------------------------------------------|------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1)                                               | BSC,ESC & HSMC                     | BSC – Basic Sciences Courses                               | Includes - Mathematics, Physics and Chemistry Subjects                                                                                         | <b>25*</b>              |
| 2)                                               |                                    | ESC - Engineering Sciences Courses                         | Includes fundamental engineering subjects.                                                                                                     | <b>24*</b>              |
| 3)                                               |                                    | HSMC – Humanities and Social Sciences including Management | Includes subjects related to Humanities, Social Sciences and Management.                                                                       | <b>12*</b>              |
| 4)                                               | PCC                                | PCC – Professional Core Courses                            | Includes core subjects related to the Parent Discipline/ Department/ Branch of Engg.                                                           | <b>48*</b>              |
| 5)                                               | PEC                                | PEC– Professional Elective Courses                         | Includes Elective subjects related to the Parent Discipline / Department / Branch of Engg.                                                     | <b>18*</b>              |
| 6)                                               | OEC                                | OEC – Open Elective Courses                                | Elective subjects which include inter-disciplinary subjects or subjects in an area outside the Parent Discipline/ Department / Branch of Engg. | <b>18*</b>              |
| 7)                                               | PWC                                | Project Work                                               | Major Project.                                                                                                                                 | <b>15*</b>              |
| 8)                                               |                                    | Industrial Training/ Mini-Project                          | Industrial Training/ Internship/ Mini-Project.                                                                                                 |                         |
| 9)                                               |                                    | Seminar                                                    | Seminar / Colloquium based on core contents related to Parent Discipline/ Department/ Branch of Engg.                                          |                         |
| 10)                                              | MC                                 | Mandatory Courses                                          | Mandatory Courses (non-credit)                                                                                                                 | <b>-</b>                |
| <b>Total Credits for UGP (B. Tech.)Programme</b> |                                    |                                                            |                                                                                                                                                | <b>160 (100%)</b>       |

- Minor variations as per AICTE guidelines

## 6. COURSE REGISTRATION

- 6.1 A 'Faculty Advisor or Counsellor' shall be assigned to each student, who advises him/her about the UGP, its Course Structure and Curriculum, Choice/Option for Subjects/Courses, based on his/her competence, progress, pre-requisites and interest.
- 6.2 Academic Section of the College invites 'Registration Forms' from students prior (before the beginning of the Semester), ensuring 'DATE and TIME Stamping'. The Registration Requests for any 'CURRENT SEMESTER' shall be completed BEFORE the commencement of SEEs (Semester End Examinations) of the 'PRECEDING SEMESTER'
- A Student can apply for Registration, which includes approval from his faculty advisor, and then should be submitted to the College Academic Section through the Head of Department (a copy of the same being retained with Head of Department, Faculty Advisor and the Student).
- 6.3 A student may be permitted to register for his/her course of CHOICE with a total of prescribed credits per semester (permitted deviation being  $\pm 12\%$ ), based on his PROGRESS and

SGPA/CGPA, and completion of the 'PRE-REQUISITES' as indicated for various courses in the Department Course Structure and Syllabus contents.

- 6.4 Choice for 'additional Courses' must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Counsellor.
- 6.5 If the Student submits ambiguous choices or multiple options or erroneous (incorrect) entries during Registration for the Course(s) under a given/specified Course Group/ Category as listed in the Course Structure, only the first mentioned Course in that Category will be taken into consideration.
- 6.6 Dropping of Courses or changing of options may be permitted, ONLY AFTER obtaining prior approval from the Faculty Advisor, 'within 15 Days of Time' from the commencement of that Semester. Course Options exercised through Registration are final and CAN NOT be changed, and CAN NOT be inter-changed; further, alternate choices will also not be considered. However, if the Course that has already been listed for Registration (by the Head of Department) in a Semester could not be offered due to any unforeseen or unexpected reasons, then the Student shall be allowed to have alternate choice - either for a new Subject (subject to offering of such a Subject), or for another existing Subject (subject to availability of seats), which may be considered. Such alternate arrangements will be made by Head of the Department, with due notification and time-framed schedule, within the FIRST WEEK from the commencement of Class-work for that Semester.

## **7. COURSES TO BE OFFERED**

- 7.1 A typical section (or class) strength for each semester shall be 60.
- 7.2 courses may be offered to the Students, only if minimum of 20 students (1/3<sup>rd</sup> of the section strength) opt for it.
- 7.2 More than ONE TEACHER may offer the SAME SUBJECT (Lab/Practical's may be included with the corresponding Theory Subject in the same Semester) in any Semester. However, selection choice for students will be based on - 'CGPA Basis Criterion' (i.e., the first focus shall be on early Registration in that Semester, and the second focus, if needed, will be on CGPA of the student).
- 7.3 If more entries for Registration of a Subject come into picture, then the concerned Head of the Department shall take necessary decision, whether to offer such a Subject/Course for TWO (or multiple) SECTIONS or NOT.
- 7.4 OPEN ELECTIVES will be offered by a department to the students of other departments.

## **8. ATTENDANCE REQUIREMENTS**

- a. A student will be eligible to appear for the End Semester Examinations, if he acquires a minimum of 75% of attendance in aggregate of all the Subjects/Courses (excluding Mandatory or Non-Credit Courses) for that Semester.
- b. Condoning of shortage of attendance in aggregate up to 10% (65% and above, and below 75%) in each Semester may be granted by the College Academic Committee on genuine and valid grounds, based on the student's representation with supporting evidence by following the govt. rules in vogue.
- c. A stipulated fee shall be payable towards condoning of shortage of attendance.
- d. Shortage of Attendance below 65% in aggregate shall in No case be condoned.
- e. A student shall not be promoted to the next semester unless he/she satisfies the attendance requirements of the current semester. The student may seek readmission for the semester when offered next. He / She shall not be allowed to register for the subjects of the semester while he/she is in detention. A student detained due to shortage of attendance, will have to repeat that semester when offered next.
- f. Students whose attendance is less than 75% are not entitled to get the scholarship / fee reimbursement in any case as per the TS Govt. Rules in force.



## **9. ACADEMIC REQUIREMENTS FOR PROMOTION / COMPLETION OF REGULAR B.TECH PROGRAM COURSE STUDY**

- 9.1 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to each Course, if he secures not less than 35% marks in the End Semester Examination, and a minimum of 40% of marks in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of Letter Grades, this implies securing P Grade or above in that Course.
- 9.2 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to – Internship, if he/she secures not less than 40% of the total marks to be awarded for each. The student would be treated as failed, if he/she - (i) does not submit a report on his/her Internship, or does not make a presentation of the same before the Evaluation Committee as per the schedule, or (ii) does not present the Seminar as required in the IV year I/II Semester, or (iii) secures less than 40% of marks in Internship evaluations. He/She may reappear once for each of the above evaluations, when they are scheduled again; if he/she fails in such 'one reappearance' evaluation also, he/she has to reappear for the same in the next subsequent semester, as per the schedule.
- 9.3 A Student will not be promoted from I Year to II Year, unless he/she fulfils the Attendance requirements
- 9.4 A Student will not be promoted from II Year to III Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a total 50% of Credits up to II Year II Semester from all the relevant regular and supplementary examinations.
- 9.5 A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a total 50% of Credits up to III Year II Semester, from all the regular and supplementary examinations.
- 9.6 After securing the necessary 160 Credits as specified for the successful completion of the entire UGP, resulting in 160 Credits for UGP performance evaluation, i.e., the performance of the Student in these 160 Credits shall alone be taken into account for the calculation of the final CGPA.  
If a Student registers for some more 'extra courses' (in the parent Department or other Departments/Branches of Engg.) other than those listed courses totalling to 160 Credits as specified in the Course Structure of his/her Department, the performances in those 'extra courses' (although evaluated and graded using the same procedure as that of the required 160 Credits) will not be taken into account while calculating the SGPA and CGPA. For such 'extra courses' registered, % marks and Letter Grade alone will be indicated in the Grade Card, as a performance measure, subject to completion of the Attendance and Academic Requirements as stated in items 8 and 9.1-9.6.
- 9.7 Students who fail to earn minimum of 160 Credits as per the Course Structure, and as indicated above, within 8 Academic Years from the Date of Commencement of their I Year shall forfeit their seats in B.Tech Programme and their admissions shall stand cancelled.  
When a Student is detained due to shortage of attendance/lack of credits in any Semester, he may be re-admitted into that Semester, as and when offered. However the regulations at the time of admissions hold good.

## **10. EVALUATION - DISTRIBUTION AND WEIGHTAGE OF MARKS**

- 10.1 The performance of a student in each Semester shall be evaluated Course-wise (irrespective of Credits assigned) with a maximum of 100 marks for Theory. The B.Tech Project Work (Major Project) will be evaluated for 200 Marks. These evaluations shall be based on 50 marks shall be evaluated through CIE (Continuous Internal Evaluation) and 150 marks shall be evaluated through SEE (Semester End Examination), and a Letter Grade corresponding to the % marks obtained shall be given.
- 10.2 For all Theory Courses as mentioned above, the distribution shall be 30 marks for CIE, and 70 marks for the SEE.
- 10.3 a) For Theory Subjects (inclusive of Minor Courses), during the Semester, there shall be two CIE examinations for 30 marks each. Each CIE examination consists of one subjective

paper for 25 marks, and assignment for 5 marks for each subject.

Question paper contains 2 Parts (Part-A and Part-B.) The distribution of marks for PART-A and PART-B will be 10 marks & 15 marks respectively for UG programs.

Pattern of the question paper is as follows:

**PART-A**

Consists of **one compulsory question** with five sub questions each carrying two marks. For the CIE-I examinations the sub question would be from first 2 ½ units and for the CIE-II examination the sub question would be from the remaining 2 ½ units.

**PART-B**

Consists of five questions (out of which students have to answer three questions) carrying five marks each. Each question there will be an “either” “or” choice (that means there will be two questions from each unit and the student should answer any one question). The questions can consist of sub questions also.

- b) The first mid-term examination shall be conducted for the first 50% of the syllabus, and the second mid-term examination shall be conducted for the remaining 50% of the syllabus.
  - c) First Assignment should be submitted before the commencement of the first mid-term examinations, and the Second Assignment should be submitted before the commencement of the second mid-term examinations. The assignments shall be specified/given by the concerned subject teacher.
  - d) If any candidate is absent for the CIE examination or those who want to improve their internal marks in any subject can opt for Computer Based Test (CBT) as and when offered. The CBT is a 45 minutes duration ONLINE exam consisting of 30 objective questions from the entire syllabus of the subject. Best marks is consider as a final marks from the average of two mid examinations or CBT examination marks. The CBT can be taken after the payment of prescribed fee. There is no Internal Improvement for the courses Machine Drawing, Production Drawing, Engineering Drawing, Engineering Graphics and practical courses.
- 10.4 For Practical Courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned for Lab/Practical End Semester Examination (SEE). Out of the 30 marks for internals, day-to-day work in the laboratory shall be evaluated for 15 marks; and for the remaining 15 marks - two internal practical tests (each of 15 marks) shall be conducted by the concerned laboratory teacher and the average of the two tests is taken into account. The SEE for Practical's shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.
  - 10.5 For the Subjects having Design and/or Drawing, (such as Engineering Graphics, Engineering Drawing, Machine Drawing, Production Drawing Practice, and Estimation), the distribution shall be 30 marks for CIE (15 marks for day-to-day work and 15 marks for internal tests) and 70 marks for SEE. There shall be two internal tests in a semester and the better of the two shall be considered for the award of marks for internal tests.
  - 10.6 **Open Elective Course:** Students can choose One Open Elective Course (OEC-I) during V Semester, one (OEC-II) during VI Semester, one (OEC-III) in VII Semester, and one (OEC-IV) in VIII Semester from the list of Open Elective Courses given. However, Students cannot opt for an Open Elective Courses offered by their own (parent) Department, if it is already listed under any category of the Subjects offered by parent Department in any Semester.
  - 10.7 There shall be an internship to be taken up during the vacation after II year II Semester (IV Semester) examination. However, the internship and its report shall be evaluated in III Year I Semester (VI Semester) SEE. The internship shall be submitted in a report form and presented before the committee. It shall be evaluated for 100 marks. The committee consists of head of the department, and a senior faculty member(s)(minimum Two) of the department. There shall be no internal marks for internship.
  - 10.8 There shall be a Comprehensive Viva in VI Semester and will be conducted SEE by through a test or a committee consisting of One External Examiner, Head of the Department and two Senior faculty members of the Department. The comprehensive viva is intended to assess the

student's understanding of the subjects he/she studied during the B.Tech course of study. The Comprehensive Viva is evaluated for 100 marks by the test or committee. There shall be no CIE for Comprehensive Viva.

- 10.9 Each Student shall start the Project Work Phase-1 during the IV B.Tech I Semester(VII Semester), as per the instructions of the Project Guide/Project Supervisor assigned by the Head of Department. Total 100 marks allotted for the Project Work Stage-I. 40% of marks shall be evaluated Project Guide/Project supervisor CIE (Continuous Internal Evaluation) based on the reports submitted and conduct presentations. Remaining 60% of marks shall be evaluated by committee comprising of the Head of the Department, project supervisor and senior faculty member from concerned department based on Viva/Seminar Presentation. He/She must secure the 40% of the marks from CIE.

10.10. **End semester examination:**

- Question paper contains 2 Parts (Part-A and Part-B) having the questions distributed equally among all units.
- The distribution of marks for PART-A and PART-B will be 20 marks & 50 marks respectively for UG programs. Pattern of the question paper is as follows:

**PART-A**

Consists of one question which are compulsory. The first question consists of ten sub-questions two from each unit and carry 2 marks each.

**PART-B**

Consists of 5 questions carrying 10 marks each. Each of these questions is from one unit and may contain sub questions. Each question there will be an "either" "or" choice (that means there will be two questions from each unit and the student should answer any one question).

- 10.11 For Mandatory Non-Credit Courses offered in a Semester, after securing  $\geq 65\%$  attendance and has secured not less than 35% marks in the SEE, and a minimum of 40% of marks in the sum total of the CIE and SEE taken together in such a course, then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses '**Satisfactory**' or '**Unsatisfactory**' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.
- 10.12. For social innovation and Engineering Exploration courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned End Semester Examination (SEE). Internal evaluations done through presentations in CIE-1 & CIE-2. The SEE for shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.

## 11. **AWARD OF DEGREE**

After a student has satisfied the requirement prescribed for the completion of the Program and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes Shown in Table.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

| Class Awarded                | Grade to be Secured          |
|------------------------------|------------------------------|
| First Class with Distinction | CGPA $\geq 8.00$             |
| First Class                  | $\geq 6.50$ to $< 8.00$ CGPA |
| Second Class                 | $\geq 5.50$ to $< 6.50$ CGPA |
| Pass Class                   | $\geq 5.00$ to $< 5.50$ CGPA |
| FAIL                         | CGPA $< 5$                   |

### a) **Improvement of Grades and Completion of the Course**

- Candidates who have passed in a theory course in a semester are allowed to appear for improvement of Grade in the next immediate supplementary examination for a

maximum of three subjects only. Candidates will not be allowed to improve grade in the Laboratory, Seminars, Internships and Project Work.

- ii) Improved grade will not be counted for the award of prizes/medals and Rank. However the previous grade will be considered for the award of prizes/medals and rank in case of toppers.
- iii) If the candidate does not show improvement in the grade, his/her previous grade will be taken into consideration.

## 12. LETTER GRADE AND GRADE POINT

- 12.1 Marks will be awarded to indicate the performance of each student in each Theory Subject, or Lab/Practical's, or Seminar, or Project, or Internship\*/Mini-Project, Minor Course etc., based on the %marks obtained in CIE+SEE (Continuous Internal Evaluation + Semester End Examination, both taken together), and a corresponding Letter Grade shall be given.  
\* For internship only SEE will be taken into consideration. There is no CIE for Internship.
- 12.2 As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades (UGC Guidelines) and corresponding percentage of marks shall be followed...

| <b>% of Marks Secured<br/>(Class Intervals)</b>               | <b>Letter Grade<br/>(UGC Guidelines)</b> | <b>Grade Points</b> |
|---------------------------------------------------------------|------------------------------------------|---------------------|
| 90% and above<br>( $\geq 90\%$ , $\leq 100\%$ )               | O<br>(Outstanding)                       | 10                  |
| Below 90% but not less than 80%<br>( $\geq 80\%$ , $< 90\%$ ) | A <sup>+</sup><br>(Excellent)            | 9                   |
| Below 80% but not less than 70%<br>( $\geq 70\%$ , $< 80\%$ ) | A<br>(Very Good)                         | 8                   |
| Below 70% but not less than 60%<br>( $\geq 60\%$ , $< 70\%$ ) | B <sup>+</sup><br>(Good)                 | 7                   |
| Below 60% but not less than 50%<br>( $\geq 50\%$ , $< 60\%$ ) | B<br>(above Average)                     | 6                   |
| Below 50% but not less than 40%<br>( $\geq 40\%$ , $< 50\%$ ) | C<br>(Average)                           | 5                   |
| Below 40%<br>( $< 40\%$ )                                     | F<br>(FAIL)                              | 0                   |
| Absent                                                        | AB                                       | 0                   |

- 12.3 A student obtaining F Grade in any Subject shall be considered 'failed' and will be required to reappear as 'Supplementary Candidate' in the End Semester Examination (SEE), as and when offered. In such cases, his Internal Marks (CIE Marks) in those Subject(s) will remain same as those he obtained earlier.
- 12.4 A Letter Grade does not imply any specific % of Marks.
- 12.5 In general, a student shall not be permitted to repeat any Subject/Course (s) only for the sake of 'Grade Improvement' or 'SGPA/CGPA Improvement'. However, he has to repeat all the Subjects/Courses pertaining to that Semester, when he is detained.
- 12.6 A student earns Grade Point (GP) in each Subject/Course, on the basis of the Letter Grade obtained by him in that Subject/Course (excluding Mandatory non-credit Courses). Then the corresponding 'Credit Points' (CP) are computed by multiplying the Grade Point with Credits for that particular Subject/Course.

**Credit Points (CP) = Grade Point (GP) x Credits .... For a Course**

- 12.7 The Student passes the Subject/Course only when he gets  $GP \geq 4$  (P Grade or above).
- 12.8 The Semester Grade Point Average (SGPA) is calculated by dividing the Sum of Credit Points ( $\Sigma CP$ ) secured from ALL Subjects/Courses registered in a Semester, by the Total Number of Credits registered during that Semester. SGPA is rounded off to TWO Decimal Places. SGPA is thus computed as

$$SGPA = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\} \dots \text{For each Semester,}$$

where 'i' is the Subject indicator index (takes into account all Subjects in a Semester), 'N' is the no. of Subjects 'REGISTERED' for the Semester (as specifically required and listed under the Course Structure of the parent Department),  $C_i$  is the no. of Credits allotted to that ix Subject, and  $G_i$  represents the Grade Points (GP) corresponding to the Letter Grade awarded for that i Subject.

#### Illustration of Computation of SGPA Computation

| Course  | Credit | Grade Letter | Grade Point | Credit Point<br>(Credit x Grade) |
|---------|--------|--------------|-------------|----------------------------------|
| Course1 | 3      | A            | 8           | $3 \times 8 = 24$                |
| Course2 | 4      | B+           | 7           | $4 \times 7 = 28$                |
| Course3 | 3      | B            | 6           | $3 \times 6 = 18$                |
| Course4 | 3      | O            | 10          | $3 \times 10 = 30$               |
| Course5 | 3      | C            | 5           | $3 \times 5 = 15$                |
| Course6 | 4      | B            | 6           | $4 \times 6 = 24$                |

Thus, **SGPA =  $139/20 = 6.95$**

- 12.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student over all Semesters considered for registration. The CGPA is the ratio of the Total Credit Points secured by a student in ALL registered Courses in ALL Semesters, and the Total Number of Credits registered in ALL the Semesters. CGPA is rounded off to TWO Decimal Places. CGPA is thus computed from the I Year Second Semester onwards, at the end of each Semester, as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered}$$

(i.e., up to and inclusive of S Semesters,  $S \geq 2$ ),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1<sup>st</sup> Semester onwards up to and inclusive of the Semester S (obviously  $M > N$ ), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters),  $C_j$  is the no. of Credits allotted to the jth Subject, and  $G_j$  represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

#### For CGPA Computation

| Semester 1                 | Semester 2                 | Semester 3                 | Semester 4                 | Semester 5                 | Semester 6                 |
|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|
| Credits : 20<br>SGPA : 6.9 | Credits : 22<br>SGPA : 7.8 | Credits : 25<br>SGPA : 5.6 | Credits : 26<br>SGPA : 6.0 | Credits : 26<br>SGPA : 6.3 | Credits : 25<br>SGPA : 8.0 |

$$\text{Thus, CGPA} = \frac{20 \times 6.9 + 22 \times 7.8 + 25 \times 5.6 + 26 \times 6.0 + 26 \times 6.3 + 25 \times 8.0}{144} = 6.73$$

- 12.10 For Merit Ranking or Comparison Purposes or any other listing, ONLY the 'ROUNDED OFF' values of the CGPAs will be used.
- 12.11 For Calculations listed in Item 12.6–12.10, performance in failed Subjects/Courses (securing F Grade) will also be taken into account, and the Credits of such Subjects/Courses will also be included in the multiplications and summations. However, Mandatory Courses will not be taken into consideration.
- 12.12 Conversion formula for the conversion of GPA into indicative percentage is  
 $[\text{CGPA Earned} - 0.50] \times 10 = \% \text{ of marks scored}$

### 13. DECLARATION OF RESULTS

Computation of SGPA and CGPA are done using the procedure listed in 12.6– 2.10.

No SGPA/CGPA is declared, if a candidate is failed in any one of the courses of a given semester.



**14. WITH HOLDING OF RESULTS**

If the student has not paid fees to College at any stage, or has pending dues against his name due to any reason what so ever, or if any case of indiscipline is pending against him, the result of the student may be withheld, and he will not be allowed to go into the next higher Semester. The Award or issue of the Degree may also be withheld in such cases.

**15. REVALUATION**

Students shall be permitted for revaluation after the declaration of end semester examination results within due dates by paying prescribed fee. After revaluation if there is any betterment in the grade, then improved grade will be considered. Otherwise old grade shall be retained.

**16. SUPPLEMENTARY EXAMINATIONS**

16.1 Supplementary examinations for the odd semester shall be conducted with the regular examinations of even semester and vice versa, for those who appeared and failed or absent in regular examinations. Such candidates writing supplementary examinations may have to write more than one examination per day.

**16.2 ADVANCED SUPPLEMENTARY EXAMINATION**

Advanced supplementary examinations will be conducted for IV year II semester after announcement of regular results.

**17. TRANSCRIPTS**

After successful completion of prerequisite credits for the award of degree a Transcript containing performance of all academic years will be issued as a final record. Duplicate transcripts will also be issued if required after the payment of requisite fee and also as per norms in vogue.

**18. RULES OF DISCIPLINE**

- 18.1 Any attempt by any student to influence the teachers, Examiners, faculty and staff of controller of Examination for undue favours in the exams, and bribing them either for marks or attendance will be treated as malpractice cases and the student can be debarred from the college.
- 18.2 When the student absents himself, he is treated as to have appeared and obtained zero marks in that course(s) and grading is done accordingly.
- 18.3 When the performance of the student in any subject(s) is cancelled as a punishment for indiscipline, he is awarded zero marks in that subject(s).
- 18.4 When the student's answer book is confiscated for any kind of attempted or suspected malpractice the decision of the Examiner is final.

**19. MALPRACTICE PREVENTION COMMITTEE**

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convenor
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice

to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

**20. TRANSITORY REGULATIONS**

Student who has discontinued for any reason, or has been detained for want of attendance or lack of required credits as specified, or who has failed after having undergone the Degree Programme, may be considered eligible for readmission to the same Subjects/Courses (or equivalent Subjects/Courses, as the case may be), and same Professional Electives/Open Electives (or from set/category of Electives or equivalents suggested, as the case may be) as and when they are offered (within the time-frame of 8 years from the Date of Commencement of his I Year I Semester).

**21. AMENDMENTS TO REGULATIONS**

The Academic Council of MLR Institute of Technology reserves the right to revise, amend, or change the regulations, scheme of examinations, and / or syllabi or any other policy relevant to the needs of the society or industrial requirements etc., without prior notice.

**22. STUDENT TRANSFERS**

There shall be no Branch transfers after the completion of Admission Process. Transfer of students from other colleges or universities are permitted subjected to the rules and regulations of TSCHE (TE Department) and JNTUH in vogue.

**23. GRADUATION DAY**

The College shall have its own Annual Graduation Day for the award of Degrees issued by the College/University.

**24. AWARD OF MEDALS**

Institute will award Medals to the outstanding students who complete the entire course in the first attempt within the stipulated time.

**25. SCOPE**

- i) Where the words “he”, “him”, “his”, occur in the write-up of regulations, they include “she”, “her”.
- ii) Where the words “Subject” or “Subjects”, occur in these regulations, they also imply “Course” or “Courses”.
- iii) The Academic Regulations should be read as a whole, for the purpose of any interpretation.
- iv) In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of the Academic Council is final.

## Academic Regulations for B. Tech. (Lateral Entry Scheme)

**(Effective for the students getting admitted into II year from the Academic Year 2018-2019 on wards)**

1. The Students have to acquire 120 credits from II to IV year of B.Tech Program (Regular) for the award of the degree.
2. Students, who fail to fulfil the requirement for the award of the degree in 6 consecutive academic years from the year of admission, shall forfeit their seat.
3. The same attendance regulations are to be adopted as that of B. Tech. (Regular)
4. **Promotion Rule:**
5. A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a total of 50% Credits up to III Year II Semester, from all the regular and supplementary examinations.
6. **Award of Class:**

After the student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes: The marks obtained for 120 credits will be considered for the calculation of CGPA and award of class shall be shown separately.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

| Class Awarded                | Grade to be Secured          |
|------------------------------|------------------------------|
| First Class with Distinction | CGPA $\geq$ 8.00             |
| First Class                  | $\geq 6.50$ to $< 8.00$ CGPA |
| Second Class                 | $\geq 5.50$ to $< 6.50$ CGPA |
| Pass Class                   | $\geq 5.00$ to $< 5.50$ CGPA |
| FAIL                         | CGPA $< 5$                   |

7. All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B.Tech (Lateral Entry Scheme).



**MALPRACTICES RULES - DISCIPLINARY ACTION FOR /IMPROPER CONDUCT IN EXAMINATIONS**

| <b>S. No</b> | <b>Nature of Malpractices / Improper Conduct</b>                                                                                                                                                                                                                                                                                                                                                                                     | <b>Punishment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (a)        | Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination) | Expulsion from the examination hall and cancellation of the performance in that subject only.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| (b)          | Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.                                                                                                                                                                                            | Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2            | Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.                                                                                                                                                                                                | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the Principal.                                                                                                                                                                                                                                                                                                                                                                               |
| 3            | Impersonates any other candidate in connection with the examination.                                                                                                                                                                                                                                                                                                                                                                 | The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practical's and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him. |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. |
| 5 | Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Cancellation of the performance in that subject.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 6 | Refuses to obey the orders of the Addl. Controller of examinations / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the addl. Controller of examinations or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the addl. Controller of examinations, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination. | In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.                                           |
| 7 | Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred for two consecutive semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. |

|    |                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8  | Possess any lethal weapon or firearm in the examination hall.                                                                                                                                             | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat.                                                                                                                                                      |
| 9  | If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8. | Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them. |
| 10 | Comes in a drunken condition to the examination hall.                                                                                                                                                     | Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.                                                                                                                                                                                                            |
| 11 | Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.                                                                                                 | Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that semester/year examinations.                                                                                                                                                                                                                                                                                                                         |
| 12 | If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the principal for further action to award suitable punishment.                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

# **COURSE STRUCTURE**

## INFORMATION TECHNOLOGY

| <b>I B.Tech - I SEMESTER</b>  |                                  |             |                |           |           |             |                                     |                |            |
|-------------------------------|----------------------------------|-------------|----------------|-----------|-----------|-------------|-------------------------------------|----------------|------------|
| Course Code                   | Course Title                     | Course Area | Hours per Week |           |           | Credits     | Scheme of Examination Maximum Marks |                |            |
|                               |                                  |             | L              | T         | P         |             | Internal (CIE)                      | External (SEE) | Total      |
| A4BS02                        | Linear Algebra and Calculus      | BSC         | 3              | 1         | 0         | 4           | 30                                  | 70             | 100        |
| A4BS08                        | Applied Physics                  | BSC         | 3              | 1         | 0         | 4           | 30                                  | 70             | 100        |
| A4EE01                        | Basic Electrical Engineering     | ESC         | 3              | 1         | 0         | 4           | 30                                  | 70             | 100        |
| A4ES02                        | Engineering Graphics and Design  | ESC         | 1              | 0         | 4         | 3           | 30                                  | 70             | 100        |
| A4BS10                        | Applied Physics Lab              | BSC         | 0              | 0         | 3         | 1.5         | 30                                  | 70             | 100        |
| A4EE02                        | Basic Electrical Engineering Lab | ESC         | 0              | 0         | 2         | 1           | 30                                  | 70             | 100        |
| A4AE63                        | Workshop Practices               | ESC         | 1              | 0         | 2         | 2           | 30                                  | 70             | 100        |
| A4HS03                        | Social Innovation                | HSMC        | 0              | 0         | 2         | 1           | 30                                  | 70             | 100        |
| <b>TOTAL</b>                  |                                  |             | <b>11</b>      | <b>03</b> | <b>13</b> | <b>20.5</b> | <b>240</b>                          | <b>560</b>     | <b>800</b> |
| Mandatory Course (Non-Credit) |                                  |             |                |           |           |             |                                     |                |            |
| A4MC04                        | Technical Seminar - I            | MC          | -              | -         | 2         | -           | 30                                  | 70             | 100        |

| <b>I B.Tech. - II SEMESTER</b> |                                               |             |                |           |           |             |                                     |                |            |
|--------------------------------|-----------------------------------------------|-------------|----------------|-----------|-----------|-------------|-------------------------------------|----------------|------------|
| Course Code                    | Course Title                                  | Course Area | Hours per Week |           |           | Credits     | Scheme of Examination Maximum Marks |                |            |
|                                |                                               |             | L              | T         | P         |             | Internal (CIE)                      | External (SEE) | Total      |
| A4BS04                         | Advanced Calculus                             | BSC         | 3              | 1         | 0         | 4           | 30                                  | 70             | 100        |
| A4BS11                         | Chemistry                                     | BSC         | 4              | 0         | 0         | 4           | 30                                  | 70             | 100        |
| A4CS01                         | Programming for Problem Solving               | ESC         | 3              | 0         | 0         | 3           | 30                                  | 70             | 100        |
| A4HS01                         | English                                       | HSMC        | 2              | 0         | 0         | 2           | 30                                  | 70             | 100        |
| A4CS02                         | Programming for Problem Solving Lab           | ESC         | 0              | 0         | 4         | 2           | 30                                  | 70             | 100        |
| A4BS12                         | Chemistry Lab                                 | BSC         | 0              | 0         | 3         | 1.5         | 30                                  | 70             | 100        |
| A4HS02                         | English Language and Communication Skills Lab | HSMC        | 0              | 0         | 2         | 1           | 30                                  | 70             | 100        |
| A4HS04                         | Engineering Exploration                       | HSMC        | 0              | 0         | 2         | 1           | 30                                  | 70             | 100        |
| <b>TOTAL</b>                   |                                               |             | <b>12</b>      | <b>01</b> | <b>11</b> | <b>18.5</b> | <b>240</b>                          | <b>560</b>     | <b>800</b> |
| Mandatory Course (Non-Credit)  |                                               |             |                |           |           |             |                                     |                |            |
| A4MC05                         | Technical Seminar-II                          | MC          | -              | -         | 2         | -           | 30                                  | 70             | 100        |

**II B.Tech.- I SEMESTER**

| Course Code                          | Course Title                    | Course Area | Hours per Week |           |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------------------------------|---------------------------------|-------------|----------------|-----------|----------|-----------|-------------------------------------|----------------|------------|
|                                      |                                 |             | L              | T         | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A4BS05                               | Probability and Statistics      | BSC         | 2              | 1         | 0        | 3         | 30                                  | 70             | 100        |
| A4CS03                               | Discrete Structures             | PCC         | 3              | 1         | 0        | 4         | 30                                  | 70             | 100        |
| A4CS04                               | Data Structures                 | PCC         | 3              | 1         | 0        | 4         | 30                                  | 70             | 100        |
| A4CS05                               | Database Management Systems     | PCC         | 3              | 0         | 0        | 3         | 30                                  | 70             | 100        |
| A4EC01                               | Electronic Devices and Circuits | ESC         | 3              | 0         | 0        | 3         | 30                                  | 70             | 100        |
| A4CS06                               | Data Structures Lab             | PCC         | 0              | 0         | 3        | 1.5       | 30                                  | 70             | 100        |
| A4CS07                               | Database Management Systems Lab | PCC         | 0              | 0         | 3        | 1.5       | 30                                  | 70             | 100        |
| <b>Total</b>                         |                                 |             | <b>14</b>      | <b>03</b> | <b>6</b> | <b>20</b> | <b>210</b>                          | <b>490</b>     | <b>700</b> |
| <b>Mandatory Course (Non-Credit)</b> |                                 |             |                |           |          |           |                                     |                |            |
| A4MC01                               | Environmental Studies           | MC          | 2              | 0         | 0        | 0         | 30                                  | 70             | 100        |

**II B.Tech.- II SEMESTER**

| Course Code                          | Course Title                              | Course Area | Hours per Week |          |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------------------------------|-------------------------------------------|-------------|----------------|----------|----------|-----------|-------------------------------------|----------------|------------|
|                                      |                                           |             | L              | T        | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A4EC02                               | Digital Logic Design                      | ESC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4CS08                               | Computer Organization and Architecture    | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| A4CS09                               | Object Oriented Programming               | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| A4CS10                               | Design and Analysis of Algorithms         | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| A4HS06                               | Business Economics and Financial Analysis | HSMC        | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4EC36                               | Electronic Devices and Digital Logic Lab  | ESC         | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| A4CS11                               | Object Oriented Programming Lab           | PCC         | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| <b>Total</b>                         |                                           |             | <b>15</b>      | <b>3</b> | <b>6</b> | <b>21</b> | <b>210</b>                          | <b>490</b>     | <b>700</b> |
| <b>Mandatory Course (Non-Credit)</b> |                                           |             |                |          |          |           |                                     |                |            |
| A4MC02                               | Gender Sensitization                      | MC          | 2              | 0        | 0        | 0         | 30                                  | 70             | 100        |

**III B.Tech.- I SEMESTER**

| Course Code                          | Course Title                              | Course Area | Hours per Week |          |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------------------------------|-------------------------------------------|-------------|----------------|----------|----------|-----------|-------------------------------------|----------------|------------|
|                                      |                                           |             | L              | T        | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A4IT01                               | Python Programming                        | PCC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4IT02                               | Automata and Compiler Design              | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| A4CS13                               | Operating Systems                         | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| PEC                                  | Professional Elective - 1                 | PEC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| OEC                                  | Open Elective - 1                         | OEC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4IT03                               | Python Programming Lab                    | PCC         | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| A4HS07                               | Advanced English Communication Skills Lab | HSMC        | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| A4IT24                               | Summer Internship                         | PWC         | 0              | 0        | 0        | 2         | -                                   | 100            | 100        |
| <b>Total</b>                         |                                           |             | <b>15</b>      | <b>2</b> | <b>6</b> | <b>22</b> | <b>210</b>                          | <b>590</b>     | <b>800</b> |
| <b>Mandatory Course (Non-Credit)</b> |                                           |             |                |          |          |           |                                     |                |            |
| A4MC03                               | Constitution of India                     | MC          | 2              | 0        | 0        | 0         | 30                                  | 70             | 100        |

**III B.Tech.- II SEMESTER**

| Course Code                          | Course Title                            | Course Area | Hours per Week |          |          | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------------------------------|-----------------------------------------|-------------|----------------|----------|----------|-----------|-------------------------------------|----------------|------------|
|                                      |                                         |             | L              | T        | P        |           | Internal (CIE)                      | External (SEE) | Total      |
| A4CS15                               | Computer Networks                       | PCC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4CS16                               | Linux Programming                       | PCC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4IT04                               | Web Technologies                        | PCC         | 3              | 1        | 0        | 4         | 30                                  | 70             | 100        |
| PEC                                  | Professional Elective - 2               | PEC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| PEC                                  | Professional Elective - 3               | PEC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| OEC                                  | Open Elective-2                         | OEC         | 3              | 0        | 0        | 3         | 30                                  | 70             | 100        |
| A4CS18                               | Linux Programming Lab                   | PCC         | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| A4IT05                               | Web Technologies Lab                    | PCC         | 0              | 0        | 3        | 1.5       | 30                                  | 70             | 100        |
| A4IT25                               | Comprehensive Viva                      | PWC         | 0              | 0        | 0        | 1         | -                                   | 100            | 100        |
| <b>Total</b>                         |                                         |             | <b>18</b>      | <b>1</b> | <b>6</b> | <b>23</b> | <b>240</b>                          | <b>660</b>     | <b>900</b> |
| <b>Mandatory Course (Non-Credit)</b> |                                         |             |                |          |          |           |                                     |                |            |
| A4MC06                               | Essence of Indian Traditional Knowledge | MC          | 2              | 0        | 0        | 0         | 30                                  | 70             | 100        |



**IV B.Tech.- I SEMESTER**

| Course Code  | Course Title              | Course Area | Hours per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|---------------------------|-------------|----------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|              |                           |             | L              | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A4CS19       | BigData Analytics         | PCC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| PEC          | Professional Elective - 4 | PEC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| PEC          | Professional Elective - 5 | PEC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| OEC          | Open Elective-3           | OEC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| A4CS20       | Big Data Analytics Lab    | PCC         | 0              | 0        | 4         | 2         | 30                                  | 70             | 100        |
| A4IT26       | Project Phase - 1         | PWC         | 0              | 0        | 8         | 4         | 100                                 | -              | 100        |
| <b>Total</b> |                           |             | <b>12</b>      | <b>0</b> | <b>12</b> | <b>18</b> | <b>250</b>                          | <b>350</b>     | <b>600</b> |

**IV B.Tech. - II SEMESTER**

| Course Code  | Course Title              | Course Area | Hours per Week |          |           | Credits   | Scheme of Examination Maximum Marks |                |            |
|--------------|---------------------------|-------------|----------------|----------|-----------|-----------|-------------------------------------|----------------|------------|
|              |                           |             | L              | T        | P         |           | Internal (CIE)                      | External (SEE) | Total      |
| A4HS05       | Operations Research       | HSMC        | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| PEC          | Professional Elective - 6 | PEC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| OEC          | Open Elective-4           | OEC         | 3              | 0        | 0         | 3         | 30                                  | 70             | 100        |
| A4IT27       | Project Phase - 2         | PWC         | 0              | 0        | 16        | 8         | 50                                  | 150            | 200        |
| <b>Total</b> |                           |             | <b>9</b>       | <b>0</b> | <b>16</b> | <b>17</b> | <b>140</b>                          | <b>360</b>     | <b>500</b> |

**PROFESSIONAL ELECTIVES**

| PROFESSIONAL ELECTIVE – 1 |                                    | PROFESSIONAL ELECTIVE - 2 |                                     |
|---------------------------|------------------------------------|---------------------------|-------------------------------------|
| Code                      | Subject                            | Code                      | Subject                             |
| A4CS21                    | Artificial Intelligence            | A4IT07                    | Object Oriented Analysis and Design |
| A4IT06                    | Principles of Programming Language | A4CS30                    | Mobile Application Development      |
| A4CS22                    | Distributed Databases              | A4CS25                    | Data Mining                         |
| A4CS23                    | Human Computer Interaction         | A4CS26                    | Software Engineering                |

| PROFESSIONAL ELECTIVE – 3 |                                    | PROFESSIONAL ELECTIVE - 4 |                                   |
|---------------------------|------------------------------------|---------------------------|-----------------------------------|
| Code                      | Subject                            | Code                      | Subject                           |
| A4CS27                    | Internet of Things                 | A4CS31                    | Software Testing                  |
| A4CS28                    | Cloud Computing                    | A4CS32                    | Cryptography and Network Security |
| A4IT08                    | Multimedia Application Development | A4IT10                    | Design Patterns                   |
| A4IT09                    | Angular JS                         | A4CS34                    | Ad-hoc and Sensor Networks        |

| PROFESSIONAL ELECTIVE – 5 |                            | PROFESSIONAL ELECTIVE - 6 |                              |
|---------------------------|----------------------------|---------------------------|------------------------------|
| Code                      | Subject                    | Code                      | Subject                      |
| A4CS35                    | Distributed Systems        | A4CS39                    | Software Project Management  |
| A4CS36                    | Soft Computing             | A4CS40                    | Machine Learning             |
| A4CS37                    | Agile Software Development | A4CS41                    | Information Retrieval System |
| A4IT23                    | Network Manage Systems     | A4CS42                    | Predictive Analytics         |

| OPEN ELECTIVE COURSE-I |             |                                             |                                         |
|------------------------|-------------|---------------------------------------------|-----------------------------------------|
| S. No.                 | Course Code | Course Name                                 | Offering Department                     |
| 1                      | A4AE17      | Fabrication Processes                       | Aeronautical Engineering                |
| 2                      | A4AE18      | Fundamentals of Avionics                    |                                         |
| 3                      | A4AE19      | Introduction to jets and rockets            |                                         |
| 4                      | A4CS43      | Fundamentals of Databases                   | Computer Science and Engineering        |
| 5                      | A4CS44      | Introduction to C++                         |                                         |
| 6                      | A4CS45      | Core Java Programming                       |                                         |
| 7                      | A4EC59      | Logic Design                                | Electronics & Communication Engineering |
| 8                      | A4EC60      | Principles of Communications                |                                         |
| 9                      | A4EC61      | Measurements and Instruments                |                                         |
| 10                     | A4EE52      | Electrical Wiring and Safety Measures       | Electrical & Electronics Engineering    |
| 11                     | A4EE53      | Electrical Materials                        |                                         |
| 12                     | A4EE54      | New trends in Electrical Energy             |                                         |
| 13                     | A4IT11      | Fundamentals of Data Structures             | Information Technology                  |
| 14                     | A4IT12      | Software Engineering Principles             |                                         |
| 15                     | A4IT13      | Computer Organization and Operating Systems |                                         |
| 16                     | A4ME60      | Elements Of Mechanical Engineering          | Mechanical Engineering                  |
| 17                     | A4ME61      | Fundamentals Of Engineering Materials       |                                         |
| 18                     | A4ME62      | Fundamentals Of Theory Of Machines          |                                         |
| 19                     | A4HS10      | Business Economics and Financial Analysis   | HS                                      |
| 20                     | A4HS11      | Basics of Entrepreneurship                  |                                         |
| 21                     | A4HS12      | Human Values and Professional Ethics        |                                         |

| OPEN ELECTIVE COURSE-II |             |                                            |                                         |
|-------------------------|-------------|--------------------------------------------|-----------------------------------------|
| S. No.                  | Course Code | Course Name                                | Offering Department                     |
| 1                       | A4AE25      | Introduction to aircraft Industry          | Aeronautical Engineering                |
| 2                       | A4AE26      | Non destructive testing Methods            |                                         |
| 3                       | A4AE27      | Fundamentals of Finite element method      |                                         |
| 4                       | A4CS46      | Introduction to Computer Networks          | Computer Science and Engineering        |
| 5                       | A4CS47      | Introduction to Linux Programming          |                                         |
| 6                       | A4CS48      | Fundamentals of Artificial Intelligence    |                                         |
| 7                       | A4EC62      | Fundamentals of Integrated Circuits        | Electronics & Communication Engineering |
| 8                       | A4EC63      | Signal Transmission through linear systems |                                         |
| 9                       | A4EC64      | Fundamentals of VLSI Design                |                                         |
| 10                      | A4EE55      | Power Plant Engineering                    | Electrical & Electronics Engineering    |
| 11                      | A4EE56      | Analysis of Linear Systems                 |                                         |
| 12                      | A4EE57      | Neural Networks and Fuzzy Logic            |                                         |
| 13                      | A4IT14      | Basics of Python Programming               | Information Technology                  |
| 14                      | A4IT15      | Human Computer Interaction                 |                                         |
| 15                      | A4IT16      | Software Testing Fundamentals              |                                         |
| 16                      | A4ME63      | Fundamentals of Mechatronics               | Mechanical Engineering                  |
| 17                      | A4ME64      | Basics Of Thermodynamics                   |                                         |
| 18                      | A4ME65      | Fundamentals of Smart Materials            |                                         |
| 19                      | A4HS13      | Advanced Entrepreneurship                  | HS                                      |

| OPEN ELECTIVE COURSE-III |             |                                                      |                                         |
|--------------------------|-------------|------------------------------------------------------|-----------------------------------------|
| S. No.                   | Course Code | Course Name                                          | Offering Department                     |
| 1                        | A4AE40      | Guidance and control of aerospace vehicles           | Aeronautical Engineering                |
| 2                        | A4AE41      | Wind tunnel Techniques                               |                                         |
| 3                        | A4AE42      | Introduction to Aerospace Technology                 |                                         |
| 4                        | A4CS49      | Soft Computing                                       | Computer Science and Engineering        |
| 5                        | A4CS50      | Introduction to Cloud Computing                      |                                         |
| 6                        | A4CS51      | Web Programming                                      |                                         |
| 7                        | A4EC65      | Introduction of Microprocessors and Microcontrollers | Electronics & Communication Engineering |
| 8                        | A4EC66      | Fundamentals of Image processing                     |                                         |
| 9                        | A4EC67      | TV Engineering                                       |                                         |
| 10                       | A4EE58      | Electrical Engineering Materials                     | Electrical & Electronics Engineering    |
| 11                       | A4EE59      | Non-Conventional Power Generation                    |                                         |
| 12                       | A4EE60      | Solar Energy and Applications                        |                                         |
| 13                       | A4IT17      | E- Commerce                                          | Information Technology                  |
| 14                       | A4IT18      | Cyber Forensics                                      |                                         |
| 15                       | A4IT19      | Discrete Structures                                  |                                         |
| 16                       | A4ME66      | Fundamentals of Operation Research                   | Mechanical Engineering                  |
| 17                       | A4ME67      | Fundamentals of Energy Sources and Applications      |                                         |
| 18                       | A4ME68      | Basics of Robotics                                   |                                         |
| 19                       | A4HS14      | Indian Ethos & Business Ethics                       | HS                                      |

| OPEN ELECTIVE-IV |             |                                                |                                         |
|------------------|-------------|------------------------------------------------|-----------------------------------------|
| S. No.           | Course Code | Course Name                                    | Offering Department                     |
| 1                | A4HS15      | Management Science                             | HS                                      |
| 2                | A4HS16      | Intellectual Property Rights                   |                                         |
| 3                | A4HS17      | Number Theory                                  |                                         |
| 4                | A4HS18      | Physics and Technology of Thin films           |                                         |
| 5                | A4HS19      | Polymer chemistry                              |                                         |
| 7                | A4CS52      | Introduction to Network Security               | Computer Science and Engineering        |
| 8                | A4CS53      | Data Analytics                                 |                                         |
| 9                | A4CS54      | .NET & C# Programming                          |                                         |
| 10               | A4EC68      | Introduction to signal Processing              | Electronics & Communication Engineering |
| 11               | A4EC69      | Introduction to mobile communication           |                                         |
| 12               | A4EC70      | Basic Embedded systems Design                  |                                         |
| 13               | A4EE61      | Instrumentation and Control                    | Electrical & Electronics Engineering    |
| 14               | A4EE62      | Energy Audit and Management Systems            |                                         |
| 15               | A4EE63      | Energy Storage Systems                         |                                         |
| 16               | A4IT20      | Introduction to Mobile Application Development | Information Technology                  |
| 17               | A4IT21      | Distributed Databases                          |                                         |
| 18               | A4IT22      | Big Data                                       |                                         |
| 19               | A4ME69      | Introduction to Material Handling              | Mechanical Engineering                  |
| 20               | A4ME70      | Non Conventional Energy Sources                |                                         |
| 21               | A4ME71      | Aspects of Heat Transfer                       |                                         |

# **I B.TECH I SEMESTER SYLLABUS**

## LINEAR ALGEBRA AND CALCULUS

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS02      | BSC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Usage of the appropriate test to find the convergence and divergence of the given series.
2. The concept of differential equations and solve them using appropriate methods.
3. Concept of Rank of a matrix, Consistency and solving system of linear equations.
4. The Rank and Nullity of vectors.
5. Concept of eigen values, eigen vectors and diagonalization of the matrix.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply the appropriate test to find the convergence and divergence of the given series.
2. Identify the different types of differential equations and solve them using appropriate methods.
3. Solve the system of linear equations using rank of a matrix.
4. Find Rank and Nullity of given vectors.
5. Diagonalize the matrix using eigen values and eigen vectors.

### SEQUENCES AND SERIES

Classes: 10

#### UNIT-I

Basic definitions of Sequences and series – Convergence and divergence –Comparison Test- Ratio Test –Raabe's Test - Cauchy's  $n^{\text{th}}$  root Test –Integral Test – Absolute and Conditional convergence – Power Series.

### ORDINARY DIFFERENTIAL EQUATIONS

Classes: 12

#### UNIT-II

Introduction- Exact and reducible to Exact differential equations -Equations not of first degree: Equations solvable for p-equations, solvable for y- equations and solvable for x and Clairaut's type. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type  $Q(x) = e^{ax}$ ,  $\sin ax$ ,  $\cos ax$ ,  $e^{ax}v(x)$ ,  $x^n v(x)$  - Method of variation of parameters.

### THEORY OF MATRICES

Classes: 15

#### UNIT-III

Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew –Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix- Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination.

### VECTOR SPACES

Classes: 12

#### UNIT-IV

The n-dimensional Vectors –Vector space – linear dependence of vectors –Basis and dimensions –linear transformations-range and kernel of a linear map – rank and nullity - rank and nullity theorem – inverse of a linear transformation-composition of linear map- Matrix associated with a linear map.

**UNIT-V****EIGEN VALUES, EIGEN VECTORS AND INNER  
PRODUCT SPACES****Classes: 8**

Eigen values and Eigen vectors of a matrix- Eigenbases - Diagonalization- Inner product space – Norm of a vector – Schwarz's Inequality – Normed vector space – Orthogonal and orthonormal sets – Gram Schmidt orthogonalization process.

**Text Books:**

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9<sup>th</sup> Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
3. V. Krishnamurthy, V.P. Mainra and J.L. Arora, An introduction to Linear Algebra, Affiliated East – West press, Reprint 2005.

**Reference Books:**

1. G.B.Thomas, calculus and analytical geometry,9<sup>th</sup> Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.

**Web references:**

1. [https://www.efunda.com/math/math\\_home/math.cfm](https://www.efunda.com/math/math_home/math.cfm)
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

**E -Text Books:**

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

**MOOCS Course:**

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

## APPLIED PHYSICS

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS08      | BSC      | 3            | 1 | 0 | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Learn the behaviour of matter waves and applications of Schrodinger wave equations in periodic potential energy of electron.
2. Understand the formation of energy bands in solids.
3. Gain the knowledge of carrier concentration and recombination process of semiconductor materials.
4. Learn the basic principles of laser and optical fiber.
5. Understand the development of nano technology and synthesis of nano materials by using different techniques.

### COURSE OUTCOMES:

The student will able to:

1. Conclude the dual nature of material particles and able to explain how moving particles are associated with its energies
2. Analyze the energy bands in solids and accordingly classify the materials
3. Evaluate the mobility of charge carrier concentration of a given semiconductor material.
4. Justify how the graded index optical fiber is more efficient than step index optical fiber in fiber optic communication system.
5. Recommend appropriate synthesis method and explain the characterization techniques.

### UNIT-I

### QUANTUM MECHANICS

Classes: 12

**Introduction to quantum physics:** Black body radiation, Planck's law, photoelectric effect, Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.

### UNIT-II

### INTRODUCTION TO ELECTRONIC MATERIALS

Classes: 10

**Band theory** - Free electron theory, Origin of Energy Band formation in Solids, Estimation of Fermi energy level, Density of states and energy band diagrams, Kronig-Penny model (to introduce origin of band gap), Energy bands in solids, E-k diagram, Direct and indirect bandgaps.

**Classification of Materials:** Conductors, Semiconductors & Insulators, Effective mass of an Electron. Fermi-Dirac Statistics (Qualitative treatment).

### UNIT-III

### SEMICONDUCTORS AND OPTOELECTRONICS

Classes: 12

**Semiconductors:** Intrinsic and extrinsic semiconductors, Dependence of Fermi level on carrier-concentration and temperature (equilibrium carrier statistics). Carrier generation and recombination, Carrier transport: diffusion and drift, Hall effect and its applications.

Semiconductors design- PN junction-diode, Zener diode, fabrication and characterization techniques- Hetero junctions and associated band-diagrams.

**Optoelectronic devices:** properties of photo detectors, solar cells, Semiconductor laser, Four-point probe measurement for carrier density.



**UNIT-IV****LASER & OPTICAL FIBER****Classes: 12**

**Laser:** Characteristics of Laser beams, Energy levels in atoms, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: gas lasers -He-Ne, solid-state lasers -ruby, Nd-YAG, Applications of Lasers.

**Fiber Optic Communication:** Structure of Optical fibers, Basic principle of fibers, Acceptance angle and Numerical aperture, Types of Optical Fibers-Step Index and Graded Index fibers; Modes of fibers-SMSI, MMSI, MMGI., Optical fiber Communication System with block diagram. Applications of fibers, fiber optic sensors – Basic principle, Intrinsic, Extrinsic sensors. Working of Pressure and Temperature Sensors.

**UNIT-V****INTRODUCTION TO ENGINEERED MATERIALS****Classes: 12**

Fundamentals of nano particles, nano scale, properties, Techniques for synthesis of nano materials - Sol-gel, Chemical vapor deposition (CVD) methods.

**Characterization of nanomaterials:** Imaging methods- SEM, TEM, Scanning Probe Microscopy: STM. Fabrication method- quantum wire, Applications of Nano materials in engineering and Biomedical fields.

**Text Books:**

1. Engineering Physics, B.K. Pandey, S. Chaturvedi – Cengage Learning
2. Haliday and Resnick, Physics – wiley
3. R. Robinett, "Quantum Mechanics", OUP Oxford, 2006. II<sup>nd</sup> Edn.
4. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IV<sup>th</sup> Edn.
5. Essentials of Nano Tecnology by Jeremy Ramsden.

**Reference Books:**

1. E. Hecht, "Optics", Pearson Education, 2008.
2. J. Singh, Semiconductor Optoelectronics: Physics and Technology, Mc Graw-Hill inc. (1995)
3. R. Robinett, "Quantum Mechanics", OUP Oxford, 2006.
4. O. 1Svelto, "Principles of Lasers", Springer Science & Business Media, 2010.
5. D. A. Neamen, "Semiconductor Physics and Devices", Times Mirror High Education Group, Chicago, 1997.

**Web References:**

1. [https://www.edx.org/course?search\\_query=semiconductor+physics](https://www.edx.org/course?search_query=semiconductor+physics)
2. <https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x>
3. <https://www.edx.org/course/physics-electronic-polymers-pep-purdue-nano600>

**E-Text Books:**

1. [http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010\\_Fall/classnotes/NanoB\\_week14.pdf](http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010_Fall/classnotes/NanoB_week14.pdf)
2. <https://www.scribd.com/document/70908178/Semiconductor-Devices-Basic-Principles-Jasprit-Singh>
3. <https://www.scribd.com/doc/105174065/Fundamentals-of-Photonics>
4. [ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20\(2ed.,%20GTP,%20Springer,%202010\)\(ISBN%20144196441X\)\(O\)\(674s\)\\_PEo\\_.pdf](ftp://nozdr.ru/biblio/kolxo3/P/PE/PEo/Thyagarajan%20K.,%20Ghatak%20A.%20Lasers..%20Fundamentals%20and%20Applications%20(2ed.,%20GTP,%20Springer,%202010)(ISBN%20144196441X)(O)(674s)_PEo_.pdf)
5. [https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser\\_2.pdf](https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf)
6. <http://www.hailienene.com/resources/nano-technology.pdf>

**MOOC Course**

1. <http://nptel.ac.in/courses/115103030/>(Four-point probe measurement for carrier density)
2. <http://nptel.ac.in/courses/115102025/> (Fundamental concepts of semiconductors)
3. <http://nptel.ac.in/courses/118104008/1> ([Fundamentals of Nano](#) technology)
4. <http://nptel.ac.in/courses/118104008/13> (Nano structures, synthesis and characterization)
5. <http://nptel.ac.in/courses/104104085/2> (Lasers and its applications )

## BASIC ELECTRICAL ENGINEERING

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4EE01      | ESC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To understand and analyze basic electric and magnetic circuits.
2. To study the working principles of electrical machines.
3. To introduce the components of low voltage electrical installations.

### COURSE OUTCOMES:

The student will able to:

1. To analyze and solve electrical circuits using network laws and theorems.
2. To analyze basic Electric and Magnetic circuits.
3. To study the working principles of Electrical Machines.
4. To introduce components of Low Voltage Electrical Installations

### UNIT-I

#### DC CIRCUITS

Classes: 10

Electrical circuit elements (R, L and C), voltage and current sources, Kirchoff current and voltage laws, analysis of simple circuits with dc excitation. Superposition, Thevenin and Norton Theorems. Time-domain analysis of first-order RL and RC circuits.

### UNIT-II

#### AC CIRCUITS

Classes: 12

Representation of sinusoidal waveforms, peak and rms values, phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three-phase balanced circuits, voltage and current relations in star and delta connections.

### UNIT-III

#### TRANSFORMERS & ELECTRICAL INSTALLATIONS

Classes: 13

Magnetic materials, BH characteristics, ideal and practical transformer, equivalent circuit, losses in transformers, regulation and efficiency. Auto-transformer and three-phase transformer connections.

Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Types of Wires and Cables, Earthing. Types of Batteries, Important Characteristics for Batteries

### UNIT-IV

#### DC MACHINES

Classes: 12

Principle and operation of DC Motor, Construction of DC machine. Types of DC motor, losses and Torque equation. DC generator construction, working principle and its EMF equation. Types of dc generators, efficiency of dc generator, problems on Emf equation. Applications.

### UNIT-V

#### AC MACHINES

Classes: 10

Generation of rotating magnetic fields, Construction and working of a three-phase induction motor, Significance of torque-slip characteristic. Loss components and efficiency, starting and speed control of induction motor. Construction, working of Single-phase induction motor. Construction and working of synchronous generators, Emf equation and problems.

### Text Books:

1. D. P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 2010.
2. D. C. Kulshreshtha, "Basic Electrical Engineering", McGraw Hill, 2009.
3. Basic Electrical Engineering by M.S. Naidu and S. Kamakshaiah TMH
4. Mehta V K, —Principles of Electrical Engineering, S. Chand & Company

### Reference Books:

1. L. S. Bobrow, "Fundamentals of Electrical Engineering", Oxford University Press, 2011.
2. E. Hughes, "Electrical and Electronics Technology", Pearson, 2010.
3. V. D. Toro, "Electrical Engineering Fundamentals", Prentice Hall India, 1989.
4. Nagsarkar T K and Sukhija M S, —Basics of Electrical Engineering, Oxford press. Basic concepts of Electrical Engineering, P.S. Subramanyam, BS Publications.

## ENGINEERING GRAPHICS & DESIGN

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ES02      | ESC      | 1            | 0 | 4 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Create awareness and emphasize the need for Engineering Drawing in various branches of engineering.
2. Enable the student with various concepts of dimensioning, conventions and standards related to engineering drawings.
3. Follow the basic drawing standards and conventions.
4. Develop skills in three-dimensional visualization of engineering component.

### COURSE OUTCOMES:

The student will able to:

- 1.

#### UNIT-I

#### INTRODUCTION

Classes: 7

Introduction to Engineering Drawing covering: Principles of Engineering Graphics and their significance, usage of drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute.

#### UNIT-II

#### DRAWING OF PROJECTIONS OR VIEWS: ORTHOGRAPHIC PROJECTION IN FIRST ANGLE PROJECTION ONLY

Classes: 10

Principles of orthographic projections – conventions – first and third angle projections. Projections of points-Projection of lines inclined to both the planes.

PROJECTIONS OF PLANES: Projections of regular planes, inclined to both planes.

#### UNIT-III

#### INTRODUCTION TO COMPUTER AIDED DRAFTING

Classes: 8

INTRODUCTION TO COMPUTER AIDED DRAFTING: Generation of points, lines, curves, polygons, simple solids, dimensioning. LAYERS: Concept of layers, working with layers, creating, display, locking, unlocking, and delete commands, Coordinate system in AutoCAD, UCS, WCS, MCS

PROJECTION OF SOLIDS-Solids inclined to both planes(Auxiliary plane method )

DEVELOPMENT OF SURFACES OF SOLIDS:Theory of development,development of lateral surface along with base.

#### UNIT-IV

#### ISOMETRIC DRAWINGS

Classes: 5

Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections; Drawing Isometric circles, Dimensioning Isometric Objects; Conversion of Isometric view to Orthographic views and Orthographic to isometric views.

**UNIT-V****3D MODELING****Classes: 4**

Types of 3D models, 3D Coordinate Systems, basic commands in 3D, PEDIT command. CREATING SOLID MODELS: creating pre-defined Solid Primitives, Dynamic UCS, methods of creating solids by - Extrude, Revolve, Swept, Loft, & Presspull, in 3Dcreating solid models, Dynamic UCS. MODIFYING 3D OBJECTS: Fillet, Chamfer, Rotate, Mirror, Array, Slicing solid Models. EDITING 3D OBJECTS: SOLVIEW, SOLDRAW, SOLPROF,

**Text Books:**

1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House
2. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
3. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication
4. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers.
5. D.M. Kulkarni,A.P.Rastogi,A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications, 2013

**Reference Books:**

1. Johle (2009), Engineering Drawing, Tata Mc Graw Hill, New Delhi, India.
2. Trymbaka Murthy (2007), Computer Aided Engineering Drawing, I.K. International Publishers, New Delhi.
3. Sham Tickoo, D. saravanan, "AutoCAD 2010 for engineers and designers" Dreamtech Press, 2010.

**Web References:**

1. [nptel.ac.in/courses/112103019/](https://nptel.ac.in/courses/112103019/)
2. [web.iitd.ac.in/~achawla/public\\_html/201/lectures/sp46.pdf](http://web.iitd.ac.in/~achawla/public_html/201/lectures/sp46.pdf)

**E-Text Books:**

1. [https://www.researchgate.net/publication/305754529\\_A\\_Textbook\\_of\\_Engineering\\_Drawing\\_A\\_Textbook\\_of](https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing_A_Textbook_of)
2. [https://www.researchgate.net/publication/305754529\\_A\\_Textbook\\_of\\_Engineering\\_Drawing\\_A\\_Textbook\\_of](https://www.researchgate.net/publication/305754529_A_Textbook_of_Engineering_Drawing_A_Textbook_of)

## APPLIED PHYSICS LAB

**I Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS10      | BSC      | -            | - | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To provide an experimental foundation for the theoretical concepts introduced in the lectures.
2. To teach how to make careful experimental observations and how to think about and draw conclusions from such data.
3. To help students understand the role of direct observation in physics and to distinguish between inferences based on theory and the outcomes of experiments.
4. To introduce the concepts and techniques which have a wide application in experimental science but have not been introduced in the standard courses.
5. To teach how to write a technical report which communicates scientific information in a clear and concise manner.

### COURSE OUTCOMES:

The student will be able to:

1. To make careful experimental observations and draw conclusions from such data.
2. To distinguish between inferences based on theory and the outcomes of experiments.
3. To write a technical report which communicates scientific information in a clear and concise manner.

## LIST OF EXPERIMENTS

### Week-1 LIGHT EMITTING DIODE (LED)

- a. Analyze the V-I characteristics of GREEN LED Source by varying input voltage from zero to two volts. Analyze the graph obtained.
- b. Analyze the results obtained for V-I characteristics of RED LED Source by varying input voltage with interval 0.2 volts.
- c. Analyze the V-I characteristics of YELLOW LED Source for the resistance of 100Ω. What is the effect of doubling the resistance on the V-I characteristics.

### Week-2 SOLAR CELL

- a. Study the V-I characteristics of a solar cell for the voltage interval of 0.4 V when the given electric bulb is at a distance of 10cm. Will there be any change if the distance is doubled?
- b. Study the P-V characteristics of a solar cell for the voltage interval of 0.3 V for four different distances of the given electric bulb. Discuss your observations.
- c. Study the P-I characteristics of a solar cell for the given electric bulb for the distances 30 cm. What is expected if the distance is doubled?

### Week-3 OPTICAL FIBER

To determine the numerical aperture and acceptance angle of an optical fiber.

### Week-4 HALL EFFECT

- a. Investigate the deflection of the carriers in the conductor under the function of the magnetic field.
- b. Based on this measurement, calculate the density of the carriers and the sign of the charges in the conductor.
- c. Verify the Hall effect in extrinsic semiconducting samples and determine the type of given semiconductor and density of majority charge carriers

#### **Week-5 ENERGY GAP OF PN JUNCTION DIODE**

- a. Using PN junction diode, determine the energy gap of a semiconductor by applying 2V of reverse bias by increasing the temperature from 30-70 degree centigrade. Analyze the obtained results.
- b. Evaluate the energy gap of Si PN junction diode by applying 1.5V of reverse bias for values of temperature 70,60,50,40,30 centigrade.
- c. Evaluate the energy gap of a semiconductor by applying 0.5V of reverse bias for temperatures at 75, 65, 55,45,35 centigrade.

#### **Week-6 THERMISTER**

- a. Study the variation of Resistance versus Temperature using Thermister. Plot graph between Resistance vs Temperature and analyze the results.
- b. Evaluate the Temperature dependent resistance of a given material by using Thermister.

#### **Week-7 LCR CIRCUIT**

- a. To determine the Resonance frequency and Quality factor of a LCR Circuit
- b. Study the Resonance frequency of an electrical circuit by varying values of inductance, Capacitance and resistors and analyze the results.

#### **Week-8 PIN PHOTO DIODE**

To determine the V-I characteristics of PIN photo diode.

#### **Week-9 RC CIRCUIT**

- a. Analyze the time constant of R-C circuit by varying Resistance and Capacitance values in a electrical circuit.
- b. Determine the time constant of a given RC circuit by plotting a graph between Charging current versus time.

#### **Week-10 TORSIONAL OSCILLATOR**

- a. Making use of a torsional oscillator of 300gms circular disc, determine the rigidity modulus of given steel wire for 65 cm & 55 cm lengths. Analyze the results obtained.
- b. Determine the rigidity modulus of the given copper wire for 50,40,30 lengths by using torsional oscillator of 400gms circular disc. Analyze the results obtained.
- c. Give your analysis of  $L \& T^2$  behavior of a torsional oscillator. You may pick your own values for the analysis.

#### **Week-11 LASER - DIFFRACTION GRATING**



- a. Using a diffraction grating element of 2500 LPI determine the wavelength of LASER source for first and second order diffraction when the distance between the screen and grating is 50cm. What is your analysis?
- b. Determine the wavelength of a LASER source for first three orders of diffraction by maintaining a distance of 30 cms between grating material and the screen. Use diffraction grating element of 15000 LPI. What differences do you observe for the three orders.

### **Week-12 MELDE'S EXPERIMENT**

- a. Determine the longitudinal frequency of tuning fork by using 100cm length of the thread by varying masses of 5gms, 10gms. Discuss your findings.
- b. Findout the transverse frequency of tuning fork by using 80cm length of the thread and by varying masses of 10gms and 40gms. Analyze the results obtained.

### **Reference Books:**

1. "Semiconductor Physics and Devices: Basic Principles" by Donald A Neamen.
2. "Optics, Principles and Applications" by K K Sharma.
3. "Principles of Optics" by M Born and E Wolf.
4. "Oscillations and Waves" by Satya Prakash and Vinay Dua.

### **Web References:**

1. <http://www.arxiv.org/pdf/1510.00032>
2. <http://www.nptel.ac.in/courses/122103010/>
3. [http://www.researchgate.net/.../276417736\\_Video\\_Presentations\\_in\\_Engineering-Ph...](http://www.researchgate.net/.../276417736_Video_Presentations_in_Engineering-Ph...)
4. <http://www.wilevindia.com/engineering-physics-theory-and-practical.html>

**Note:** Students can perform any 8 experiments

## BASIC ELECTRICAL ENGINEERING LAB

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credit | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|--------|---------------|-----|-------|
|             |          | L            | T | P |        | CIE           | SEE | Total |
| A4EE02      | ESC      | -            | - | 2 | 1      | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Get an exposure to common electrical components and their ratings.
2. Make electrical connections by wires of appropriate ratings.
3. Understand the usage of common electrical measuring instruments.
4. Understand the basic characteristics of transformers and electrical machines.

### COURSE OUTCOMES:

1. Get an exposure to basic electrical laws.
2. Demonstrate different types of electrical circuits to different excitations.
3. Calculate the relation between the basic electrical parameters.
4. Analyze the basic characteristics of transformers and electrical machines.

## LIST OF EXPERIMENTS

### Week-1 INTRODUCTION AND USE OF MEASURING INSTRUMENTS & SAFETY PRECAUTIONS

To study the usage of electrical instruments and the required precautions to be taken.

### Week-2 KIRCHOFF'S LAWS( KVL & KCL)

To verify kvl and kcl

### Week-3 SUPERPOSITION THEOREM

To verify superposition theorem

### Week-4 THEVENIN'S AND NORTON'S THEOREM

To obtain equivalent circuit of a complex network

### Week-5 STEADY STATE AND TRANSIENT RESPONSE OF R-L, R-C & R-L-C CIRCUITS

To find the steady and transient response of r-l, r-c & r-l-c circuits for step input.

### Week-6 OPEN CIRCUIT, SHORT CIRCUIT & LOAD TEST ON SINGLE PHASE TRANSFORMER

To calculate the efficiency of single phase transformer.

### Week-7 CUT OUT VIEW OF DC MACHINE

Demonstration on constructional and cut out view of dc machine

**Week-8      CUT OUT VIEW OF INDUCTION MOTOR**

Demonstration on constructional and cut out view of single phase induction motor

**Week-9      MAGNETIZATION CHARACTERISTICS OF DC SHUNT GENERATOR**

To find the magnetization characteristics of dc shunt generator

**Week-10      BRAKE TEST ON DC SHUNT MOTOR**

To find the torque-speed characteristics of dc shunt motor.

**Week-11      THREE PHASE TRANSFORMER CONNECTIONS**

To calculate the relation between phase and line voltages

**Week-12      BRAKE TEST ON 3-PHASE INDUCTION MOTOR**

To find the torque-slip characteristics of induction motor

**Reference Books:**

1. Department Lab Manual
2. Chakrabarthi, "Circuit Theory", Dhanpat Rai Publications, 6<sup>th</sup> Edition, 2006
3. V K Mehta, Rohit Mehta, "Principles of Electrical Machines", S Chand Publications, 1<sup>st</sup> Edition, 2006
4. I Nagrath & DP Kothari, "Electrical Machines", McGraw Hill Education Publications, 4<sup>th</sup> Edition, 2010.

**Web References:**

1. <http://www.ee.iitkgp.ac.in>
2. <http://www.citchennai.edu.in>

## WORKSHOP PRACTICES

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4AE63      | ESC      | 1            | - | 2 | 2       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

Student will

1. Get the hands on experience on various trades.
2. Perform various machining operations.
3. Capable to make useful products using one or more operations.

### COURSE OUTCOMES:

Student will be able to:

1. Fabricate components with their own hands
2. Get practical knowledge of the dimensional accuracies and tolerances possible with different manufacturing processes
3. Assemble different components
4. Produce small devices of their interest

### LIST OF EXPERIMENTS

| WEEKS         | BASIC TRADES                                | BASIC MANUFACTURING                                 |
|---------------|---------------------------------------------|-----------------------------------------------------|
|               | <b>Fitting</b>                              | <b>Machine Shop</b>                                 |
| <b>Week 1</b> | Filing Four Sides of Work piece             | Facing & Step Turning on Lathe                      |
| <b>Week 2</b> | L Fit                                       | Milling and Drilling                                |
|               | <b>Carpentry</b>                            | <b>Black Smithy</b>                                 |
| <b>Week 3</b> | Half Lap Joint                              | Convert round rod to S-hook                         |
| <b>Week 4</b> | Dove Tail Joint                             | Convert round rod to Chisel                         |
|               | <b>Tin Smithy</b>                           | <b>Casting</b>                                      |
| <b>Week 5</b> | Tin Smithy- Prepare a Rectangular Tray      | Preparation of Mould Cavity for Multi Piece Pattern |
| <b>Week 6</b> | Prepare A Square Tin                        | Casting of Simple pattern                           |
|               | <b>Electrical</b>                           | <b>Welding Shop</b>                                 |
| <b>Week 7</b> | House Wiring Parallel and Series Connection | Lap/Butt joint Using Arc Welding                    |
| <b>Week 8</b> | House Wiring Two Way Switch                 | Lap/Butt joint Using Gas Welding                    |
|               | <b>Electronics</b>                          | <b>Plastic Moulding &amp; Glass Cutting</b>         |
| <b>Week 9</b> | Soldering Parallel Connection               | Injection moulding of Simple Components             |

|                |                             |                   |
|----------------|-----------------------------|-------------------|
| <b>Week 10</b> | Soldering Series Connection | Glass Cutting     |
| <b>Week 11</b> | Revision/Practice           | Revision/Practice |

**Text Books:**

1. Workshop Manual by P. Kannaiah and K. L. Narayana.
2. Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II.

**Reference Books:**

1. HajraChoudhury S.K., HajraChoudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. H. S. Bawa, "Workshop Practice", Tata McGraw-Hill Publishing Company Limited, New Delhi, (2007).
3. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4<sup>th</sup>edition,2002.

**E-Text Books:**

1. <https://blogpuneet.files.wordpress.com/2013/07/introduction-to-basic-manufacturing-processes-and-workshop-technology.pdf>
2. <https://soaneemrana.org/onewebmedia/Manufacturing%20Processes%20By%20H.N.%20Gupta.pdf>

**MOOC Course:**

1. <https://www.class-central.com/course/edx-fundamentals-of-manufacturing-processes-7224>

## SOCIAL INNOVATION

### I Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS03      | HSMC     | -            | - | 2 | 1       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand social innovation concepts and approaches.
2. Understand the community problems, social and economical change.
3. Identify new and unaddressed social needs.
4. Analysis of social innovation disclosures in different sectors.
5. Design innovative solutions with Social impact through application of new models of leadership, collective intelligence and creativity techniques.

### UNIT-I INTRODUCTION TO SOCIAL INNOVATION

**Classes: 8**

Core definitions, core elements and common features of social innovation, a topology of social innovations, history of social innovation, social and economic change, Swachh Bharat, Unnat Bharat Abhiyan, National Service Scheme (NSS).

### UNIT-II INTERACTION AND ENGAGEMENT WITH SOCIETY

**Classes: 8**

Engage with community, interact with them to understand the community problems, Understanding social and economical change – individuals, organizations and movements.

### UNIT-III PROCESS OF SOCIAL INNOVATION

**Classes: 8**

Understanding the pain/need, description and problem definition, social and economic constraints for affordable and appropriate technology.

### UNIT-IV SOCIAL INNOVATION ACROSS FOUR SECTORS IN INDIA AND GLOBAL SCENARIO

**Classes: 8**

The four sectors – the non-profit sector, public sector, the private sector, the informal sector, links between and cross sectors.

### UNIT-V SOCIAL INNOVATION – CASE STUDIES

**Classes: 8**

Designing and implementing social innovations, report writing and documentation, presentation of the case studies with a focus on impact and vision on society.

### Text Books:

1. The Power of Social Innovation: How Civic Entrepreneurs Ignite Community Networks for Good 1st Edition by Stephen Goldsmith, Michael R. Bloomberg, Gigi Georges, Tim Glynn Burke.
2. The Open Book of Social Innovation: Ways to Design, Develop and Grow Social Innovation Paperback – March, 2010 by Robin Murray, Julia Caulier-Grice, Geoff Mulgan.

**Reference Books:**

1. Social innovator series: ways to design, develop and grow social innovation, the open book of social innovation by robin murray julie caulier-grice geoff mulgan.
2. The International Handbook on Social Innovation: Collective Action, Social Learning and Transdisciplinary Research Paperback by Frank Moulaert , Diana MacCallum.
3. Guide to Social Innovation by Johannes HAHN and Laszlo ANDOR

**Web References:**

1. [http://s3platform.jrc.ec.europa.eu/documents/20182/84453/Guide\\_to\\_Social\\_Innovation.pdf](http://s3platform.jrc.ec.europa.eu/documents/20182/84453/Guide_to_Social_Innovation.pdf)
2. <https://www.si-drive.eu/wp-content/uploads/2016/12/SI-DRIVE-CA-short-2016-11-30-Druckversion.pdf>

**E-Text Books:**

1. <https://epdf.tips/the-power-of-social-innovation-how-civic-entrepreneurs-ignite-community-networks.html>
2. <https://youngfoundation.org/wp-content/uploads/2012/10/The-Open-Book-of-Social-Innovationg.pdf>
3. <http://www.idmais.org/desislab/wp-content/media/social.pdf>

**MOOC Course**

1. <https://iversity.org/en/courses/social-innovation-101-en>
2. <https://www.class-central.com/tag/social%20innovation>
3. <https://www.mooc-list.com/tags/social-innovation>

# **I B.TECH II SEMESTER SYLLABUS**



## ADVANCED CALCULUS

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS04      | BSC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Evaluation of length, areas & volumes of different curves of revolution.
2. The partial derivatives of several variable functions.
3. Concept and application of Laplace transforms.
4. Fourier series for periodic functions.
5. Numerical techniques.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate the improper integrals using beta and gamma functions.
2. Find the Maxima and Minima of several variable functions.
3. Solve the differential equations using Laplace transform techniques.
4. Find the Fourier series of the periodic functions.
5. Apply various numerical techniques to solve differential equations.

### UNIT-I Classes: 11 BETA GAMMA FUNCTIONS AND APPLICATIONS OF DEFINITE INTEGRALS

Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.

Application of definite integrals: Lengths, evaluate surface areas and volumes of revolution of curves (only in Cartesian co-ordinates).

### UNIT-II Classes: 11 CALCULUS OF SEVERAL VARIABLES

Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians -functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints)

### UNIT-III Classes: 12 LAPLACE TRANSFORMS

Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by  $t^n$ - Division by  $t$  – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.

### UNIT-IV Classes: 10 FOURIER SERIES

Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Even Odd periodic continuation-Half range Fourier sine and cosine expansions.

**UNIT-V****NUMERICAL TECHNIQUES****Classes: 08****ROOT FINDING TECHNIQUES :**

Bisection method-Regula falsi method, Iteration method and Newton Raphson method.

**NUMERICAL INTEGRATION :**

Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule.

**NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS:** Taylor's series method – Euler's - modified Euler's Method – Runge-Kutta method.

**Text Books:**

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9<sup>th</sup> Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36<sup>th</sup> Edition, 2010.

**Reference Books:**

1. G.B.Thomas, calculus and analytical geometry, 9<sup>th</sup> Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics, Laxmi publications, 2008.
3. E.L.Ince, Ordinary differential Equations, Dover publications, 1958.

**Web references:**

1. [https://www.efunda.com/math/math\\_home/math.cfm](https://www.efunda.com/math/math_home/math.cfm)
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

**E -Text Books:**

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

**MOOCS Course:**

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

## CHEMISTRY

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS11      | BSC      | 4            | - | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

#### Student will be able to:

1. Bring adaptability to the concepts of chemistry and to acquire the required skills to become a perfect engineer.
2. To impart the basic knowledge of atomic, molecular and electronic modifications which makes the student to understand the technology based on them.
3. To acquire the knowledge of electrochemistry, corrosion and water treatment which are essential for the Engineers and in industry.
4. To acquire the skills pertaining to spectroscopy and to apply them for medical and other fields.
5. To impart the knowledge of stereochemistry and synthetic aspects useful for understanding reaction pathways.

### COURSE OUTCOMES:

The basic concepts included in this course will help the student to gain:

1. The knowledge of atomic, molecular and electronic changes, band theory related to conductivity.
2. The required principles and concepts of electrochemistry, corrosion and in understanding the problem of water and its treatments.
3. The required skills to get clear concepts on basic spectroscopy and application to medical and other fields.
4. The knowledge of configurational and conformational analysis of molecules and reaction mechanisms.

### UNIT-I MOLECULAR STRUCTURE AND THEORIES OF BONDING Classes: 12

**Atomic and Molecular orbitals:** Linear Combination of Atomic Orbitals (LCAO), molecular orbitals of diatomic molecules, molecular orbital energy level diagrams of  $N_2$ ,  $O_2$  and  $F_2$  molecules.  $\pi$  molecular orbitals of butadiene and benzene.

**Crystal Field Theory (CFT):** Salient Features of CFT – Crystal Field Splitting of transition metal ion d- orbitals in Tetrahedral, Octahedral and square planar geometries. Band structure of solids and effect of doping on conductance.

### UNIT-II WATER AND ITS TREATMENT Classes: 14

Introduction – hardness of water – Causes of hardness - Types of hardness: temporary and permanent – expression and units of hardness – Estimation of hardness of water by complexometric method. Potable water and its specifications. Steps involved in treatment of water – Disinfection of water by chlorination and ozonization. Boiler feed water and its treatment – Calgon conditioning, Phosphate conditioning and Colloidal conditioning. External treatment of water – Ion exchange process. Desalination of water – Reverse osmosis. Numerical problems.

### UNIT-III ELECTROCHEMISTRY AND CORROSION Classes: 14

Electro chemical cells – electrode potential, standard electrode potential, types of electrodes – calomel, Quinhydrone and glass electrode. Nernst equation Determination of pH of a solution by using quinhydrone and glass electrode. Electrochemical series and its applications. Numerical problems. Potentiometric titrations. Batteries – Primary (Lithium cell) and secondary batteries (Lead –

acid storage battery).

**Corrosion:** Causes and effects of corrosion – theories of chemical and electrochemical corrosion – mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection– Sacrificial anode and impressed current cathodic methods. Surface coatings – metallic coatings – Hot dipping, metal cladding and electro plating (copper plating).

#### UNIT-IV **STEREOCHEMISTRY, REACTION MECHANISM AND SYNTHESIS OF DRUG MOLECULES** **Classes: 10**

Introduction to representation of 3-dimensional structures, Structural and stereoisomers, configurations, symmetry and chirality. Enantiomers, diastereomers, optical activity and Absolute configuration. Conformational analysis of n-butane. Substitution reactions: Nucleophilic substitution reactions: Mechanism of SN<sub>1</sub>, SN<sub>2</sub> reactions. Electrophilic and nucleophilic addition reactions: Addition of HBr to propene. Markownikoff and anti Markownikoff's additions. Grignard additions on carbonyl compounds. Elimination reactions: Dehydro halogenation of alkylhalides. Saytzeff rule. Oxidation reactions: Oxidation of alcohols using KMnO<sub>4</sub> and chromic acid. Reduction reactions: reduction of carbonyl compounds using LiAlH<sub>4</sub> & NaBH<sub>4</sub>. Hydroboration of olefins. Structure, synthesis and pharmaceutical applications of Aspirin.

#### UNIT-V **SPECTROSCOPIC TECHNIQUES AND APPLICATIONS** **Classes: 10**

Principles of spectroscopy, selection rules and applications of electronic spectroscopy, vibrational and rotational spectroscopy. Basic concepts of Nuclear magnetic resonance Spectroscopy, chemical shift. Introduction to Magnetic resonance imaging. Fluorescence and its applications in medicine.

#### Text Books:

1. Engineering Chemistry 14<sup>th</sup> edition by P.C.Jain & M.Jain; Dhanpat Rai Publishing Company (P) Ltd., New Delhi.
2. Fundamentals of Molecular Spectroscopy 5<sup>th</sup> edition by C.N. Banwell Mc.Graw-Hills book company.

#### Reference Books:

1. Organic Chemistry 7<sup>th</sup> Edition, Robert Thornton Morrison, Robert Neilson Boyd, Saibal Kanti Bhattacharjee Pearson Education Singapore Pvt.Ltd.
2. Engineering Chemistry (NPTEL Web-book), 11<sup>th</sup> edition by B.L. Tembe, Kamaluddin and M.S. Krishnan.
3. Physical Chemistry 11<sup>th</sup> edition by P.W. Atkins OUP Oxford.

#### Web References:

1. <https://www.scribd.com/document/23180395/Engineering-Chemistry-Unit-I-Water-Treatment>
2. [https://chem.libretexts.org/Core/Inorganic\\_Chemistry/Descriptive\\_Chemistry/Periodic\\_Trends\\_of\\_Elemental\\_Properties/Periodic\\_Properties\\_of\\_the\\_Elements](https://chem.libretexts.org/Core/Inorganic_Chemistry/Descriptive_Chemistry/Periodic_Trends_of_Elemental_Properties/Periodic_Properties_of_the_Elements)
3. <https://www.khanacademy.org/science/biology/chemistry--of-life/chemical-bonds-and-reactions/v/intermolecular-forces-and-molecular-bonds>
4. <https://study.com/academy/lesson/the-relationship-between-free-energy-and-the-equilibrium-constant.html>

#### E-Text Books:

1. <http://www.freebookcentre.net/Chemistry/Chemistry-Books-Online.html>
2. [https://www.sdu.dk/en/om\\_sdu/institutter\\_centre/fysik\\_kemi\\_og\\_farmaci](https://www.sdu.dk/en/om_sdu/institutter_centre/fysik_kemi_og_farmaci)

#### MOOC Course:

1. <http://nptel.ac.in/courses/122101001/34>
2. <https://ocw.mit.edu/courses/chemistry/>

## PROGRAMMING FOR PROBLEM SOLVING

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS01      | ESC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES

1. To impart basic knowledge about simple algorithms for arithmetic and logical problems.
2. To understand how to write a program, syntax and logical errors.
3. To enable them how to implement conditional branching, iteration and recursion.
4. To understand how to decompose a problem into functions and synthesize a complete program.
5. To enable them to use arrays, pointers, strings and structures in solving problems.
6. To understand how to solve problems related to matrices, Searching and sorting.
7. To make them to understand the use files to perform read and write operations.

### COURSE OUTCOMES

At the end of the course, student will be able to:

1. Formulate simple algorithms for arithmetic and logical problems.
2. Test and execute the programs and correct syntax and logical errors.
3. Implement conditional branching, iteration and recursion.
4. Decompose a problem into functions and synthesize a complete program.
5. Use arrays, pointers, strings and structures to formulate algorithms and programs.
6. Apply programming to solve problems related to matrices, Searching and sorting.
7. Use files to perform read and write operations.

### UNIT-I

#### INTRODUCTION

**Classes: 12**

**Introduction to Programming:** Computer system, components of a computer system, computing environments, computer languages, creating and running programs, Algorithms, flowcharts.

**Introduction to C language:** History of C, basic structure of C programs, process of compiling and running a C program, C tokens, keywords, identifiers, constants, strings, special symbols, variables, data types, I/O statements.

### UNIT-II

#### OPERATORS, EXPRESSIONS AND CONTROL STRUCTURES

**Classes: 15**

**Operators and expressions:** Operators, arithmetic, relational and logical, assignment operators, increment and decrement operators, bitwise and conditional operators, special operators, operator precedence and associativity, evaluation of expressions, type conversions in expressions.

**Control structures:** Decision statements; if and switch statement; Loop control statements: while, for and do while loops, jump statements, break, continue, goto statements.

### UNIT-III

#### ARRAYS AND FUNCTIONS

**Classes: 17**

**Arrays:** Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays, Basic Algorithms: Searching, Basic Sorting Algorithms- Bubble sort, Insertion sort and Selection sort.

**Functions:** User defined and built-in Functions, storage classes, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, Quick sort or Merge sort.

### UNIT-IV

#### STRINGS AND POINTERS

**Classes: 10**

**Strings:** Arrays of characters, variable length character strings, inputting character strings, character library functions, string handling functions.

**Pointers:** Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, functions returning pointers, Dynamic memory allocation.

### UNIT-V

#### STRUCTURES AND FILE HANDLING

**Classes: 10**

**Structures and unions:** Structure definition, initialization, accessing structures, nested structures, arrays of structures, structures and functions, self referential structures, unions, typedef, enumerations.

**File handling:** command line arguments, File modes, basic file operations read, write and append, example programs

### Text Books:

1. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3<sup>rd</sup> edition, 2017.
2. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 6<sup>th</sup> Edition, 2012.

### Reference Books:

1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2<sup>nd</sup> Edition, 1988.
2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2<sup>nd</sup> Edition, 2003.
3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4<sup>th</sup> Edition, 2014.
4. R. S. Bichkar, "Programming with C", Universities Press, 2<sup>nd</sup> Edition, 2012.
5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2<sup>nd</sup> Edition, 2006.
6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4<sup>th</sup> Edition, 2014.
7. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3<sup>rd</sup> Edition, 2014.

### Web References:

1. <https://www.bfoit.org/itp/Programming.html>
2. <https://www.khanacademy.org/computing/computer-programming>
3. <https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0>
4. <https://www.edx.org/course/introduction-computer-science-harvardx-cs50x>

### E-Text Books:

1. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>
2. <http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/>
3. <http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

### MOOC Course:

1. [https://onlinecourses.nptel.ac.in/noc18\\_cs33/preview](https://onlinecourses.nptel.ac.in/noc18_cs33/preview)
2. <https://www.alison.com/courses/Introduction-to-Programming-in-c>
3. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm>

## ENGLISH

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS01      | HSMC     | 2            | - | - | 2       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Improve language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

### COURSE OUTCOMES:

Students should be able to:

1. Use English Language effectively in spoken and written forms.
2. Comprehend the given texts and respond appropriately.
3. Communicate confidently in various contexts and different cultures.
4. Acquire basic proficiency in English including reading and listening comprehension, writing and speaking skills.

### UNIT-I THE RAMAN EFFECT

**Classes: 10**

Chapter entitled 'The Raman Effect' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press.

V- The concept of Word Formation, Root Words and their use in English

G- Types of Sentences, Parts of Speech, Nouns, Pronouns, Adjectives

R- Reading and its importance

W- Sentence Structures, Punctuation, Writing Introductions and Conclusions

### UNIT-II LETTER WRITING

**Classes: 10**

Letter Writing.

V- Synonyms and Antonyms, Standard Abbreviations

G- Verbs, Modal Auxiliaries, Adverbs, Prepositions, Conjunctions

R- Improving Comprehension Skills, Reading and Comparing Two Articles

W- Writing Paragraphs, Letter Writing- Letters of Request, Apology and Complaint- Letter of Application with Resume.

### UNIT-III BLUE JEANS

**Classes: 10**

Chapter entitled 'Blue Jeans' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press.

V- Prefixes and Suffixes, Idioms and Phrasal verbs

G- Articles, Tenses and its forms

R- Sub skills of Reading- Skimming and Scanning

W- Essay writing and Describing Objects, Places and Events

#### **UNIT-IV WHAT SHOULD YOU BE EATING**

**Classes: 10**

Chapter entitled 'What Should You Be Eating' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press.

V- One word Substitutes, Words often confused

G- Active and passive Voice Subject Verb Agreement (Concord)

R- Reading Comprehension- Intensive and Extensive

W- Technical Report Writing, E-mail writing, Picture Essay

#### **UNIT-V HOW A CHINESE BILLIONAIRE BUILT HER FORTUNE Classes: 10**

Chapter entitled 'How a Chinese Billionaire Built Her Fortune' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press.

V- Technical Vocabulary and their Usage, Misplaced Modifiers, Redundancies

G- Direct and Indirect Speech, Degrees of Comparison, Common Errors in English

R- Effective Reading and Exercises for Practice

W- Memo, Précis and Resume Writing

#### **Text Books:**

1. Sudarshan, N. P. and Savitha, C. (2018). English for Engineers. Cambridge University Press.
2. Wren & Martin. (2017). High School English Grammar and Composition Book. S Chand Publishing.

#### **Reference Books:**

1. Murphy, R. (2015). Essential Grammar in Use. Cambridge University Press.
2. Wood, F.T. (2007). Remedial English Grammar. Macmillan.
3. Swan, M. (2016). Practical English Usage. Oxford University Press.
4. Kumar, S and Lata, P. (2018). Communication Skills. Oxford University Press.
5. Zinsser, William. (2001). On Writing Well. Harper Resource Book.
6. Hamp-Lyons, L. (2006). Study Writing. Cambridge University Press.
7. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.

#### **Web References:**

1. <https://www.oxfordlineenglish.com>
2. <https://www.bbclearningenglish.com>
3. <https://www.learnenglish.britishcouncil.org>
4. <https://www.fluentu.com/english>

#### **E-Text Books:**

1. [https://www.uop.edu/jo/download/research/members/oxford\\_guide\\_to\\_english\\_grammar.pdf](https://www.uop.edu/jo/download/research/members/oxford_guide_to_english_grammar.pdf)
2. <http://www.espressoenglish.net/wp-content/uploads/2012/07/Free-Grammar-Ebook-Level-2.pdf>
3. <https://update24hour.com/2018/03/wren-and-martin-english-grammar-pdf>

#### **MOOC Course**

1. <http://www.cambridgeenglish.org/learning-english/free-resources/mooc/>
2. <https://www.britishcouncil.org/tr/en/english/mooc>
3. <https://mooc.com>



## PROGRAMMING FOR PROBLEM SOLVING LAB

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS02      | ESC      | -            | - | 4 | 2       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To understand how to formulate the algorithms for simple problems
2. To be able to translate given algorithms to a working and correct program
3. To make them understand how to correct syntax errors as reported by the compilers
4. To be able to identify and correct logical errors encountered at run time
5. To understand how to write iterative as well as recursive programs
6. To enable them to represent data in arrays, strings and structures
7. To impart the knowledge of declare pointers of different types and their usage.
8. To understand how to create, read and write to and from simple text files.

### COURSE OUTCOMES:

At the end of the course, student will be able to

1. Formulate the algorithms for simple problems
2. Translate given algorithms to a working and correct program
3. Correct syntax errors as reported by the compilers
4. Identify and correct logical errors encountered at run time
5. Write iterative as well as recursive programs
6. Represent data in arrays, strings and structures and manipulate them through a program
7. Declare pointers of different types and use them in defining self-referential structures.
8. Create, read and write to and from simple text files.

### LIST OF EXPERIMENTS

#### Week-1 INTRODUCTION TO LINUX COMMANDS

1. Basic Linux commands
2. Write a C program to use printf() and scanf() functions
3. Write C programs to implement basic arithmetic operations – sum, average, product, difference, quotient and remainder of given numbers etc.

#### Week-2 OPERATORS AND EVALUATION OF EXPRESSIONS

1. Write a C program to check whether a number is even or odd using ternary operator.
2. Write a C program to perform the addition of two numbers without using +operator.
3. Write a C program to evaluate the arithmetic expression  $((a + b / c * d - e) * (f - g))$ . Read the values a, b, c, d, e, f, g from the standard input device.
4. Write a C program to find the sum of individual digits of a 3 digit number.
5. Write a C program to read the values of x and y and print the results of the following expressions in one line:
  - i.  $(x + y) / (x - y)$
  - ii.  $(x + y)(x - y)$

#### Week-3 CONDITIONAL STATEMENTS

1. Write a C program to find largest and smallest of given numbers.
2. Write a C program to find roots of a quadratic equation.
3. Write a C program which takes two integer operands and one operator from the user(+,-,\*,/,%) use switch)

**Week-4    LOOPING STATEMENTS**

1. Write a C program to find Sum of individual digits of given integer
2. Write a C program to generate first n terms of Fibonacci series
3. Write a C program to generate prime numbers between 1 and n

**Week-5    LOOPING STATEMENTS**

1. Write a C Program to find the Sum of Series  $SUM=1-x^2/2! +x^4/4!-x^6/6!+x^8/8!-x^{10}/10!$
2. Write a C program to generate Pascal's triangle.
3. Write a C program to generate pyramid of numbers.

```

                1
              1 3
            1 3 5
          1 3 5 3 1
```

**Week-6    ARRAYS**

1. Write a C Program to implement following sorting methods
  - i. Bubble sort
  - ii. Selection sort
  - iii. Insertion sort
2. Write a C program to find largest and smallest number in a list of integers

**Week-7    ARRAYS**

1. Write a C program
  - i. To add two matrices
  - ii. To multiply two matrices
2. Write a C program to find Transpose of a given matrix

**Week-8    FUNCTIONS**

1. Write a C program to find the factorial of a given integer using functions
2. Write a C program to find GCD of given integers using functions
3. Write a C Program to find the power of a given number using functions

**Week-9    RECURSION**

1. Write a C Program to find binary equivalent of a given decimal number using recursive functions.
2. Write a C Program to print Fibonacci sequence using recursive functions.
3. Write a C Program to find LCM of 3 given numbers using recursive functions

**Week-10    STRINGS**

1. Write a C program using functions to
  - a. Insert a sub string into a given main string from a given position
  - b. Delete n characters from a given position in a string
2. Write a C program to determine if given string is palindrome or not

**Week-11 POINTERS AND STRUCTURES**

1. Write a C program to print 2-D array using pointers
2. Write a C program to allocate memory dynamically using memory allocation functions (malloc, calloc, realloc, free)
3. Write a C Program using functions to
  - a. Reading a complex number
  - b. Writing a complex number
  - c. Add two complex numbers
  - d. Multiply two complex numbersNote: represent complex number using structure.

**Week-12 FILES**

1. Write a C program to copy one file to other
2. Write a C program to copy one file to other
3. Write a C Program to merge two files into a third file

**Text Books:**

1. Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13<sup>th</sup> Edition, 2012.
2. Oualline Steve, "Practical C Programming", O'Reilly Media, 3<sup>rd</sup> Edition, 1997.

**Reference Books:**

1. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2<sup>nd</sup> Edition, 2015.
2. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3<sup>rd</sup> Edition, 2004.
3. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1<sup>st</sup> Edition, 1994.

**Web References:**

1. <http://www.sanfoundry.com/c-programming-examples>
2. <http://www.geeksforgeeks.org/c>
3. <http://www.cprogramming.com/tutorial/c>
4. <http://www.cs.princeton.edu>

## CHEMISTRY LAB

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS12      | BSC      | -            | - | 3 | 1.5     | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. Estimation of hardness and chloride content in water to check its suitability for drinking purpose.
2. To determine the rate constant of reactions from concentrations as a function of time.
3. The measurement of physical properties like adsorption and viscosity.
4. To synthesize the drug molecules and check the purity of organic molecules by thin layer chromatographic (TLC) technique.

#### COURSE OUTCOMES:

The course should enable the students to:

1. Determination of parameters like hardness and chloride content in water.
2. Estimation of rate constant of a reaction from concentration – time relationships.
3. Determination of physical properties like adsorption and viscosity.
4. Calculation of  $R_f$  values of some organic molecules by TLC technique.

### LIST OF EXPERIMENTS

|                |                                                                                           |
|----------------|-------------------------------------------------------------------------------------------|
| <b>Week-1</b>  | DETERMINATION OF TOTAL HARDNESS OF WATER BY COMPLEXOMETRIC METHOD USING EDTA              |
| <b>Week-2</b>  | ESTIMATION OF AN HCL BY CONDUCTOMETRIC TITRATIONS                                         |
| <b>Week-3</b>  | ESTIMATION OF ACETIC ACID BY CONDUCTOMETRIC TITRATIONS                                    |
| <b>Week-4</b>  | ESTIMATION OF HCL BY POTENTIOMETRIC TITRATIONS                                            |
| <b>Week-5</b>  | ESTIMATION OF ACETIC ACID BY POTENTIOMETRIC TITRATIONS                                    |
| <b>Week-6</b>  | DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE             |
| <b>Week-7</b>  | SYNTHESIS OF ASPIRIN                                                                      |
| <b>Week-8</b>  | THIN LAYER CHROMATOGRAPHY CALCULATION OF $R_f$ VALUES. EG ORTHO AND PARA NITRO PHENOLS    |
| <b>Week-9</b>  | VERIFICATION OF FREUNDLICH ADSORPTION ISOTHERM-ADSORPTION OF ACETIC ACID ON CHARCOAL      |
| <b>Week-10</b> | DETERMINATION OF VISCOSITY OF CASTOR OIL AND GROUND NUT OIL BY USING OSTWALD'S VISCOMETER |
| <b>Week-11</b> | DETERMINATION OF SURFACE TENSION OF A GIVEN LIQUID USING STALAGMOMETER                    |
| <b>Week-12</b> | SYNTHESIS OF THIOL RUBBER                                                                 |

- Week-13** DETERMINATION OF CHLORIDE CONTENT OF WATER USING ARGENTOMETRIC METHOD
- Week-14** DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE

#### Reference Books:

1. Senior practical physical chemistry, B.D. Khosla, A. Gulati and V. Garg (R. Chand & Co., Delhi).
2. An introduction to practical chemistry, K.K. Sharma and D. S. Sharma (Vikas publishing, N. Delhi).
3. Vogel's text book of practical organic chemistry 5th edition.
4. Text book on Experiments and calculations in Engineering chemistry – S.S. Dara.

#### Web References:

1. <http://amrita.olabs.edu.in/?sub=73&brch=8&sim=153&cnt=2>
2. [https://chem.libretexts.org/Textbook\\_Maps/Analytical\\_Chemistry\\_Textbook\\_Maps/Map%3A\\_Analytical\\_Chemistry\\_2.0\\_\(Harvey\)/11\\_Electrochemical\\_Methods/11.2%3A\\_Potentiometric\\_Methods](https://chem.libretexts.org/Textbook_Maps/Analytical_Chemistry_Textbook_Maps/Map%3A_Analytical_Chemistry_2.0_(Harvey)/11_Electrochemical_Methods/11.2%3A_Potentiometric_Methods)
3. [http://fch.upol.cz/skripta/fcc\\_and\\_zvem\\_english/FCH/Adsorption%20of%20oxalic%20acid%20on%20activated%20charcoal.htm](http://fch.upol.cz/skripta/fcc_and_zvem_english/FCH/Adsorption%20of%20oxalic%20acid%20on%20activated%20charcoal.htm)
4. <https://www.askiitians.com/iit-jee-chemistry/physical-chemistry/electrolytic-conductance-molar-conductance-and-specific-conductance.aspx>
5. <https://owlcation.com/stem/tlc-thin-layer-chromatography-Principle-Procedure>

## ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS02      | HSMC     | 0            | 0 | 2 | 1       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Facilitate computer-assisted multi-media instruction enabling individualized and independent language learning.
2. Enhance English language skills, communication skills and to practice soft skills.
3. Improve fluency and pronunciation intelligibility by providing an opportunity for practice in speaking.
4. Train students in different interview and public speaking skills such as JAM, debate, role play, group discussion etc.
5. Instill confidence and make them competent enough to express fluently and neutralize their mother tongue influence.

### COURSE OUTCOMES:

The course should enable the students to:

1. Better perception of nuances of English language through audio- visual experience
2. Neutralization of accent for intelligibility
3. Take part in group activities
4. Speaking skills with clarity and confidence which in turn enhances their employability

## LIST OF ACTIVITIES

### Week-1 GENERAL INTRODUCTIONS AND FORMAL GREETINGS

- a. Introductions and greetings in formal and informal situations
- b. Worksheets to extract information
- c. Questionnaires to enquire about the expressions used during formal introductions
- d. Ice Breaking activity by preparing and asking five questions each
- e. Creation of dialogues using greetings, leave- taking and introductions

### Week-2 JAM- JUST A MINUTE

- a. Strengthen the ability to analyze a topic and logical organization of thoughts.
- b. Logically starting with introductory sentence, points of discussion and closing sentence.
- c. Practicing to speak within one minute
- d. Activity based on JAM on a familiar topic
- e. Planning and executing JAM considering the parameters

### Week-3 PHONETICS

- a. Speech sounds and their prominence in pronunciation
- b. Understanding and practicing word stress
- c. Neutralizing the accent and practicing the right intonation
- d. Knowing the differences between different accents
- e. Increase fluency with the help of Phonetics.

### Week-4 LISTENING SKILLS

- a. Developing good listening skills for effective communication
- b. Enhancing listening skills through audio tracks and oral conversation
- c. Empathizing others point of view while they speak
- d. Incorporating non verbal communications while listening
- e. Improving overall performance listening to the audio tracks

#### **Week-5      SITUATIONAL DIALOGUES AND GIVING DIRECTIONS**

- a. Creating dialogues in any given situations
- b. Framing and choosing appropriate words to frame the dialogues in any situation
- c. Guiding and giving directions using appropriate expressions
- d. Activities on how to make polite requests, offers, rejections etc
- e. Practicing to speak confidently in different situations

#### **Week-6      ROLE PLAY**

- a. Understanding a Role play and its procedure
- b. Planning and Executing a Role Play accordingly
- c. Practicing to get into the role and perform within stipulated time
- d. Activities based on Role Play with different situations
- e. Performing a Role Play considering the parameters

#### **Week-7      GROUP DISCUSSIONS**

- a. Understanding a Group Discussion (GD) and its procedure
- b. Following the rules of a GD
- c. Planning and Executing a GD within the stipulated time
- d. Activities based on GD
- e. Performing a GD considering the parameters

#### **Week-8      DEBATE**

- a. Understanding the procedure of a Debate
- b. Planning and executing a Debate following its rules
- c. Strengthen the ability to analyze a topic and logical organization of thoughts.
- d. Logically arranging the arguments
- e. Performing a Debate considering the parameters

#### **Week-9      TELEPHONIC ETIQUETTES**

- a. Understanding basic Telephonic Etiquettes
- b. The approach one needs to follow while making and answering a call
- c. Making a formal telephonic conversation
- d. Activities based on modulating voice and tone
- e. Interpersonal skills required to overcome rude and hostile behavior

#### **Week-10    PRESENTATION SKILLS**

- a. Planning a Presentation
- b. Enhancing skills required for making effective presentations
- c. Usage of different tools that help us to give effective presentations
- d. Executing a presentation effectively
- e. Activities based on presentations

**Week-11 ORAL PRESETATIONS AND EXTEMPORE**

- a. Planning an oral presentation or an Extempore
- b. Preparing good PPT
- c. Using appropriate body language in public speaking domain
- d. Planning and Executing oral presentation
- e. Activities based on oral presentations and extempore

**Week-12 INTERVIEW SKILLS**

- a. Preparing to succeed in Interviews
- b. Preparing a strong Resume for interviews
- c. Practicing different techniques to overcome nervousness in interviews
- d. Using appropriate body language in interviews
- e. Activities based on Interviews skills

**Week-13 INFORMATION TRANSFER**

- a. Extracting Information Transfer from different kinds of representation
- b. Reading and decoding the information given in various types
- c. Representing the information in charts or graphs in a written document
- d. Developing writing skills from these aspects
- e. Activity on transferring given data into graphs or charts for presentation skills

**Reference Books:**

1. E. Suresh Kumar. A Handbook for English Language Laboratories (with CD) Revised Edition
2. Normal Whitby. Business Benchmark. Cambridge University Press( with CD) 2<sup>nd</sup> Edition
3. Liz Hamp-Lyons and Ben Heasley. Study Writing. Cambridge University Press. 2006.
4. Sanjay Kumar and Pushp Lata. Communication Skills. Oxford University Press. 2011.
5. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press
6. Raman Sharma, Technical Communications, Oxford Publication, London, 2004.
7. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004.
8. T. Balasubramanian, A Text book of English Phonetics for Indian Students, Macmillan, 2008.
9. Edgar Thorpe, Winning at Interviews, Pearson Education, 2006.
10. J. Sethi et al, A Practical Course in English Pronunciation (with CD), Prentice Hall of India, 2005.
11. Hari Mohan Prasad, How to Prepare for Group Discussions and Interviews, Tata McGraw Hill, 2006

**Websites:**

1. <https://www.britishcouncil.org>
2. <https://www.bbc.co.uk>
3. <https://www.grammarly.com>
4. <https://www.fluentu.com>
5. <https://www.cambridgeenglish.org/exams-and-tests/business-preliminary>
6. <https://www.cambridgeenglish.org/exams-and-tests/business-vantage>



## ENGINEERING EXPLORATION

### I Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS04      | HSMC     | -            | - | 2 | 1       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the Engineering attributes and Ethics.
2. Identify the community problem and its stakeholder.
3. Examine required specifications and gap in existing and required product.
4. Build sustaining interactions among people that create social value by transforming ideas into tangible products, services, or initiatives.
5. Develop skills to work collaboratively, reports and progress updates throughout the lifecycle of the project.

### UNIT-I INTRODUCTION TO ENGINEERING AND ENGINEERING EXPLORATION Classes: 10

Engineering Projects in Community Service, Design Thinking Process-Empathize, Define, Ideate, Prototype, Test.

**Engineering Ethics:** Introduction to ethics, moral values, significance of professional ethics, code of conduct for engineers, identify ethical dilemmas in different tasks of engineering, applying moral theories and codes of conduct for resolution of ethical dilemmas.

### UNIT-II PROBLEM IDENTIFICATION Classes: 10

Authentic need in the community or society. Identify a real user or stake holder, Interaction with Stakeholders, Viewpoints, Interviewing, Scenario.

### UNIT-III SPECIFICATION DEVELOPMENT Classes: 10

Clear and measurable requirements, criteria for success, Identifying relevant benchmarks, identifying the gap between the available and required products, requirements documentation.

### UNIT-IV CONCEPTUAL DESIGN Classes: 10

Ideation-generated multiple ideas, evaluation of ideas, systems model, Architectural Design, prototype development, testing – real/simulated users, feedback.

### UNIT-V PROJECT MANAGEMENT Classes: 10

Importance of team work, importance of project life cycle, project management, tools, various tools used in electronics documentation, importance of communication, usage of communication media.

### Text Books:

1. Software Engineering: A Practitioner's Approach, Roger S. Pressman, 7<sup>th</sup> Edition, Mc Graw Hill Education (India) Pvt. Ltd.
2. Software Engineering, Sommerville Ian, 7<sup>th</sup> Edition, Pearson Education.
3. EPICS Design Process

[https://sharepoint.ecn.purdue.edu/epics/teams/Public%20Documents/EPICS\\_Design\\_Process.pdf](https://sharepoint.ecn.purdue.edu/epics/teams/Public%20Documents/EPICS_Design_Process.pdf)

4. Examples of good practice in Special Needs Education & Community Based Programs, UNESCO PRESS.
5. Project Management, GRY r. Heerkens, McGraw-Hill

#### **Web References:**

1. <http://www.purdue.edu/epics>
2. <http://epics.ieee.org/>
3. <https://www.uninettunouniversity.net/en/epics.aspx>

#### **E-Text Books:**

1. [http://www.uoitc.edu.ig/images/documents/informatics-institute/exam\\_materials/Software%20Engineering%20\(9th%20Edition\)%20by%20Ian%20Sommerville.pdf](http://www.uoitc.edu.ig/images/documents/informatics-institute/exam_materials/Software%20Engineering%20(9th%20Edition)%20by%20Ian%20Sommerville.pdf)
2. <https://engineering.purdue.edu/EPICS/k12/resources/1.6%20Teacher%20Toolbox%20EPICS%20High%20Design%20Process%20and%20Cycle.pdf>
3. [https://launchschool.com/books/agile\\_planning/read/epics\\_and\\_stories](https://launchschool.com/books/agile_planning/read/epics_and_stories)  
<http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

#### **MOOC Course:**

1. <https://www.mooc-list.com/tags/design-thinking>
2. <https://www.class-central.com/tag/design%20thinking>

## **II B.TECH I SEMESTER SYLLABUS**

## PROBABILITY AND STATISTICS

### II Year I Semester: CSE/IT

| Course Code: | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------|----------|--------------|---|---|---------|---------------|-----|-------|
|              |          | L            | T | P |         | CIE           | SEE | Total |
| A4BS05       | BSC      | 2            | 1 | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. The concepts of discrete and continuous random variables, the probability distribution and density function.
2. Evaluation of marginal and conditional distribution of multiple random variables.
3. The concept of correlation and regression to find covariance.
4. Evaluation of the given data for appropriate test of hypothesis.
5. Analyzing the data for variance.

### COURSE OUTCOMES:

After learning the contents of this paper the student must be able to

1. Describe the continuous random variables and moments of Discrete and continuous random variables.
2. Calculate probabilities and derive the marginal and conditional distribution of bivariate random variables.
3. Apply the concept of correlation and regression to find covariance.
4. Evaluate the given data for appropriate test of hypothesis.
5. Test the data for analysis of variance.

### UNIT-I SINGLE RANDOM VARIABLES

**Classes: 10**

Basic definitions of probability- Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation, moments about origin, central moments, skewness, Kurtosis. Moment generating function of probability distribution.

### UNIT-II PROBABILITY DISTRIBUTIONS

**Classes: 12**

Binomial, Poisson, Normal, exponential and Gamma distributions -their Properties.

Moment generating functions of the above distributions and hence find the mean and variance.

Joint probability distributions- Joint probability mass /density function, Marginal probability, mass / density functions.

### UNIT-III CORRELATION & REGRESSIONSAMPLING DISTRIBUTIONS

**Classes: 14**

Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression and multiple correlation & regression

**Sampling:** Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance.

Parameter estimation- Point estimation and interval estimation

**UNIT-IV TESTING OF HYPOTHESIS - I****Classes: 14**

**Testing of hypothesis:** Null hypothesis, Alternate hypothesis, Type I & Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test.

**Large sample tests:**

- (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances)
- (ii) Tests of significance of difference between sample S.D and population S.D.
- (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.

**UNIT-V TESTING OF HYPOTHESIS-II****Classes: 10**

Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit.

**Text Books:**

1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.
3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

**Reference Books:**

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

**Web references:**

1. [https://www.efunda.com/math/math\\_home/math.cfm](https://www.efunda.com/math/math_home/math.cfm)
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

**E -Text Books:**

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

**MOOCS Course:**

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

## DISCRETE STRUCTURES

**II Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS03      | PCC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To help students understand discrete and continuous mathematical structures
2. To impart basics of relations and functions
3. To facilitate students in applying principles of Recurrence Relations to calculate generating Functions and solve the Recurrence relations
4. To acquire knowledge in graph theory

### COURSE OUTCOMES:

**At the end of the course, student will be able to**

1. Analyze and examine the validity of argument by using propositional and predicate calculus
2. Apply basic counting techniques to solve the combinatorial problems
3. Apply sets relations and digraphs to solve applied problems
4. Solve the given recurrence relation using different methods such as substitution, Generating function and Characteristic roots equation.

### UNIT-I MATHEMATICAL LOGIC

**Classes: 11**

Statements and notations, Connectives, Well formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving. Quantifiers, Universal quantifiers. Predicates: Predicative logic, Free & Bound variables.

### UNIT-II RELATIONS

**Classes: 16**

Introduction to set theory, Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function, Composition of functions, Recursive Functions

### UNIT-III ELEMENTARY COMBINATORICS

**Classes: 12**

Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.

### UNIT-IV RECURRENCE RELATION

**Classes: 11**

Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions, The method of Characteristics roots, Solution of Inhomogeneous Recurrence Relation.

### UNIT-V GRAPHS

**Classes: 10**

Basic Concepts, Isomorphism and Sub graphs, Trees and their properties, Spanning Trees- DFS, BFS, Minimal Spanning Trees- Prim's, Kruskal's Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.

**Text Books:**

1. Discrete Mathematics for computer scientists & Mathematicians, *J.L. Mott, A. Kandel, T.P. Baker* PHI
2. Discrete Mathematical Structures With Applications to Computer Science, JP Tremblay, R Manohar

**Reference Books:**

1. Logic and Discrete Mathematics, *Grass Man & Trembley*, Pearson Education.

## DATA STRUCTURES

### II Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS04      | PCC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. Impart the basic concepts of data structures and algorithms.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching and hashing.
6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

### COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Evaluate algorithms in terms of time and memory complexity.
2. Formulate new solutions for problems or improve existing code using data structures and algorithms.
3. Implement basic data structures such as arrays, linked lists, stacks and queues.
4. Solve problem involving graphs, trees and heaps
5. Apply Algorithms for solving problems like sorting, searching, and hashing.
6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

### UNIT-I INTRODUCTION TO DATA STRUCTURES

**Classes: 12**

**Basic concepts-** Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations.

**Introduction to Linear and Non Linear data structures-**Singly Linked Lists-Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion. Representation of single, two dimensional arrays, sparse matrices-array and linked representations.

### UNIT-II STACKS AND QUEUES

**Classes: 10**

**Stacks-**Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation, recursion implementation,

**Queues-**Queue ADT, definition and operations ,array and linked Implementations in C, Circular queues-Insertion and deletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.

### UNIT-III TREES AND GRAPHS

**Classes: 14**

**Trees** – Terminology, Representation of Trees, Binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals, Threaded binary trees,

**Max Priority Queue-**ADT-implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from a Max Heap. Graphs , Introduction, Definition, Terminology, Graph ADT, Graph Representations- Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.

### UNIT-IV SEARCHING AND SORTING

**Classes: 12**



**Searching-** Linear Search, Binary Search, Static Hashing-Introduction, hash tables, hash functions, Overflow Handling.

**Sorting-** Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.

## UNIT-V      BINARY SEARCH TREES

**Classes: 12**

**Search Trees-** Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees- Definition and Examples, Insertion into an AVL Tree ,B-Trees, Definition, B-Tree of order m, operations-Insertion and Searching, Introduction to Red-Black and Splay Trees(Elementary treatment-only Definitions and Examples), Comparison of Search Trees.

**Pattern matching algorithm-** The Knuth-Morris-Pratt algorithm, Tries (examples only).

### Text Books:

1. "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and SusanAnderson-Freed, Universities Press.

### Reference Books:

1. Algorithms, Data Structures, and Problem Solving with C++", Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company
2. "How to Solve it by Computer", 2nd Impression by R. G. Dromey, Pearson Education

### Web References:

1. <https://hackr.io/tutorials/learn-data-structures-algorithms>
2. <https://www.geeksforgeeks.org/fundamentals-of-algorithms/>
3. <https://www.udemy.com/introduction-to-algorithms-and-data-structures-in-c/>
4. <https://leetcode.com>

### E-Text Books:

1. <http://www.freetechbooks.com/algorithm-analysis-and-design-t1030.html>
2. <http://www.freetechbooks.com/algorithmic-problem-solving-t373.html>
3. <http://www.freetechbooks.com/algorithms-and-data-structures-the-basic-toolbox-t871.html>

### MOOC Course:

1. <https://www.coursera.org/specializations/data-structures-algorithms>
2. [https://onlinecourses.nptel.ac.in/noc16\\_cs06/preview](https://onlinecourses.nptel.ac.in/noc16_cs06/preview)

## DATABASE MANAGEMENT SYSTEMS

### II Year I Semester: CSE

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS05      | PCC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. **Discuss** the basic database concepts, applications, data models, schemas and instances.
2. **Design** Entity Relationship model for a database.
3. **Demonstrate** the use of constraints and relational algebra operations.
4. **Describe** the basics of SQL and construct queries using SQL
5. **Understand** the importance of normalization in databases.
6. **Demonstrate** the basic concepts of transaction processing and concurrency control.
7. **Understand** the concepts of database storage structures and identify the access techniques.

### COURSE OUTCOMES:

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Apply SQL queries to interact with Database
3. Apply normalization on database design to eliminate anomalies
4. Analyze database transactions and can control them by applying ACID properties
5. Analyze physical database storage system of database.

### UNIT-I

#### INTRODUCTION

**Classes: 09**

**INTRODUCTION:** Introduction and applications of DBMS, Purpose of data base, Data Independence, Database System architecture- Levels, Database users and DBA.

**DATABASE DESIGN:** Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.

### UNIT-II

#### RELATIONAL MODEL AND SQL

**Classes: 15**

**THE RELATIONAL MODEL:** Introduction to the relational model, Integrity constraints over relations, Enforcing integrity constraints, Querying relational data, Logical database design: E-R to relational, Introduction to views, Destroying/altering tables and views.

**RELATIONAL ALGEBRA AND CALCULUS:** Relational algebra operators, relational calculus - Tuple and domain relational calculus.

**SQL:** Basics of SQL, DDL,DML,DCL, structure – creation, alteration, defining constraints – Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions – numeric, date, string functions, set operations, sub-queries, correlated sub-queries, Use of group by, having, order by, join and its types, Exist, Any, All . Transaction control commands – Commit, Rollback, Save point, cursors, stored procedures, Triggers.

### UNIT-III

#### SCHEMA REFINEMENT

**Classes: 11**

**SCHEMA REFINEMENT AND NORMAL FORMS:** Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, properties of decompositions, normalization, schema refinement in database design.

**TRANSACTIONS MANAGEMENT:** Transaction concept, transaction state, concurrent executions, Serializability, recoverability, testing for serializability.

#### **UNIT-IV                      TRANSACTION AND CONCURRENCY CONTROL                      Classes: 11**

**CONCURRENCY CONTROL AND RECOVERY SYSTEM:** Concurrency control, lock based protocols, time-stamp based protocols, validation based protocols, multiple granularity, and deadlock handling. Recovery system - failure classification, storage structure, recovery and atomicity, log based recovery, shadow paging, recovery with concurrent transactions, failure with loss of non-volatile storage, advanced recovery techniques, remote backup systems.

#### **UNIT-V                      QUERY PROCESSING & QUERY OPTIMIZATION                      Classes: 11**

**QUERY PROCESSING & QUERY OPTIMIZATION:** Overview, measures of query cost, selection operation, sorting, join, evaluation of expressions.

**OVERVIEW OF STORAGE AND INDEXING:** Tree structured indexing - intuition for tree indexes, indexed sequential access method (ISAM), B+ Trees - a dynamic tree structure.

#### **Text Books:**

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011.
2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007.
3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.

#### **Reference Books:**

1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009.
2. Scott Urman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ", 6th edition, Tata McGraw Hill, 2010
3. S.K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006.
4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addison wesley, 2007

#### **Web References:**

1. <http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database>
2. [https://www.w3schools.com/sql/sql\\_create\\_table.asp](https://www.w3schools.com/sql/sql_create_table.asp)
3. [http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm\\_export=print](http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print)
4. <http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf>
5. <http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf>

#### **E-Text Books:**

1. <http://www.freebookcentre.net/Database/Free-Database-Systems-Books-Download.html>
2. <http://www.ddegjust.ac.in/studymaterial/mca-3/ms-11.pdf>

#### **MOOC Course:**

1. <https://www.mooc-list.com/tags/dbms-extensions>
2. [https://onlinecourses.nptel.ac.in/noc18\\_cs15/preview](https://onlinecourses.nptel.ac.in/noc18_cs15/preview)

## ELECTRONIC DEVICES AND CIRCUITS

### II Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4EC01      | ESC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To learn principle of operation, construction and characteristics of various electronic devices.
2. To study operation and characteristics of Rectifiers with filters.
3. To understand the analysis of small signal low frequency amplifiers.
4. To provide the concepts involved in design of electronic Circuits.

### COURSE OUTCOMES:

After completing this course, the student will be able to:

1. Identify, formulate & solve Engineering problems by making use of modern software/hardware tools.
2. Analyze behaviour of electronic devices.
3. Design and analyze various rectifiers, filter circuits.
4. Design an amplifier circuit with proper biasing techniques (BJT and FET).
5. Analyze Field effect transistor (FET) and mosfet characteristics

#### UNIT-I P-N Junction Diode :

**Classes: 12**

PN JUNCTION DIODE: Operation of PN junction Diode: No bias, forward bias and reverse bias, diode current equation (qualitative treatment), volt-ampere (V-I) characteristics, temperature dependence of V-I characteristics, ideal versus practical diode, static and dynamic resistances, diode equivalent circuits, breakdown mechanisms in semiconductor diodes, zener diode characteristics.

#### UNIT-II Rectifiers and Filters :

**Classes: 10**

The P-N junction as a Rectifier, Half wave Rectifier, Full wave Rectifier, Bridge Rectifier, Harmonic components in a Rectifier Circuit, Inductor Filters, Capacitor Filters, L Section Filters,  $\pi$ - Section Filters, Comparison of Filters, Voltage Regulation using Zener Diode.

#### UNIT-III Bipolar Junction Transistor and its Biasing

**Classes: 14**

BIPOLAR JUNCTION TRANSISTOR (BJT): BJT construction, operation, symbol, transistor current components, input & output characteristics of a transistor in CB, CE and CC configurations  
TRANSISTOR BIASING AND STABILIZATION: Need for biasing, operating point, DC and AC load lines, stability factor, fixed bias circuit, collector to base bias circuit, self bias circuit.

#### UNIT-IV BJT Amplifiers And Field Effect Transistor

**Classes: 14**

BJT AMPLIFIERS: Operation of CE amplifier, Operation of RC coupled amplifier, Operation of Class-A power amplifier, Operation of Class-B Push-Pull power amplifier

FIELD EFFECT TRANSISTOR: Junction field effect transistor (construction, principle of operation, symbol), volt-ampere characteristics of JFET, MOSFETS (construction, principle of operation,

symbol), volt- ampere characteristics of MOSFETS in enhancement and depletion modes

## **UNIT-V      Feedback Amplifiers & Oscillators**

**Classes: 10**

**FEEDBACK AMPLIFIERS:** Feedback concepts, types of feedback circuits (block diagram representation), general characteristics of negative feedback amplifier **OSCILLATORS:** Barkhausen criterion, RC Phase shift oscillator using BJT, General form of LC oscillators, Hartley oscillator, Colpitts oscillator and Crystal oscillator.

### **Text Books:**

1. Jacob Milliman, Christos C .Halkias, Satyabrata Jit (2015), *Electronic Devices and Circuits*, 3rd edition, Tata McGraw Hill, New Delhi.
2. G. K. Mittal (1999), *Electronic Devices and Circuits*, 22nd edition, Khanna Publications, New Delhi.
3. Robert Boylestad, Lowis Nashelsky (1993), *Electronic Devices and Circuit Theory*, 5th edition, Prentice Hall of India, New Delhi, India.

### **Reference Books:**

1. David. A. Bell (1986), *Electronic Devices and Circuits*, 4th edition, Prentice Hall of India, New Delhi.
2. S. Shalivahanan, N. Suresh Kumar, A. Vallavaraj (2007), *Electronic Devices and Circuits*, 3rd edition, McGraw Hill, New Delhi, India.
3. Theodore. F. Bogart Jr, Jeffrey S. Beasley, Guillermo Rico (2004), *Electronic Devices and Circuits*, 6th edition Pearson Education, India.

### **Web References:**

1. <https://unacademy.com/course/electronic-devices-gate-ece/JTIAAKX1>
2. <https://freevideolectures.com/course/2261/basic-electronics-and-lab/2>
3. <https://unacademy.com/lesson/build-in-potential-and-depletion-width/ALHF5QVM>

### **E-Text Books:**

1. [http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-\(PDF-313p\).html](http://www.freebookcentre.net/electronics-ebooks-download/Electronic-Devices-and-Circuits-(PDF-313p).html)
2. <https://www.goodreads.com/book/show/25345857-electronic-devices-and-circuits>
3. <https://thebookee.net/el/electronic-devices-and-circuits-by-jb-gupta-pdf>

### **MOOC Course:**

1. <https://www.edx.org/>
2. <https://www.coursera.org/learn/electronics>

## DATA STRUCTURES LAB

### II Year I Semester: CSE

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS06      | PCC      | -            | - | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Ability to identify the appropriate data structure for given problem.
2. Design and analyze the time and space complexity of algorithm or program.
3. Effectively use compilers include library functions, debuggers and trouble shooting.
4. Write and execute programs using data structures such as arrays, linked lists to implement stacks, queues.
5. Write and execute programs in C to implement various sorting and searching.
6. Write and execute programs using data structures such as arrays, linked lists to implement trees, graphs, hash tables and search trees.

### COURSE OUTCOMES:

The course should enable the students to:

1. Use appropriate data structure for given problem.
2. To analyze the time and space complexity of algorithm or program.
3. Use compilers include library functions, debuggers and trouble shooting.
4. Execute programs using data structures such as arrays, linked lists to implement stacks and queues.
5. Execute write programs in C to implement various sorting and searching.
6. Execute programs using data structures such as arrays, linked lists to implement trees, graphs, hash tables and search trees.

## LIST OF EXPERIMENTS

### Week-1 SINGLE LINKED LIST

Write a C program that uses functions to perform the following:

- a) Create a singly linked list of integers.
- b) Delete a given integer from the above linked list.
- c) Display the contents of the above list after deletion.

### Week-2 DOUBLE LINKED LIST

Write a C program that uses functions to perform the following:

- a) Create a doubly linked list of integers.
- b) Delete a given integer from the above doubly linked list.
- c) Display the contents of the above list after deletion.

### Week-3 INFIX TO POSTFIX CONVERSION

Write a C program that uses stack operations to convert a given infix expression into its postfix Equivalent, Implement the stack using an array.

### Week-4 DOUBLE ENDED QUEUE

Write C programs to implement a double ended queue ADT using

- i) array and
- ii) doubly linked list

### Week-5 BINARY SEARCH TREES USING RECURSION

Write a C program that uses functions to perform the following:

- a) Create a binary search tree of characters.
- b) Traverse the above Binary search tree recursively in Post-order.

#### **Week-6 BINARY SEARCH TREES USING NON-RECURSION**

Write a C program that uses functions to perform the following:

- a) Create a binary search tree of integers.
- b) Traverse the above Binary search tree non recursively in in-order.

#### **Week-7 SORTING**

Write C programs for implementing the following sorting methods to arrange a list of integers in Ascending order :

- a) Insertion sort
- b) Merge sort

#### **Week-8 SORTING**

Write C programs for implementing the following sorting methods to arrange a list of integers in ascending order:

- a) Quick sort
- b) Selection sort

#### **Week-9 AVL-TREES**

1. Write a C program to perform the following operations on AVL:
  - a) Insertion into an AVL.
  - b) Display elements of AVL Tree
2. Write a C program for implementing Heap sort algorithm for sorting a given list of integers in ascending order

#### **Week-10 HASHING**

Write a C program to implement all the functions of a dictionary (ADT) using hashing.

#### **Week-11 PATTERN MATCHING ALGORITHM**

Write a C program for implementing Knuth-Morris- Pratt pattern matching algorithm.

#### **Week-12 GRAPH TRAVERSAL ALGORITHMS**

Write C programs for implementing the following graph traversal algorithms:

- a)Depth first traversal   b)Breadth first traversal

#### **Text Books:**

1. C and Data Structures, Prof. P.S.Deshpande and Prof. O.G. Kakde, Dreamtech Press.
2. Data structures using C, A.K.Sharma, 2nd edition, Pearson.
3. Data Structures using C, R.Thareja, Oxford University Press.

#### **Web References:**

1. <http://www.sanfoundry.com/data-structures-examples>
2. <http://www.geeksforgeeks.org/c>
3. <http://www.cs.princeton.edu>

## DATABASE MANAGEMENT SYSTEMS LAB

### II Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credit | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|--------|---------------|-----|-------|
|             |          | L            | T | P |        | CIE           | SEE | Total |
| A4CS07      | PCC      | -            | - | 3 | 1.5    | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Use the basics of SQL and construct queries using SQL in database creation and interaction
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
4. Analyze and Select storage and recovery techniques of database system.

### COURSE OUTCOMES:

The course should enable the students to:

1. Apply the basic concepts of Database Systems and Applications.
2. Develop an ER model for a given database.
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.
5. Analyze and Select storage and recovery techniques of database system.

### LIST OF EXPERIMENTS

#### Week-1 INTRODUCTION TO E R MODEL

Student should decide on a case study and formulate the problem statement.

#### Week-2 CONCEPTUAL DESIGNING USING ER DIAGRAMS

Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.) Note: Student is required to submit a document by drawing ER Diagram to the Lab teacher.

#### Week-3 CONVERTING ER MODEL TO RELATIONAL MODEL

Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys) Note: Student is required to submit a document showing the database tables created from ER Model.

#### Week-4 NORMALIZATION

Normalization -To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form

#### Week-5 CREATION OF TABLES USING SQL

Creation of Tables using SQL- Overview of using SQL tool, Data types in SQL, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables

#### Week-6 DML COMMANDS

Practicing DML commands- Insert, Select, Update, Delete.



**Week-7    QUERIES**

Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, CONSTRAINTS etc.

**Week-8        SUB QUERIES**

Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi)..

**Week-9        AGGREGATE FUNCTIONS**

Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.

**Week-10      TRIGGERS**

Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.

**Week-11      PROCEDURES**

Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure.

**Week-12      CURSORS**

Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

**Reference Books:**

1. Database System Concepts, by Silberschatz, Sudarshan, and Korth, 6th edition.
2. Database management System by RaghuRamaKrishna, 3rd edition

**Web References:**

1. <http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database>
2. [https://www.w3schools.com/sql/sql\\_create\\_table.asp](https://www.w3schools.com/sql/sql_create_table.asp)
3. [http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm\\_export=print](http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print)
4. <http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf>
5. <http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf>

## ENVIRONMENTAL STUDIES

### II Year I Semester: CSE

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4MC01      | MC       | 2            | - | - | 0       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understanding the importance of ecological balance for sustainable development.
2. Understanding the impacts of developmental activities and mitigation measures.
3. Understanding the environmental policies and regulations

### COURSE OUTCOMES:

1. Based on this course, the Engineering graduate will understand /evaluate / develop technologies on the basis of ecological principles and environmental regulations which in turn helps in sustainable development

### UNIT-I Ecosystems

Classes: 02

Definition, Scope, and Importance of ecosystem. Classification, structure, and function of an ecosystem, Food chains, food webs, and ecological pyramids. Flow of energy, Biogeochemical cycles, Bioaccumulation, Bio-magnification, ecosystem value, services and carrying capacity, Field visits.

### UNIT-II Natural Resources: Classification of Resources

Classes: 02

Living and Non-Living resources, **water resources**: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. **Mineral resources**: use and exploitation, environmental effects of extracting and using mineral resources, **Land resources**: Forest resources, **Energy resources**: growing energy needs, renewable and non renewable energy sources, use of alternate energy source, case studies.

### UNIT-III Biodiversity And Biotic Resources

Classes: 03

Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Hot spots of biodiversity. Field visit. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity act.

### UNIT-IV Environmental Pollution and Control Technologies

Classes: 05

**Environmental Pollution**: Classification of pollution, **Air Pollution**: Primary and secondary pollutants, Automobile and Industrial pollution, Ambient air quality standards. **Water pollution**: Sources and types of pollution, drinking water quality standards. **Soil Pollution**: Sources and types, Impacts of modern agriculture, degradation of soil. **Noise Pollution**: Sources and Health hazards, standards, **Solid waste**: Municipal Solid Waste management, composition and characteristics of e-Waste and its management. **Pollution control technologies**: Wastewater Treatment methods: Primary, secondary and Tertiary.

Overview of air pollution control technologies, Concepts of bioremediation. **Global Environmental Issues and Global Efforts:** Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification. International conventions / Protocols: Earth summit, Kyoto protocol, and Montréal Protocol. NAPCC-GoI Initiatives.

#### **UNIT-V Environmental Policy, Legislation & EIA**

**Classes: 04**

Environmental Protection act, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management and handling rules, biomedical waste management and handling rules, hazardous waste management and handling rules. EIA: EIA structure, methods of baseline data acquisition. Overview on Impacts of air, water, biological and Socio-economical aspects. Strategies for risk assessment, Concepts of Environmental Management Plan (EMP). **Towards Sustainable Future:** Concept of Sustainable Development Goals, Population and its explosion, Crazy Consumerism, Environmental Education, Urban Sprawl, Human health, Environmental Ethics, Concept of Green Building, Ecological Foot Print, Life Cycle assessment (LCA), Low carbon life style.

#### **Text Books:**

1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
2. Environmental Studies by R. Rajagopalan, Oxford University Press.

#### **Reference Books:**

1. Environmental Science: towards a sustainable future by Richard T. Wright. 2008 PHL Learning Private Ltd. New Delhi.
2. Environmental Engineering and science by Gilbert M. Masters and Wendell P. Ela. 2008 PHI Learning Pvt. Ltd.
3. Environmental Science by Daniel B. Botkin & Edward A. Keller, Wiley INDIA edition.
4. Environmental Studies by Anubha Kaushik, 4<sup>th</sup> Edition, New age international publishers.
5. Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications.
6. Introduction to Environmental Science by Y. Anjaneyulu, BS.Publications.

## **II B.TECH II SEMESTER SYLLABUS**

## DIGITAL LOGIC DESIGN

### II Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4EC02      | ESC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand numerical and character representations in digital logic including ASCII and error detecting and correcting codes.
2. Design combinational and sequential logic circuits
3. Optimize combinational and sequential logic circuits.
4. Analyze a memory cell and apply for organizing larger memories

### COURSE OUTCOMES:

1. Understand the different switching algebra theorems and apply them for logic functions.
2. Define the Karnaugh map for a few variables and perform an algorithmic reduction of logic functions.
3. Define the following combinational circuits: buses, encoders/decoders, (de)multiplexers, exclusive-ORs, comparators, arithmetic-logic units.
4. Understand the bistable element and the different latches and flip-flops.
5. Understand sequential circuits, like counters and shift registers.

### UNIT-I NUMBER THEORY AND LOGIC SIMPLIFICATION

**Classes: 10**

Number systems like binary, octal, hexadecimal and r's and r-1's complements, Review of Boolean Algebra and De Morgan's Theorem, SOP & POS forms, Canonical forms, Karnaugh maps up to 6 variables, Binary codes, Code Conversion.

### UNIT-II COMBINATIONAL LOGIC DESIGN

**Classes: 10**

MSI devices like Comparators, Multiplexers, Encoder, Decoder, seven-segment Display, Half and Full Adders, Subtractors, Serial and Parallel Adders, BCD Adder, and ALU.

### UNIT-III SEQUENTIAL LOGIC DESIGN

**Classes: 14**

Ripple and Synchronous counters, Shift registers, Finite state machines, Design of synchronous FSM, Algorithmic State Machines charts. Designing synchronous circuits like Pulse train generator, Pseudo Random Binary Sequence generator, Clock generation.

### UNIT-IV LOGIC FAMILIES AND SEMICONDUCTOR MEMORIES

**Classes: 14**

TTL NAND gate, Specifications, Noise margin, Propagation delay, fan-in, fan-out, Tristate TTL, ECL, CMOS families and their interfacing, Memory elements like SRAM and DRAM, Concept of Programmable logic devices like PAL and PLA.

### UNIT-V VLSI DESIGN FLOW

**Classes: 12**

Introduction to HDL, Data types and objects, different modeling styles in VHDL, Dataflow, Behavioural and Structural Modeling, Synthesis and Simulation VHDL constructs and programs for combinational and sequential circuits including test-bench programming.

**Text Books:**

1. R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition, 2009.
2. Douglas Perry, "VHDL", Tata McGraw Hill, 4th edition, 2002.

**Reference Books:**

1. W.H. Gothmann, "Digital Electronics- An introduction to theory and practice", PHI, 2nd edition ,2006.
2. D.V. Hall, "Digital Circuits and Systems", Tata McGraw Hill, 1989
3. Charles Roth, "Digital System Design using VHDL", Tata McGraw Hill 2nd edition 2012.

**Web References:**

1. [www.wikipedia.org](http://www.wikipedia.org)
2. [www.pa.msu.edu](http://www.pa.msu.edu)
3. [www.tutorvista.com](http://www.tutorvista.com)
4. [www.globalspec.com](http://www.globalspec.com)
5. [www.ee.bilkent.edu.tr](http://www.ee.bilkent.edu.tr)

**E-Text Books:**

1. <http://www.site.uottawa.ca/~petriu/Digital-Logic.pdf>
2. [http://uav.ece.nus.edu.sg/~bmchen/courses/EG1108\\_Digital.pdf](http://uav.ece.nus.edu.sg/~bmchen/courses/EG1108_Digital.pdf)
3. <http://info.iet.unipi.it/~luigi/biomedica/sito/cosc205.pdf>

**MOOC Course:**

1. [https://onlinecourses.nptel.ac.in/noc18\\_ee33/preview](https://onlinecourses.nptel.ac.in/noc18_ee33/preview)
2. [https://onlinecourses.nptel.ac.in/noc18\\_ee34/preview](https://onlinecourses.nptel.ac.in/noc18_ee34/preview)

## COMPUTER ORGANIZATION AND ARCHITECTURE

**II Year II Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS08      | PCC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Analyze basic components and understand the Architecture and organization of a computer
2. Demonstrate key skills of constructing cost-effective computer systems
3. Apply register transfer and micro programmed control operations.
4. Analyze memory and I/O systems.

### COURSE OUTCOMES:

At the end of the course students will be able to:

1. Identify various components of computer and their interconnection
2. Identify basic components and design of the CPU: the ALU and control unit.
3. Compare and select various Memory devices as per requirement.
4. Compare various types of IO mapping techniques
5. Critique the performance issues of cache memory and virtual memory

### UNIT-I STRUCTURE OF COMPUTERS

**Classes: 10**

**STRUCTURE OF COMPUTERS:** Computer types, Functional units, Basic operational concepts, Von Neumann Architecture, Bus Structures, Software, Performance, Multiprocessors and Multicomputer, Data representation, Fixed and Floating point, Error detection and Hamming codes.

**COMPUTER ARITHMETIC:** Addition and Subtraction, Multiplication and Division algorithms, Floating-point Arithmetic Operations, Decimal arithmetic operations

### UNIT-II BASIC COMPUTER ORGANIZATION AND DESIGN

**Classes: 12**

**BASIC COMPUTER ORGANIZATION AND DESIGN:** Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control, Memory-Reference Instructions, Input-Output and interrupt. Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC) Reduced Instruction Set Computer (RISC), CISC vs RISC

### UNIT-III REGISTER TRANSFER AND MICRO-OPERATIONS

**Classes: 12**

**REGISTER TRANSFER AND MICRO-OPERATIONS:** Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit.

**MICRO-PROGRAMMED CONTROL:** Control Memory, Address Sequencing, Micro-Program, Design of Control Unit

### UNIT-IV MEMORY SYSTEM

**Classes: 10**

**MEMORY SYSTEM:** Memory Hierarchy, Semiconductor Memories, RAM(Random Access Memory), Read Only Memory (ROM), Types of ROM, Cache Memory, Performance considerations, Virtual

memory, Paging, Secondary Storage, RAID.

## **UNIT-V     INPUT OUTPUT**

**Classes:10**

**INPUT OUTPUT:** I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA.

**MULTIPROCESSORS:** Characteristics of multiprocessors, Interconnection structures, Inter Processor Arbitration, Inter processor Communication and Synchronization, Cache Coherence

### **Text Books:**

1. M. Morris Mano (2006), Computer System Architecture, 3rd edition, Pearson/PHI, India.
2. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill

### **Reference Books:**

1. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition, McGraw Hill, New Delhi, India.
2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey.
3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc,



## OBJECT ORIENTED PROGRAMMING

### II Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |           |
|-------------|----------|--------------|---|---|---------|---------------|-----|-----------|
|             |          | L            | T | P |         | C             | CIE | SEE Total |
| A4CS09      | PCC      | 3            | 1 | - | 4       |               | 30  | 70 100    |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance concepts for reusing the program.
3. Demonstrate on the multi-tasking by using multiple threads.
4. Develop data-centric applications using JDBC.
5. Understand the basics of java console and GUI based programming

### COURSE OUTCOMES:

At the end of the course students will be able to:

1. Use object oriented programming concepts to solve real world problems.
2. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
3. Use multithreading concepts to develop inter process communication.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).
5. Build the internet-based dynamic applications using the concept of applets.

### UNIT-I JAVA BASICS

Classes: 09

**JAVA BASICS:** Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method String and String Buffer Classes, Using Java API Document.

### UNIT-II INHERITANCE , INTERFACES AND PACKAGES

Classes: 09

**INHERITANCE AND POLYMORPHISM:** Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Dynamic method dispatch, Usage of final keyword.

**PACKAGES AND INTERFACES:** Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces.

**I / O STREAMS:** Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, File Handling.

### UNIT-III EXCEPTION HANDLING AND MULTITHREADING

Classes: 09

**EXCEPTION HANDLING:** Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes. **MULTI THREADING:** Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication.

**UNIT-IV FILES ,COLLECTION FRAMEWORK AND JDBC****Classes: 09**

**Collection Framework in Java** - Introduction to Java Collections, Overview of Java Collection framework, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calendar and Properties

**Connecting to Database** - JDBC Type 1 to 4 drives, connecting to a database, querying a database and processing the results, updating data with JDBC.

**UNIT-V EVENT HANDLING AND APPLETS****Classes: 09**

**Event handling** - Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model,

Examples: handling a button click, handling mouse events, Adapter classes.

**Applets** - Inheritance hierarchy for applets, differences between applets and applications, life cycle of an applet, passing parameters to applets, applet security issues.

**Text Books:**

1. Herbert Schildt and Dale Skrien, "Java Fundamentals – A comprehensive Introduction", McGraw Hill, 1<sup>st</sup> Edition, 2013.
2. Herbert Schildt, "Java the complete reference", McGraw Hill, Osborne, 7<sup>th</sup> Edition, 2011.
3. T.Budd, "Understanding Object- Oriented Programming with Java", Pearson Education, Updated Edition (New Java 2 Coverage), 1999.

**Reference Books:**

1. P.J.Dietel and H.M.Dietel , "Java How to program", Prentice Hall, 6<sup>th</sup> Edition, 2005.
2. P.Radha Krishna , "Object Oriented programming through Java", CRC Press, 1<sup>st</sup> Edition, 2007.
3. S.Malhotra and S. Choudhary, "Programming in Java", Oxford University Press, 2<sup>nd</sup> Edition, 2014 .

**Web References:**

1. <https://www.geeksforgeeks.org/java/>
2. <https://stackify.com/oops-concepts-in-java/>
3. [https://www.tutorialspoint.com/java/java\\_overview.htm](https://www.tutorialspoint.com/java/java_overview.htm)
4. <https://www.javatpoint.com/java-oops-concepts>

**E-Text Books:**

1. <http://iiti.ac.in/people/~tanimad/JavaTheCompleteReference.pdf>
2. [www.portcity.edu.bd/ELibrary/CSE/javaprogramming.pdf](http://www.portcity.edu.bd/ELibrary/CSE/javaprogramming.pdf)
3. <https://www.pdfdrive.net/head-first-javapdf-e18943750.html>
4. <https://www.amazon.com/Introduction-Java-Programming-Comprehensive-International>

**MOOC Course:**

1. <https://www.mooc-list.com/tags/java>
2. <https://www.class-central.com/tag/java>
3. <https://www.quora.com/What-are-the-best-MOOCs-for-learning-Java>

## DESIGN AND ANALYSIS OF ALGORITHMS

**II Year II Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS10      | PCC      | 3            | 1 | - | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To demonstrate performance of algorithms with respect to time and space complexity.
2. To explain graph and tree traversals.
3. To explain the concepts greedy method and dynamic programming. Applying for several applications like knapsack problem, job sequencing with deadlines, and optimal binary search tree, TSP and so on respectively.
4. To illustrate the methods of backtracking and branch bound techniques to solve the problems like n-queens problem, graph colouring and TSP respectively.
5. To familiarize the concepts of deterministic and non-deterministic algorithms.

### COURSE OUTCOMES:

At the end of this course students will be able to:

1. Identify various Time and Space complexities of various algorithms
2. Understand Tree Traversal method and Greedy Algorithms
3. Apply Dynamic Programming concept to solve various problems
4. Apply Backtracking, Branch and Bound concept to solve various problems
5. Implement different performance analysis methods for non deterministic algorithms

### UNIT-I INTRODUCTION

**Classes: 10**

Characteristics of algorithm. Analysis of algorithm: Asymptotic analysis of complexity bounds – best, average and worst-case behavior; Performance measurements of Algorithm, Time and space trade-offs, Analysis of recursive algorithms through recurrence relations: Substitution method, Recursion tree method and Masters' theorem.

### UNIT-II FUNDAMENTAL ALGORITHMIC STRATEGIES – Part I

**Classes: 12**

**DIVIDE AND CONQUER:** General method, applications-analysis of binary search, quick sort, merge sort, AND OR Graphs.

**GREEDY METHOD:** Heuristics –characteristics, Applications-job sequencing with deadlines, 0/1 knapsack problem, minimum cost spanning trees, Single source shortest path problem, Bin Packing.

### UNIT-III FUNDAMENTAL ALGORITHMIC STRATEGIES – Part II

**Classes: 16**

**DYNAMIC PROGRAMMING:** General method, applications - optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem, Reliability design, Bin Packing.

**BACKTRACKING:** Heuristics –characteristics, Applications- n-queen problem, Sum of subsets problem, Graph coloring, 0/1knapsack problem, and Hamiltonian cycles.

**BRANCH AND BOUND:** General method, applications - travelling sales person problem, 0/1 knapsack problem- LC branch and bound solution, FIFO branch and bound solution.

#### UNIT-IV GRAPH AND TREE ALGORITHMS

Classes: 12

GRAPHS (Algorithm and Analysis): Breadth first search and traversal, Depth first search and traversal, Spanning trees, connected components and bi-connected components, Articulation points, Shortest path algorithms, Transitive closure, Topological sorting, Network Flow Algorithm.

#### UNIT-V TRACTABLE AND INTRACTABLE PROBLEMS

Classes: 10

Computability of Algorithms, Computability classes – P, NP, NP-complete and NP-hard. Cook's theorem, Standard NP-complete problems and Reduction techniques, Approximation algorithms, Randomized algorithms, Class of problems beyond NP – P SPACE.

#### Text Books:

1. Introduction to Algorithms, 4TH Edition, Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, MIT Press/McGraw-Hill.
2. Fundamentals of Algorithms – E. Horowitz et al.

#### Reference Books:

1. Algorithm Design, 1ST Edition, Jon Kleinberg and ÉvaTardos, Pearson.
2. Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Michael T Goodrich and Roberto Tamassia, Wiley.
3. Algorithms -- A Creative Approach, 3RD Edition, UdiManber, Addison-Wesley, Reading, MA.

#### Web References:

1. <https://www.hackerrank.com/domains/algorithms>
2. <https://discuss.codechef.com/questions/48877/data-structures-and-algorithms>
3. <http://openclassroom.stanford.edu/MainFolder/CoursePage.php?course=IntroToAlgorithms>
4. [https://www.tutorialspoint.com/design\\_and\\_analysis\\_of\\_algorithms/design\\_and\\_analysis\\_of\\_algorithms\\_tutorial.pdf](https://www.tutorialspoint.com/design_and_analysis_of_algorithms/design_and_analysis_of_algorithms_tutorial.pdf)
5. <http://nptel.ac.in/courses/106101060/>

#### E-Text Books:

1. <http://www.trips-to-morocco.com/introduction-to-algorithms-3rd-edition-mit-press-english.pdf>
2. <https://comsci.files.wordpress.com/2015/12/horowitz-and-sahani-fundamentals-of-computer-algorithms-2nd-edition.pdf>
3. [https://doc.lagout.org/science/0\\_Computer%20Science/2\\_Algorithms/Algorithm%20Design\\_%20Foundations%2C%20Analysis%2C%20and%20Internet%20Examples%20%5BGoodrich%20%26%20Tamassia%202001%5D.pdf](https://doc.lagout.org/science/0_Computer%20Science/2_Algorithms/Algorithm%20Design_%20Foundations%2C%20Analysis%2C%20and%20Internet%20Examples%20%5BGoodrich%20%26%20Tamassia%202001%5D.pdf)

#### MOOC Course:

1. [https://onlinecourses.nptel.ac.in/noc17\\_cs27/preview](https://onlinecourses.nptel.ac.in/noc17_cs27/preview)
2. <https://www.coursera.org/courses?languages=en&query=Algorithm+design+and+analysis>

## BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

### II Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS06      | HSMC     | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

### COURSE OUTCOMES:

At the end of the course, the student will

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Develop an understanding of
4. Analyze how capital budgeting decisions are carried out.
5. Understanding the framework for both manual and computerized accounting process
6. Know how to analyze and interpret the financial statements through ratio analysis.

### UNIT-I Introduction & Demand Analysis:

**Classes: 10**

Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

### UNIT-II Production & Cost Analysis:

**Classes: 10**

Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.

### UNIT-III Markets & New Economic Environment

**Classes: 12**

Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.

#### **UNIT-IV     Capital Budgeting:**

**Classes: 14**

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).

#### **UNIT-V     Introduction to Financial Accounting & Financial Analysis:**

**Classes: 14**

Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.

#### **Text Books:**

1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

#### **Reference Books:**

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey & Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

## ELECTRONIC DEVICES AND DIGITAL LOGIC DESIGN LAB

### II Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4EC36      | ESC      | -            | - | 3 | 1.5     | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. Design the semiconductor devices based circuits and analyze its parameters
2. Design of oscillators and amplifiers
3. Study of logic gates and verify their truth tables.
4. Implementation of the given Boolean function using logic gates.
5. Verification of K-Map.
6. Design the combinational and sequential circuits.

#### COURSE OUTCOMES:

The course should enable the students to:

1. Calculate various parameters of devices from their characteristics.
2. Understanding of the fundamental concepts and techniques used in digital electronics.
3. To understand and examine the structure of various number systems and its application in digital design.
4. The ability to understand, analyze and design various combinational and sequential circuits.
5. Ability to identify basic requirements for a design application and propose a cost effective solution.
6. To develop skill to build, and troubleshoot digital circuits.

### LIST OF EXPERIMENTS

#### Week-1

1. Characteristics of PN Junction diode
2. Characteristics of Zener diode
3. Ripple Factor and Load Regulations of Half-wave Rectifier with and without filters
4. Ripple Factor and Load Regulations of Full-wave Rectifier with and without filters
5. Input and Output characteristics of Transistor in Common Emitter configuration

#### Week-2

1. Ripple Factor and Load Regulations of Half-wave Rectifier with and without filters
2. Ripple Factor and Load Regulations of Full-wave Rectifier with and without filters

#### Week-3

1. Input and Output characteristics of Transistor in Common Emitter configuration

#### Week-4

1. Drain and Transfer Characteristics of Junction Field Effect Transistor (JFET)

#### Week-5

1. Gain and Frequency response of Common Emitter Amplifier

#### Week-6

1. Gain and Frequency response of Feedback Amplifier (Voltage series or current series)

#### Week-7

1. Hartley and Colpitts Oscillator

#### Week-8

1. RC phase shift Oscillator

**Week-9**

1. Realization of Logic gates using discrete components
2. Binary Adders ,

**Week-10**

1. Subtractors and Comparators

**Week-11**

1. Multiplexers
2. Decoders

**Week-12**

1. Flip-Flops

**Week-13**

1. Counters

**Week-14**

1. Shift Registers

**Reference Books:**

1. R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition, 2009.
2. W.H. Gothmann, "Digital Electronics- An introduction to theory and practice", PHI, 2nd edition ,2006.
3. D.V. Hall, "Digital Circuits and Systems", Tata McGraw Hill, 1989

**Web References:**

1. <https://www.youtube.com/watch?v=M0mx8S05v60&list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm>
2. <https://www.youtube.com/watch?v=CeD2L6KbtVM&list=PL018B3BB2E6FE781D>



## OBJECT ORIENTED PROGRAMMING LAB

### II Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS11      | PCC      | -            | - | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Practice object-oriented programs and build java applications.
2. Implement java programs for establishing interfaces.
3. Implement sample programs for developing reusable software components.
4. Create database connectivity in java and implement GUI applications.

### COURSE OUTCOMES:

The course should enable the students to:

1. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
2. Understand the use of different exception handling mechanisms and concept of multithreading for robust and efficient application development.
3. Understand and implement concepts on file streams and operations in java programming for a given application programs.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

## LIST OF EXPERIMENTS

### Week-1 BASIC PROGRAMS

- a. Try debug step by step with small program of about 10 to 15 lines which contains at least one if else condition and a for loop.
- b. Write a java program that prints all real solutions to the quadratic equation  $ax^2+bx+c=0$ . Read in a, b, c and use the quadratic formula.
- c. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non recursive functions.

### Week-2 MATRICES, OVERLOADING, OVERRIDING

- a. Write a java program to multiply two given matrices.
- b. Write a java program to implement method overloading and constructors overloading.
- c. Write a java program to implement method overriding.

### Week-3 PALINDROME, ABSTRACT CLASS

- a. Write a java program to check whether a given string is palindrome.
- b. Write a java program for sorting a given list of names in ascending order.
- c. Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes only the method print Area () that prints the area of the given shape.

**Week-4    INTERFACE**

Write a program that creates a user interface to perform integer division. The user enters two numbers in the text fields, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception Display the exception in a message dialog box.

**Week-5    MULTITHREADING**

- a. Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number.
- b. Write a java program that correct implements of producer consumer program.

**Week-6    FILES**

- a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.
- b. Write a java program that displays the number of characters, lines and words in a text file.
- c. Write a java program that reads a file and displays the file on the screen with line number before each line.

**Week-7    FILES**

- a. Suppose that table named table.txt is stored in a text file. The first line in the file is the header, and the remaining lines correspond to rows in the table. The elements are separated by commas. Write a java program to display the table using labels in grid layout.
- b. Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.

**Week-8    JAVA PROGRAM WITH DATABASE**

- a. Write a java program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (/t). It takes a name or phone number as input and prints the corresponding other value from the hash table. **Hint: Use hash tables.**
- b. Implement the above program with database instead of a text file.

**Week-9    FILES**

- a. Write a java program that takes tab separated data (one record per line) from a text file and insert them into a database.
- b. Write a java program that prints the metadata of a given table.

**Week-10    LINKED LIST**

- a. Write a Java program for the following:
  - i. Create a doubly linked list of elements.
  - ii. Delete a given element from the above list.
  - iii. Display the contents of the list after deletion.

**Week-11      EVENTS**

- a. Write a java program that handles all mouse events and shows the event name at the center of the window when a mouse event is fired. Use adapter classes.
- b. Write a java program to demonstrate the key event handlers.

**Week-12      APPLET**

- a. Develop an applet that displays a simple message
- b. Develop an applet to perform integer addition. Pass Parameters from applet

**Reference Books:**

1. P. J. Deitel, H. M. Deitel, "Java for Programmers", Pearson Education, PHI, 4<sup>th</sup> Edition, 2007.
2. P. Radha Krishna, "Object Oriented Programming through Java", Universities Press, 2<sup>nd</sup> Edition, 2007
3. Bruce Eckel, "Thinking in Java", Pearson Education, 4<sup>th</sup> Edition, 2006.
4. Sachin Malhotra, Saurabh Chaudhary, "Programming in Java", Oxford University Press, 5<sup>th</sup> Edition, 2010.

**Web References:**

1. [http://vlab.co.in/ba\\_labs\\_all.php?id=2](http://vlab.co.in/ba_labs_all.php?id=2)
2. <http://www.javatpoint.com/java-programs>
3. <http://introcs.cs.princeton.edu/java/10elements/>

## GENDER SENSITIZATION

**IVSemester: Common to all branches**

| Course Code | Category | Hours / Week |    |    | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|----|----|---------|---------------|-----|-------|
|             |          | L            | T  | P  |         | CIE           | SEE | Total |
| A4MC02      | MC       | 2            | -- | -- | 0       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To provide a critical perspective on the socialization of men and women.
2. To introduce students to information about some key biological aspects of genders. To expose the students to debates on the politics and economics of work.
3. To help students reflect critically on gender violence.
4. To expose students to more egalitarian interactions between men and women.

### COURSE OUTCOMES:

1. Students will have developed a better understanding of important issues related to gender in contemporary India.
2. Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film.
3. Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.
4. Students will acquire insight into the gendered division of labour and its relation to politics and economics.
5. Men and women students and professionals will be better equipped to work and live together as equals.
6. Students will develop a sense of appreciation of women in all walks of life.
7. Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

### UNIT-I UNDERSTANDING GENDER

**Classes: 03**

**Gender:** Why Should We Study It? (*Towards a World of Equals*: Unit -1) **Socialization:** Making Women, Making Men (*Towards a World of Equals*: Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.

### UNIT-II GENDER AND BIOLOGY

**Classes: 03**

**Missing Women:** Sex Selection and Its Consequences (*Towards a World of Equals*: Unit -4) Declining Sex Ratio. Demographic Consequences.

**Gender Spectrum:** Beyond the Binary (*Towards a World of Equals*: Unit -10) Two or Many? Struggles with Discrimination.

### UNIT-III GENDER AND LABOUR

**Classes: 03**

**Housework:** the Invisible Labour (*Towards a World of Equals*: Unit -3) "My Mother doesn't Work." "Share the Load."

**Women's Work:** Its Politics and Economics (*Towards a World of Equals*: Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.

### UNIT-IV ISSUES OF VIOLENCE

**Classes: 04**

**Sexual Harassment:** Say No! (*Towards a World of Equals*: Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading:

“Chupulu”.

**Domestic Violence:** Speaking Out (*Towards a World of Equals*: Unit -8)

Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading: New Forums for Justice.

Thinking about Sexual Violence (*Towards a World of Equals*: Unit -11)

Blaming the Victim-“I Fought for my Life....” - Additional Reading: The Caste Face of Violence.

## **UNIT-V GENDER: CO - EXISTENCE**

**Classes: 03**

**Just Relationships:** Being Together as Equals (*Towards a World of Equals*: Unit -12)

Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers.

Additional Reading: Rosa Parks-The Brave Heart.

### **Text Books:**

“*Towards a World of Equals: A Bilingual Textbook on Gender*”, A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by **Telugu Akademi, Hyderabad, Telangana State, 2015.**

### **Reference Books:**

1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012
2. Abdulali Sohaila. “*I Fought For My Life...and Won.*” Available online at:  
<http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/>

### **Web references:**

1. <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/>

## **III B.TECH I SEMESTER SYLLABUS**

## PYTHON PROGRAMMING

### III Year I Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT01      | PCC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To Describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methods through use of these in the context of a Python application.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

### UNIT – I INTRODUCTION

Classes: 10

Introduction, python overview. Basic data types. Control Flows, Tuples, Lists More advanced data types (dictionary, string), file I/O, functions

### UNIT – II MODULE AND PACKAGES

Classes: 10

Module and Packages, Object oriented programming, multi-processing and multi-threading

### UNIT – III SERIALIZATION WITH PICKLE

Classes: 15

Serialization with pickle, Network programming with python, debugging with pdb, python unit testing, Exceptions, sorting, advanced function: map, filter, and reduce, Introduction to standard libraries (pandas, Turtle, numpy, os).

### UNIT – IV NETWORK PROGRAMMING WITH PYTHON

Classes: 12

Serialization with pickle, Network programming with python, debugging with pdb, python unit testing

**UNIT – V****SELECTED TOPICS****Classes: 12**

**Selected topics:** DB programming (Basic operation and connectivity), Web development (Django with python), Performance optimizations.

**Text Books:**

1. Learning Python, by Shroff Pub & Dist., O'relly publications, Publication Year: 2013

**Reference Books:**

1. Python Programming for Beginners: Python Programming Language by Joseph Joyner



## AUTOMATA AND COMPILER DESIGN

### III Year I Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT02      | PCC      | 3            | 1 | 0 | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Construct an insight of fundamental principles of designing compilers
2. Illustrating different phases of compilation.
3. Describe the steps and algorithms used by language translators and features.
4. Enumerating top down and bottom up parsing techniques used in compilation process.
5. Learning the effectiveness of optimization.
6. Introducing the syntax directed translation and type checking.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the concept of abstract machines and their power to recognize the languages.
2. Analyze phases of compilation, particularly lexical analysis, parsing, semantic analysis and code generation.
3. Construct parsing tables for different types of parsing techniques and syntax directed translations.
4. Apply code optimization techniques to different programming languages.
5. Generate object code for natural language representations.

### UNIT – I      FORMAL LANGUAGE AND REGULAR EXPRESSIONS      Classes: 08

**Formal Language and Regular Expressions:** Languages, Definition: regular expressions, Finite Automata-DFA, NFA. Conversion of regular expression to NFA, Conversion of NFA to DFA, Equivalence of NFA and DFA.

### UNIT – II      INTRODUCTION TO COMPILERS      Classes: 15

**INTRODUCTION TO COMPILERS:** Definition of compiler, interpreter and its differences, the phases of a compiler, pass and phases of translation, bootstrapping, LEX-lexical analyzer generator.

**PARSING:** Parsing, role of parser, context free grammar, derivations, parse trees, ambiguity, elimination of left recursion, left factoring, classes of parsing, top down parsing - LL(1) grammars.

### UNIT – III      BOTTOM UP PARSING      Classes: 11

**BOTTOM UP PARSING:** Definition of bottom up parsing, handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing, LR grammars, LR parsers-simple LR, canonical LR (CLR) and Look Ahead LR (LALR) parsers, YACC-automatic parser generator.

**UNIT – IV      SYNTAX DIRECTED TRANSLATION****Classes: 12**

**SYNTAX DIRECTED TRANSLATION:** Syntax directed definition, construction of syntax trees, S-Attributed and L-Attributed definitions.

**INTERMEDIATE CODE GENERATION:** intermediate forms of source programs– abstract syntax tree, polish notation and three address code, types of three address statements and its implementation, syntax directed translation into three-address code, translation of simple statements.

**TYPE CHECKING:** Definition of type checking, type expressions, type systems, static and dynamic checking of types, specification of a simple type checker, equivalence of type expressions, type conversions.

**UNIT – V      RUN TIME ENVIRONMENTS****Classes: 12**

**RUN TIME ENVIRONMENTS:** Source language issues, Storage organization, storage-allocation Strategies, parameter passing, and symbol tables.

**CODE OPTIMIZATION:** basic blocks and flow graphs, optimization of basic blocks, principal sources of optimization, directed a cyclic graph (DAG) representation of basic block.

**CODE GENERATION:** Machine dependent code generation, object code forms, peephole optimization.

**Text Books:**

1. K. L. P Mishra, N. Chandrashekar (2003), Theory of computer science- Automata Languages and computation, 2nd edition, Prentice Hall of India, New Delhi, India.
2. Compilers Principles, Techniques and Tools Aho, Ullman, Ravisethi, Pearson Education.

**Reference Books:**

1. Introduction to Theory of computation. Sipser, 2nd Edition, Thomson., 2009.
2. Modern Compiler Construction in C, Andrew W. Appel Cambridge University Press.
3. Kenneth C. Loudon (1997), Compiler Construction– Principles and Practice, 1st edition, PWS Publishing.
4. Elements of Compiler Design, A. Meduna, Auerbach Publications, Taylor and Francis Group.
5. Principles of Compiler Design, V. Raghavan, TMH.

## OPERATING SYSTEMS

### III Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS13      | PCC      | 3            | 1 | 0 | 4       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

To learn

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as a resource Manager
3. To impart various scheduling policies of OS
4. To teach the different memory management techniques.

#### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify and analyze the different structures and services of operating system.
2. Compare various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
5. Analyze the memory management approaches of operating systems.

#### UNIT – I OPERATING SYSTEMS OVERVIEW

Classes: 12

**OPERATING SYSTEMS OVERVIEW:** Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems.

**OPERATING SYSTEMS STRUCTURES:** Operating system services and systems calls, system programs, operating system structure, operating systems generations.

#### UNIT – II PROCESS MANAGEMENT

Classes: 12

**PROCESS MANAGEMENT:** Process concepts, process state, process control block, scheduling queues, process scheduling, multithreaded programming, threads in UNIX, comparison of UNIX and windows.

**CONCURRENCY AND SYNCHRONIZATION:** Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris), atomic transactions, Comparison of UNIX and windows.

#### UNIT – III DEADLOCKS

Classes: 12

**DEADLOCKS:** System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

**MEMORY MANAGEMENT:** Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing, case study - UNIX.

**UNIT – IV      FILE SYSTEM****Classes: 12**

**FILE SYSTEM:** Concept of a file, access methods, directory structure, file system mounting, file sharing, protection. **File system implementation:** file system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance, comparison of UNIX and windows.

**UNIT – V      MASS STORAGE STRUCTURE****Classes: 10**

**MASS STORAGE STRUCTURE:** overview of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure.

**I/O:** Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance.

**Text Books:**

1. Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), Operating System Principles, 7th edition, Wiley India Private Limited, New Delhi.

**Reference Books:**

1. Stallings (2006), Operating Systems, Internals and Design Principles, 5th edition, Pearson Education, India.
2. Andrew S. Tanenbaum (2007), Modern Operating Systems, 2nd edition, Prentice Hall of India, India
3. Deitel & Deitel (2008), Operating systems, 3rd edition, Pearson Education, India.

## ARTIFICIAL INTELLIGENCE (PROFESSIONAL ELECTIVE – 1)

**III Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS21      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To learn the difference between optimal reasoning vs human like reasoning
2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. To learn different knowledge representation techniques
4. To understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Possess the ability to formulate an efficient problem space for a problem expressed in English.
2. Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
3. Possess the skill for representing knowledge using the appropriate technique
4. Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing

### UNIT – I

**Classes: 12**

**Introduction:** History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A\*, Constraint Satisfaction, Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning

### UNIT – II

**Classes: 12**

**Logic Concepts and Logic Programming:** Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming.

**Knowledge Representation:** Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.

### UNIT – III

**Classes: 12**

**Expert System and Applications:** Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools.

**Uncertainty Measure** – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster - Shafer Theory.

#### UNIT – IV

**Classes: 12**

**Machine-Learning Paradigms:** Introduction, Machine Learning Systems, Supervised and Unsupervised Learning, Inductive Learning, Learning Decision Trees, Deductive Learning, Clustering, Support Vector Machines.

**Artificial Neural Networks:** Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.

#### UNIT – V

**Classes: 09**

**Advanced Knowledge Representation Techniques:** Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, and Universal Networking Knowledge.

#### Text Books:

1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011
2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004

#### Reference Books:

1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009.
2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson.
3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI.
4. Artificial Intelligence by George Fluger Pearson fifth edition.

## PRINCIPLES OF PROGRAMMING LANGUAGE (PROFESSIONAL ELECTIVE – 1)

**III Year I Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT06      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To familiarize various programming methodologies
2. To explain merits and demerits of various programming patterns
3. To gain knowledge of new programming practices
4. To familiarize various debugging techniques.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe syntax and semantics of programming language design and implementation.
2. Compare programming languages and assess programming languages.
3. Use formal description for a programming language and assess using evaluators.
4. Apply different programming paradigms: analyze the principles of imperative, object-oriented, functional and logic programming.
5. Design a new programming language in principle.

### UNIT – I

**Classes: 12**

**Preliminary Concepts:** concepts of programming languages, Programming domains, Language Evaluation Criteria, influences on Language design, Language categories, Programming Paradigms – Imperative, Object Oriented, functional Programming, Logic Programming, Programming Language Implementation – Compilation and Virtual Machines, programming environments.

### UNIT – II

**Classes: 08**

**Syntax and Semantics:** General Problem of describing Syntax and Semantics, formal methods of describing syntax - BNF, EBNF for common programming languages features, parse trees, ambiguous grammars, attribute grammars, denotation semantics and axiomatic semantics for common programming language features.

### UNIT – III

**Classes: 18**

**Data types:** Introduction, primitive, character, user defined, array, associative, record, union, pointer and reference types, design and implementation uses related to these types, Names, Variable, concept of binding, type checking, strong typing, type compatibility, named constants, variable initialization.

**UNIT – IV****Classes: 09**

**Expressions and Statements:** Arithmetic relational and Boolean expressions, Short circuit evaluation mixed mode assignment, Assignment Statements, Control Structures – Statement Level, Compound Statements, Selection, Iteration, Unconditional Statements, guarded commands.

**UNIT – V****Classes: 12**

**Subprograms and Blocks:** Fundamentals of sub-programs, Scope and lifetime of variable, static and dynamic scope, Design issues of subprograms and operations, local referencing environments, parameter passing methods, overloaded sub-programs, generic sub-programs, parameters that are sub-program names, design issues for functions user defined overloaded operators, co routines.

**Text Books:**

1. Concepts of Programming Languages Robert .W. Sebesta 6/e, Pearson Education.
2. Programming Languages – Loudon, Second Edition, Thomson.

**Reference Books:**

1. Programming languages –Ghezzi, 3/e, John Wiley
2. Programming Languages Design and Implementation – Pratt and Zelkowitz, Fourth Edition  
PHI/Pearson Education
3. Programming languages –Watt, Wiley Dreamtech
4. LISP Patric Henry Winston and Paul Horn Pearson Education.
5. Programming in PROLOG Clocksin, Springer



## DISTRIBUTED DATABASES (PROFESSIONAL ELECTIVE – 1)

**III Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS22      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To understand the theoretical and practical aspects of the database technologies.
2. To understand the need for distributed database technology to tackle deficiencies of the centralized database systems.
3. To introduce the concepts and techniques of distributed database including principles, architectures, design, implementation and major domain of application.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the introductory distributed database concepts and its structures.
2. Describe terms related to distributed object database design and management.
3. Produce the transaction management and query processing techniques in DDBMS.
4. Relate the importance and application of emerging database technology.

**Pre-requisites:** Basic concept of DBMS and RDBMS.

### UNIT – I

**Classes: 12**

**Introduction:** Features of Distributed versus Centralized Databases, Principles of Distributed Databases, Levels Of Distribution Transparency, Reference Architecture for Distributed Databases, Types of Data Fragmentation, Integrity Constraints in Distributed Databases, Distributed Database Design

### UNIT – II

**Classes: 14**

**Query Processing:** Translation of Global Queries to Fragment Queries, Equivalence transformations for Queries, Transforming Global Queries into Fragment Queries, Distributed Grouping and Aggregate Function Evaluation, Parametric Queries. Optimization of Access Strategies, A Framework for Query Optimization, Join Queries, General Queries

### UNIT – III

**Classes: 14**

**Transaction Management and Concurrency control:** The Management of Distributed Transactions, A Framework for Transaction Management, Supporting Atomicity of Distributed Transactions, Concurrency Control for Distributed Transactions, Architectural Aspects of Distributed Transactions, Concurrency Control, Foundation of Distributed Concurrency Control, Distributed Deadlocks, Concurrency Control based on Timestamps, Optimistic Methods for Distributed Concurrency Control.

**UNIT – IV****Classes: 12**

**Reliability And Security In The Distributed Databases:** Reliability, Basic Concepts, Non blocking Commitment Protocols, Reliability and concurrency Control, Determining a Consistent View of the Network, Detection and Resolution of Inconsistency, Checkpoints and Cold Restart, Distributed Database Administration, Catalog Management in Distributed Databases, Authorization and Protection

**UNIT – V****Classes: 12**

**Distributed Object Database Management Systems:** Architectural Issues, Alternative Client/Server Architectures, Cache Consistency, Object Management, Object Identifier Management, Pointer Swizzling, Object Migration, Distributed Object Storage, Object Query Processing, Object Query Processor Architectures, Query Processing Issues, Query Execution, Transaction Management, Transaction Management in Object DBMSs, Transactions as Objects

**Text Books:**

1. Distributed Databases - Principles and Systems; Stefano Ceri; Guiseppe Pelagatti; Tata McGraw Hill; 1985.
2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia Database System Concepts; Korth & Sudarshan; TMH
3. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall.

**Reference Books:**

1. Data Base Management System; Leon & Leon; Vikas Publications
2. Introduction to Database Systems; Bipin C Desai; Galgotia

## HUMAN COMPUTER INTERACTION (PROFESSIONAL ELECTIVE – 1)

**III Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A4CS23      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain Computer components functions regarding interaction with human
3. Demonstrate Understanding of Interaction between the human and computer components.
4. Use Paradigms
5. Implement Interaction design basics
6. Use HCI in the software process

### UNIT – I

**Classes: 11**

**Introduction:** Importance of user Interface – definition, importance of good design, Benefits of good design, A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics- Principles of user interface.

### UNIT – II

**Classes: 16**

**Design process** – Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, and understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.

**UNIT – III****Classes: 07**

**Screen Designing:** Windows – New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.

**UNIT – IV****Classes: 04**

**Windows & Components:** Software tools – Specification methods, interface – Building Tools.

**UNIT – V****Classes: 08**

**Software tools:** Interaction Devices – Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.

**Text Books:**

1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech.
2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.

**Reference Books:**

1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEALG, PEARSON.
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

## PYTHON PROGRAMMING LAB

### III Year I Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT03      | PCC      | 0            | 0 | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Enabling Knowledge: effectively apply knowledge of Python to new situations. Critical Analysis: examine and consider accurately and objectively any topic, evidence, or situation.
2. Analyze requirements of software systems for the purpose of determining the suitability of implementing in Python;
3. Analyze and model requirements and constraints for the purpose of designing and implementing software systems in Python;
4. Evaluate and compare designs of such systems on the basis of specific requirements and constraints.
5. Problem Solving: analyze problems and synthesize suitable solutions.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

## LIST OF EXPERIMENTS

### Week 1:

A) Write a Python program to print the following string in a specific format (see the output).

*Sample String* : "Twinkle, twinkle, little star, How I wonder what you are! Up above the world so high, Like a

diamond in the sky. Twinkle, twinkle, little star, How I wonder what you are"

*Output :*

```
Twinkle, twinkle, little star,
How I wonder what you are!
Up above the world so high,
Like a diamond in the sky.
Twinkle, twinkle, little star,
How I wonder what you are
```

- B) Write a Python program to get the Python version you are using.

### Week 2:

- A) Write a Python program to display the current date and time.

*Sample Output:*

Current date and time : 2014-07-05 14:34:14

- B) Write a Python program which accepts the radius of a circle from the user and compute the area.

### Week 3:

- A) Write a Python program which accepts the user's first and last name and print them in reverse order with a space between them.

- B) Write a Python program which accepts a sequence of comma-separated numbers from user and

generate a list and a tuple with those numbers.

*Sample data :* 3, 5, 7, 23

*Output :*

List : ['3', '5', '7', '23']

Tuple : ('3', '5', '7', '23')

### Week 4:

- A) Write a Python program to accept a filename from the user and print the extension of that.

*Sample filename :* abc.java

*Output :* java

- B) Write a Python program to display the first and last colors from the following list.

color\_list = ["Red", "Green", "White", "Black"]

### Week 5:

- A) Write a Python program to display the examination schedule. (extract the date from exam\_st\_date).

exam\_st\_date = (11, 12, 2014)

Sample Output : The examination will start from : 11 / 12 / 2014

- B) Write a Python program that accepts an integer (n) and computes the value of n+nn+nnn.

### Week 6:

- A) Write a Python program to print the documents (syntax, description etc.) of Python built-in function(s).

- B) Write a Python program to print the calendar of a given month and year.

*Note :* Use 'calendar' module.

### Week 7:

- A) Write a Python program to print the following here document.

*Sample string :* a string that you "don't" have to escape

This is a ..... multi-line

heredoc string -----> example

- B) Write a Python program to calculate number of days between two dates.

*Sample dates :* (2014, 7, 2), (2014, 7, 11)

*Expected output :* 9 days

**Week8:**

- A) Write a Python program to get the volume of a sphere with radius 6.
- B) Write a Python program to get the difference between a given number and 17, if the number is greater than 17 return double the absolute difference.

**Week 9:**

- A) write a program to find sum of two numbers using class and methods
- B) write a program to read 3 subject marks and display pass or failed using class and object.

**Week 10:**

- A) A. Demonstrate a python code to sort the list L=[15,3,11,7,21] based on their remainder as dividing from 7?
- B) Write a python program to write the content "hi python programming" for the existing File

**Week11:**

- A) Write a program to double a given number using lambda()?
- B) Write a program for filter() to filter only even numbers from a given list.  
myList=[1,2,3,4,5,6]

**Week 12:**

- A) Write a python code to set background color and pic using turtle graphics?
- B) Write a python code to read a csv file using pandas module?

**Week 13:**

- A) Write a python code to serialize and desterialize the data by using pickle module?
- B) Write a python code to establish a communication between client and server?

**Week 14:**

- A) Write a python code to perform addition using functions with pdb module?
- B) Write a python code to insert student information into database with DB connectivity?

**Text Books:**

1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013.
2. Python Programming for Beginners: Python Programming Language by Joseph Joyner.

## ADVANCED ENGLISH COMMUNICATION SKILLS LAB

### III Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4HS07      | HSMC     | 0            | 0 | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To improve the students' fluency in English, through a well-developed vocabulary and enable them to listen to English spoken at normal conversational speed by educated English speakers and respond appropriately in different socio-cultural and professional contexts.
2. Further, they would be required to communicate their ideas relevantly and coherently in writing. To prepare all the students for their placements

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Gathering ideas and information to organize ideas relevantly and coherently.
2. Engaging in debates.
3. Participating in group discussions.
4. Facing interviews.
5. Writing project / research reports/technical reports.
6. Making oral presentations.
7. Writing formal letters.
8. Transferring information from non-verbal to verbal texts and vice-versa.
9. Taking part in social and professional communication.

### LIST OF EXPERIMENTS

The following course content to conduct the activities is prescribed for the Advanced Communication Skills (ACS) Lab:

1. Listening for writing short answers, identifying topic, context, function, etc.
2. Activities on Fundamentals of Inter-personal Communication and Building Vocabulary-Starting a conversation - responding appropriately and relevantly - using the right body language - Role Play in different situations & Discourse Skills- using visuals - Synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, business vocabulary, analogy, idioms and phrases, collocations & usage of vocabulary.
3. Activities on Reading Comprehension-General Vs Local comprehension, reading for facts, guessing meanings from context, scanning, skimming, inferring meaning, critical reading & effective googling, understanding sentence structure/ error identification.
4. Functional/Communicative Task e.g. giving instructions, explaining a development, asking for comments, requesting information, agreeing to requests Correspondence: e.g. explaining, apologizing, reassuring, and complaining. Report: describing, summarizing. Proposal: describing summarizing, recommending, persuading. Activities on Writing Skills - Structure and presentation of different types of writing - letter writing/Resume writing/ e-correspondence/ Technical report writing/ Portfolio writing - planning for writing - improving one's writing.



5. Format and Focus on Conversation between the interlocutor and each candidate-Giving personal information. Talking about present circumstances, past experiences and future plans, expressing opinions, speculating etc.-A 'mini presentation' by each candidate on a business theme- Organizing a larger unit of discourse-Giving information and expressing and justifying opinions-Two-way conversation between candidates followed by further prompting from the interlocutor -Expressing and justifying opinions, speculating, comparing and contrasting agreeing and disagreeing etc.
6. Activities on Presentation Skills - Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations through posters/projects/reports/e-mails/assignments etc.
7. Activities on Group Discussion and Interview Skills - Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process, pre-interview planning, opening strategies, answering strategies, interview through tele-conference & video-conference and Mock Interviews.

#### **Minimum Requirement:**

The Advanced Communication Skills (ACS) Laboratory shall have the following infra-structural facilities to accommodate at least 35 students in the lab:

- Spacious room with appropriate acoustics.
- Round Tables with movable chairs
- Audio-visual aids
- LCD Projector
- Public Address system
- P - IV Processor, Hard Disk - 80 GB, RAM-512 MB Minimum, Speed - 2.8 GHZ
- T. V, a digital stereo & Camcorder
- Headphones of High quality

#### **Suggested Software:**

The software consisting of the prescribed topics elaborated above should be procured and used.

- Oxford Advanced Learner's Compass, 3<sup>rd</sup> Edition
- DELTA's key to the Next Generation TOEFL Test: Advanced Skill Practice.
- Lingua TOEFL CBT Insider, by Dreamtech
- TOEFL & GRE (KAPLAN, AARCO & BARRONS, USA, Cracking GRE by CLIFFS)
- The following software from 'train2success.com'~ Preparing for being Interviewed ~ Positive Thinking ~ Interviewing Skills ~ Telephone Skills ~ Time Management

#### **Books Recommended:**

1. Technical Communication by Meenakshi Raman & Sangeeta Sharma, Oxford University Press 2009.
2. Advanced Communication Skills Laboratory Manual by Sudha Rani, D, Pearson Education 2011.
3. Technical Communication by Paul V. Anderson. 2007. Cengage Learning pvt. Ltd. New Delhi.
4. Business and Professional Communication: Keys for Workplace Excellence. Kelly M. Quintanilla & Shawn T. Wahl. Sage South Asia Edition. Sage Publications. 2011.
5. The Basics of Communication: A Relational Perspective. Steve Duck & David T. McMahan. Sage South Asia Edition. Sage Publications. 2012.
6. English Vocabulary in Use series, Cambridge University Press 2008.
7. Management Shapers Series by Universities Press(India)Pvt Ltd., Himayalnagar,

- Hyderabad 2008.
8. Handbook for Technical Communication by David A. McMurrey & Joanne Buckley. 2012. Cengage Learning.
  9. Communication Skills by Leena Sen, PHI Learning Pvt Ltd., New Delhi, 2009.
  10. Handbook for Technical Writing by David A McMurrey & Joanne Buckley CENGAGE Learning 2008.
  11. Job Hunting by Colm Downes, Cambridge University Press 2008.
  12. Master Public Speaking by Anne Nicholls, JAICO Publishing House, 2006.
  13. English for Technical Communication for Engineering Students, Aysha Vishwamohan, Tata Mc Graw-Hi 2009.
  14. Books on TOEFL/GRE/GMAT/CAT/ IELTS by Barron's/DELTA/Cambridge University Press.
  15. International English for Call Centres by Barry Tomalin and Suhashini Thomas, Macmillan Publishers, 2009.

#### **DISTRIBUTION AND WEIGHT AGE OF MARKS:**

Advanced Communication Skills Lab Practical's:

1. The practical examinations for the ACS Laboratory practice shall be conducted as per the University norms prescribed for the core engineering practical sessions
2. For the English Language lab sessions, there shall be continuous evaluation during the year for 25 sessional marks and 50 End Examination marks. Of the 25 marks, 15 marks shall be awarded for day to-day work and 10 marks to be awarded by conducting Internal Lab Test(s). The End Examination shall be conducted by the teacher concerned, by inviting the External Examiner from outside. In case of the non-availability of the External Examiner, other teacher of the same department can act as the External Examiner

**Mini Project:** As a part of Internal Evaluation.

1. Seminar/ Professional Presentation
  2. A Report on the same has to be prepared and presented
- \* Teachers may use their discretion to choose topics relevant and suitable to the needs of students.
- \* Not more than two students to work on each mini project.
- \* Students may be assessed by their performance both in oral presentation and written report.

#### **Learning Outcomes:**

1. Accomplishment of sound vocabulary and its proper use contextually.
2. Flair in Writing and felicity in written expression.
3. Enhanced job prospects.
4. Effective Speaking Abilities

## CONSTITUTION OF INDIA (MANDATORY COURSE NON-CREDIT)

**III Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4MC03      | MC       | 2            | 0 | 0 | 0       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy ,its institutions and processes at the local,state and union levels.
5. Explain the functions and responsibilities of election commission of india and union public service commission.

### UNIT – I HISTORY OF MAKING OF THE INDIAN CONSTITUTION Classes: 06

**History of Making of the Indian Constitution:** Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

### UNIT – II PHILOSOPHY OF THE INDIAN CONSTITUTION Classes: 13

**Philosophy of the Indian Constitution:** Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.

**UNIT – III****UNION GOVERNMENT****Classes: 10**

**Union Government:** Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.

**UNIT – IV****STATE GOVERNMENT****Classes: 12**

**State Government:** State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)

**UNIT – V****ELECTION COMMISSION****Classes: 09**

**Election Commission:** Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

**Text Books:**

1. M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice – Hall of India Pvt. Ltd. New Delhi
2. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012
3. The constitution of India, P.M.Bakshi, Universal Law Publishing Co.,
4. The Constitution of India, 1950 (Bare Act), Government Publication.
5. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.

**Reference Books:**

1. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014.
2. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015.
3. Indian constitution at work, NCERT
4. SubashKashyap, Indian Constitution, National Book Trust
5. J.A. Siwach, Dynamics of Indian Government & Politics
6. D.C. Gupta, Indian Government and Politics
7. H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal Law Publication)
8. J.C. Johari, Indian Government and Politics Hans

9. J. Raj Indian Government and Politics

**E-Resources:**

1. [nptel.ac.in/courses/109104074/8](https://nptel.ac.in/courses/109104074/8)
2. [nptel.ac.in/courses/109104045/](https://nptel.ac.in/courses/109104045/)
3. [nptel.ac.in/courses/101104065/](https://nptel.ac.in/courses/101104065/)
4. [www.hss.iitb.ac.in/en/lecture-details](http://www.hss.iitb.ac.in/en/lecture-details)
5. [www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution](http://www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution)

## **III B.TECH II SEMESTER SYLLABUS**

## COMPUTER NETWORKS

### III Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS15      | PCC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To introduce the fundamental s of various types of computer networks
2. To demonstrate the TCP/IP and OSI models with merits and demerits
3. To explore the various layers of OSI model
4. To introduce UDP and TCP models

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify computer networks and its components.
2. Identify the different types of network topologies and protocols.
3. Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer.
4. Select and use various sub netting and routing mechanisms.
5. Design a network diagram for a given scenario.

### UNIT – I

**Classes: 12**

**INTRODUCTION:** Network applications, network hardware, network software, reference models: OSI, TCP/IP, Internet, Connection oriented network - X.25, frame relay.

**THE PHYSICAL LAYER:** Theoretical basis for communication, guided transmission media, wireless transmission, the public switched telephone networks, mobile telephone system.

### UNIT – II

**Classes: 13**

**THE DATA LINK LAYER:** Design issues, error detection and correction, elementary data link protocols, sliding window protocols.

**THE MEDIUM ACCESS SUBLAYER:** Channel allocations problem, multiple access protocols, Ethernet, Data Link Layer switching, Wireless LAN, Broadband Wireless, Bluetooth

### UNIT – III

**Classes: 08**

**THE NETWORK LAYER:** Network layer design issues, routing algorithms, Congestion control algorithms, Internetworking, the network layer in the internet (IPv4 and IPv6), Quality of Service.

### UNIT – IV

**Classes: 08**

**THE TRANSPORT LAYER:** Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP.

**UNIT – V****Classes: 12**

**THE APPLICATION LAYER:** Domain name system, electronic mail, World Wide Web: architectural overview, dynamic web document and http.

**APPLICATION LAYER PROTOCOLS:** Simple Network Management Protocol, File Transfer Protocol, Simple Mail Transfer Protocol, Telnet.

**Text Books:**

1. Computer networks-Andrew S Tanenbaum, 4th edition, pearson education Data communication and networking-Behrouz. A. Forouzan , fifth edition, TMH, 2013

**Reference Books:**

1. Behrouz A. Forouzan (2006), Data communication and Networking, 4<sup>th</sup> Edition, Mc Graw-Hill, India.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Pearson Education, India.



## LINUX PROGRAMMING

### III Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS16      | PCC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Understand basic principles of Linux Internals.
2. To learn Linux process control and shell programming.
3. Explain the basic Methods on which the Linux kernel is built upon.
4. To facilitate students in understanding process.
5. To facilitate students in understanding Inter process communication.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand how to work with Linux commands and how to write Shell Scripts.
2. Apply fundamental knowledge of Linux Internals in Real time scenarios.
3. Demonstrate tools and interfaces to successfully develop new features of the kernel.
4. Design client server application to support communication interfaces.

### UNIT – I INTRODUCTION & LINUX UTILITIES

Classes: 10

**Linux Utilities:** File handling utilities, Security by file permissions, process utilities, Disk utilities, networking commands, filters, and text processing utilities, back up utilities.

**Sed-** scripts, operation, addresses, commands, applications, **awk-** execution, fields and records, scripts, operations, patterns, actions, functions, using system commands in awk.

**Working with bourne again shell(bash):** Introduction, shell responsibilities, pipes and redirection, output redirection, here documents, running a shell script, the shell as programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples, interrupt processing, functions, debugging shell scripts

### UNIT – II FILES

Classes: 08

**Files:** File Concept, File System Structure, Inodes, File Attributes, File types, Library functions, the standard I/O and formatted I/O in C, stream errors, kernel support for files, System calls, file descriptors, low level file access – File structure related system calls(File APIs), file and record locking, file and directory management – Directory file APIs, Symbolic links & hard links

### UNIT – III PROCESS & SIGNALS

Classes: 08

**Process:** Process concept, process identifiers, process structure, process table, viewing processes, system processes, process scheduling, Process creation: waiting for a process, process termination, zombie processes, orphan process, process APIs.

**Signals:** Introduction to signals, signal generation and handling, kernel support for signals, signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions.

### UNIT – IV IPC

Classes: 12

**Inter Process Communication:** Introduction to IPC, IPC between processes on a single computer system, IPC between processes on different systems, pipes-creation, IPC between related processes using unnamed pipes, FIFOs-creation, IPC between unrelated processing using FIFOs( named pipes), differences between unnamed and named pipes, popen and pclose library functions.

**Message queues:** kernel; support for message queues, unix system V APIs for messages, client/server example.

**Semaphores-** kernel support for semaphores, unix system V APIs for semaphores, file locking with semaphores.

## UNIT – V      SOCKETS

Classes: 10

**Sockets:** Introduction to Berkeley Sockets, IPC over a network, Client-Server model, Socket address structures (Unix domain and Internet domain), socket system calls for connection oriented protocols and connectionless protocol, Example-client/server programs-Single server-client connection, Multiple simultaneous clients, Socket options-setsocket and fcntl system calls, Comparison of IPC mechanisms

### Text Books:

1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3<sup>rd</sup> edition, Pearson Education, New Delhi, India.
2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg.Thomson

### Reference Books:

1. Linux System Programming, *Robert Love*, O'Reilly, SPD.
2. Advanced Programming in the UNIX environment, 2nd Edition, *W.R.Stevens*, Pearson Education.
3. UNIX Network Programming, *W.R. Stevens*, PHI.
4. UNIX for Programmers and Users, 3rd Edition, *Graham Glass*, *King Ables*, Pearson Education
5. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3<sup>rd</sup> Edition, 2014.

### Web References:

1. <https://www.bfoit.org/itp/Programming.html>
2. <https://www.khanacademy.org/computing/computer-programming>
3. <https://www.edx.org/course/programming-basics-iitbombayx-cs101-1x-0>
4. <https://www.edx.org/course/introduction-computer-science-harvardx-cs50x>

### E-Text Books:

1. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>
2. <http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/>
3. <http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

### MOOC Course:

1. <https://www.alison.com/courses/Introduction-to-Programming-in-c>
2. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm>

## WEB TECHNOLOGIES

### III Year II Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT04      | PCC      | 3            | 1 | 0 | 4       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To teach students the basics of server side scripting using PHP
2. To explain web application development procedures
3. To impart servlet technology for writing business logic
4. To facilitate students to connect to databases using JDBC
5. To familiarize various concepts of application development using JSP

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create web pages using PHP
2. Identify the difference between the HTML PHP and XML documents.
3. Identify the engineering structural design of XML and parse tree
4. Analyze the difference between and PHP and XML.
5. Understand the concept of JAVA SCRIPTS.
6. Identify the difference between the JSP and Servlet.
7. Design web application using MVC architecture
8. Understand the JSP and Servlet concepts.
9. Apply JDBC and ODBC technologies to create database connectivity

### UNIT – I

Classes: 08

#### CLIENT SIDE SCRIPTING

Introduction to JavaScript: JavaScript language – declaring variables, scope of variables functions, event handlers (on click, on submit etc.), Document Object Model, Form validations. Simple AJAX applications.

### UNIT – II

Classes: 10

#### XML

Introduction to XML, Defining XML tags, their attributes and values, Document type definition, XML Schemas, Document Object model, XHTML

#### Parsing XML Data

DOM and SAX parsers in java

### UNIT – III

Classes: 22

#### INTRODUCTION TO PHP

Declaring variables, data types, arrays, strings, operations, expressions, control structures, functions, Reading data from web form controls like Text Boxes, radio buttons, lists etc., Handling File Uploads, Connecting to database (My SQL as reference), executing simple queries, handling results, Handling sessions and cookies.

#### FILE HANDLING IN PHP

File operations like opening, closing, reading, writing, appending, deleting etc. on text and binary files, listing directories.

## **UNIT – IV**

**Classes: 10**

### **INTRODUCTION TO SERVLETS**

Common Gateway Interface (CGI), Lifecycle of a Servlets, deploying a Servlets, The Servlets API, Reading Servlets parameters, Reading initialization parameters, Handling Http Request & Responses, Using Cookies and sessions, connecting to a database using JDBC.

## **UNIT – V**

**Classes: 10**

### **INTRODUCTION TO JSP**

The Anatomy of a JSP Page, JSP Processing, Declarations, Directives, Expressions, Code Snippets, implicit objects, Using Beans in JSP Pages, Using Cookies and session tracking, connecting to database in JSP.

### **Text Books:**

1. Dietel and Nieto (2008), Internet and World wide Web How to Program, 4<sup>th</sup> edition, PHI/Pearson Education Asia, New Jersey.
2. Java Script step by step, Suehring Steve, PHI publications.

### **Reference Books:**

1. Head first servlet and jsp.
2. Database programming with JDBC and JAVA, O'reilly publications.
3. Beginning ASP.NET 4.5 in C# and VB.NET, IvarSpanjaars , WROX publications.
4. C. Bates(2002), Web Programming building Internet Applications, 2<sup>nd</sup> edition, WILEY Dream Tech, New Delhi, India.

## OBJECT ORIENTED ANALYSIS AND DESIGN

### (PROFESSIONAL ELECTIVE – II)

**III Year II Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT07      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

To Learn

1. Describe the activities in the different phases of the object-oriented development life cycle.
2. State the advantages of object-oriented modeling vis-à-vis structured approaches.
3. Compare and contrast the object-oriented model with the E-R and EER models.
4. Model a real-world application by using a UML class diagram.
5. Provide a snapshot of the detailed state of a system at a point in time using a UML (Unified Modeling Language) object diagram.
6. Recognize when to use generalization, aggregation, and composition relationships.
7. Specify different types of business rules in a class diagram.

#### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze the conceptual model of UML for the given system specifications.
2. Apply Rational rose tool for modeling the requirements.
3. State the advantages of object-oriented modeling vis-à-vis structured approaches.
4. Design an interface by applying various classes in Java.

#### UNIT – I

**Classes: 08**

**INTRODUCTION TO UML:** Importance of modeling, principles of modeling, object oriented modeling, conceptual model of the UML, Architecture, Software Development Life Cycle.

#### UNIT – II

**Classes: 12**

**BASIC STRUCTURAL MODELING:** Classes, Relationships, common Mechanisms, and diagrams. Advanced Structural Modeling: Advanced classes, advanced relationships, Interfaces, Types and Roles, Packages. Class & Object Diagrams: Terms, concepts, modeling techniques for Class & Object Diagrams.

#### UNIT – III

**Classes: 08**

**BASIC BEHAVIORAL MODELING-I:** Interactions, Interaction diagrams.

**BASIC BEHAVIORAL MODELING-II:** Use cases, Use case Diagrams, Activity Diagrams.

**UNIT – IV****Classes: 08**

**ADVANCED BEHAVIORAL MODELING:** Events and signals, state machines, processes and Threads, time and space, state chart diagrams.

**ARCHITECTURAL MODELING:** Component, Deployment, Component diagrams and Deployment diagrams.

**UNIT – V****Classes: 08**

Patterns and Frameworks, Artifact Diagrams. Case Study: The Unified Library application

**Text Books:**

1. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Pearson Education 2nd Edition.
2. Object-Oriented Analysis and Design with the Unified Process By John W. Satzinger, Robert B Jackson and Stephen D Burd, Cengage Learning.

**Reference Books:**

1. Meilir Page-Jones: Fundamentals of Object Oriented Design in UML, Pearson Education.
2. Pascal Roques: Modeling Software Systems Using UML2, WILEY-Dreamtech India Pvt. Ltd.
3. Atul Kahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
4. Mark Priestley: Practical Object-Oriented Design with UML, TMH.
5. Applying UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, Craig Larman, Pearson Education.

## MOBILE APPLICATION DEVELOPMENT (PROFESSIONAL ELECTIVE – II)

**III Year II Semester: CSE/IT**

| Course Code   | Category   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|---------------|------------|--------------|---|---|---------|---------------|-----|-------|
| <b>A4CS30</b> | <b>PEC</b> | L            | T | P | C       | CIE           | SEE | Total |
|               |            | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To facilitate students to understand android SDK
2. To help students to gain a basic understanding of Android application development
3. To inculcate working knowledge of Android Studio development tool

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. **Describe** the components and **compare** different mobile application models/architectures and patterns.
2. **Apply** the mobile development framework knowledge to develop an application for android devices.
3. **Use the various layout structures to design** an Android application with rich uses interactive interfaces
4. **Develop** an application that focus on user experience design, native data handling and background tasks and notification
5. **Develop** applications for Android Devices and deploy in mobile stores(ex:google playstore)

### UNIT – I

**Classes: 08**

**INTRODUCTION TO ANDROID:** Setting up Android Studio Development Environment, Creating an Example Android App in Android Studio, Creating and Configuring the Android Studio AVD Emulator, Android Architecture, The Anatomy of an Android Application

### UNIT – II

**Classes: 12**

**ANDROID VIEW, VIEWGROUP AND LAYOUTS:** Android Views: Text View, Edit Text, Button, Checkbox, Radio Button, Image Button, Progress Bar, Spinner, View Groups and Layouts: Linear Layout, Relative Layout, Table Layout, Frame Layout, Scroll View, Web View and Grid View, Event Handling.

### UNIT – III

**Classes: 10**

**ACTIVITY AND INTENTS:** Understanding Activities, Activity Lifecycle and Intents: Explicit Intent and Implicit Intents, Animating User Interfaces: Image Rotation and Interpolators, Working with Menu

**UNIT – IV****Classes: 12**

**ANDROID USER INTERFACE DESIGN ESSENTIALS:** Fragments, Android Floating Action Button, Working with AppBar Creating a Tabbed Interface, an Android List View and Recycler View, Implementing an Android Navigation Drawer, Android Broadcast Intents and Broadcast Receivers, Services

**UNIT – V****Classes: 12**

**ANDROID DATABASE, TESTING AND PUBLISHING APPS:** Android SQLite Database, Introduction to Firebase, Firebase User Authentication, reading and writing data to Firebase database, Shared Preferences, Content Providers, Testing and Debugging, Publishing and Deploying Android application to the world.

**Text Books:**

1. Neil Smyth, "Android Studio 3.0 Development Essentials – Android 8 Edition", Payload Media, 3<sup>rd</sup> Edition, 2017.
2. Lauren Darcey and Shane Conder, "Android Wireless Application Development", Pearson Education, 2nd ed. (2011)
3. Neil Smyth, "Firebase Essentials – Android Edition", Payload Media, 1<sup>st</sup> Edition, 2017.

**Reference Books:**

1. Reto Meier, "Professional Android 2 Application Development", Wiley India Pvt Ltd
2. Mark L Murphy, "Beginning Android", Wiley India Pvt Ltd
3. Android Application Development All in one for Dummies by Barry Burd,

**Web References:**

1. <https://developer.android.com/>
2. <https://google-developer-training.gitbooks.io/android-developer-fundamentals-course-concepts/content/en/>
3. <https://google-developer-training.gitbooks.io/android-developer-fundamentals-course-practicals/content/en/>

**E-Text Books:**

1. <https://developer.android.com/>
2. <http://enos.itcollege.ee/~jpoial/allalaadimised/reading/Android-Programming-Cookbook.pdf>

**MOOC Course:**

1. <https://www.coursera.org/specializations/android-app-development>



## DATA MINING (PROFESSIONAL ELECTIVE – II)

**III Year II Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS25      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Learn data mining concepts understand association rules mining.
2. Discuss classification algorithms learn how data is grouped using clustering techniques.
3. To develop the abilities of critical analysis to data mining systems and applications.
4. To implement practical and theoretical understanding of the technologies for data mining
5. To understand the strengths and limitations of various data mining models

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Perform the pre processing of data and apply mining techniques on it.
2. Identify the association rules, classification and clusters in large data sets.
3. Solve real world problems in business and scientific information
4. Classify web pages, extracting knowledge from the web

### UNIT – I

**Classes: 12**

**INTRODUCTION TO DATA MINING:** Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values

**PREPROCESSING:** Data Quality, Major Tasks in Data Pre-processing, Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization, Measures of Similarity and Dissimilarity- Basics

### UNIT – II

**Classes: 10**

**ASSOCIATION RULES:** Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, The Partition Algorithms, FP-Growth Algorithms, Compact Representation of Frequent Item Set- Maximal Frequent Item Set, Closed Frequent Item Set.

### UNIT – III

**Classes: 12**

**CLASSIFICATION:** Problem Definition, General Approaches to solving a classification problem ,Evaluation of Classifiers , Classification techniques, Decision Trees-Decision tree Construction , Methods for Expressing attribute test conditions, Measures for Selecting the Best Split, Algorithm for Decision tree Induction ; Naive-Bayes Classifier, Bayesian Belief Networks; K- Nearest neighbor classification-Algorithm and Characteristics.

**UNIT – IV****Classes: 12**

**CLUSTERING:** Problem Definition, Clustering Overview, Evaluation of Clustering Algorithms, Partitioning Clustering-K-Means Algorithm, K-Means Additional issues, PAM Algorithm; Hierarchical Clustering-Agglomerative Methods and divisive methods, Basic Agglomerative Hierarchical Clustering Algorithm, Specific techniques, Key Issues in Hierarchical Clustering, Strengths and Weakness; Outlier Detection.

**UNIT – V****Classes: 12**

**WEB AND TEXT MINING:** Introduction, web mining, web content mining, web structure mining, we usage mining, Text mining –unstructured text, episode rule discovery for texts, hierarchy of categories, text clustering.

**Text Books:**

1. Data Mining: Concepts and Techniques - Jiawei Han, Micheline Kamber, Jian Pei (2012), 3<sup>rd</sup> edition, Elsevier, United States of America.
2. Introduction to Data Mining, Pang-Ning Tan, Vipin Kumar, Michael Steinbanch, Pearson Education
3. Data mining Techniques and Applications, Hongbo Du Cengage India Publishing

**Reference Books:**

1. Data Mining Techniques, Arun K Pujari, 3rd Edition, Universities Press.
2. Data Mining Principles & Applications – T.V Sveresh Kumar, B.Esware Reddy, Jagadish S Kalimani, Elsevier.
3. Data Mining, Vikaram Pudi, P Radha Krishna, Oxford University Press

## SOFTWARE ENGINEERING (PROFESSIONAL ELECTIVE – II)

### III Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS26      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To familiarize with basic Software engineering methods and practices, and its applications.
2. To explain layered technology in software engineering
3. To teach software metrics and software risks.
4. To familiarize with software requirements and the SRS documents.
5. To facilitate students in software design

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand software development life cycle and select appropriate model suited for diverse software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document
3. Conceptualize the system through design with emphases on architectural modeling and user interface
4. Classify software testing strategies and recommend testing techniques during the construction of software.
5. Examine the application of metrics and software tools during software development

### UNIT – I

**Classes: 12**

**INTRODUCTION TO SOFTWARE ENGINEERING:** The Evolving nature of software engineering, Changing nature of software engineering, Software engineering Layers, The Software Processes, Software Myths.

**PROCESS MODELS:** A Generic Process Model, Waterfall Model, Incremental Process Models, Evolutionary Process Models, Spiral Model, The Unified Process, Personal and Team Process Models, the Capability Maturity Model Integration (CMMI).

### UNIT – II

**Classes: 12**

**REQUIREMENTS ENGINEERING:** Functional and Non-Functional Requirements, The Software requirements Document, Requirements Specification, requirements Engineering, Requirements Elicitation and Analysis, Requirement Validation, Requirement Management, System Modeling: Context Models, Interaction Models, Structural Models, Behavioral Model, Model-Driven Engineering.

**DESIGN CONCEPTS:** The Design Process, Design Concepts, The Design Models, Architectural Design: Software Architecture, Architectural Genres, Architectural Styles.

**UNIT – III****Classes: 10**

**DESIGN AND IMPLEMENTATION:** Design Patterns, Implementation Issues, Open Source Development. User Interface Design: The Golden Rules, User Interface Analysis and Design, Interface Analysis, Interface Design Steps, Design Evaluation.

**SOFTWARE TESTING STRATEGIES:** A Strategic approach to Software Testing, Strategic Issues, Test Strategies for Conventional Software, Validation Testing, System Testing, The Art of Debugging, White-Box Testing, Black Box Testing.

**UNIT – IV****Classes: 12**

**PRODUCT METRICS:** A Frame Work for Product Metrics, Metrics for the Requirements Model, Metrics for Design Model, Metrics for Source Code, Metrics for Testing.

**PROCESS AND PROJECT METRICS:** Metrics in the Process and Project Domains, Software Measurements, Metrics for Software Quality, Risk Management: Risk versus Proactive Risk Strategies, Software Risks, Risk Identification, Risk Projection, Risk Refinements, Risk Mitigation Monitoring and Management (RMMM), The RMMM Plan.

**UNIT – V****Classes: 10**

**OVERVIEW OF QUALITY MANAGEMENT AND PROCESS IMPROVEMENT:** Overview of SEI - CMM, ISO 9000, CMMI, PCMM, TQM and Six Sigma.

**OVERVIEW OF CASE TOOLS:** Software tools and environments: Programming environments; Project management tools; Requirements analysis and design modeling tools; testing tools; Configuration management tools;

**Text Books:**

1. Roger S. Pressman (2011), Software Engineering, A Practitioner's approach, 7<sup>th</sup> edition, McGraw Hill International Edition, New Delhi.
2. Sommerville (2001), Software Engineering, 9<sup>th</sup> edition, Pearson education, India.

**Reference Books:**

1. K. K. Agarwal, Yogesh Singh (2007), Software Engineering, 3rd edition, New Age International Publishers, India.
2. Lames F. Peters, Witold Pedrycz(2000), Software Engineering an Engineering approach, John Wiely & Sons, New Delhi, India.
3. Shely Cashman Rosenblatt (2006), Systems Analysis and Design, 6<sup>th</sup> edition, Thomson Publications, India.

## INTERNET OF THINGS (PROFESSIONAL ELECTIVE – III)

**III Year II Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS27      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To understand the fundamentals of Internet of Things.
2. To learn about the basics of IOT protocols.
3. To build a small low cost embedded system using Raspberry Pi and Arduino board.
4. To apply the concept of Internet of Things in the real world scenario.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Design a portable IoT using Arduino/ equivalent boards and relevant protocols.
2. Develop web services to access/control IoT devices.
3. Analyze various protocols for IoT
4. Deploy an IoT application.
5. Analyze applications of IoT in real time scenario

### UNIT – I INTRODUCTION TO IoT

**Classes: 08**

**Introduction to IoT:** Definition, Characteristics of IOT, IOT Architectural view, Physical design of IOT, Logical design of IOT, IOT Enabling technologies, Application of IOT.

### UNIT – II IoT ARCHITECTURE

**Classes: 10**

**Domain specific IoT's:** Introduction, Home Automation, Cities, Environment, Energy, retail, Logistics, Agriculture, Industry, Health.

**Machine-to-machine (M2M):** Introduction, Difference between IOT and M2M, SDN (software defined networking) and NFV (network function virtualization) for IOT.

### UNIT – III IoT PROTOCOLS & BUILDING IoT WITH ARDUINO

**Classes: 09**

**Developing Internet of things:** Introduction, IoT platforms design methodology, Need for IoT systems management

### UNIT – IV BUILDING IoT WITH RASPBERRY PI

**Classes: 09**

IoT Physical devices & End points: Basic building blocks of IoT device, About Board, Raspberry Pi Interfaces, Other IoT devices.

### UNIT – V CASE STUDIES AND REAL – WORLD APPLICATIONS

**Classes: 09**

Case Studies Illustrating IoT design: Home automation, Weather Monitoring, Agriculture, Cities

### Text Books:

1. Vijay Madiseti and Arshdeep Bahga, "Internet of things(A-Hand-on-Approach)" 1st Edition, Universal Press

### Reference Books:

1. Rajkamal,"Internet of Things", Tata McGraw Hill publication
2. Hakima Chaouchi "The Internet of Things: Connecting Objects", Wiley pcation.
3. Charless Bell "MySQL for the Internet of things", Apress publications.
4. Francis dacosta "Rethinking the Internet of things:A scalable Approach to
5. connecting everything", 1st edition, Apress publications 2013.
6. Donald Norris"The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi and BeagleBone Black", McGraw Hill publication.

## CLOUD COMPUTING (PROFESSIONAL ELECTIVE – III)

**III Year II Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS28      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To inculcate the concepts of distributed computing
2. To familiarize the concepts of cloud computing and services
3. To explain cloud platform and types of cloud
4. To explain resource management in cloud

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the fundamental principles of distributed computing
2. Create virtual machines and virtual templates.
3. Create Cloud platform using Virtual machines
4. Identify suitable business models of cloud computing

### UNIT – I

**Classes: 10**

**INTRODUCTION TO VIRTUALIZATION AND TECHNOLOGIES:** Introduction to Virtualization: Definition, Objectives, Characteristics, Benefits of virtualization, Taxonomy of virtualization technologies, Pros and cons of virtualization. Virtualization Technologies: VMware, Hyper-V, Zen and virtual iron.

### UNIT – II

**Classes: 10**

**FUNDAMENTAL CLOUD COMPUTING AND MODELS:** Cloud Computing: Origin and influences, Basic concepts and terminology, Goals and benefits, Risks and challenges. Cloud Models, roles and boundaries, Cloud characteristics, Cloud delivery models, Cloud deployment models.

### UNIT – III

**Classes: 10**

**CLOUD COMPUTING MECHANISMS AND ARCHITECTURE:** Cloud-Enabling Technology: Broadband networks and internet architecture, Data center technology, Virtualization technology, Web technology, Multitenant technology, Service technology. Cloud Architectures: Architecture - Workload distribution, Resource pooling, Dynamic scalability, Elastic resource capacity, Service load balancing, Cloud bursting, Elastic disk provisioning, Redundant storage.

### UNIT – IV

**Classes: 10**

**CLOUD SECURITY AND DISASTER RECOVERY:** Cloud Security: Data, Network and host security, Cloud security services and cloud security possible solutions. Cloud Disaster Recovery: Disaster recovery planning, Disasters in the cloud, Disaster management, Capacity planning and cloud scale.

**UNIT – V****Classes: 10**

**CLOUD CASE STUDIES:** Case Studies: Software-as-a-Service (SaaS) - Salesforce.com, Facebook; Platform-as-a-Service (PaaS) - Google App Engine, MS-Azure and IBM Bluemix; Infrastructure-as-a-Service (IaaS) - Amazon EC2, Amazon S3 and Netflix.

**Text Books:**

1. Thomas Erl and Ricardo Puttini Cloud Computing Concepts, Technology and Architecture, Pearson, 2013.
2. Ivanka Menken and Gerard Blokdijs, Cloud Computing Virtualization Specialist Complete Certification Kit-Study Guide Book, Lightning Source, 2009

**Reference Books:**

1. Barrie Sosinsky, Cloud Computing Bible, Wiley India Pvt Ltd, 2011.
2. Rajkumar Buyya, James Broberg and Andrzej Goscinski, Cloud Computing Principles and Paradigms, John Wiley and Sons, 2011.
3. John W. Rittinghouse and James F. Ransome, Cloud Computing Implementation, Management and Security, CRC Press, Taylor & Francis Group, 2010.



## MULTIMEDIA APPLICATION DEVELOPMENT (PROFESSIONAL ELECTIVE – III)

**III Year II Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT08      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To understand the definition of multimedia
2. To understand and differentiate text, image, video & audio.
3. To describe the ways in which multimedia information is captured, processed, and rendered
4. Introduce multimedia quality of service (QoS) and to compare subjective and objective methods of assessing user satisfaction and multicast protocols to provide QoS guarantees
5. Discuss privacy and copyright issues in the context of multimedia

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe different realizations of multimedia tools and the way in which they are used
2. Analyze the structure of the tools in the light of low-level constraints imposed by the adoption of various QoS schemes (i.e. bottom up approach)
3. Analyze the effects of scale and use on both presentation and lower-level requirements (i.e. top down approach)
4. State the properties of different media streams;
5. Compare and contrast different network protocols and to describe mechanisms for providing QoS guarantees in the network.

### UNIT – I

**Classes: 12**

**Introduction:** Definitions - Brief history of Multimedia; its market; content and copyright –public Domain, establishment of Copyright, fair use, multimedia copyright issues; resources for multimedia developers – Uses of multimedia - Making multimedia : Stages of a project

### UNIT – II

**Classes: 14**

**Hardware Macintosh Versus Windows Platform** – Connections – SCSI – IDE – EIDE – ULTRA – IDE – ATA – ULTRA - ATA - Memory and Storage Devices - Input Devices - Output Hardware – Communication Devices

**Basic Software Tools** :Text Editing - Word Processing - OCR Software - Painting and Drawing Tools - 3D Modeling and Animation Tools - Image Editing - Sound Editing – Animation – Video - Digital Movie tools - Movie Editors - Compressing Movie Files

**UNIT – III****Classes: 14**

**Text :** Fonts – Designing – Choosing -Menus for Navigation - Buttons for Interaction – Fields for Readings - HTML Documents - Symbols and Icons – Animating - Fonts Foundries – Managing Fonts - Character sets and Alphabets - Mapping Text – Fontographer - Hypermedia Structures – Hypertext tools

**Sound :** Power of sound - Multimedia System Sound - MIDI Versus Digital Audio - Preparing Digital Audio Files - Making MIDI Audio - Audio File Formats - Sound for the World Wide Web – Adding Sound to Your Multimedia Project - Toward Professional Sound - The Red Book Standard – Space Considerations - Production Tips - Audio Recording - Keeping Track of Your Sounds - Testing and Evaluation

**UNIT – IV****Classes: 12**

**Introduction:** The Bandwidth Bottleneck - Internet Services - MIME-Types - World Wide Web and HTML – Dynamic Web Pages and XML - Multimedia on the Web.

Tools for the World Wide Web: Web Servers - Web Browsers - Web Page Makers and Site Builders - Plug ins and Delivery Vehicles - Text – Images –Sound - Animation, Video and Presentation - Beyond HTML - 3D Worlds,designing for the World Wide Web..

**UNIT – V****Classes: 08**

**Multimedia File Handling:** Compression & De compression - Data & file formats standards - Digital voice, Audio, video – Video image and Animation - Full motion video - storage and retrieval Technologies

**Text Books:**

1. Multimedia making it work Tay Vaughan Tata McGrawHill, Delhi
2. Multimedia – Technology and applications David Hillman Galgotia Publications, Delhi

## ANGULAR JS

### (PROFESSIONAL ELECTIVE – III)

**III Year II Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT09      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

To learn

1. Explore to reduce the code to build user interface applications
2. Describe Client Side MVC
3. Understanding single page applications development
4. Understanding concept of interactive and attractive interface development techniques
5. Explore services and server communication.

#### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply basic programming to implement dynamic web pages and applications
2. Apply the knowledge of the design patterns to control the application
3. Apply the knowledge to control events handling and error handling
4. Apply the knowledge to filter the data more effective way to present
5. Construct services and server communication.

#### UNIT-I

#### JAVASCRIPT YOU NEED TO KNOW

**Classes: 12**

JavaScript Primer: Including Scripts on a Page, Statements, Functions, Parameters and Return Values, Types and Variables, Primitive Types, JavaScript Operators, Equality vs. Identity, Pre- vs. Post-Increment, Working with Objects: Creating Objects, Reading and Modifying an Object's Properties, Adding Methods to Objects, Enumerating Properties, Control Flow, Conditional Statements, Working with Arrays, Array Literals, Enumerating and Modifying Array Values: Callbacks, JSON

#### UNIT-II

#### THE BASICS OF ANGULARJS, INTRODUCTION TO MVC

**Classes: 12**

Why We Need Frameworks: What Is a Framework, Downloading and Installing AngularJS, Browser Support, Your First AngularJS Application: Declarative vs. Procedural Programming, Directives and Expressions: What Is a Directive, What Are Expressions. Design Patterns Model View Controlled, A Separation of Concerns: Why MVC Matters, MVC the AngularJS Way

#### UNIT-III

#### FILTERS AND MODULES, DIRECTIVES

**Classes: 12**

Introduction to Filters, Built-in Filters: The Number Filter, The Date Filter, The *limitTo* Filter. AngularJS Modules: What Is a Module. Bootstrapping AngularJS, Creating a Custom Filter. The

Basics of Directives, Using Directives, Built-in Directives, Event-Handling Directives, Using the API Documentation, Creating a Custom Directive

**UNIT-IV****WORKING WITH FORMS****Classes: 12**

HTML Forms Overview: The *form* Element, The *input* Element: button, submit, text, checkbox, password, radio. The *textarea* Element, The *select* Element, The *label* Element. Model Binding, AngularJS Forms, Validating Forms.

**UNIT-V****SERVICES AND SERVER COMMUNICATION****Classes: 12**

Using Services: The *\$window* Service, The *\$location* Service, The *\$document* Service. Why Use Services, Creating Services: Promises. Server Communication, Handling Returned Data: Accessing Returned Data, Handling Errors

**Text Books:**

- 1 Andrew Grant — "Beginning Angular JS", A Press 2014.

**Reference Books:**

1. Green, Brad – "Angular JS", O'Reilly 2013.
2. Adam Freeman, — "Pro AngularJS ", A Press 2014

## LINUX PROGRAMMING LAB

### III Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS18      | PCC      | 0            | 0 | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Apply the basic commands to handle the Linux Environment
2. Use the Shell/C scripting constructs to modify the file system content.
3. Implement the process management concepts using C language.
4. Design Client-Server Applications using Sockets and TCP/UDP protocols.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Write and use the Shell Scripts in managing Linux Environment
2. Construct C Scripts to handle the File system in Linux.
3. Implement the IPC Mechanisms in Linux operating system using C language
4. Build and Analyze the Client-Server Environment using various protocols.

### LIST OF EXPERIMENTS

1. Linux Commands & Introduction of Shell
2. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
3. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
4. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
5. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.
6. Write a shell script that accepts a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
7. Write a shell script to list all of the directory files in a directory.
8. Write an awk script to count the number of lines in a file that do not contain vowels.
9. Write an awk script to find the number of characters, words and lines in a file.
10. Write a c program that makes a copy of a file using standard I/O and system calls.
11. Implement in C the following UNIX commands using System calls.  
A. cat      B. mv
12. Write a C Program to list files in a directory.
13. Write a C program to emulate the UNIX ls -l command.
14. Write a C program to list for every file in a directory, its inode number and file name.
15. Write a C program to create a child process and allow the parent to display "parent" and the child to display "child" on the screen.
16. Write a C program to create a Zombie process.
17. Write a C program that illustrates how an orphan is created.
18. Write C programs that illustrate communication between two unrelated processes using named pipe (FIFO).

19. Write a C program in which a parent writes a message to a pipe and the child reads the message.
20. Write a C program that illustrates suspending and resuming processes using signals.
21. Write Client server program in C for connection oriented communication between server and client process using unix domain sockets to perform the following
22. Client process to send a message to server process. The server receives the message, reverses it and send back to the client will then display the message to the standard output device.
23. Write Client server program in C for connection oriented communication between server and client process using internet domain sockets.

#### TEXT BOOKS:

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH,2006.

#### REFERENCES:

1. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones,Wrox, Wiley India Edition,rp-2008
2. Linux System Programming, Robert Love, O'Reilly, SPD.
3. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
4. Unix Network Programming ,W.R.Stevens,PHI.
5. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education.

## WEB TECHNOLOGIES LAB

### III Year II Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT05      | PCC      | 0            | 0 | 3 | 1.5     | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To teach students the basics of server side scripting using PHP
2. To explain web application development procedures
3. To impart Servlets technology for writing business logic
4. To facilitate students to connect to databases using JDBC
5. To familiarize various concepts of application development using JSP

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create web pages using PHP
2. Identify the difference between the HTML PHP and XML documents.
3. Identify the engineering structural design of XML and parse tree
4. Analyze the difference between and PHP and XML.
5. Understand the concept of JAVA SCRIPTS.
6. Identify the difference between the JSP and Servlets.
7. Design web application using MVC architecture
8. Understand the JSP and Servlet concepts.

## LIST OF EXPERIMENTS

### Note:

1. Apache, MySQL and PHP for the Lab Experiments. Though not mandatory, encourage the use of Eclipse platform wherever applicable
  2. The list suggests the minimum program set. Hence, the concerned staff is requested to add more problems to the list as needed
1. Install the following on the local machine
    - i. Apache Web Server (if not installed)
    - ii. Tomcat Application Server locally
    - iii. Install MySQL (if not installed)
    - iv. Install PHP and configure it to work with Apache web server and MySQL (if not already configured)
  2. Write an HTML page including any required Javascript that takes a number from one text field in the range of 0 to 999 and shows it in range of 0 to 999 and shows it in another text field in words. If the number is out of range, it should show "out of range" and if it is not a number, it should show "not a number" message in the result box.
  3. Write an HTML page that has one input, which can take multi-line text and a submit button. Once the user clicks the submit button, it should show the number of characters, words and lines in the text entered using an alert message. Words are separated with while space and lines are separated with new line character.
  4. Write an HTML page that contains a selection box with a list of 5 countries. When the user selects a country, its capital should be printed next to the list. Add CSS to customize the properties of the font of the capital (color, bold and font size).

5. Create an XML document that contains 10 users information. Write a Java program, which takes User Id as input and returns the user details by taking the user information from the XML document using (a) DOM Parser and (b) SAX parser.
6. Implement the following web applications using (a) PHP, (b) Servlets and (c) JSP:
  - i. A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in Database and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.
  - ii. Modify the above program to use an xml file instead of database.
  - iii. Modify the above program to use AJAX to show the result on the same page below the submit button.
  - iv. A simple calculator web application that takes two numbers and an operator (+, -, /, \* and %) from an HTML page and returns the result page with the operation performed on the operands.
  - v. Modify the above program such that it stores each query in a database and checks the database first for the result. If the query is already available in the DB, or it computes the result and returns it after storing the new query and result on DB.
  - vi. A web application takes a name as input and on submit it shows a hello<name>page where<name> is taken from the request. It shows the start time at the right top corner of the page and provides a logout button. On clicking this button, it should show a logout page with Thank You <name> message with the duration of usage (hint: Use session to store name and time).
  - vii. A web application that takes name and age from an HTML page. If the age is less than 18, it should send a page with "Hello <name>, you are not authorized to visit this site" message, where <name> should be replaced with the entered name. Otherwise it should send "Welcome <name> to the site" message.
  - viii. A web application for implementation: The user is first served a login page which takes user's name and password. After submitting the details the server checks these values against the data from a database and takes the following decisions. If name and password doesn't match, then serves "password mismatch" page. If name is not found in the database, serves a registration page, where user's full name is asked and on submitting the full name, it stores, the login name, password and full name in the database (hint: use session for storing the submitted login name and password)
  - ix. A web application that lists all cookies stored in the browser on clicking "List Cookies" button. Add cookies if necessary.

#### TEXT BOOKS:

1. Web Technologies, Uttam K Roy, Oxford University Press
2. The Complete Reference PHP – Steven Holzner, Tata McGraw-Hill

#### REFERENCE BOOKS:

1. Web Programming, building internet applications, Chris Bates 2<sup>nd</sup> edition, Wiley Dremtech
2. Java Server Pages – Hans Bergsten, SPD O'Reilly
3. Java Script, D.Flanagan, O'Reilly, SPD.
4. Beginning Web Programming-Jon Duckett WROX.
5. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson.
6. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson.



## ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE

### III Year II Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4MC06      | MC       | 2            | 0 | 0 | 0       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

To facilitate the students with the concepts of Indian traditional knowledge and to make them understand the Importance of roots of knowledge system.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Upon completion of the course, the students are expected to:
2. Understand the concept of Traditional knowledge and its importance
3. Know the need and importance of protecting traditional knowledge.
4. Know the various enactments related to the protection of traditional knowledge.
5. Understand the concepts of Intellectual property to protect the traditional knowledge.

### UNIT – I

Classes: 18

**Introduction to traditional knowledge:** Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, the physical and social contexts in which traditional knowledge develop, the historical impact of social change on traditional knowledge systems. Indigenous Knowledge (IK), characteristics, traditional knowledge vis-à-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge vis-à-vis formal knowledge

### UNIT – II

Classes: 08

**Protection of traditional knowledge:** The need for protecting traditional knowledge Significance of TK Protection, value of TK in global economy, Role of Government to harness TK.

### UNIT – III

Classes: 12

**Legal frame work and TK: A:** The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001

(PPVFR Act); B: The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016. Geographical indicators act 2003.

#### UNIT – IV

Classes: 12

**Traditional knowledge and intellectual property:** Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Certain non IPR mechanisms of traditional knowledge protection, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge, global legal FORA for increasing protection of Indian Traditional Knowledge.

#### UNIT – V

Classes: 12

**Traditional knowledge in different sectors:** Traditional knowledge and engineering, Traditional medicine system, TK and biotechnology, TK in agriculture, Traditional societies depend on it for their food and healthcare needs, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of TK. 139.

#### Text Books:

1. Traditional Knowledge System in India, by Amit Jha, 2009.
2. Traditional Knowledge System and Technology in India by Basanta Kumar Mohanta and Vipin Kumar Singh, Pratibha Prakashan 2012.

#### Reference Books:

1. Traditional Knowledge System in India by Amit Jha Atlantic publishers, 2002
2. "Knowledge Traditions and Practices of India" Kapil Kapoor<sup>1</sup>, Michel Danino<sup>2</sup>

#### E-Resources:

1. <https://www.youtube.com/watch?v=LZP1StpYEPM>
2. <http://nptel.ac.in/courses/121106003>

# **IV B.TECH I SEMESTER SYLLABUS**

## BIG DATA ANALYTICS

**IV Year I Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS19      | PCC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. **Compare** various file systems and **use** an appropriate file system for storing different types of data.
2. **Demonstrate** the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. **Apply** the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. **Connect** to web data sources for data gathering, **Integrate** data sources with hadoop components to process streaming data.
5. **Tabulate** and **examine** the results generated using hadoop components

### UNIT – I

**Classes: 15**

**INTRODUCTION TO BIG DATA:** Data and its importance, Big Data - definition, implications of Big Data, addressing Big Data implications using Hadoop, Hadoop Ecosystem

#### HADOOP ARCHITECTURE:

Hadoop Storage : HDFS, Hadoop  
 Processing : Map Reduce Framework  
 Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, TaskTracker

**HDFS-HADOOP DISTRIBUTED FILE SYSTEM:** Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE: File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

### UNIT – II

**Classes: 10**

**MAPREDUCE PROGRAMMING MODEL:** Introduction to Map Reduce Programming model to process Big Data, key features of Map Reduce, Map Reduce Job skeleton, Introduction to Map Reduce API, Hadoop Data Types, Develop Map Reduce Job using Eclipse, built a Map Reduce Job export it as a java archive(.jar file).

**MAPREDUCE JOB LIFE CYCLE:** Understanding Mapper, Combiner, Partitioner, Shuffle & Sort and Reduce phases of Map Reduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

**UNIT – III****Classes: 12**

**INTRODUCTION TO PIG:** Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage.

**PIG LATIN LANGUAGE –SEMANTICS –DATA TYPES IN PIG:** Pig Latin Basics, Key words, Pig Data types, Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell, expressions, Data processing operators, using Built in functions.

**WRITING PIG SCRIPTS USING PIG LATIN:** Writing pig scripts and saving them text editor, running pig scripts from command line.

**UNIT – IV****Classes: 10**

**INTRODUCATION TO HIVE:** Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write.

**HIVE QL DATA TYPES, SEMANTICS:** Introduction to Hive QL (Query Language), Language semantics, Hive Data Types.

**HIVE DDL, DML AND HIVE SCRIPTS:** Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and command line.

**UNIT – V****Classes: 12**

**SQOOP:** Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS.

**FLUME:** Introduction to Flume agent, understanding Flume components Source, Channel and Sink.

**OOZIE:** Introduction to Oozie, Understanding work flow Management.

**Text Books:**

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Chris Eaton, Dirk deroos et al. , "Understanding Big data ", McGraw Hill, 2012.
3. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.

**Reference Books:**

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.

## SOFTWARE TESTING (PROFESSIONAL ELECTIVE-4)

**IV Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS31      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. An ability to design and conduct a software test process for a software testing project and identify the needs of software test automation.
2. An ability understand and identify various software testing problems, and solve these problems by designing and selecting software test models, criteria, strategies, use software testing methods and modern software testing tools for their testing projects and basic understanding and knowledge of contemporary issues in software testing, such as component-based software testing problems.
3. An ability to use various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects.
4. Analyze different approaches to software testing and quality assurance, and select optimal solutions for different situations and projects.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply fundamental knowledge of Testing in Real time scenarios.
2. Test a simple application.
3. Understand and Applying the Techniques in Software Development Life cycle
4. Implement various communication methods and skills to communicate with their teammates to conduct their practice-oriented software testing projects.

### UNIT – I

**Classes: 12**

**INTRODUCTION:** Purpose of testing-Dichotomies-Software Testing Principles- Bugs, consequences of bugs, Taxonomy of bugs -The Tester's Role in a Software Development Organization-Black box testing and white box testing- Defects -Cost of defects- Defect Classes- Defect Examples, software testing life cycle.

### UNIT – II

**Classes: 10**

**Flow graphs and Path testing:** Basics concepts of path testing-predicates-path predicates and achievable paths- path sensitizing- path instrumentation, application of path testing. Transaction Flow Testing: Transaction flows- transaction flow testing techniques, Dataflow testing: Basics of dataflow testing - strategies in data flow testing – application of data flow testing.

### UNIT – III

**Classes: 12**

**Test Case Design Strategies:** Using Black Box Approach to Test Case Design - Random Testing – Requirements based testing – Boundary Value Analysis – Equivalence Class Partitioning–

Compatibility testing – User documentation testing – Domain testing. Alpha, Beta Tests – Usability and Accessibility testing – Configuration testing - Compatibility testing – Testing the documentation.

#### UNIT – IV

Classes:12

**Paths, Path products and Regular expressions:** Path products & path expression- reduction procedure- applications- regular expressions & flow anomaly detection. Logic Based Testing: Overview decision tables-path expressions, k-v charts. State, state graphs and transition testing: state graphs, good and bad state graphs, state testing, testability tips.

#### UNIT – V

Classes: 12

**Graph Matrices and applications:** motivational over view, matrix of graph, relations, power of matrix, node reduction algorithm. People and organizational issues in testing: Organization structures for testing teams – testing services - Test Planning – Test Plan Components – Test Plan Attachments – Locating Test Items – test management – test process- bug detection life cycle.

#### Text Books:

1. “Software Testing – Principles and Practices”, Srinivasan Desikan and Gopalaswamy Ramesh, Pearson education, 2006.
2. “Software Testing Techniques”, Boris Beizer, 2nd Edition, Van Nostrand Reinhold, New York, 1990

#### Reference Books:

1. “Software Testing”, Ron Patton, Second Edition, Sams Publishing, Pearson education, 2007
2. “Software Testing – Effective Methods, Tools and Techniques”, Renu Rajani, Pradeep Oak, Tata McGraw Hill, 2004.
3. “Software Testing in the Real World – Improving the Process”, Edward Kit, Pearson Education, 1995.
4. “Foundations of Software Testing – Fundamental algorithms and techniques”, Aditya P. Mathur, Dorling Kindersley (India) Pvt. Ltd., Pearson Education, 2008

## CRYPTOGRAPHY AND NETWORK SECURITY (PROFESSIONAL ELECTIVE-4)

### IV Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS32      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To provide deeper understanding into cryptography, its application to network security, threats/vulnerabilities to networks and countermeasures.
2. To explain various approaches to Encryption techniques, strengths of Traffic Confidentiality, Message Authentication Codes.
3. To familiarize Digital Signature Standard and provide solutions for their issues.
4. To familiarize with cryptographic techniques for secure (confidential) communication of two parties over an insecure (public) channel; verification of the authenticity of the source of a message

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify basic security attacks and services
2. Use symmetric and asymmetric key algorithms for cryptography
3. Design a security solution for a given application
4. Analyze Key Management techniques and importance of number Theory.
5. Understanding of Authentication functions the manner in which Message Authentication Codes and Hash Functions works.
6. To examine the issues and structure of Authentication Service and Electronic Mail Security

### UNIT – I

Classes: 12

**INTRODUCTION:** Security trends, The OSI Security Architecture, Security Attacks, Security Services and Security Mechanisms, A model for Network security.

**CLASSICAL ENCRYPTION TECHNIQUES:** Symmetric Cipher Modes, Substitute Techniques, Transposition Techniques, Stenography

### UNIT – II

Classes: 12

**BLOCK CIPHER AND DATA ENCRYPTION STANDARDS:** Block Cipher Principles, Data Encryption Standards, the Strength of DES, Block Cipher Design Principles.

**ADVANCED ENCRYPTION STANDARDS:** Evaluation Criteria for AES, the AES Cipher.

**MORE ON SYMMETRIC CIPHERS:** Multiple Encryption, Triple DES, Block Cipher Modes of Operation, Stream Cipher and RC4.



**UNIT – III****Classes: 14**

**PUBLIC KEY CRYPTOGRAPHY AND RSA:** Principles Public key crypto Systems the RSA algorithm, Key Management, Diffie Hellman Key Exchange.

**MESSAGE AUTHENTICATION AND HASH FUNCTIONS:** Authentication Requirement, Authentication Function, Message Authentication Code, Hash Function, Security of Hash Function and MACs.

**HASH AND MAC ALGORITHM:** Secure Hash Algorithm, Whirlpool, HMAC, CMAC. **DIGITAL SIGNATURE:** Digital Signature, Authentication Protocol, Digital Signature Standard.

**UNIT – IV****Classes: 12**

**AUTHENTICATION APPLICATION:** Kerberos, X.509 Authentication Service, Public Key Infrastructure. **EMAIL SECURITY:** Pretty Good Privacy (PGP) and S/MIME.

**IP SECURITY:** Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management.

**UNIT – V****Classes: 10**

**WEB SECURITY:** Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET), Intruders, Viruses and related threats.

**FIREWALL:** Firewall Design principles, Trusted Systems.

**Text Books:**

1. William Stallings (2006), Cryptography and Network Security: Principles and Practice, 4th edition, Pearson Education, India.
2. William Stallings (2000), Network Security Essentials (Applications and Standards), Pearson Education, India.

**Reference Books:**

1. Charlie Kaufman (2002), Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall of India, New Delhi.
2. Atul Kahate (2008), Cryptography and Network Security, 2nd edition, Tata Mc Grawhill, India.
3. Robert Bragg, Mark Rhodes (2004), Network Security: The complete reference, Tata Mc Grawhill, India.

## DESIGN PATTERNS (PROFESSIONAL ELECTIVE-4)

**IV Year I Semester: IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4IT01      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. The aim of the course is to appreciate the idea behind Design Patterns in handling common problems faced during building an application
2. This course covers all pattern types from creational to structural, behavioral to concurrency and highlights the scenarios when one pattern must be chosen over others.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Create software designs that are scalable and easily maintainable
2. Understand the best use of Object Oriented concepts for creating truly OOP programs
3. Use creational design patterns in software design for class instantiation
4. Use structural design patterns for better class and object composition
5. Use behavioral patterns for better organization and communication between the objects
6. Use refactoring to compose the methods for proper code packaging
7. Use refactoring to better organize the class responsibilities of current code

### UNIT – I

**Classes: 10**

**Introduction:** What is a design pattern? design patterns in Smalltalk MVC, Describing Design Patterns, The Catalog of Design Patterns, Organizing the Catalog, How Design Patterns Solve Design Problems, How to Select a Design Pattern, How to Use a Design Pattern.

### UNIT – II

**Classes: 15**

**Designing a Document Editor:** Design Problems, Document Structure, Formatting, Embellishing the User Interface, and Supporting Multiple Look-and-Feel Standards, Supporting Multiple Window Systems, User Operations Spelling Checking and Hyphenation, Summary.

### UNIT – III

**Classes: 09**

**Creational Patterns:** Abstract Factory, Builder, Factory Method, Prototype, Singleton, Discussion of Creational Patterns.

### UNIT – IV

**Classes: 08**

**Structural Pattern:** Adapter, Bridge, Composite, Decorator, Façade, Flyweight, Proxy

**UNIT – V****Classes: 12**

**Behavioral Patterns:** Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, State, Strategy, Template Method, Visitor.

**Text Books:**

1. Design Patterns by Erich Gamma, Pearson Education.

**Reference Books:**

1. Pattern's in Java, Vol –I, Mark Grand, Wiley Dream Tech.
2. Patterns in Java, Vol-II, Mark Grand, Wiley Dream Tech.
3. Java Enterprise Design Patterns Vol-III, Mark Grand, Wiley Dream Tech.
4. Head First Design Patterns, Eric Freeman, O'reilly publications

## AD-HOC AND SENSOR NETWORKS (PROFESSIONAL ELECTIVE-4)

**IV Year I Semester: CSE/IT**

| Course Code   | Category   | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|---------------|------------|--------------|---|---|---------|---------------|-----|-------|
| <b>A4CS34</b> | <b>PEC</b> | L            | T | P | C       | CIE           | SEE | Total |
|               |            | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Describe the concepts of adhoc wireless networks.
2. Analyze different routing protocols of mobile adhoc networks.
3. Apply the energy management policies in routing algorithms.
4. Implement protocols for location based QoS.
5. Design and simulate sensor networks and evaluate performance.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the different types of Ad-hoc networks.
2. Differentiate various types of Routing Protocols and apply right protocol wherever it is necessary.
3. Analyze energy management issues in Ad-hoc networks for better network formation.
4. Identify the Sensor network Applications and differentiate various MAC Protocols.

#### UNIT – I      **Mobile Ad-Hoc Networking with a View of 4G Wireless**      **Classes: 10**

**Mobile Ad-Hoc Networking with a View of 4G Wireless:** Imperatives and Challenges, Off-the-Shelf Enables of Ad Hoc Networks, IEEE 802.11 in Ad Hoc Networks: Protocols, Performance and Open Issues, Scatternet Formation in Bluetooth Networks, Antenna Beam forming and Power Control for Ad Hoc Networks.

#### UNIT – II      **Routing Protocols in Ad Hoc Networks**      **Classes: 10**

**Routing Protocols in Ad Hoc Networks:** Topology Control in Wireless Ad Hoc Networks, Broadcasting and Activity Scheduling in Ad Hoc Networks, Location Discovery, Mobile Ad Hoc Networks (MANETs): Routing Technology for Dynamic, Wireless Networking, Routing Approaches in Mobile Ad Hoc Networks.

#### UNIT – III      **Energy Management in Ad hoc Wireless Networks**      **Classes: 12**

**Energy Management in Ad hoc Wireless Networks:** Energy-Efficient Communication in Ad Hoc Wireless Networks, Ad Hoc Networks Security, Self-Organized and Cooperative Ad Hoc Networking, Simulation and Modeling of Wireless, Mobile, and Ad Hoc Networks, Modeling Cross-Layering Interaction Using Inverse Optimization, Algorithmic Challenges in Ad Hoc Networks.

**Energy Management in Ad hoc Wireless Networks:** Introduction, need for energy management in ad hoc networks, battery management schemes-overview of battery characteristic, device dependent schemes.

#### UNIT – IV      **Introduction and Overview of Wireless Sensor Networks**      **Classes: 10**

**Introduction and Overview of Wireless Sensor Networks:** Applications of Wireless Sensor Networks, Examples of Category 1 WSN Applications, Another Taxonomy of WSN Technology. Basic Wireless Sensor Technology: Sensor Node Technology, Sensor Taxonomy, WN Operating Environment, WN Trends.

**UNIT – V      Wireless Sensor Networks Technology and Medium  
Access Control    Protocols**

**Classes: 10**

**Wireless Transmission Technology and Systems:** Radio Technology Primer, Available Wireless Technologies.

**Medium Access Control Protocols for Wireless Sensor Networks:** Fundamentals of MAC Protocols, MAC Protocols for WSNs, Sensor-MAC Case Study, IEEE 802.15.4 LR-WPANs Standard Case Study.

**Text Books:**

1. **“Adhoc and Sensor Networks”** by Stefano Basagni, Silvia Giordano, Ivan Stojmenvic. IEEE Press, A John Wiley & Sons, Inc., Publication 2004.
2. **Wireless Sensor Networks**, Kazem Sohraby, Daniel Minoli, Taieb Znati. A John Wiley & Sons, Inc., Publication 2007
3. C. Siva Ram Murthy and B. S. Manoj, Ad Hoc Wireless Networks-Architectures and Protocols, New Delhi: Pearson Education, 2013.

**Reference Books:**

1. Feng Zhao and Leonidas Guibas, Wireless Sensor Networks. Noida: Morgan Kaufman Publishers, 2004.
2. C. K. Toh, Ad Hoc Mobile Wireless Networks. New Delhi: Pearson Education, 2002.
3. Thomas Krag and SebastinBuettrich, Wireless Mesh Networking. Mumbai: O'Reilly Publishers, 2007

**Web References:**

1. <https://www.elsevier.com/journals/ad-hoc-networks/1570-8705/guide-for-authors>
2. [https://en.wikipedia.org/wiki/Wireless\\_ad\\_hoc\\_network](https://en.wikipedia.org/wiki/Wireless_ad_hoc_network)
3. [www.oldcitypublishing.com/journals/ahsw-n-home/](http://www.oldcitypublishing.com/journals/ahsw-n-home/)

**E-Text Books:**

1. <https://ebooks.benthams-science.com/book/9781608050185/>
2. <https://books.google.co.in> › Computers › Networking › General

**MOOC Course:**

1. [nptel.ac.in/courses/106105160/](https://nptel.ac.in/courses/106105160/)
2. [https://onlinecourses.nptel.ac.in/noc18\\_cs09/](https://onlinecourses.nptel.ac.in/noc18_cs09/)
3. <https://swayam.gov.in/course/4408-wireless-adhoc-and-sensor-networks>

## DISTRIBUTED SYSTEMS (PROFESSIONAL ELECTIVE-5)

### IV Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS35      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To layout foundations of Distributed Systems.
2. To introduce the idea of middleware and related issues
3. To understand in detail the system level and support required for distributed system
4. To understand the issues involved in studying data and design of distributed algorithms

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the advantages and challenges in designing distributed algorithms for different primitives like mutual exclusion, deadlock detection, agreement, etc.
2. Design and develop distributed programs using sockets and RPC/RMI.
3. Differentiate between different types of faults and fault handling techniques in order to implement fault tolerant systems.
4. Analyze different algorithms and techniques for the design and development of distributed systems subject to specific design and performance constraints.

### UNIT – I

**Classes: 09**

**Introduction:** Definition of Distributed Systems, Goals, Types of Distributed Systems.

**Characterization of Distributed Systems:** Introduction, Examples of Distributed Systems, Resource Sharing and the Web, Challenges.

**System Models:** Introduction, Architectural Models, Fundamental Models.

### UNIT – II

**Classes: 13**

**Time and Global States:** Introduction, Clocks Events and Process States, Synchronizing Physical Clocks, Logical Time and Logical Clocks, Global States, Distributed Debugging.

**Coordination and Agreement:** Introduction, Distributed Mutual Exclusion, Elections, Multicast Communication, Consensus and Related Problems.

### UNIT – III

**Classes: 13**

**Inter Process Communication:** Introduction, the API for the Internet Protocols, External Data Representation and Marshalling, Client-Server Communication, Group Communication, and Case Study: IPC in UNIX.

**Distributed Objects and Remote Invocation:** Introduction, Communication between Distributed Objects, Remote Procedure Call, Events and Notifications, Case Study: JAVA RMI.

**UNIT – IV****Classes: 15**

**Distributed File Systems:** Introduction, File Service Architecture, and Case Study 1: Sun Network File System, Case Study 2: The Andrew File System.

**Name Services:** Introduction, Name Services and the Domain Name System, Directory Services, Case Study of the Global Name Services.

**Distributed Shared Memory:** Introduction, Design and Implementation Issues, Sequential Consistency and IVY case study, Release Consistency, Munin Case Study, Other Consistency Models.

**UNIT – V****Classes: 12**

**Transactions and Concurrency Control:** Introduction, Transactions, Nested Transactions, Locks, Optimistic Concurrency Control, Timestamp Ordering, Comparison of Methods for Concurrency Control.

**Distributed Transactions:** Introduction, Flat and Nested Distributed Transactions, Atomic Commit Protocols, Concurrency Control in Distributed Transactions, Distributed Deadlocks, Transaction Recovery.

**Text Books:**

1. Distributed Systems, Concepts and Design, George Koulouris, J Dollimore and Tim Kindberg, Pearson Education, 41 Edition. 2009.

**Reference Books:**

1. Distributed Systems, Principles and Paradigms, Andrew S. Tanenbaum, Maarten Van Steen, 2nd Edition, PHI.
2. Distributed Systems, an Algorithm Approach, Sukumar Ghosh, Chapman Hall/CRC, Taylor & Fransis Group, 2007.

## SOFT COMPUTING (PROFESSIONAL ELECTIVE-5)

**IV Year I Semester: CSE/IT**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS36      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To teach basic neural networks, fuzzy systems, and optimization algorithms concepts and their relations
2. To provide knowledge of Neuron model, and Applications of NN to discuss their work
3. To provide the graduate the better understanding of Fuzzy Logic and Evolutionary Computations

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. To understand neural network (NN) paradigms
2. Apply different supervised learning algorithms on given data
3. Understand feedback neural networks & self organizing feature map
4. To learn fuzzy logic To have a knowledge of evolutionary computations, genetic algorithm(GA), evolutionary programming, classifier systems, genetic programming parse trees, mathematical foundation of GA variants of GA

### UNIT – I

**Classes: 12**

**BASICS OF ARTIFICIAL NEURAL NETWORK:** Characteristics of Neural Networks, Structure and working of a biological neural network, Artificial neural network: terminology, models of neurons: McCulloch Pitts model, Perceptron model, Adaline model, topology, Basic learning laws.

**FUNCTIONAL UNITS FOR ANN FOR PATTERN RECOGNITION TASK:** Pattern recognition problem, Basic functional units, PR by functional units.

### UNIT – II

**Classes: 12**

**FEEDFORWARD NEURAL NETWORKS:**

**SUPERVISED LEARNING - I:** Perceptrons - Learning and memory, Learning algorithms, Error correction and gradient decent rules, Perceptron learning algorithms.

**SUPERVISED LEARNING-II:** Back propagation, Multilayered network architectures, Back propagation learning algorithm, Example applications of feed forward neural networks.

### UNIT – III

**Classes: 15**

**FEEDBACK NEURAL NETWORKS & SELF ORGANIZING FEATURE MAP:** Introduction, Associative learning, Hopfield network, Error performance in Hopfield networks, simulated annealing, Boltzmann machine and Boltzmann learning, state transition diagram and false minima problem, stochastic update, simulated annealing, Boltzmann machine, bidirectional associative memory, bam stability analysis. Self organization, generalized learning laws, competitive learning, vector quantization, self organizing feature map, applications of self organizing feature map.



**UNIT – IV****Classes: 12**

**FUZZY LOGIC:** Fuzzy set theory, crisp sets, operations on crisp set, fuzzy sets, fuzzy versus crisp, operations, fuzzy relations, crisp relations, properties. Fuzzy logic Application: Fuzzy Control of Blood Pressure.

**UNIT – V****Classes: 13**

**FUZZY LOGIC IN DATABASE AND INFORMATION SYSTEMS:** Fuzzy Information, Fuzzy Logic in database Systems, Fuzzy Relational data Models, operations in Fuzzy Relational data Models, Design theory for Fuzzy Relational databases, Fuzzy information Retrieval and Web search, Fuzzy Object Oriented databases.

**GENETIC ALGORITHMS:** Introduction to Genetic Algorithms, Evolutionary Algorithms.

**Text Books:**

1. Satish Kumar (2004), Neural Networks A classroom Approach, Tata McGraw Hill Publication, New Delhi.
2. Lotfi A. Zadeh(1997), Soft computing and Fuzzy Logic, World Scientific Publishing Co., Inc. River Edge, NJ, USA.

**Reference Books:**

1. B. Yegnanarayana (2006), Artificial Neural Networks, Prentice Hall of India, New Delhi, India.
2. John Yen, Reza Langari(2006), Fuzzy Logic, Pearson Education, New Delhi, India.
3. S. Rajasekaran, Vijayalakshmi Pari (2003), Neural networks, Fuzzy Logic and Genetic Algorithms Synthesis and Applications, Prentice Hall of India, New Delhi, India.

## AGILE SOFTWARE DEVELOPMENT (PROFESSIONAL ELECTIVE-5)

### IV Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS37      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To familiarize the students with the features of agile development.
2. To familiarize the students with basic Agile Estimation & Planning.
3. To facilitate students in understanding the Tracking Agile Projects.
4. To facilitate students in differentiating tools.
5. To felicitate students in Testing

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Define the common characteristics of an agile development process.
2. List and contrast state of the practice agile methodologies.
3. Contrast agile software development process models and plan driven process models.
4. Identify software project characteristics that would be suitable for an agile process.
5. Identify software project characteristics that would not be suitable for an agile process.
6. Plan and implement a small scale software project using the Scrum process methodology.

### UNIT – I

**Classes: 10**

**History of Agile Methodologies, Agile and Lean Software Development:** Basics and Fundamentals, Extreme Programming, Scrum, Agile and Scrum Principles Agile Manifesto, Twelve Practices of XP

### UNIT – II

**Classes: 10**

Agile Estimation & Planning, Agile Requirements, Agile techniques to improve requirements, User Stories, Backlog Management: scrum product backlog, product backlog, Agile Architecture

### UNIT – III

**Classes: 14**

**Part-1:** Tracking Agile Projects: Time tracking agile projects, Tracking agile development project progress

**Part-2:** Lean Software Development, Agile model driven development, Agile Risk management

### UNIT – IV

**Classes: 10**

Agile Project Tools, Enterprise Agility, Dynamic Systems Development Method, Continuous Integration (CI).

### UNIT – V

**Classes: 10**

Team Dynamics and Collaboration, Enterprise agile project team, Agile Testing, Scaling Agile for Large Projects, Best Practices for Achieving Agile at Scale

**Text Books:**

1. Agile Development with Scrum, Ken Schwaber & Mike Beedle, Prentice Hall, 2001
2. Integrating Agile Development in the Real World, Peter Schuh, Charles River Media, 2005

**Reference Books:**

1. Agile Software Development – The Cooperative Game (2nd Edition), Alistair Cockburn, 2007
2. Succeeding With Agile, Software Development Using Scrum, Mike Cohn, Addison Wesley, 2010

## NETWORK MANAGE SYSTEMS (PROFESSIONAL ELECTIVE-5)

### IV Year I Semester: IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A4IT23      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Demonstrate the need for interoperable network management
2. Understand general concepts and architecture behind standards based network management
3. Understand concepts and terminology associated with SNMP and TMN
4. Appreciate network management as a typical distributed application
5. Understand Advanced Information Processing Techniques such as Distributed Object Technologies, Software Agents and Internet Technologies used for network management

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. After the completion of course, the student will able to
2. Acquire the knowledge about network management standards (OSI and TCP/IP)
3. Acquire the knowledge about various network management tools and the skill to use them in monitoring a network
4. Analyze the challenges faced by Network managers
5. Evaluate various commercial network management systems and open network management systems.
6. Analyze and interpret the data provided by NMS and take suitable actions.

### UNIT – I

Classes: 15

**Introduction:** Analogy of Telephone Network Management, Data and Telecommunication Network Distributed computing Environments, TCP/IP- Based Networks: The Internet and Intranets.

**Communications Protocols and Standards-** Communication Architectures, Protocol Layers and Services; Case Histories of Networking and Management – Challenges of Information Technology Managers.

### UNIT – II

Classes: 15

**Network Management:** Goals, Organization, and Functions, Goal of Network Management, Network Provisioning, Network Operations and the NOC, Network Installation and Maintenance; Network and System Management, Network Management System platform, Current Status and Future of Network Management.

**UNIT – III****Classes: 19**

**Basic Foundations: Standards, Models, and Language:** Network Management Standards, Network Management Model, Organization Model, Information Model, Communication Model, Abstract Syntax Notation one:–ASN.1- Terminology, Symbols, and Conventions, Objects and Data Types, Object Names, An Example of ASN.1 from ISO 8824; Encoding Structure, Macros, Functional Model.

**UNIT – IV****Classes: 21**

**SNMPv1 Network Management:** Organization and Information models: Managed Network, The History of SNMP Management, Internet Organizations and standards, The SNMP Model, The Organization Model, and Information Model.

**Communication and Function models:** SNMP Communication Model – The SNMP Architecture, Administrative Model, SNMP Specifications, SNMP Operations, Functional Model

**SNMP Management:** SNMPv2 and SNMPv3 System Architecture

**UNIT – V****Classes: 16**

**SNMP Management – RMON:** Remote Monitoring, RMON SMI and MIB, RMON1- RMON1 Textual Conventions, RMON1 Groups and Functions, Relationship Between Control and Data Tables, RMON1 Common and Ethernet Groups, RMON Token Ring Extension Groups, RMON2 – The RMON2 Management Information Base, RMON2 Conformance Specifications; ATM Remote Monitoring, A Case Study of Internet Traffic Using RMON.

**Text Books:**

1. Mani Subramanian, “Network Management Principles and Practice”, 2nd Edition, Pearson Education, 2010.

**Reference Books:**

1. Morris, “Network management”, 1st Edition, Pearson Education, 2008.
2. Mark Burges, “Principles of Network System Administration”, 1st Edition, Wiley Dream Tech, 2008.

## BIG DATA ANALYTICS LAB

### IV Year I Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS20      | PCC      | 0            | 0 | 4 | 2       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics for Business
3. To introduce the tools, technologies & programming languages this is used in day to day analytics cycle.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Connect to Hadoop cluster, experiment with various Linux and HDFS commands to store data.
2. Apply the knowledge of Map Reduce programming to process the stored data in HDFS.
3. Make use of database operations to store results in tables and generate reports.
4. Connect to web data sources for data gathering, Integrate data sources with Hadoop components to process streaming data.
5. Generate reports using data visualization tools.

## LIST OF EXPERIMENTS

### Week 1

- i) Perform setting up and installing Vmware for Hadoop and Linux.
- ii) Basic Linux Commands

### Week 2

- i) Run basic HDFS shell commands

### Week 3

- i) Implement the following file management tasks in Hadoop:
  - Adding files and directories
  - Retrieving files
  - Deleting files and directories.

**Hint:** A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.

### Week 4

- i) Write the steps to export JAR using eclipse.
- ii) Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm.

**Week 5**

- i) Write a Map Reduce program that mines weather data.  
(Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented).
- ii) Implement Matrix Multiplication with Hadoop Map Reduce

**Week 6**

- i) Run Pig and perform basic PIG commands.

**Week 7**

- i) Write Pig Latin scripts to sort, group, join, project, and filter your data.

**Week 8**

- i) Run HIVE and perform basic HIVE commands to create a table and enter data into tables.

**Week 9**

- i) Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

**Week 10**

- i) Use CDH and HUE to analyze data and generate reports for sample datasets

**Week 11**

- i) Importing and exporting Data in HDFS using Sqoop from MySql database

## **IV B.TECH II SEMESTER SYLLABUS**



## OPERATIONS RESEARCH

### IV B. Tech II Semester: Common for CSE/IT

| Course Code: | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------|----------|--------------|---|---|---------|---------------|-----|-------|
|              |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS05       | HSMC     | 3            | 0 | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Explain optimum utilization of resources, effort and implement the decision effectively.
2. Understand scientific systematic approach involved and provide a good intellectual support for decision making.
3. Discuss substantial experience in taking timely management decisions and for corrective measures.
4. Analyzing the behavior of the system for the purpose of improving its performance.
5. Demonstrate reliability of solution obtained from a model depends on the validity of the model in representing the real systems.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Applying a different linear programming problem technique which has a broad experience in finding the optimum profit.
2. Apply the knowledge of the course in solving real life problems.
3. Calculate the knowledge that tries to optimize total return by maximizing the profit and minimizing the cost or loss
4. Analyze the best (optimum) decisions relative to largest possible portion of the total organization.
5. Analyze queuing models and characteristic features of a queuing system.

### UNIT-I LINEAR PROGRAMMING

Classes: 10

Linear Programming-Simplex methods, Primal & Dual Problem Sensitivity Analysis.

### UNIT-II TRANSPORTATION AND ASSIGNMENT PROBLEMS

Classes: 08

Transportation problem – Methods of finding Initial Basic Feasible Solution, Test for optimality ( MODI method), Assignment Problem –Hungarian Method ,Maximization case in assignment problem, unbalanced assignment problem.

### UNIT-III GAME THEORY

Classes: 08

Decision theory-Decision under various conditions. Game Theory-Minimum & maximum strategies.  
Application of linear programming

#### **UNIT-IV INVENTORY MODELS**

**Classes: 08**

Stochastic inventory models-Single & multi period models with continuous & discrete demands.

Cartesian co-ordinates).

#### **UNIT-V QUEING MODELS**

**Classes: 08**

Simulations-Monte Carlo simulation, generation of random numbers & simulation. Queuing models-M.M.1 & M/M/S system cost consideration

#### **Text Books:**

1. H.A.Taha, — “Operations Research ”, Prentice Hall of India, 1999, Sixth Edition.
2. J.K.Sharma, — “Operations Research Theory and applications”, Macmillian

#### **Reference Books:**

1. R.Panner Selvam – “Operation Research”, PHI Learning Private Limited.
- 2.Hillier and Lieberman, — “Operations Research ”, Holden Day, 1986

## SOFTWARE PROJECT MANAGEMENT (PROFESSIONAL ELECTIVE – VI)

### IV Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS39      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. The main goal of software development projects is to create a software system with a predetermined functionality and quality in a given time frame and with given costs.
2. For achieving this goal, models are required for determining target values and for continuously controlling these values.
3. This course focuses on principles, techniques, methods & tools for model-based engagement of software projects.
4. Assurance of product quality and process adherence (quality assurance), as well as experience-based creation & improvement of models (process management).

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understanding the specific roles within a software organization as related to project and process management
2. Understanding the basic infrastructure competences (e.g., process modeling and measurement)
3. Understanding the basic steps of project planning, project management. Quality assurance, and process management and their relationships
4. Describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project.
5. Compare and differentiate organization structures and project structures.
6. Implement a project to manage project schedule, expenses and resources with the application of suitable project management tools.

### UNIT – I

Classes: 10

**CONVENTIONAL SOFTWARE MANAGEMENT:** The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics. Pragmatic software cost estimation.

**IMPROVING SOFTWARE ECONOMICS:** Reducing Software product size, Improving software processes, improving team effectiveness. Improving automation, Achieving required quality, peer inspections.

### UNIT – II

Classes: 12

**THE OLD WAY AND THE NEW WAY** - The principles of conventional software engineering. Principles of modern software management, transitioning to an iterative process.

**ARTIFACTS OF THE PROCESS:** The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

**UNIT – III****Classes: 13**

**WORK FLOWS OF THE PROCESS:** Software Process Workflow, Inter Trans Workflows.

**CHECKPOINTS OF THE PROCESS:** Major Mile Stones, Minor Milestones, Periodic status assessments.

**ITERATIVE PROCESS PLANNING:** Work breakdown structures, planning guidelines, cost and scheduled estimating, Interaction, planning process, Pragmatic planning.

**UNIT – IV****Classes: 09**

**PROCESS AUTOMATION:** Automation Building blocks, Project Control and Process instrumentation: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation, Tailoring the Process: Process discriminants.

**UNIT – V****Classes: 13**

**PROJECT CONTROL AND PROCESS INSTRUMENTATION:** The seven core Metrics, Management indicators, and quality indicators. life cycle expectations pragmatic Software Metrics, Metrics automation. Tailoring the Process: Process discriminates, Example.

**FUTURE SOFTWARE PROJECT MANAGEMENT:** Modern Project Profiles Next generation Software economics modern Process transitions.

**Case Study:** The Command Centre Processing and Display System, Replacement (CCPDS. R).

**Text Books:**

1. Software Project Management. Walker Royce, Pearson Education.
2. Software Project Management, Bob Hughes & Mike Cotterell, fourth edition, Tate McGraw Hd.

**Reference Books:**

1. Applied Software Project Management, Andrew Stelbman & Jennifer Greene, O'Reilly. 2006
2. Head First PMP, Jennifer Greene & Andrew Stelman, O'Reilly. 2007
3. Software Engineering Project Management. Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004.
4. A Guide to Project Management, Jim Highsmith. Pearson education, 2004
5. The art of Project management. Scott Berkun. O'Reilly, 2005.
6. Software Project Management in Practice. Pankaj Jalote. Pearson Education, 2002.

## MACHINE LEARNING (PROFESSIONAL ELECTIVE – VI)

### IV Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS40      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To introduce the fundamental concepts of machine learning and its applications
2. To learn the classification, clustering and regression machine learning algorithms
3. To understand the methods of solving real life problems using the machine learning techniques

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Illustrate the basic concepts of machine learning
2. Implement the classification, clustering and regression algorithms
3. Design and implement a method for solving real life problem using a suitable machine learning technique
4. Combine the evidence from two or more models/methods for designing a system.

### UNIT – I

Classes: 10

**INTRODUCTION AND BAYESIAN DECISION THEORY:** Machine perception - feature extraction - classification, clustering and regression - design cycle - types of learning, Bayesian decision theory - classifiers, discriminant functions, and decision surfaces - univariate and multivariate normal densities - Bayesian belief networks.

### UNIT – II

Classes: 12

**COMPONENT ANALYSIS AND HIDDEN MARKOV MODELS:** Principal component analysis - Linear discriminant analysis - Independent component analysis. Expectation-maximization algorithm - hidden Markov models: Evaluation - decoding - learning.

### UNIT – III

Classes: 13

**CLASSIFICATION ALGORITHMS:** Perceptron and back propagation neural network - radial basis function neural network - probabilistic neural network - k-nearest-neighbour rule. Support vector machine: Training - multiclass generalizations. Decision trees: classification and regression tree - random forest.

**UNIT – IV****Classes: 13**

**CLUSTERING AND REGRESSION ALGORITHM:** k-means clustering - fuzzy k-means clustering - Gaussian mixture models - auto associative neural network. Regression analysis - support vector regression.

**UNIT – V****Classes: 12**

**COMBINING MULTIPLE LEARNERS:** Generating diverse learners - model combination schemes - voting - error-correcting output codes - bagging - boosting - mixture of experts revisited - stacked generalization - fine-tuning an ensemble - cascading.

**Text Books:**

1. R. O. Duda, E. Hart, and D.G. Stork, "Pattern classification", Second edition, John Wiley & Sons, Singapore, 2003.
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Third Edition, 2014.
3. C. M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.

**Reference Books:**

1. M. Mohri, A. Rostamizadeh, and A. Talwalkar, "Foundations of Machine Learning", MIT Press, 2012.
2. Kevin P. Murphy, "Machine Learning: A Probabilistic Perspective", MIT Press, 2012.

## INFORMATION RETRIEVAL SYSTEM (PROFESSIONAL ELECTIVE – VI)

### IV Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A4CS41      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To Learn

1. To learn about the relationships to data base management systems and libraries.
2. To understand the important concepts, algorithms, and data/file structures
3. To learn about clusters, searching techniques and visualization.
4. To expose the students to find the searching algorithms and Evaluation systems.
5. To facilitate design, and implementation of Information Retrieval (IR) systems.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the retrieval information systems and its capabilities.
2. Analyze and apply appropriate data structures and indexing concepts.
3. Use the various searching techniques for improving the information visualization.
4. Apply the searching algorithms to evaluate information systems.
5. Understand and use the multimedia information retrieval of query languages.

### UNIT – I

Classes: 08

**INTRODUCTION TO INFORMATION RETRIEVAL SYSTEMS:** Definition, Objectives, Functional Overview, Relationship to Database Management Systems, Digital Libraries and Data Warehouses.

**INFORMATION RETRIEVAL SYSTEM CAPABILITIES:** Search, Browse and Miscellaneous.

### UNIT – II

Classes: 12

**CATALOGING AND INDEXING:** Objectives, Indexing Process, Automatic Indexing, Information Extraction. **DATA STRUCTURES:** Introduction, Stemming Algorithms, Inverted file structures, N-gram data structure, and PAT data structure, Signature file structure, Hidden Markov Models.

**AUTOMATIC INDEXING:** Classes of Automatic Indexing, Statistical Indexing, Natural Language, concept, Indexing, Hypertext Linkages.

### UNIT – III

Classes: 12

**DOCUMENT AND TERM CLUSTERING:** Introduction, Thesaurus Generation, Item Clustering,

Hierarchy of Clusters.

**USER SEARCH TECHNIQUES:** Search Statements and Binding, Similarity Measures and Ranking, Relevance Feedback, Selective Dissemination of Information Search, Weighted Searches of Boolean Systems, Searching the Internet and Hypertext.

**INFORMATION VISUALIZATION:** Introduction, Cognition and Perception, Information Visualization Technologies.

#### UNIT – IV

Classes: 12

**TEXT SEARCH ALGORITHMS:** Introduction, Software Text Search Algorithms, Hardware Text Search Systems.

**INFORMATION SYSTEM EVALUATION:** Introduction, Measures used in System Evaluation, Measurement Example -TREC results.

#### UNIT – V

Classes: 14

**MULTIMEDIA INFORMATION RETRIEVAL:** Models and Languages, Data Modeling, Query Languages, Indexing and Searching.

**LIBRARIES AND BIBLIOGRAPHICAL SYSTEMS:** Online IR Systems, OPACs, Digital Libraries.

#### Text Books:

1. Gerald J. Kowalski, Mark T. Maybury (2000), Information Storage and Retrieval Systems: Theory and Implementation, 2nd edition, Springer International Edition, USA.
2. Information Retrieval: Algorithms and Heuristics By David A Grossman and Ophir Frieder, 2<sup>nd</sup> Edition, Springer.

#### Reference Books:

1. Database Systems, 6th edition, RamezElmasri, ShamkantB.Navathe, Pearson Education, 2013.
2. Introduction to Database Systems, C.J.Date, Pearson Education.
3. Fundamentals of Relational Database Management Systems, S. Sumathi, S. Esakkirajan, Springer.



## PREDICTIVE ANALYTICS

### (PROFESSIONAL ELECTIVE – VI)

#### IV Year II Semester: CSE/IT

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS42      | PEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of predictive analysis
3. To introduce linear regression, time series concepts
4. To introduce the tools, technologies, programming languages which are used in day to day analytics cycle.

#### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify basic terminology of Data Analytics
2. Compare learning with classification
3. Develop the knowledge skill and competences using tools and training
4. Analyze the importance of Analytics in business perspective.

#### UNIT – I

Classes: 14

**INTRODUCTION TO PREDICTIVE ANALYTICS & LINEAR REGRESSION (NOS 2101):** What and Why Analytics, Introduction to Tools and Environment, Application of Modelling in Business, Databases & Types of data and variables, Data Modelling Techniques, Missing imputations etc. Need for Business Modelling, Regression— Concepts, Blue property- assumptions- Least Square Estimation, Variable Rationalization, and Model Building etc.

#### UNIT – II

Classes: 10

**LOGISTIC REGRESSION (NOS 2101):** Model Theory, Model fit Statistics, Model Conclusion, Analytics applications to various Business Domains etc.

Regression Vs Segmentation --- Supervised and Unsupervised Learning , Tree Building ---

Regression, Classification, Over fitting, pruning and complexity, Multiple Decision Trees etc.

#### UNIT – III

Classes: 08

**DEVELOP KNOWLEDGE, SKILL AND COMPETENCES (NOS 9005):** Introduction to Knowledge skills & competences, Training & Development, Learning & Development, Policies and Record keeping, etc.

**UNIT – IV****Classes: 10**

**TIME SERIES METHODS / FORECASTING, FEATURE EXTRACTION (NOS 2101):** ARIMA, Measures of Forecast Accuracy, STL approach, Extract features from generated model as Height, Average, Energy etc and Analyze for prediction.

**UNIT – V****Classes: 14**

**WORKING WITH DOCUMENTS (NOS 0703):** Standard Operating Procedures for documentation and Knowledge sharing, Defining purpose and scope documents, Understanding structure of documents – case studies, articles, white papers, technical reports, minutes of meeting etc., Style and format, Intellectual Property and Copyright, Document preparation tools – Visio, PowerPoint, Word, Excel etc., Version Control, Accessing and updating corporate knowledge base, Peer review and feedback.

**Text Books:**

1. Student's Handbook for Associate Analytics-III

**Reference Books:**

1. Gareth James Daniela Witten Trevor Hastie Robert Tibshirani. An Introduction to Statistical Learning with Applications in R

## OPEN ELECTIVES OFFERED BY AERO DEPARTMENT

| OE1    |                                  | OE2    |                                       | OE3    |                                            |
|--------|----------------------------------|--------|---------------------------------------|--------|--------------------------------------------|
| A4AE17 | Fabrication Processes            | A4AE25 | Introduction to aircraft Industry     | A4AE40 | Guidance and control of aerospace vehicles |
| A4AE18 | Fundamentals of Avionics         | A4AE26 | Non destructive testing Methods       | A4AE41 | Wind tunnel Techniques                     |
| A4AE19 | Introduction to jets and rockets | A4AE27 | Fundamentals of Finite element method | A4AE42 | Introduction to Aerospace Technology       |

## FABRICATION PROCESSES

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE17      | OEC      | 3            | 1 | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To develop competency in understanding major fabrication processes used for aircraft production.
2. To have a broad knowledge in choosing a suitable fabrication process for manufacturing aircraft components.

### COURSE OUTCOMES:

The students should be able to

1. Exemplify the major metal casting processes used in aircraft industry.
2. Exemplify the major metal joining processes used in aircraft industry.
3. Exemplify hot & cold working of metals including high energy rate forming.
4. Exemplify the major surface finishing processes used in aircraft industry.
5. Exemplify the major sheet metal forming processes used in aircraft industry.
6. Suggest suitable molding process for plastics for the given application

### UNIT-I

### METAL CASTING PROCESSES

**Classes: 12**

Sand casting – Sand moulds – Types of patterns – Pattern materials – Pattern allowances – Types of Moulding sand- Properties – Core making – Moulding machines – Types of moulding machines – Melting furnaces – Working principle of Special casting processes – Shell, investment casting – Lost Wax process –Centrifugal casting –Sand Casting defects.

### UNIT-II

### METAL JOINING TECHNIQUES

**Classes: 12**

Fusion welding processes – Types of Gas welding – Equipment's used – Flame characteristics – Filler and Flux materials – Arc welding equipment – Electrodes – Coating and specifications – Principles of Resistance welding – Spot/butt, seam welding – Percussion welding – Gas metal arc welding – Flux cored – Submerged arc welding –Electro slag welding – TIG welding – Principle and application of special welding processes – Plasma arc welding – Thermit welding – Electron beam welding – Friction welding –Weld defects – Brazing and soldering process.

### UNIT-III

### BULK DEFORMATION AND SURFACE FINISHING PROCESSES

**Classes: 12**

Hot working and cold working of metals – Forging processes – Open and close die forging – Characteristics of the process – Types of Forging Machines – Typical forging operations – Rolling of metals –Types of Rolling mills –Tube piercing – Defects in rolled parts – Principles of Extrusion – Types of Extrusion – Hot and Cold extrusion – Principle of rod and wire drawing – Equipments used - surface roughness and measurements - grinding - honing - lapping - polishing - burnishing - thermal spray coating.

### UNIT-IV

### SHEET METAL FORMING PROCESSES

**Classes: 12**

Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics

**UNIT-V****FORMING AND SHAPING OF PLASTICS****Classes: 12**

Types of plastics – Characteristics of forming and shaping processes – Moulding of Thermoplastics – Working principles– Injection moulding – Plunger and screw machines – Blow moulding –Extrusion – Typical industrial applications – Thermoforming – Processing of Thermosets – Working principles and typical applications – Compression moulding – Transfer moulding – Bonding of Thermoplastics – Induction and Ultrasonic methods.

**Text Books:**

1. Hajra Choudhury , Elements of Workshop Technology, Vol. I and II, Media Promoters Pvt. Ltd., Mumbai, 2007
2. Serope Kalpak Jain, Steven R, Schmid, Manufacturing Engineering and Technology, Pearson Education, Inc. 4th Edition, 2009

**Reference Books:**

1. Elements of Manufacturing Processes, B.S. Magendran Parashar & R.K. Mittal, Prentice Hall of India, 2008
2. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010.
3. A text book of production technology, P.c. Sharma, S. Chand and Company, 2010
4. Manufacturing Process – Begman, John Wiley& Sons, VIII Edition, 1999

**Web References:**

<https://www.velaction.com/fabrication/>  
<https://www.boconline.co.uk/en/.../fabrication-processes/fabrication-processes.html>

**E-Text Books:**

1. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010.
2. Manufacturing Process – Begman, John Wiley& Sons, VIII Edition, 1999

**MOOC Course:**

<https://nptel.ac.in/courses/112107144/>  
<https://nptel.ac.in/courses/112105127/>  
<https://nptel.ac.in/courses/112104195/>  
<https://nptel.ac.in/courses/112107078/>

## FUNDAMENTALS OF AVIONICS

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE18      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. Overview on Aviation using Electronics
2. Basic understanding about major electronics systems used for communication
3. Basic understanding about major devices, display and flight controls used in aircraft

### COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 To explain the instrumentation used in avionics.
- 2 To classify various ranges of the communication techniques used in aircraft.
- 3 To distinguish between network systems, controlling parts & surfaces
- 4 To compare various principles of navigation systems
- 5 To build phenomena of auto pilot control system

### UNIT-I BASICS & FLIGHT DECK AND DISPLAY SYSTEMS Classes: 8

**BASICS:** Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles.

**FLIGHT DECK AND DISPLAY SYSTEMS:** Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, Electronic instrumentation systems.

### UNIT-II COMMUNICATION SYSTEMS Classes: 10

**AUDIO AND COMMUNICATION SYSTEMS:** Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.

### UNIT-III FREQUENCY RANGING SYSTEM Classes: 12

**RANGING AND LANDING SYSTEMS:** VHF Omniscope, VOR receiver principles, distance measuring equipment, principles of operation, Instrument landing system, and localizer and glide slope.

**POSITIONING SYSTEM:** Global positioning system principles, triangulation, position accuracy, applications in aviation

### UNIT-IV NAVIGATION SYSTEM Classes: 12

**INERTIAL NAVIGATION SYSTEM:** Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform.

**SURVEILLANCE SYSTEM:** ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system

### UNIT-V AUTO FLIGHT SYSTEM Classes: 8

**AUTO FLIGHT SYSTEM:** Automatic flight control systems fly by wire and fly by light technologies, flight director systems, flight management systems..

### **Text Books:**

1. N. S. Nagaraja(1996),Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi.
2. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance.

### **Reference Books:**

1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India.
2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi.
3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London

### **Web References:**

<https://www.britannica.com/technology/avionics>

<https://www.pdc.com/aviation/>

### **E-Text Books:**

Advanced Avionics Handbook(FAA)

Digital Avionics: A Computing Perspective - Elisabeth A. Strunk John C. Knight

### **MOOC Course**

<https://www.canvas.net/browse/erau/courses/aviation-101>

## INTRODUCTION TO JETS AND ROCKETS

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE19      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems.

- 1 Illustrate an overview of aerospace propulsion system.
- 2 Identify the foundation in fundamentals of thermodynamics.
- 3 Compare the ideal components and characteristics of jet engine
- 4 Interpret the performance of nozzles
- 5 Simplify the ideal performance analysis of rocket engines.
- 6 Select appropriate fuel for aerospace application.

### COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Explain the complexity in working of various engines
- 2 Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis
- 3 Analyze the process involved in individual components
- 4 Compare the nozzles with various operating conditions.
- 5 Determine Equations of motion in free space, Tsiolkovsky's equation.
- 6 Classify the types of fuel in aviation and aerospace engineering.

### UNIT-I INTRODUCTION TO AEROSPACE PROPULSION Classes: 12

What is Propulsion system? Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.

### UNIT-II PRINCIPLES OF JET PROPULSION Classes: 12

Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit- significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.

### UNIT-III RAMJET, SCRAMJET ENGINES AND NOZZLES Classes: 12

Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.

### UNIT-IV ROCKET THEORY Classes: 12

Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiolkovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems



**UNIT-V****PROPELLANT ROCKETS****Classes: 12**

Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.

**Text Books:**

1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2<sup>nd</sup>Edt
2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993.
3. Fuels and Combustion, Sharma, S.P. and Mohan .C., Tata McGraw Hill Publishing Co, Ltd., 1984

**Reference Books:**

- 1.The Jet Engine – Rolls Royce
2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.

**Web References:**

<http://engineeringvideolectures.com/course/24>

<http://engineeringvideolectures.com/course/28>

**E-Text Books:**

Jet Propulsion (3rd ed. - Nicholas Cumpsty  
A Simple Guide To Understanding Jet Engines

**MOOC Course**

<https://online.stanford.edu/courses/aa283-aircraft-and-rocket-propulsion>

## INTRODUCTION TO AIRCRAFT INDUSTRY

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE25      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems.

- 1 Familiarize students with major types of aircraft industry.
- 2 Illustrate the role of regulatory bodies and its business context
- 3 Exemplify the organisational structure of airlines and airports
- 4 Illustrate the importance of air navigation services and its types.
- 5 Familiarize students with air safety and security as a means for ensuring passenger safety.

### COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Discuss the types of aerospace industry and evolution of flying machine to airline alliances.
- 2 Explain standardisation and the importance of regulatory bodies.
- 3 Outline the organisational structure of airlines and airports.
- 4 Demonstrate major types of air navigation services and air traffic control services.
- 5 Illustrate the major processes involved in air safety and security.

### UNIT-I Classes: 12

#### AIRCRAFT INDUSTRY OVERVIEW

Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.

### UNIT-II Classes: 12

#### REGULATORY AND BUSINESS CONTEXT

Regulatory bodies, Introduction to standardisation, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometre, yield etc.), Industry characteristics of passenger airlines.

### UNIT-III Classes: 12

#### AIRLINES AND AIRPORTS

Introduction to airlines, Organisational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organisational culture of an airline  
Introduction to airports – Airport personnel, Processing passengers and freight, Security

### UNIT-IV Classes: 12

#### AIR NAVIGATION SERVICES

Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers

**UNIT-V****AIR SAFETY AND SECURITY****Classes: 12**

Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).

**Text Books:**

1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom.
2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.
3. Shevell (2004), Fundamentals of Flight, 2nd edition, Pearson Education Limited, New Delhi
4. Allan Seabridge, Ian Moir (2008), Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, 3rd edition, John Willey & Sons, New York.

**Reference Books:**

1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, Apress, New York
2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modeling Applications in the Airline Industry, Ashgate Publishing Ltd, England.
3. 3.. Kermode, A.C. (1989), Flight without Formulae, 5th edition, Pearson Education Limited, New Delhi.
4. Raymer Daniel (2002), Aircraft Design: A Conceptual Approach, AIAA Publisher, USA.

**Web References:**

<https://www.investopedia.com/features/industryhandbook/airline.asp>

[http://web.mit.edu/airlines/analysis/analysis\\_airline\\_industry.html](http://web.mit.edu/airlines/analysis/analysis_airline_industry.html)

<https://www.ukessays.com/essays/commerce/an-introduction-to-the-airline-industry-business-essay.php>

**E-Text Books:**

Introduction to the Airline Industry (IATA Aviation Distance Learning) Kindle Edition, IATA

Commercial Aviation 101, Greg Gayden.

**MOOC Course**

<https://www.iata.org/training/courses/Pages/airline-industry-introduction-talg50.aspx>

## NON DESTRUCTIVE TESTING METHODS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE26      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To impart knowledge about the non-destructive testing methods.
2. To provide knowledge on the selection of NDT methods for application in engineering industries.

#### COURSE OUTCOMES:

1. Recognize various non-destructive techniques for engineering industries.
2. Select appropriate non-destructive technique for defects detection in manufactured/operating parts.
3. Perform inspection using major non-destructive testing methods.

#### UNIT-I

#### SURFACE TECHNIQUES

**Classes: 12**

Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - ASME, ASTM, AWS, BIS, SAE standard sample specifications - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - properties of liquid penetrants and developers - advantages and limitations of LPT - applications of LPT.

#### UNIT-II

#### MAGNETIC PARTICLE TESTING

**Classes: 12**

Introduction to magnetic particle testing (MPT) - magnetization methods - dry particle and wet fluorescent particle techniques - demagnetization - advantages and limitations of MPT - magnetic flux leakage testing - principle, instrumentation and applications of electromagnetic induction techniques and eddy current testing (ECT) method.

#### UNIT-III

#### ULTRASONIC TESTING

**Classes: 12**

Introduction to ultrasonic testing (UT) - characteristics of ultrasonic waves - principle of UT – UT probes - UT inspection methods (pulse echo, transmission and phased array techniques, PAUT) - types of scanning and displays - application of UT for welded parts.

#### UNIT-IV

#### RADIOGRAPHY TESTING

**Classes: 12**

Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of X-rays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) - inverse square law - exposure charts - penetrameters - safety issues.

#### UNIT-V

#### SPECIAL TECHNIQUES

**Classes: 12**

Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - pressure and leak detection - LASER shearography - acoustic holography.

#### Text Books:

1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012.
2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.

### Reference Books:

1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001
2. W.T. Mc Gonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.
3. Louis Cartz, "Non-Destructive Testing", ASM International, Metals Park Ohio, US, 2007.
4. Barry Hull and Vernon John, "Non Destructive Testing", ELBS/Macmillan, Hampshire, UK, 2015.

### Web References:

- <https://inspectioneering.com/tag/nondestructive+testing>
- <https://www.element.com/materials-testing-services/non-destructive-testing-and-inspection>
- <https://www.twi-global.com/technical-knowledge/faqs/what-is-non-destructive-testing>

### E-Text Books:

1. Nondestructive Testing Methods and New Applications by Mohammed Omar, Intech, 2012.
2. Non-Destructive Testing by S Ramachandran, A Anderson and T Rajasanthosh Kumar, Airwalk Publications, 2017, Kindle Edition.

### MOOC Course

- <https://nptel.ac.in/courses/113106070/>

## FUNDAMENTALS OF FINITE ELEMENT METHODS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE27      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To understand the concept of numerical analysis of structural components.
2. To understand the basic knowledge of Isoparametric Element Formulation.
3. To impart knowledge on both ANSYS, NASTRAN.

### COURSE OUTCOMES:

1. Demonstrate an understanding of the principles and theories of finite element method.
2. Understand the concepts of discretization, approximations, mathematical tools, FEM tools and error control
3. Analyze 2D problems using bar, beam and truss elements.
4. Demonstrate the basic concepts of isoparametric elements and its application in aircraft.
5. Apply Finite element methods for heat transfer and dynamic loading problems.

### UNIT-I INTRODUCTION TO FEM & 1-D PROBLEMS Classes: 12

Basic concept, historical background, application of FEM, general description, comparison of FEM with other methods. Basic equations of elasticity, Stress, strain relations, Strain, Displacement relations. ONE DIMENSIONAL PROBLEM: Finite element modelling coordinates and shape functions. Potential Energy approach: Assembly of Global stiffness matrix and load vector. Finite element equations, Treatment of boundary conditions.

### UNIT-II ANALYSIS OF BEAMS Classes: 12

Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, deflection, stresses.

### UNIT-III 2-D PROBLEMS Classes: 12

CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems.

### UNIT-IV STEADY STATE HEAT TRANSFER ANALYSIS Classes: 12

One dimensional analysis of a fin and two dimensional analysis of thin plate.

### UNIT-V DYNAMIC ANALYSIS Classes: 12

Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar.

### Text Books:

1. R. Tirupathi Chandra patla (2011), Introduction to Finite Elements in Engineering, 4<sup>th</sup> edition, Pearson Education, India.
2. S. S. Rao (2012), The Finite Element Methods in Engineering, 5<sup>th</sup> edition, Elsevier, USA.
3. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1<sup>st</sup> edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India

### Reference Books:

1. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India.
2. J. N. Reddy (2010), An introduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
3. Kenneth H. Huebner, Donald L. rDdewhirst, Douglas E. Smith , Ted G. Byrom (2009), The Finite
4. Element Methodfor Engineers, 3 edition, John Wiley & sons (ASIA) Pvt. Ltd., New York.

### Web References:

<https://nptel.ac.in/courses/112104116/>

### E-Text Books:

[http://web.mit.edu/kjb/www/Books/FEP\\_2nd\\_Edition\\_4th\\_Printing.pdf](http://web.mit.edu/kjb/www/Books/FEP_2nd_Edition_4th_Printing.pdf)

### MOOC Course

<https://nptel.ac.in/courses/112104193/>

## GUIDANCE AND CONTROL OF AEROSPACE VEHICLES

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE40      | OEC      | 3            | 1 | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To introduce the concepts of Navigation, guidance and control
2. To familiarize with various ways in which aerospace vehicles are guided and controlled
3. The dynamic objectives which students also learn to achieve by designing flight control systems.
4. Familiarize with the control principles of rockets and missiles
5. To give insight into the manoeuvres of the space craft

### COURSE OUTCOMES:

At the end of the course the students are able to:

1. Formulate the navigational equations of the space vehicle
2. Describe the guidance of the vehicle with state feed back
3. Explain the automatic control and guidance of the aircraft
4. Evaluate the control techniques of the rockets and missiles
5. Describe major manoeuvres of the space aircraft.

#### UNIT-I

#### NAVIGATION

**Classes: 13**

Introduction, Basic Principles and Definitions; Dead reckoning and Position Fixing, Celestial, Radio, Inertial Navigation; Principle and Construction of Accelerometers, Mechanical Gyros and Ring Laser Gyros, Inertial Measurement Units, Navigation Equations, Sensor Error Models, Kalman Filter, Attitude Heading Reference System, GPS, Terrain Reference Navigation.

#### UNIT-II

#### GUIDANCE

**Classes: 11**

Optimal Terminal Guidance of Interceptors, Optimal Terminal Guidance - planar and non-planar, Robust and Adaptive Guidance, Guidance with State Feedback, Guidance with Normal Acceleration Input, Minimum Energy Orbital Transfer.

#### UNIT-III

#### GUIDANCE AND CONTROL OF AIRCRAFT

**Classes: 13**

Powered Flying Controls, Helicopter Flight Controls, Fly-by-Wire Flight Control, Control laws, Redundancy and Failure Survival, Digital Implementation, Fly-by-Light Flight Control, Auto Pilot, Flight Management Systems, Unmanned Aerial Vehicle.

#### UNIT-IV

#### CONTROL TECHNIQUES/ CONTROL OF ROCKETS AND MISSILES

**Classes: 12**

Open-loop and Closed Loop Control Systems, Multi-variable Optimization, Optimal Control of Dynamic Systems, Hamiltonian and Minimum Principle and Jacobi-Bellman Equation, Linear Time-Varying System with Quadratic Performance Index..

#### UNIT-V

#### CONTROL OF SPACECRAFT

**Classes: 12**

Launch of Satellite/ Spacecraft, Terminal Control of Spacecraft Attitude, Optimal Single-Axis Rotation of Spacecraft, Multi-axis Rotational Manoeuvres of Spacecraft, Spacecraft Control Torques, Rocket Thrusters, Reaction Wheels, Momentum Wheels and Control Moment Gyros, Torque.



### Text Books:

1. Tewari, A.—Advanced Control of Aircraft, Spacecraft and RocketsII, John Wiley & Sons, Ltd, Chichester, UK, 2011
2. Nelson R. C - Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York, 2007.
3. R.P.G Collinson —Introduction to Avionics SystemsII, Springer; 3rd ed. edition, 2011.

### Reference Books:

1. Noton,M. —Spacecraft navigation and Guidancell, Springer-Verlag, Germany, 1998
2. Mc. Cormic 2. B. W - Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA, 2010.
3. Richard H. Battin —An Introduction to the Mathematics and Methods of AstrodynamicsII, AIAA, 1999.

### Web References:

<https://nptel.ac.in/courses/101106043/6>

<https://nptel.ac.in/courses/101106042/>

### E-Text Books:

<https://www.amazon.com/Guidance-Control-Aerospace-Vehicles-Cornelius/dp/1258807521>

### MOOC Course

[https://swayam.gov.in/nd1\\_noc19\\_ae10](https://swayam.gov.in/nd1_noc19_ae10)

<https://www.mooc-list.com/course/abc-flight-mechanics-how-do-planes-fly-futurelearn>

## WIND TUNNEL TECHNIQUES

### VII Semester: OPEN ELECTIVE - III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE41      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To interpret the basic concepts of measurement of forces and moments on models during the wind tunnel testing.
2. To understand the application of various types of wind tunnels.
3. To learn the basic measurement procedure involving wind tunnel testing.

#### COURSE OUTCOMES:

1. Ability to understand basics of aerodynamics and to identify the type of wind tunnel
2. Ability to develop and understand flow visualization techniques over model
3. Ability to understand concepts of low speed and high speed wind tunnels
4. Ability to understand measurement and balancing of loads on model
5. Ability to understand the different types of equipment's for measuring pressure and velocity

#### UNIT-I

#### WIND TUNNELS

**Classes: 12**

Wind Tunnel, layouts and nomenclature, Types of Wind Tunnels – continuous and intermittent -closed circuit and open circuit - closed jet and open jet test section – application. Special purpose tunnels - Smoke Tunnels – Water Tunnels – Spin tunnel, automobile wind tunnel and environmental wind tunnel Important parameters of flow similarity. types of flow similarities for compressible and incompressible flows Model power consideration.

#### UNIT-II

#### FLOW VISUALIZATION TECHNIQUES

**Classes: 12**

Path – Streak – Stream and Timelines; Techniques: Smoke, Tuft, Streaks, Surface oil flow. Pressure measurements: Manometers – U-Tube, Inclined and Precession. Bourdon Gauge and Pressure Transducer – Strain Gauge, Semi conductor – Absolute and Differential. Velocity Measurements: Pivot Tube – Static and Total. Calibration of test section: Test section flow calibration and Boundary Layers.

#### UNIT-III

#### MEASUREMENTS OF FORCES AND MOMENTS

**Classes: 12**

Forces, moments and Reference Frames – Balances – Internal and External - Requirements and Specifications – Fundamentals of Model Installations. Boundary correction, types of blockages: 2-d.

#### UNIT-IV

#### HIGH SPEED WIND TUNNELS

**Classes: 12**

Supersonic Wind Tunnels and - Classification - Runtime - Compressors - Charging Times - nozzle Mass Flows - Starting Loads - Model Size – Calibration.

Hypersonic Wind Tunnels: Classification – Runtime – Vacuum Tanks – Vacuum pumps – Evacuation Times. Shock Tube: Driver – driven – Vacuum Pumps – Diaphragm.

#### UNIT-V

#### HIGH SPEED FLOW VISUALIZATIONS AND MEASUREMENTS

**Classes: 12**

Schlieren and Shadow Graph – Pressure sensitive Paints – Temperature sensitive Paints – Force Measurements – Strain Gauge Balances – Pressure Measurements.

#### Text Books:

1. Rae, W.H. and Pope, A. —Low Speed Wind Tunnel Testingll, John Wiley Publication, 1999
2. Pope, A., and Goin, L., —High Speed wind Tunnel Testingll, John Wiley Publication , 1999
3. Pope, J B Barlow —low speed wind tunnel testing — 3 edition j.w publication

### Reference Books:

1. John D. Anderson, Jr., "Fundamentals of Aerodynamics", Third edition, McGraw-Hill publications, 2001
2. E L Houghton and PW Carpenter, "Aerodynamics for Engineering students", Fourth edition, Edward Arnold publications, 1993.
3. L.M Miline Thomson, — Theoretical Aerodynamics II, 1996 McGraw-Hill, New Delhi.
4. R. Halmshaw (1991), Non-Destructive Testing, 2nd edition, Edward Arnold, New York

### Web References:

[wingsofaero.in/introduction-to-wind-tunnel-techniques](http://wingsofaero.in/introduction-to-wind-tunnel-techniques)  
<https://www.keyshone.com/wind-tunnel-techniques-lecture-notes-for-aero/>

### E-Text Books:

<https://www.scribd.com/document/324894849/wind-tunnel-technique-notes-for-aeronautical-engineers-tp-1-pdf>  
[https://www.academia.edu/.../john\\_d.\\_anderson\\_jr.\\_fundamentals\\_of\\_aerodynamics\\_](https://www.academia.edu/.../john_d._anderson_jr._fundamentals_of_aerodynamics_)

### MOOC Course

<https://nptel.ac.in/courses/101106040>

## INTRODUCTION TO AEROSPACE TECHNOLOGY

### VII Semester: OPEN ELECTIVE - III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4AE42      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To introduce the basic concepts of Aerospace technology and the current developments in the field.
2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.

### COURSE OUTCOMES:

Students should be able to

1. Compare the atmosphere conditions of different altitudes for spacecraft system
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in a satellite mission
5. Design lighter than air vehicle using Catia software

### UNIT-I HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT Classes: 12

Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments

### UNIT-II AERODYNAMICS AND FLIGHT VEHICLE PROPULSION Classes: 12

Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.

### UNIT-III FLIGHT VEHICLE PERFORMANCE AND STABILITY Classes: 12

Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes

### UNIT-IV SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION Classes: 12

Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity

### UNIT-V INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION Classes: 12

Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.

**Text Books:**

1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0.
2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082

**Reference Books:**

1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017.
2. Ajoykumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.

**Web References:**

<https://www.grc.nasa.gov/www/k-12/airplane/index.html>

**E-Text Books:**

[https://www.researchgate.net/publication/259576416\\_Fundamentals\\_of\\_Aerospace\\_Engineering](https://www.researchgate.net/publication/259576416_Fundamentals_of_Aerospace_Engineering)  
[http://www.adl.gatech.edu/extrovert/Ebooks/ebook\\_Intro.pdf](http://www.adl.gatech.edu/extrovert/Ebooks/ebook_Intro.pdf)

**MOOC Course**

[https://onlinecourses.nptel.ac.in/noc18\\_ae07/](https://onlinecourses.nptel.ac.in/noc18_ae07/)

## OPEN ELECTIVES OFFERED BY CSE DEPARTMENT

| OE1    |                                 | OE2    |                                         |
|--------|---------------------------------|--------|-----------------------------------------|
| A4CS43 | Fundamentals of Databases       | A4CS46 | Introduction to Computer Networks       |
| A4CS44 | Introduction to C++             | A4CS47 | Introduction to Linux Programming       |
| A4CS45 | Core Java Programming           | A4CS48 | Fundamentals of Artificial Intelligence |
| OE3    |                                 | OE4    |                                         |
| A4CS49 | Soft Computing                  | A4CS52 | Introduction to Network Security        |
| A4CS50 | Introduction to Cloud Computing | A4CS53 | Data Analytics                          |
| A4CS51 | Web Programming                 | A4CS54 | .NET & C# Programming                   |

## FUNDAMENTALS OF DATABASES

### III B. TECH- I SEMESTER - OPEN ELECTIVE – I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4CS43      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. To understand the basic database concepts, data models including the architecture and operation of the relational data model.
2. To make a study of Structured Query Language (SQL) and construct simple and moderately advanced database queries using SQL.
3. To conceptualize relational database design and depict a database system using ER diagram.
4. To know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure.
5. To develop a database application using commercial databases

#### COURSE OUTCOMES:

1. Able to design effective database schema for realistic applications.
2. Design a database using ER modelling.
3. Make use the basic concepts of database systems in database design.
4. Apply SQL queries to interact with database.
5. Apply normalization on database design to eliminate anomalies.

#### UNIT-I INTRODUCTION

Classes: 12

Introduction :Relational model, File System vs. Database System -- Data Models – Database System Architecture – Relational Databases - Relational Model - Keys -- Integrity Constraints - Relational Algebra

#### UNIT-II SQL FUNDAMENTALS

Classes: 10

SQL FUNDAMENTALS :SQL fundamentals - SQL Data Definition - Basic Structure of SQL Queries - Additional Basic Operations - Set Operations - Aggregate Functions - Nested Subqueries - Join Operations - Views

#### UNIT-III DATABASE DESIGN

Classes: 15

DATABASE DESIGN : Entity-Relationship Model – ER Diagrams - Mapping ER Model to Relational Model - Functional Dependencies  
Normalization - First Normal Form - Second Normal Form - Third Normal Form

#### UNIT-IV TRANSACTION MANAGEMENT

Classes: 13

TRANSACTION MANAGEMENT: Transaction Concepts - Transaction Recovery – ACID Properties - Concurrency Control – Need for Concurrency Control – Serializability - Locking Protocols – Two Phase Locking – Deadlock – Recovery

#### UNIT-V APPLICATION DEVELOPMENT

Classes: 10

APPLICATION DEVELOPMENT :Commercial Databases - Oracle/MySQL - PL/SQL – Stored Procedures - Functions - Triggers - Database Security using SQL commands

#### TEXT BOOKS:

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, “Database System Concepts”, Sixth Edition, Tata McGraw Hill, 2010.

#### REFERENCE BOOKS:

1. C.J.Date, A.Kannan, S.Swamynathan, "An Introduction to Database Systems", Eighth Edition, Pearson Education, 2006.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Sixth Edition, Pearson Education, 2010.
3. Raghu Ramakrishnan, Johannes Gehrke "Database Management Systems", Fourth Edition, Tata McGraw Hill, 2010.
4. G.K.Gupta, "Database Management Systems", Tata McGraw Hill, 2011.
5. Carlos Coronel, Steven Morris, Peter Rob, "Database Systems: Design, Implementation and Management", Ninth Edition, Cengage Learning, 2011
6. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3<sup>rd</sup> Edition, 2014.

#### WEB REFERENCES:

1. <http://iips.icci.edu.iq/images/exam/databases-ramaz.pdf>
2. [https://www.w3schools.com/sql/sql\\_intro.asp](https://www.w3schools.com/sql/sql_intro.asp)

#### E-TEXT BOOKS:

1. <http://www.freebookcentre.net/Database/dbCategory.html>

#### MOOC Course

1. <https://swayam.gov.in/course/1322-fundamentals-of-database-system>

#### OUTCOMES:

**The course should enable the students to:**

1. List and explain the fundamental concepts of a relational database system.
2. Understand and use DML, DDL and DCL to query, update, and manage a database.
3. Convert the ER-model to relational tables, populate relational database and formulate SQL Queries on data.
4. Design database using E-R modeling and apply normalization techniques over it.
5. Manage the transactions that happens in a database.
6. Design and build a simple database system and demonstrate competence with the Fundamental tasks involved with modelling, designing, and implementing a DBMS



## INTRODUCTION TO C++

### III B. TECH- I SEMESTER - OPEN ELECTIVE – I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS44      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To familiarize the students with the different techniques pertaining to problem solving skills.
2. To familiarize the students with basic concepts of object oriented programming.
3. To facilitate students in understanding the syntax and semantics of the “C++” language as well as basic constructs offered by the language
4. To facilitate students in solving real world problems using Object Oriented approach.
5. To impart the concepts of File handling techniques
6. To familiarize students with Generic Programming

#### COURSE OUTCOMES:

Students will be able to

1. Understand different paradigms in solving the problem and the different concepts of object oriented programming. Analyze the importance of Procedure Oriented over Object Oriented Programming.
2. Analyze the performance of OOP system with POP in terms of the complexity.
3. Understand the key concepts of OOP. Apply OOP concepts and design a new system. Analyze various Inheritance & polymorphism techniques to design a system.
4. Understand file handling mechanism in C++ and design data base applications.
5. Understand Templates in C++ and Design C++ programs using Templates and Analyze the performance of OOP & POP program in terms of reusability.

#### UNIT-I

Classes: 10

Introduction to Object Oriented Programming, Structure of C++ program, Tokens, Keywords, Identifiers and Constants, Basic Data Types, Operators and Type Conversions.

**Functions:** Returning values from functions. Reference arguments, Overloaded function, Inline function, Default arguments, returning by reference.

#### UNIT-II

Classes: 12

**Object and Classes:** Specifying a Class, Member function, Nesting of Member functions, Arrays in a Class, Memory Allocation for Objects, Static Members and Member Functions, Array of Objects, Friend Functions, Pointers to Members.

**Constructors and Destructors:** Parameterized Constructor, Copy Constructor, Dynamic Constructor, Destructors.

#### UNIT-III

Classes: 12

**Operator Overloading:** Overloading unary, binary operators, data conversion, pitfalls of operators overloading and conversion keywords, Explicit and Mutable.

**Inheritance:** Derived class and Base class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Virtual Base Classes, Abstract Classes, Constructor in Derived Classes.

#### UNIT-IV

Classes: 13

**Pointer:** Introduction, Pointers to Objects, this Pointer, Pointers to Derived Class, Virtual Functions, Pure Virtual Functions.

**Streams and Files:** C++ Streams, Stream Classes, Unformatted IO Operations, Formatted Console IO Operations, Managing Output with Manipulators.

#### UNIT-V

Classes: 13

**Files:** Classes for File Handling, File Pointer, Updating a File, Error Handling in Files.

**Templates and Exception Handling:** Class Templates, Function Templates, Exception Handling Mechanism, And Components of STL: Containers, Algorithms, Iterators.

**TEXT BOOKS:**

1. Object Oriented Programming with C++, *E Balaguruswamy*, Third Edition, and McGraw Hill Publications.

**REFERENCE BOOKS:**

1. Object Oriented Programming in C++ by *Robert Lafore* Techmedia Publication.
2. Object Oriented Programming in C++ *Saurav Sahay* Oxford University Press.

## CORE JAVA PROGRAMMING

### III B. TECH- I SEMESTER - OPEN ELECTIVE – I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS45      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. Learning principles of object oriented programming paradigm including abstraction, encapsulation, inheritance and polymorphism.
2. Understand fundamentals of programming such as variables, conditional and iterative execution, methods etc.
3. Understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
4. Learning the concept of inheritance to create new classes from existing one & Design the classes needed given a problem specification;
5. Understand the concept of packages and interfaces.
6. Learning how to detect exceptions and to handle strings & Implement the designed classes using the object oriented programming language
7. Learn how to test, verify, and debug object-oriented programs; and Learning about multithreading and multitasking.
8. Learn & demonstrate the concept of event handling, JDBC Connectivity used in GUI.
9. Create Applications using object-oriented principles with concepts of JAVA

#### COURSE OUTCOMES:

1. Analyze the necessity for Object Oriented Programming paradigm over structured programming and become familiar with the fundamental concepts in OOP like encapsulation, Inheritance and Polymorphism
2. Demonstrate an ability to design and develop java programs, analyze, and interpret object oriented data and report results.
3. Demonstrate an ability to design an object oriented system, AWT components and multithreaded processes as per needs and specifications..
4. Demonstrate an ability to visualize and work on laboratory and multidisciplinary tasks like console and windows applications both for stand alone and Applets programs.
5. Demonstrate skills to use latest object oriented programming language and software like java to analyze OOP problems
6. Develop confidence for self education and ability for life-long learning needed for advanced java technologies
7. Able to participate and succeed in competitive examinations like GATE, Engineering services, recruitment interviews etc.

#### UNIT-I

Classes: 11

**Oops concepts-** Procedural and objects oriented programming paradigms, classes and objects, Data abstraction, encapsulation, polymorphism, inheritance, benefits of inheritance.

**Java Programming-** History of java, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expression, type conversion and casting, control flow- block scope, condition statements, loops, break, and continue statements, simple java stand alone programs, arrays, constructors, methods, parameter passing, static keyword, access control, this pointer, overloading methods and constructor, recursion, garbage collection, exploring string class

#### UNIT-II

Classes: 12

**Inheritance-** Inheritance hierarchies super and sub classes, member access rules, super keyword, final keyword, the Object class.

**Polymorphism-** dynamic binding, method overriding, abstract classes and methods. Interface- Interfaces vs. Abstract classes, defining an interface, implementing interfaces, implementing multiple inheritance using interfaces, extending interfaces.

**Inner classes-** Uses of inner classes, local inner classes, anonymous inner classes, static inner

classes.

### UNIT-III

Classes: 12

**Packages-** Defining, Creation and Accessing Packages, Understanding CLASSPATH, importing packages. **Exception handling-** Types of errors, benefits of exception handling, classification of exceptions- exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception sub classes.

**Files-** streams- byte streams, character streams, text input/output, binary input/output, File management

### UNIT-IV

Classes: 13

**Multithreading-** Difference between multiprocessing and multithreading, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem.

**AWT CONTROLS:** The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, ScrollBar. Working with Frame class, Colour, Fonts and layout managers.

### UNIT-V

Classes: 12

**GUI Programming with Java-** Introduction to Swing, Hierarchy for Swing components, Swing vs. AWT, Containers- JFrame, JApplet, JDialog, JPanel, JButton, JLabel, JTextField, JTextArea

**Event Handling-** Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model, Examples: handling a button click, handling mouse events, Adapter classes.

### TEXT BOOKS:

1. Java Fundamentals – A comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH
2. Java A Beginner's Guide Sixth Edition , Herbert Schildt

### REFERENCE BOOKS:

1. Java for Programmers, P.J. Deitel and H.M. Deitel, Pearson education (OR) Java: How to Program P.J. Deitel and H.M. Deitel, PHI.
2. Object Oriented Programming through Java, P. Radha Krishna, Universities Press.
3. Thinking in Java, Bruce Eckel, Pearson Education
4. Programming in Java, S. Malhotra and S. Choudary, Oxford Univ. Press.

## INTRODUCTION TO COMPUTER NETWORKS

### III B. TECH- II SEMESTER - OPEN ELECTIVE – II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS46      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To introduce the fundamentals of various types of computer networks
2. To demonstrate the TCP/IP and OSI models with merits and demerits
3. To explore the various layers of OSI model
4. To introduce UDP and TCP models
5. To explore routing and application layer protocols

#### COURSE OUTCOMES:

By the end of this course, student will be able to

1. Define and describe functionality and services offered at various layers of OSI and TCP/IP
2. Analyse the internal implementation of Data link layer, Network Layer, Transport Layer and
3. Application layer.
4. Implement the various routing protocols, congestion control protocols.
5. Analyse and solve problems with respect to various layers of reference model.

#### UNIT-I

Classes: 12

**INTRODUCTION TO COMPUTER NETWORKS:** Uses of Computer Networks, Network Hardware, Network Software, Reference Models (OSI, TCP/IP), Example Networks. **THE PHYSICAL LAYER-** Basis for Data Communication, Guided Transmission Media, Wireless Transmission, Communication Satellites, Digital Modulation And Multiplexing, The Public Switched Telephone Network.

#### UNIT-II

Classes: 12

**THE DATA LINK LAYER-** Data Link Layer Design Issues, Error Detection and Correction, Elementary Data Link Protocols, Sliding Window Protocols. The Channel Allocation Problem, Multiple Access Protocols.

#### UNIT-III

Classes:12

**THE NETWORK LAYER-** Virtual Circuit and Datagram Networks, Inside A Router, The Internet Protocol (IP): Forwarding and Addressing in The Internet, Routing Algorithms, Routing in The Internet, Broadcast And Multicast Routing.

#### UNIT-IV

Classes: 12

**TRANSPORT LAYER:** Transport-Layer Services, Multiplexing and Demultiplexing, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control, TCP Congestion Control.

#### UNIT-V

Classes: 12

**APPLICATION LAYER:** Principles of Network Applications, The Web and HTTP, File Transfer: FTP, Electronic Mail In The Internet, DNS—The Internet's Directory Service, Peer-To-Peer Application.

#### TEXT BOOKS:

1. Andrew S Tanenbaum David J Wetherall, Computer Networks, 5th Edition, Pearson Education.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education, India.

**REFERENCE BOOKS:**

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers.

## INTRODUCTION TO LINUX PROGRAMMING

### III B. TECH- II SEMESTER - OPEN ELECTIVE – II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS47      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To teach principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linux commands, Scripts and filters.
2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts.
3. To impart fundamentals of file concepts kernel support for file, File structure related system calls (file API's).
4. To facilitate students in understanding Inter process communication.
5. To facilitate students in understanding semaphore and shared memory.  
To facilitate students in understanding process

#### COURSE OUTCOMES:

1. Ability to use various Linux commands that are used to manipulate system operations at admin level and a prerequisite to pursue job as a Network administrator.
2. Ability to write Shell Programming using Linux commands.
3. Ability to design and write application to manipulate internal kernel level Linux File System.
4. Ability to develop IPC-API's that can be used to control various processes for synchronization.
5. Ability to develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.

#### UNIT-I

Classes: 12

**INTRODUCTION TO LINUX AND LINUX UTILITIES:** A brief history of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor.

Linux commands- PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, lp, od, tar, gzip, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp, telnet, rlogin. Text Processing utilities and backup utilities , tail, head , sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, cpio

#### UNIT-II

Classes: 13

**Introduction to Shells:** Linux Session, Standard Streams, Redirection, Pipes, Tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization.

**Filters:** Filters and Pipes, Concatenating files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files

#### UNIT-III

Classes: 12

**Grep:** Operation, grep Family, Searching for File Content.

**Sed :** Scripts, Operation, Addresses, commands, Applications, grep and sed.

**UNIX FILE STRUCTURE:** Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers.

**File Management :** File Structures, System Calls for File Management – create, open, close, read, write, lseek, link, symlink, unlink, stat, fstat, lstat, chmod, chown, Directory API – opendir, readdir, closedir, mkdir, rmdir, umask.

**UNIT-IV****Classes: 12**

**PROCESS AND SIGNALS:** Process, process identifiers, process structure: process table, viewing processes, system processes, process scheduling, starting new processes: waiting for a process, zombie processes, orphan process, fork, vfork, exit, wait, waitpid, exec, signals functions, unreliable signals, interrupted system calls, kill, raise, alarm, pause, abort, system, sleep functions, signal sets. File locking: creating lock files, locking regions, use of read and write with locking, competing locks, other lock commands, deadlocks.

**UNIT-V****Classes: 11**

**INTER PROCESS COMMUNICATION:** Pipe, process pipes, the pipe call, parent and child processes, and named pipes: fifos, semaphores: semget, semop, semctl, message queues: msgget, msgsnd, msgrcv, msgctl, shared memory: shmget, shmat, shmdt, shmctl, ipc status commands.

**INTRODUCTION TO SOCKETS:** Socket, socket connections - socket attributes, socket addresses, socket, connect, bind, listen, accept, socket communications.

**TEXT BOOKS:**

3. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3<sup>rd</sup> edition, Pearson Education, New Delhi, India.
4. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg. Thomson

**REFERENCE BOOKS:**

6. Linux System Programming, Robert Love, O'Reilly, SPD.
7. Advanced Programming in the UNIX environment, 2nd Edition, W.R.Stevens, Pearson Education.
8. UNIX Network Programming, W.R. Stevens, PHI.  
UNIX for Programmers and Users, 3rd Edition, Graham Glass, King Ables, Pearson Education



## FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

### III B. TECH- II SEMESTER - OPEN ELECTIVE – II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS48      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. To learn the difference between optimal reasoning vs human like reasoning
2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. To learn different knowledge representation techniques
4. To understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

#### COURSE OUTCOMES:

1. Possess the ability to formulate an efficient problem space for a problem expressed in English.
2. Possess the ability to select a search algorithm for a problem and characterize its time and space complexities.
3. Possess the skill for representing knowledge using the appropriate technique
4. Possess the ability to apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing

#### UNIT-I Introduction

Classes:10

Introduction, History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A\*, Constraint Satisfaction. Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning

#### UNIT-II Logic Concepts and Logic Programming & Knowledge Representation

Classes:12

Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.

#### UNIT-III Expert System and Applications & Uncertainty Measure – Probability Theory

Classes:13

Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools. Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster-Shafer Theory.

#### UNIT-IV Machine-Learning Paradigms & Artificial Neural Networks

Classes:13

. Machine-Learning Paradigms: Introduction. Machine Learning Systems. Supervised and Unsupervised Learning. Inductive Learning. Learning Decision Trees, Deductive Learning. Clustering, Support Vector Machines. Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.

#### UNIT-V Advanced Knowledge Representation Techniques

Classes:12

Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, Universal Networking Knowledge.

**TEXT BOOKS:**

1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011
2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004

**REFERENCE BOOKS:**

1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009.
2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson.
3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI.
4. Artificial Intelligence by George Fluger Pearson fifth edition.

## SOFT COMPUTING

### IV B. TECH- I SEMESTER - OPEN ELECTIVE – III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS49      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. Basics of ANN and its learning algorithms.
2. Fuzzy principles and relations.
3. Genetic algorithms and its applications.
4. Hybrid systems and usage of MATLAB toolbox.

#### COURSE OUTCOMES:

1. To understand neural network (NN) paradigms
2. To learn fuzzy logic, to have a knowledge of evolutionary computations, genetic algorithm(GA), evolutionary programming, classifier systems, genetic programming parse trees, mathematical foundation of GA variants of GA.

#### UNIT-I NEURAL NETWORKS

Classes: 14

Fundamentals of Neural Networks–History-Architectures-Learning methods-XOR problem-Delta rule-derivation-Back propagation-applications-parameters inBPN-Associative memory–Hetero associative-BAM-energy function-problems-applications of associative memories-ART1-ART2-applications of adaptive Networks.

#### UNIT-II FUZZY LOGIC

Classes: 10

Fuzzy set theory–crisp sets–fuzzysets–crisp relations–Fuzzy relations–Fuzzy systems-Crisp logic–predicate logic–fuzzy logic-fuzzy based systems-Defuzzification methods–applications.

#### UNIT-III GENETIC ALGORITHMS

Classes: 14

Fundamentals of GA–creation of offsprings–encoding–fitness function-reproduction–crossover-Insertion & deletion-mutation-bitwise operators–applications.

#### UNIT-IV HYBRID SYSTEMS

Classes: 12

Hybrid systems–Neuro Fuzzy–Neuro Genetic–fuzzy Genetic hybrids-GA based weight determination and applications-fuzzy BPN–simplified fuzzy ARTMAP.

#### UNIT-V FPROGRAMMING USING MATHLAB

Classes: 10

Using Neural Network toolbox–Using Fuzzy Logic toolbox-Using Genetic Algorithm & directed search toolbox.

#### Text Books:

1. Rajasekaran.S and VijayalakshmiPai.G.A, “Neural Networks, Fuzzy Logic and Genetic Algorithms”, PHI, 2011

#### Reference Books:

1. Timothy J.Ross, “Fuzzy Logic with Engineering applications”, John Wiley and Sons, 2010.
2. Jang.J.S.R, Sun.C.T, Mizutani.E, “Neuro fuzzy and Soft Computing”, PHI Learning Pvt. Ltd., 2012.
3. Davis E.Goldberg, “Genetic Algorithms: Search, Optimization and Machine Learning”, Addison Wesley, N.Y., 1989.

### Web References:

1. [http://www.myreaders.info/html/soft\\_computing.html](http://www.myreaders.info/html/soft_computing.html)
2. [https://en.wikipedia.org/wiki/Soft\\_computing](https://en.wikipedia.org/wiki/Soft_computing)

### E-Text Books:

1. <http://freecomputerbooks.com/Introduction-to-Soft-Computing.html>
2. <https://www.shroffpublishers.com/books/9789351106159/>
3. <https://bookboon.com/en/introduction-to-soft-computing-ebook>

### MOOC Course:

1. [https://onlinecourses.nptel.ac.in/noc18\\_cs13/course](https://onlinecourses.nptel.ac.in/noc18_cs13/course)
2. <https://swayam.gov.in/course/4574-introduction-to-soft-computing>

## INTRODUCTION TO CLOUD COMPUTING

### IV B. TECH- I SEMESTER - OPEN ELECTIVE – III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS50      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. **Understand** the Virtualization paradigms
2. **Learn** the Cloud Computing fundamentals and its importance to various organizations.
3. **Analyze** the concepts of IaaS, PaaS, SaaS, Public and Private Clouds.
4. **Develop** applications in cloud security.

#### COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Understand the main concepts, key technologies, strengths, and limitations of virtualization and cloud computing and the possible applications for state-of-the-art cloud computing.
2. Describe the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc.
3. Analyze the core issues of cloud computing such as security, privacy, and interoperability.
4. Identify problems, analyze, and evaluate various cloud computing solutions.

#### UNIT-I INTRODUCTION

**Classes: 12**

System Modeling, Clustering And Virtualization: Distributed system models and enabling technologies, computer clusters for scalable parallel computing, virtual machines and virtualization of clusters and data centers.

#### UNIT-II FOUNDATIONS

**Classes: 15**

Foundations: Introduction to cloud computing, migrating into cloud, enriching the integration of service paradigm for cloud era, the enterprise cloud computing paradigm.

#### UNIT-III INTER PROCESS COMMUNICATION

**Classes: 17**

Inter Process Communication: Infra Structure As Service (IAAS) & Platform And Software Service(PAAS/SAAS): Virtual machine provisioning and migration services, on the management of virtual machines for cloud infrastructure, enhancing cloud computing environments using a cluster as service.

Secure distributed data storage in cloud computing. Aneka, comet cloud, T-systems, work flow engine for clouds, understanding scientific applications for cloud environments

#### UNIT-IV MONITORING, MANAGEMENT AND APPLICATIONS

**Classes: 10**

Monitoring, Management And Applications : An Architecture for federated cloud computing ,SLA management in cloud computing, performance prediction for HPC on clouds, best practices in Architecting cloud applications in the AWS cloud, building content delivery networks using clouds, resource cloud mashups.

#### UNIT-V GOVERNANCE AND CASE STUDIES:

**Classes: 10**

Governance and case studies: organizational readiness and change management in cloud age, data security in cloud, legal issues in cloud computing, achieving production readiness for cloud services.

#### TEXT BOOKS:

1. Cloud computing: principles and paradigms by rajkumar buyya, James Broberg and Andrzej M.Goscinski,Wiley,2011.
2. Distributed and cloud computing , kai Hwang, Geoffery C.fox, jack J.dongarra, Elsevier, 2012 .

**REFERENCE BOOKS:**

1. Cloud Computing :A practical approach, Anthony T.veite, Toby J.veite, Robert Isenpeter, Tata McGraw Hill,2011.
2. Enterprise Cloud Computing, Gautam Shroff, Cambridge University press,2010.
3. Cloud computing: implementation, management and security, john W .Ritting house ,James F. Ransome ,CRC press,rp2012.
4. Cloud applications architectures: building Applications and infrastructure in the cloud,George Reese , O reilly,SPD,rp2011.
5. Cloud security and privacy: An Enterprise perspective on Risks and compliance,im Mather, Subra Kumaraswamy, Shahed Latif,Oreilly,SPD,rp2011

**E-Text Books:**

1. <https://www.amazon.com/Cloud-Computing-Principles-Rajkumar-Buyya/dp/0470887990>
2. <https://eniac2017.files.wordpress.com/2017/03/distributed-and-cloud-computing.pdf>

**MOOC Course:**

4. [https://onlinecourses.nptel.ac.in/noc18\\_cs33/preview](https://onlinecourses.nptel.ac.in/noc18_cs33/preview)
5. <https://www.alison.com/courses/Introduction-to-cloud-computing>
6. <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-cloud-computing-january-iap-2014/index.htm>

## WEB PROGRAMMING

### IV B. TECH- I SEMESTER - OPEN ELECTIVE – III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS51      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To teach students the basics of server side scripting using HTML
2. To explain web application development procedures
3. To impart servlet technology for writing business logic
4. To facilitate students to connect to databases using JDBC
5. To familiarize various concepts of application development using XML

#### COURSE OUTCOMES:

1. Students will be able to use HTML, CSS to develop web pages.
2. Students will be able to apply concepts of Java.
3. Students will be able to build JDBC connectivity.
4. Students will be able to identify the basic building blocks of XML and describe the applications of web services.
5. Students will be able to write server side programming using applets.

#### UNIT-I SCRIPTING

Classes: 11

Web page Designing using HTML, Scripting basics- Client side and server side scripting. Java Script- Object, names, literals, operators and expressions- statements and features-events - windows - documents - frames - data types - built-in functions- Browser object model - Verifying forms.-HTML5- CSS3- HTML 5 canvas - Web site creation using tools.

#### UNIT-II JAVA

Classes: 11

Introduction to object oriented programming-Features of Java – Data types, variables and arrays – Operators – Control statements – Classes and Methods – Inheritance. Packages and Interfaces – Exception Handling – Multithreaded Programming –Input/Output – Files – Utility Classes – String Handling.

#### UNIT-III JDBC

Classes: 12

JDBC Overview – JDBC implementation – Connection class – Statements – Catching Database Results, handling database Queries. Networking– Inet Address class – URLclass- TCP sockets - UDP sockets, Java Beans –RMI.

#### UNIT-IV APPLETS

Classes: 14

Java applets- Life cycle of an applet – Adding images to an applet – Adding sound to an applet, Passing parameters to an applet, Event Handling, Introducing AWT: Workingwith Windows Graphics and Text. Using AWT Controls, Layout Managers and Menus.Servlet – life cycle of a servlet, The Servlet API, Handling HTTP Request and Response, using Cookies, Session Tracking. Introduction to JSP.

#### UNIT-V XML AND WEB SERVICES

Classes: 12

Xml – Introduction-Form Navigation-XML Documents- XSL – XSLT- Web services-UDDI-WSDL-Java web services – Web resources.

#### TEXT BOOKS:

1. Java™ : The Complete Reference, Seventh Edition, Herbert Schildt
2. J2EE: The complete Reference by James Keogh
3. Java EE and HTML5 Enterprise Application Development (Oracle Press) by John Brock (Author), Arun Gupta (Author), Geertjan Wielenga (Author)
4. Struts: The Complete Reference, 2nd Edition by James Holmes
5. ASP.NET 4 Unleashed by Stephen Walther (Author), Kevin Scott Hoffman• (Author), Nate Dudek (Author)
6. Microsoft Visual C# 2013 Step by Step by John Sharp•



## INTRODUCTION TO NETWORK SECURITY

### IV B. TECH- II SEMESTER - OPEN ELECTIVE – IV (HUMANITIES)

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS52      | HSMC     | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. Able to demonstrate the basic components of security and mathematics of cryptography.
2. Able to demonstrate various traditional symmetric key ciphers
3. Analyze DES and AES its variants.
4. Able to demonstrate private and public-key cryptosystem, Digital signature and attacks on it.

#### COURSE OUTCOMES:

At the end of the course students able to

1. Identify and explain various types of security threats/vulnerabilities attacks, consequences and security mechanisms
2. Apply various Cryptographic and Authentication techniques for secure transmission of data.
3. Compare different security algorithms with respect feasibility, cost and security service
4. Choose the most suitable security mechanisms based on the nature of data and complexity of algorithm.

#### UNIT-I INTRODUCTION

Classes: 12

Introduction: Security Goals, Attacks, Services and Mechanism, Techniques, Mathematics of Cryptography, Integer arithmetic, Modular Arithmetic, Matrices, Linear Congruence, Primes, Primality Testing, Factorization, Chinese Remainder Theorem.

Traditional Symmetric key Ciphers: Introduction, Substitution Ciphers, Transposition Ciphers, Streams and Block Ciphers.

#### UNIT-II SYMMETRIC CIPHERS

Classes: 12

**Advanced Encryption Standards:** AES Structure, AES Transformation Function, AES Key Expansion, AES Implementation.

**Data Encryption Standard:** Introduction, DES Structure, DES Analysis, Multiple DES, Security of DES, Differential Cryptanalysis, Linear Cryptanalysis of DES, Encipherment Using Modern Symmetric Key Ciphers Use of Modern Block Ciphers, Use of Stream Ciphers

#### UNIT-III ASYMMETRIC CIPHERS, DATA INTEGRITY ALGORITHMS

Classes: 12

**Asymmetric Key Cryptography and Message Integrity:** Introduction, RSA Cryptosystem, Message Integrity, Message Authentication Cryptographic Hash Functions, Introduction, SHA-512, Digital Signature Comparison, Attacks on Digital Signature.

**User Authentication Protocols:** Remote User Authentication Principles, Remote User Authentication using Symmetric Encryption, Kerberos

#### UNIT-IV MUTUAL TRUST

Classes: 10

**Key Management And Distribution :** Symmetric key Distribution Using Symmetric and Asymmetric Encryption, Distribution of Public Key, X.509 Certificate , Public Key Infrastructure.

**Network Access Control :** Network Access Control, Extensible Authentication Protocol, IEEE802.1,

#### UNIT-V NETWORK SECURITY

Classes: 10

**Transport-Level Security:** Transport-Level Security, HTTPS, Secure Shell(SSH)

**Wireless Network Security:** Web Security ,Mobile Device Security, IEEE802.11 Wireless LAN Overview, IEEE802.11i Wireless LAN Security

#### Text Books:

1. "Cryptography and Network Security" by Behrouz A. Forouzan, Special Indian Edition, 2008
2. "Cryptography and Network Security" by William Stallings, 6th Edition, Pearson Education/PHI, 2014

#### Reference Books:

1. "Cryptography and Network Security" by Atul Kahate, Tata McGraw Hill, 2003.
2. "RSA Security's official guide to cryptography" by Steve Burnett, Stephen Paine, RSA Press, Tata McGraw Hill Edition, 2001

#### Web References:

1. [www.open.edu/openlearn/science.../network-security/content-section---references](http://www.open.edu/openlearn/science.../network-security/content-section---references)  
[www.cse.iitm.ac.in/~chester/courses/16e\\_cns/slides/01\\_Introduction.pdf](http://www.cse.iitm.ac.in/~chester/courses/16e_cns/slides/01_Introduction.pdf)  
[williamstallings.com/Crypto/Crypto4e.html](http://williamstallings.com/Crypto/Crypto4e.html)
2. [pubs.sciepub.com/iteces/3/1/1/](http://pubs.sciepub.com/iteces/3/1/1/)
3. <https://www.sanfoundry.com/best-reference-books-cryptography-network-security/>
4. <https://dl.acm.org/citation.cfm?id=1824151>
5. [web.mit.edu/6.857/OldStuff/Fall03/references.html](http://web.mit.edu/6.857/OldStuff/Fall03/references.html)

#### E-Text Books:

1. Online resources of Stallings's Cryptography and Network Security. Principles and Practice  
<http://williamstallings.com/Crypto3e.html>
2. Online resources of Menezes, van Oorschot, Vanstone's Handbook of Applied Cryptography  
available at <http://www.cacr.math.uwaterloo.ca/hac/>

#### MOOC Course:

1. <https://www.mooc-list.com/tags/network-security>
2. [nptel.ac.in/courses/106105031/](http://nptel.ac.in/courses/106105031/)
3. [https://onlinecourses.nptel.ac.in/noc18\\_cs07/course](https://onlinecourses.nptel.ac.in/noc18_cs07/course)

## DATA ANALYTICS

### IV B. TECH- II SEMESTER - OPEN ELECTIVE – IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4CS53      | HSMC     | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To explore the fundamental concepts of data analytics
2. Formulate and use appropriate models of data analysis to solve hidden solutions to business-related challenges.

#### COURSE OUTCOMES:

1. Students will demonstrate the ability to use technical skills in Descriptive and prescriptive modelling to support business decision-making
2. Apply the knowledge of classification and regression techniques for various data analysis.
3. Examine the results of Classification for further predictions.
4. Make use of regression techniques for predicting future outcomes on various data
5. Illustrate the concepts of data analytics for data visualization and bigdata analysis.

#### UNIT-I

Classes: 10

**Descriptive Statistics**, Introduction to the course, Descriptive Statistics, Probability Distributions , Inferential Statistics, Inferential Statistics through hypothesis tests, Permutation & Randomization Test

#### UNIT-II

Classes: 12

**Regression & ANOVA**, Regression , ANOVA(Analysis of Variance), Machine Learning: Introduction and Concepts, Differentiating algorithmic and model based frameworks.  
Regression: Ordinary Least Squares, Ridge Regression, Lasso Regression, K Nearest Neighbours  
Regression & Classification

#### UNIT-III

Classes: 13

**Supervised Learning with Regression and Classification techniques -1**, Bias-Variance Dichotomy, Model Validation Approaches, Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis Regression and Classification Trees, Support Vector Machines

#### UNIT-IV

Classes: 12

Supervised Learning with Regression and Classification techniques -2, Ensemble Methods: Random Forest Neural Networks, Deep learning, Unsupervised Learning and Challenges for Big Data Analytics Clustering Associative Rule Mining Challenges for big data analytics.

#### UNIT-V

Classes: 10

**Prescriptive analytics**, Creating data for analytics through designed experiments, Creating data for analytics through Active learning, Creating data for analytics through Reinforcement learning

#### TEXT BOOKS:

1. Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: springer, 2009.
2. Montgomery, Douglas C., and George C. Runger. Applied statistics and probability for engineers. John Wiley & Sons, 2010

#### REFERENCE BOOKS:

1. Data Analytics- Models and Algorithms for Intelligent Data Analysis, Author: Thomas Runkler  
ISBN-10: 3834825883
2. Data Analytics, Anil Maheshwari, McGraw Hill Education; First edition, 9352604180.

**.NET & C# PROGRAMMING****IV B. TECH- II SEMESTER - OPEN ELECTIVE – IV**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
| A4CS54      | PEC      | L            | T | P | C       | CIE           | SEE | Total |
|             |          | 3            | - | - | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

This course enables to student to know:

1. To impart basic concepts of .NET Framework Components.
2. To familiarize the C# Core Concepts.
3. To facilitate students in understanding OOP Concepts.
4. To facilitate students in understanding Exception Handling.
5. To impart the concepts of Delegates, Events.

**COURSE OUTCOMES:**

At the end of the course, students will

1. Explain the purpose of the .net framework, and understand how to use c# and visual studio 2010 to build .net framework applications.
2. Understand the syntax of basic c# programming constructs. Create and use new types (enumerations, classes, and structures), and understand the differences between reference types and value types.
3. Use inheritance to create new reference types. control the visibility and lifetime of members in a type.
4. Manage the lifetime of objects and control the use of resources, catch, handle and throw exceptions.
5. Understanding the .net delegate type & event handling.

**UNIT-I****Classes: 10**

**Introduction** - The Philosophy of .NET, An Initial Look at the .NET Platform, Introducing the Building Blocks of the .NET Platform (the CLR, CTS, and CLS), An Overview of .NET Assemblies, Understanding the Common Type System, Understanding the Common Language Specification, Understanding the Common Language Runtime, The Assembly/Namespace/Type Distinction, The Platform-Independent Nature of .NET.

**UNIT-II****Classes: 13**

**Core C# Programming Constructs, Part I** - The Anatomy of a Simple C# Program, An Interesting Aside: Some Additional Members of the System.Environment Class, The System.Console Class, System Data Types and Corresponding C# Keywords, Working with String Data, Narrowing and Widening Data Type Conversions, Understanding Implicitly Typed Local Variables, C# Iteration Constructs, Decision Constructs and the Relational/Equality Operators.

**Core C# Programming Constructs, Part II** - Methods and Parameter Modifiers, Understanding C# Arrays, Understanding the enum Type, Understanding the Structure Type, Understanding Value Types and Reference Types, Understanding C# Nullable Types.

**UNIT-III****Classes: 15****Object-Oriented Programming with C#**

**Understanding Encapsulation** - Introducing the C# Class Type, Understanding Constructors, The Role of the this Keyword, Understanding the static Keyword, Defining the Pillars of OOP, C# Access Modifiers, The First Pillar: C#'s Encapsulation Services, Understanding Automatic Properties, Understanding Object Initialization Syntax, Working with Constant Field Data, Understanding Partial Types.

**Understanding Inheritance and Polymorphism** - The Basic Mechanics of Inheritance, The Second Pillar of OOP: The Details of Inheritance, Programming for Containment/Delegation, The Third Pillar of OOP: C#'s Polymorphic Support, Understanding Base Class/Derived Class Casting Rules, The Master Parent Class: System.Object.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------|
| <b>UNIT-IV</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | <b>Classes: 12</b> |
| <b>Understanding Structured Exception Handling</b> - Ode to Errors, Bugs, and Exceptions, The Role of .NET Exception Handling, The Simplest Possible Example, Configuring the State of an Exception, System-Level Exceptions (System.SystemException), Application-Level Exceptions (System.ApplicationException), Processing Multiple Exceptions, Who Is Throwing What?, The Result of Unhandled Exceptions.                                                                                                                                                                                                                                                                                                                            |  |                    |
| <b>UNIT-V</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | <b>Classes: 14</b> |
| <b>Working with Interfaces</b> - Understanding Interface Types, Defining Custom Interfaces, Implementing an Interface, Invoking Interface Members at the Object Level, Interfaces As Parameters, Interfaces As Return Values, Arrays of Interface Types, Implementing Interfaces Using Visual Studio, Explicit Interface Implementation, Designing Interface Hierarchies, The IEnumerable and IEnumerator Interfaces, The ICloneable Interface, The IComparable Interface.<br><b>Delegates, Events</b> - Understanding the .NET Delegate Type, The Simplest Possible Delegate Example, Sending Object State Notifications Using Delegates, Understanding Generic Delegates, Understanding C# Events, Understanding C# Anonymous Methods. |  |                    |
| <b>TEXT BOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                    |
| 1. Andrew Troelsen, Pro C-Hash and the .NET 4 Platform, 5th edition, Springer (India) Private Limited, New Delhi, India., 2010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                    |
| <b>REFERENCE BOOKS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |                    |
| 1. E. Balagurusamy (2004), Programming in C#, 5 th edition, Tata McGraw-Hill, New Delhi, India.<br>2. Herbert Schildt (2004), The Complete Reference: C#, Tata McGraw-Hill, New Delhi, India.<br>3. Simon Robinson, Christian Nagel, Karli Watson, Jay GI (2006), Professional C#, 3rd edition, Wiley & Sons, India.                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                    |

## OPEN ELECTIVES OFFERED BY ECE DEPARTMENT

| OE1    |                                                      | OE2    |                                            |
|--------|------------------------------------------------------|--------|--------------------------------------------|
| A4EC59 | Logic Design                                         | A4EC62 | Fundamentals of Integrated Circuits        |
| A4EC60 | Principles of Communications                         | A4EC63 | Signal Transmission through linear systems |
| A4EC61 | Measurements and Instruments                         | A4EC64 | Fundamentals of VLSI Design                |
| OE3    |                                                      | OE4    |                                            |
| A4EC65 | Introduction of Microprocessors and Microcontrollers | A4EC68 | Introduction to signal Processing          |
| A4EC66 | Fundamentals of Image processing                     | A4EC69 | Introduction to mobile communication       |
| A4EC67 | TV Engineering                                       | A4EC70 | Basic Embedded systems Design              |

## LOGIC DESIGN

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC59      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Explain how to design a digital circuit.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Study the characteristics and performance of digital circuits built using various MOS technologies

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain how digital circuit of large complexity can be built in a methodological way, starting from Boolean logic and applying a set of rigorous techniques.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Study the characteristics and performance of digital circuits built using various MOS technologies

### UNIT-I

### BOOLEAN ALGEBRA

**Classes: 09**

Boolean Algebra: Axiomatic Definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates, IC Digital Logic Families

### UNIT-II

### SIMPLIFICATION OF BOOLEAN FUNCTIONS

**Classes: 09**

Simplification of Boolean Functions: The Map Method, Two & Three Variable Maps, Four Variable Map, Five & Six Variable Maps, Product of Sum Simplification, NAND & NOR Implementation, Two-Level Implementations, Don't-Care Conditions, Tabulation Method, Determination of Prime Implicants, Selection of Prime Implicants.

### UNIT-III

### COMBINATIONAL LOGIC

**Classes: 09**

Combinational Logic: Design Procedure, Adders, Subtractors, Code Conversion, Multilevel NAND Circuits, Multilevel NOR Circuits, Exclusive-Or and Equivalence Functions.

Combinational Logic with MSI And LSI: Binary Parallel Adder, Decimal Adder, Magnitude Comparator, Decoders, Multiplexers, ROM, PLA.



**UNIT-IV****SEQUENTIAL CIRCUIT DESIGN AND ANALYSIS****Classes: 09**

Sheet metal characteristics – Typical shearing operations, bending and drawing operations – Formability of sheet metal – Working principle and application of special forming processes – Hydro forming – Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics

**UNIT-V****DIGITAL INTEGRATED CIRCUITS****Classes: 09**

Digital Integrated Circuits: Introduction, Bi-Polar Transistor Characteristics, Integrated Injection Logic, Transistor-Transistor Logic (TTL), Emitter-Coupled Logic (ECL), MOS, Complementary MOS.

**Text Books:**

1. M. Morris Mano, —Digital Design II, 2nd Edition, 1997, PHI. (Unit I, II, III & IV).

**Reference Books:**

1. Zvi Kohavi, —Switching and Automata Theory II, 2nd Edition, 1978, McGraw-Hill.
2. Thomas L. Floyd —Digital Fundamentals II, 10th Edition, Pearson Education India.
3. Stephen Brown, Zvonko Vrsanec, II Fundamentals of Digital Logic with Verilog Design II, 2nd Edition, McGraw-Hill.

**Web References:**

1. <http://www.freebookcentre.net/electronics-ebooks-download/logic design.html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-logic designs-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/logic design-Lecture-Notes.html>

**E-Text Books:**

1. <http://gen.lib.rus.ec/book/logic design>
2. <http://www.faadooengineers.com/threads/logicdesign-by-RAMESH-GANOKAR-ebook-pdf-download>
3. <https://e.edim.co/123389964/logic design.pdf>

**MOOC Course:**

1. <https://www.mooc-list.com/tags/logic design>
2. <https://www.coursera.org/courses?query=logicdesign>

## PRINCIPLES OF COMMUNICATIONS

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC60      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Describe the basic concepts of Communication technology
2. Explain the Models of Communication
3. Compare Analog and Digital Signal transmission
4. Introduce the principle about satellite communication
5. Explain the about multiple accessing schemes
6. Describe the analog and digital modulation techniques

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe the basic concepts of Communication technology
2. Explain the Models of Communication
3. Compare Analog and Digital Signal transmission
4. Introduce about satellite communication
5. Explain the about multiple accessing schemes
6. Describe the analog and digital modulation techniques

### UNIT-I

#### FUNDAMENTALS OF ANALOG COMMUNICATION

Classes: 10

Principles of amplitude modulation, AM envelope, frequency spectrum and bandwidth, modulation index and percent modulation, AM Voltage distribution, AM power distribution, Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation, Frequency analysis of angle modulated waves. Bandwidth requirements for Angle modulated waves.

### UNIT-II

#### DIGITAL COMMUNICATION

Classes: 09

Introduction, Shannon limit for information capacity, digital amplitude modulation, frequency shift keying, FSK bit rate and baud, FSK transmitter, BW consideration of FSK, FSK receiver, phase shift keying binary phase shift keying QPSK, Quadrature Amplitude modulation, bandwidth efficiency, carrier recovery – squaring loop, Costas loop, DPSK.

### UNIT-III

#### DIGITAL TRANSMISSION

Classes: 10

Introduction, Pulse modulation, PCM, PCM sampling, sampling rate, signal to quantization noise rate, companding analog and digital percentage error, delta modulation, adaptive delta modulation, differential pulse code modulation, pulse transmission – Inter symbol interference, eye patterns.

### UNIT-IV

#### SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES:

Classes: 08

Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, processing gain, FH spread spectrum, multiple access techniques wireless communication, TDMA and CDMA in wireless communication systems, source coding of speech for wireless communications.

**UNIT-V****SATELLITE AND OPTICAL COMMUNICATION****Classes: 08**

Satellite Communication Systems ,Keplers Law, LEO and GEO Orbits, footprint, Link model Optical Communication Systems-Elements of Optical Fiber Transmission link, Types, Losses, Sources and Detectors

**Text Books:**

1. Wayne Tomasi, —Advanced Electronic Communication Systems, 6/e, Pearson Education, 2007.
2. Simon Haykin, —Communication Systems, 4th Edition, John Wiley & Sons., 2001.

**Reference Books:**

1. H.Taub, D L Schilling, G Saha , Principles of Communication, 3/e, 2007.
2. B.P.Lathi, Modern Analog And Digital Communication systems, 3/e, Oxford University Press, 2007
3. Blake, —Electronic Communication Systems, Thomson Delmar Publications, 2002.
4. Martin S.Roden, —Analog and Digital Communication System, 3rd Edition, PHI, 2002.
5. B.Sklar, Digital Communication Fundamentals and Applications, 2/e Pearson Education 2007.

**Web References:**

[https://books.google.co.in/books/about/Principles\\_Of\\_Communication.html?id=6Zunu4Acfg8C](https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C)

**E-Text Books:**

[https://books.google.co.in/books/about/Principles\\_Of\\_Communication.html?id=6Zunu4Acfg8C](https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C)

**MOOC Course**

[https://onlinecourses.nptel.ac.in/noc18\\_ee26/preview](https://onlinecourses.nptel.ac.in/noc18_ee26/preview)

## MEASUREMENTS AND INSTRUMENTATION

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC61      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students:

1. To understand the measuring methods and instruments of electrical quantities.
2. To understand, design aspects and performance criterion of measuring instruments.
3. To understand the working principle of various transducers.
4. To aware the students about the advances in instrumentation.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply knowledge of electronic instruments for measurement of electrical quantities.
2. Understand the principles and operations of instruments.
3. Select and use latest hardware for measurements.
4. Identify the various instruments for various measurements
5. Use the instruments in laboratory and real life

### UNIT-I MEASUREMENT CONCEPTS Classes: 10

**Measurement concepts :** Quantities of Measurement- Performance Characteristics – Static and dynamic characteristics – units and standards of measurements – errors in measurements- accuracy and precision, statistical analysis – moving coil, moving iron types Instruments –DC Voltmeter multi-meter & Calibrations

### UNIT-II DIGITAL INSTRUMENTS Classes: 08

**Digital instruments:** Digital multimeters – Digital frequency meter-digital tacho meter-digital PH meter-digital phase meter-digital capacitance meter-Microprocessor based instruments- IEEE 488 bus

### UNIT-III SIGNAL GENERATORS CATHODE RAY OSCILLOSCOPES Classes: 08

**Signal generators:** Function generators – pulse and square wave generators, RF signal generators – Sweep generators – Frequency synthesizer

**Cathode ray oscilloscopes:** Basic block diagram – operation-applications

**special Oscilloscopes:** delayed time base oscilloscopes, sampling oscilloscope

### UNIT-IV MEASUREING INSTRUMENTS AND BRIDGES Classes: 09

**Measuring instruments:** Introduction-field strength meter –Q- meter- LCR Bridge-Transistor tester

**Bridge:** Wheatstone's bridges for resistance measurements-Maxwell's bridge for inductance measurements-Schering's bridge for capacitance measurements measurement

### UNIT-V TRANSDUCERS & DATA ACQUISITION SYSTEMS Classes: 10

**Transducer:** Introduction-electrical transducers-selecting a transducers-active and passive transducers with examples

**Data acquisition systems:** Elements of a digital data acquisition system – interfacing of transducers – multiplexing –data loggers –computer controlled instrumentation

### Text Books:

1. Albert D.Helfrick and William D.Cooper – Modern Electronic Instrumentation and Measurement Techniques, Pearson / Prentice Hall of India, 2007.
2. Ernest O. Doebelin, Measurement Systems- Application and Design, TMH, 2007.

### Reference Books:

1. Joseph J.Carr, Elements of Electronics Instrumentation and Measurement, Pearson Education, 2003.
2. Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition,Prentice Hall of India, 2003.
3. David A. Bell, Electronic Instrumentation and measurements, Prentice Hall of India Pvt Ltd, 2003.
4. B.C. Nakra and K.K. Choudhry, Instrumentation, Measurement and Analysis, 2nd Edition, TMH, 2004.
5. James W. Dally, William F. Riley, Kenneth G. McConnell, Instrumentation for Engineering Measurements, 2nd Edition, John Wiley, 2003.

### Web References:

- 1.<https://www.sciencedirect.com/science/book/9780123819604>
2. <https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>

### E-Text Books:

- 1.[https://books.google.co.in/books/about/Electronics\\_Measurements\\_And\\_Instrumenta.html?id=m\\_xVRJRYCi-QC](https://books.google.co.in/books/about/Electronics_Measurements_And_Instrumenta.html?id=m_xVRJRYCi-QC)
- 2.<https://sway.com/V1Uclj1JCiLhqCcb>

### MOOC Course

- 1.<https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>
- 2.<https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/>
- 3.<https://www.coursera.org/courses?query=measurements%20and%20instrumentat>

## FUNDAMENTALS OF INTEGRATED CIRCUITS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC62      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students:

1. To understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. To learn the concepts of modeling of Delay techniques and MOS layers.
3. To learn the concepts of Technology Scaling of MOS transistors.
4. To understand the concepts of testing of combinational and sequential circuits

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze the basic fabrication steps of semiconductor devices.
2. Analyze silicon oxidation and photolithography.
3. Analyze various etching techniques.
4. Analyze the Ion implantation process and metallization.
5. Evaluate various IC technologies such as Bipolar, MOSFET and MESFET.

#### UNIT-I INTEGRATED CIRCUITS AND OPERATIONAL AMPLIFIER Classes: 09

Semiconductor materials, Semiconductor Devices, Semiconductor process technology, Basic fabrication steps. Crystal Growth: Silicon Crystal Growth from melt, Silicon Float-Zone Process, GaAs Crystal Growth Techniques, Material Characterization.

#### UNIT-II SILICON OXIDATION Classes: 09

Thermal oxidation, Impurity Redistribution during oxidation, masking properties of silicon dioxide, oxide quality, oxide thickness characterization, Photolithography.

#### UNIT-III ETCHING Classes: 09

Wet chemical etching, Dry etching. Diffusion: Basic Diffusion Process, Extrinsic Diffusion, Lateral Diffusion.

#### UNIT-IV ION IMPLANTATION Classes: 09

Range of Implanted ions, Implant Damage and annealing, Implantation-related Process. Film Deposition: Epitaxial growth techniques, Structures and defects in epitaxial layers, Dielectric deposition, Poly-silicon deposition, Metallization.

#### UNIT-V PROCESS INTEGRATION Classes: 09

Passive Components, Bipolar Technology, MOSFET technology, MESFET technology, MEMS technology. IC Manufacturing: Electrical testing, Packaging.

### Text Books:

1. Gary S. May, Simon M. Sze, "Fundamentals of Semiconductor Fabrication", John Wiley Inc., 2004.

### Reference Books:

1. C.Y. Chang and S.M.Sze (Ed), "ULSI Technology", McGraw Hill Companies Inc, 1996.
2. S.K. Ghandhi, "VLSI Fabrication Principles", John Wiley Inc., New York, 1983.
3. S.M. Sze (Ed), "VLSI Technology", 2nd Edition, McGraw Hill, 1988
4. The Science and Engineering of Microelectronic Fabrication, Stephen Cambell, Oxford University Press, 2001

### Web References:

1. <https://banavathvalyaiitmadrass.files.wordpress.com/2015/12/vlsi-fabrication-principles-silicon-and-gallium-arsenide-1994-ghandi-wiley-2e.pdf>

### E-Text Books:

1. [rku.ac.in/soe/wp-content/uploads/2015/04/VL204\\_CMOSDeviceFabrication.pdf](http://rku.ac.in/soe/wp-content/uploads/2015/04/VL204_CMOSDeviceFabrication.pdf)
2. [https://books.google.co.in/books/.../The\\_Science\\_and\\_Engineering\\_of\\_Microelectronics](https://books.google.co.in/books/.../The_Science_and_Engineering_of_Microelectronics)

### MOOC Course

1. [nptel.ac.in/courses/117103066/35](http://nptel.ac.in/courses/117103066/35)
2. <https://www.edx.org/course/circuits-electronics-1-basic-circuit-mitx-6-002-1x-0>

## SIGNALS TRANSMISSION THROUGH LINEAR SYSTEMS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC63      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students:

1. To understand various fundamental characteristics of signals and systems.
2. To study the importance of transform domain.
3. To analyze and design various systems.
4. To study the effects of sampling.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze various signals and systems by applying Laplace and Z-transforms.
2. Apply Fourier series, Fourier transform and inverse Fourier transform techniques to find magnitude and phase spectrum of various signals.
3. Apply convolution and correlation operations to various signals and systems.
4. Design IIR and FIR analog and digital filters.
5. Analyze multirate Digital Signal Processing.

### UNIT-I

#### TIME DOMAIN SIGNALS

Classes: 09

Introduction: Definitions of signals and systems, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sync, Signum, and Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.

### UNIT-II

#### SPECTRAL DOMAIN REPRESENTATION OF SIGNALS

Classes: 09

Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transforms, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.

### UNIT-III

#### CONVOLUTION AND CORRELATION

Classes: 09

Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.

### UNIT-IV

#### DIGITAL FILTERS

Classes: 09

IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method.

FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.



**UNIT-V****MULTI RATE DIGITAL SIGNAL PROCESSING****Classes: 09**

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

**Text Books:**

1. John G. Proakis, Dimitris G. Manolakis , Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi ,Signals, Systems & Communications ,BS Publications 2013 .
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

**Reference Books:**

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

**Web References:**

1. [http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-\(T.-OHaver\).html](http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-OHaver).html)
2. <https://www.sciencedirect.com/science/article/pii/B9780123965028000012>
3. Applied Digital Signal Processing (1st Edition)

**E-Text Books:**

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. [http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-\(Digital-Signal-Processing\).html](http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html)

**MOOC Course**

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

## FUNDAMENTALS OF VLSI DESIGN

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC64      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students:

1. To understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. To designing of stick diagrams and layouts for MOS transistors.
3. To learn the concepts of modeling of Delay techniques and MOS layers.
4. To learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. Designing of stick diagrams and layouts for MOS transistors.
3. Learn the concepts of modeling of Delay techniques and MOS layers.
4. Learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

### UNIT-I Classes: 09

#### REVIEW OF MICROELECTRONICS AND INTRODUCTION TO MOS TECHNOLOGY

Introduction MOS and related VLSI technology – NMOS-CMOS-BICMOS-GaAs Technologies – thermal aspects of processing – production of E beam masks.

### UNIT-II Classes: 09

#### MOS AND BICMOS CIRCUIT DESIGN PROCESS

MOS layers – stick diagrams – design rules and layout – 2m meter – 1.2 m meter CMOS rules – Layout diagrams – Symbolic diagrams.

### UNIT-III Classes: 09

#### BASIC CIRCUIT CONCEPTS

Sheet resistance – Area capacitance of layers – delay unit – wiring capacitances – choice of layers.

### UNIT-IV Classes: 09

#### SCALING OF MOS CIRCUITS

Scaling of MOS circuits: Scaling models– Scaling function for device parameters – Limitation of Scaling

Subsystem design process: Architectural issues – switch logic – examples of structural design (Combinational logic)– design of ALU subsystem – commonly used storage elements – aspects of design rules.

### UNIT-V Classes: 09

#### TEST AND TESTABILITY

Test and Testability: Design for testability built in self test (BIST) – teaching combinational logic – testing sequential logic – practical design for test guide lines – scan design techniques – etc.

### **Text Books:**

1. Basic VLSI design by Douglas A, Pucknell, Kamran Eshraghian, Prantice Hall, 1996 3rd edition.

### **Reference Books:**

1. Mead, C.A and Conway, L.A., Introduction to VLSI Systems, Wesley – Wesley.

### **Web References:**

1. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi .html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-vlsi-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/vlsi design-Lecture-Notes.html>

### **E-Text Books:**

1. <http://gen.lib.rus.ec/book/vlsi design>
2. <http://www.faadooengineers.com/threads/vlsi design -by-kamran eshragghan-ebook-pdf-download>
3. <https://e.edim.co/123389964/vlsi design.pdf>

### **MOOC Course**

1. <https://www.mooc-list.com/tags/vlsi design>
2. <https://www.coursera.org/courses?query=vlsi design>

## INTRODUCTION OF MICROPROCESSORS AND MICROCONTROLLERS

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC65      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

#### UNIT-I

#### MICROPROCESSOR ARCHITECTURE

Classes: 09

Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization, Signal descriptions of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine.

#### UNIT-II

#### INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086

Classes: 09

Instruction set, Assembler directives, Addressing modes, programming model, Programs involving logical, branch and call instructions, Sorting.

#### UNIT-III

#### INTERFACING WITH 8086

Classes: 09

8255 PPI, various modes of operation and interfacing to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086, DMA controller (8257).

#### UNIT-IV

#### INTRODUCTION TO MICROCONTROLLERS

Classes: 09

Overview of 8051 microcontroller, Architecture, I/O ports, Memory organization, addressing modes and instruction set of 8051, Simple programs.

#### UNIT-V

#### 8051 REAL TIME CONTROL

Classes: 09

Programming Timer interrupts, programming external hardware interrupts, Programming the serial communication interrupts, programming 8051 timers and Counters.

### Text Books:

1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
2. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning, 2010
3. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
4. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

### Reference Books:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,
3. Microcontrollers and application, Ajay. V. Deshmukh, TMGH, 2005.

### Web References:

1. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
2. <http://coen.boisestate.edu/sml00/sml00-courses/ece-332-microprocessors-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

### E-Text Books:

1. <http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
2. <http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
3. [https://e.edim.co/123389964/The\\_8051\\_Microcontroller\\_Architecture\\_Programming\\_And\\_Applications.pdf](https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf)
4. [https://e.edim.co/123389964/A.K.\\_Ray\\_and\\_K.M.\\_Bhurchandi-Advanced\\_Microprocessors\\_and\\_Peripherals\\_3e-Tata\\_Mcgraw\\_Hill.pdf](https://e.edim.co/123389964/A.K._Ray_and_K.M._Bhurchandi-Advanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf)

### MOOC Course

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

## FUNDAMENTALS OF IMAGE PROCESSING

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC66      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Know the Fundamentals of Digital Image Processing.
2. Apply the concepts in real time applications
3. Develop a code for various image processing concepts

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Know the Fundamentals of Digital Image Processing.
2. Apply the concepts in real time applications
3. Develop a code for various image processing concepts

### UNIT-I

### INTRODUCTION OF IMAGE PROCESSING

**Classes: 10**

Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures.

### UNIT-II

### IMAGE ENHANCEMENT

**Classes: 12**

Image Enhancements and Filtering-Gray level transformations, histogram equalization and specifications, pixel-domain smoothing filters – linear and order-statistics, pixel-domain sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass

### UNIT-III

### IMAGE SEGMENTATION

**Classes: 12**

Image Segmentation- Detection of discontinuities, edge linking and boundary detection, thresholding – global and adaptive, region-based segmentation.

Wavelets and Multi-resolution image processing- Uncertainty principles of Fourier Transform, Time-frequency localization, continuous wavelet transforms, wavelet bases and multi-resolution analysis, wavelets and Subband filter banks, wavelet packets.

### UNIT-IV

### IMAGE COMPRESSION

**Classes: 08**

Image Compression-Redundancy–inter-pixel and psycho-visual; Lossless compression – predictive, entropy; Lossy compression- predictive and transform coding; Discrete Cosine Transform; Still image compression standards – JPEG and JPEG-2000.

### UNIT-V

### VIDEO PROCESSING

**Classes: 12**

Fundamentals of Video Coding- Inter-frame redundancy, motion estimation techniques – full-search, fast search strategies, forward and backward motion prediction, frame classification – I, P and B; Video sequence hierarchy – Group of pictures, Elements of a video encoder and decoder; Video coding standards – MPEG and H.26X.

Video Segmentation- Temporal segmentation–shot boundary detection, hard-cuts and soft-cuts; spatial segmentation – motion-based; Video object detection and tracking.

**Text Books:**

1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008
2. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India.2nd edition 2004

**Reference Books:**

1. Murat Tekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015

**Web References:**

<http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf>

**E-Text Books:**

[https://books.google.co.in/books/about/Digital\\_Image\\_Processing.html?id=a62xQ2r\\_f8wC](https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC)

**MOOC Course:**

<https://nptel.ac.in/courses/117105079/>

## TELEVISION ENGINEERING

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC67      | OEC      | 4            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Know the fundamentals of Television transmitters and receivers
2. Understand different signal processing step monochrome television,
3. Introduce color television transmitters and receivers.
4. Remember the concepts of speech recognition and synthesis
5. Know the functions of the TV

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Know the fundamentals of Television transmitters and receivers
2. Understand different signal processing step monochrome television,
3. Introduce color television transmitters and receivers.
4. Remember the concepts of speech recognition and synthesis
5. Know the functions of the TV circuits

### UNIT-I

#### INTRODUCTION

**Classes: 14**

TV transmitter and receivers, synchronization. Geometric form and aspect ratio, image continuity, interlaced scanning, picture resolution Composite video signal, TV standards. Camera tubes: image Orthicon Plumbicon, vidicon, silicon Diode Array vidicon, Comparison of camera tubes Monochrome TV camera,

**TV Signal Transmission and Propagation:** Picture Signal transmission positive and negative modulation, VSB transmission, sound signal transmission, standard channel BW, TV transmitter, TV signal propagation interference, TV broadcast channels, TV transmission Antennas.

### UNIT-II

#### MONOCHROME TV RECEIVER

**Classes: 12**

RF tuner, IF subsystem, video amplifier, sound section, sync separation and processing, deflection circuits, scanning circuits, AGC, noise cancellation, video and inter carrier sound signal detection, vision IF subsystem of Black and White receivers, Receiver sound system: FM detection, FM Sound detectors, and typical applications.

### UNIT-III

#### SYNC SEPARATION AND DETECTION

**Classes: 12**

TV Receiver Tuners, Tuner operation, VHF and UHF tuners, digital tuning techniques, remote control of receiver functions. Sync Separation, AFC and Deflection Oscillators: Synchronous separation, k noise in sync pulses, separation of frame and line sync pulses. AFC, single ended AFC circuit, Deflection Oscillators, deflection drive IOs, Receiver Antennas, Picture Tubes.

### UNIT-IV

#### SPEECH RECOGNITION

**Classes: 10**

Large Vocabulary Continuous Speech Recognition: Architecture of large vocabulary continuous speech recognition system – acoustics and language models – n-grams, context dependent sub-word units;



Applications and present status.

## UNIT-V

## SPEECH SYNTHESIS

Classes: 12

**Color Receiver:** Introduction to colour receiver, Electron tuners, IF subsystem, Y-signal channel, Chroma decoder, Separation of U & V Color, Phasors, synchronous demodulators, Sub carrier generation, raster circuits.

**Digital TV:** Introduction to Digital TV, Digital Satellite TV, Direct to Home Satellite TV, Digital TV Transmitter, Digital TV Receiver, Digital Terrestrial TV, LCD TV, LED TV, CCD Image Sensors, HDTV.

### Text Books:

1. Television and Video Engineering- A.M.Dhake, 2nd Edition.
2. Modern Television Practice Principles, Technology and Service R .R.Gallatin, New Age International Publication, 2002.
3. Monochrome and Colour TV- R.R. Gulati, New Age International Publication, 2002.

### Reference Books:

1. Colour Television Theory and Practice-S.P.Bal, TMH, 1994.
2. Basic Television and Video Systems-B.Grob and C.E.Herndon, McGraw Hill, 1999.

### Web References:

<http://www.sasurieengg.com/e-course-material/ECE/IV-Year%20Sem%207/EC2034%20Television%20and%20Video%20Engineering.pdf>

### E-Text Books:

[https://books.google.co.in/books?id=AvQAa5Zfuj0C&printsec=frontcover&dq=tv+by+rr+gulati&source=gbs\\_similarbooks\\_s&cad=1#v=onepage&q=tv%20by%20rr%20gulati&f=false](https://books.google.co.in/books?id=AvQAa5Zfuj0C&printsec=frontcover&dq=tv+by+rr+gulati&source=gbs_similarbooks_s&cad=1#v=onepage&q=tv%20by%20rr%20gulati&f=false)

### MOOC Course

<https://nptel.ac.in/courses/109103018/34>

## INTRODUCTION TO SIGNAL PROCESSING

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC68      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The course should enable the students to:

1. Identify various signals and systems.
2. Understand various transformation techniques and apply them to find spectral domain representation of various signals.
3. Apply convolution and correlation operations between various signals and systems.
4. Develop the digital filter designs.
5. Understand Multirate digital signal processing concepts.

#### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify various signals and systems.
2. Understand various transformation techniques and apply them to find spectral domain representation of various signals.
3. Apply convolution and correlation operations between various signals and systems.
4. Develop the digital filter designs.
5. Understand Multirate digital signal processing concepts.

#### UNIT-I

#### TIME DOMAIN SIGNALS

Classes: 09

Introduction, Definitions of signal and system, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sinc, Signum, and Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.

#### UNIT-II

#### SPECTRAL DOMAIN REPRESENTATION OF SIGNALS

Classes: 09

Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transform, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.

#### UNIT-III

#### CONVOLUTION AND CORRELATION

Classes: 09

Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.

#### UNIT-IV

#### DIGITAL FILTERS

Classes: 09

IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method.

FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.

#### UNIT-V

#### MULTIRATE DIGITAL SIGNAL PROCESSING

Classes: 09

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

**Text Books:**

1. John G. Proakis, Dimitris G. Manolakis, Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi , Signals, Systems & Communications ,BS Publications 2013 .
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

**Reference Books:**

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

**Web References:**

1. [http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-\(T.-OHaver\).html](http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-OHaver).html)
2. <https://www.sciencedirect.com/science/article/pii/B9780123965028000012>
3. Applied Digital Signal Processing (1st Edition)

**E-Text Books:**

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. [http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-\(Digital-Signal-Processing\).html](http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html)

**MOOC Course:**

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

## INTRODUCTION TO MOBILE COMMUNICATION

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC69      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand impairments due to multi path fading channel.
2. Understand the fundamental techniques to overcome the different fading effects.
3. Understand Co-channel and Non Co channel interference
4. Familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand impairments due to multi path fading channel.
2. Understand the fundamental techniques to overcome the different fading effects.
3. Understand Co-channel and Non Co channel interference
4. Familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.

### UNIT-I INTRODUCTION TO CELLULAR MOBILE RADIO SYSTEMS Classes: 11

Limitations of Conventional Mobile Telephone Systems, Basic Cellular Mobile System, First, Second, Third and Fourth Generation Cellular Wireless Systems, Uniqueness of Mobile Radio Environment- Fading -Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time. Fundamentals of Cellular Radio System Design: Concept of Frequency Reuse, Co-Channel Interference, Co-Channel Interference Reduction Factor, Desired C/I From a Normal Case in a Omni Directional Antenna System, System Capacity, Trunking and Grade of Service, Improving Coverage and Capacity in Cellular Systems- Cell Splitting, Sectoring, Microcell Zone Concept.

### UNIT-II CO-CHANNEL INTERFERENCE Classes: 09

Measurement Of Real Time Co-Channel Interference, Design of Antenna System, Antenna Parameters and Their Effects, Diversity Techniques-Space Diversity, Polarization Diversity, Frequency Diversity, Time Diversity. Non-Co-Channel Interference: Adjacent Channel Interference, Near End Far End Interference, Cross Talk, Effects on Coverage and Interference by Power Decrease, Antenna Height Decrease, Effects of Cell Site Components.

### UNIT-III MULTIPLE ACCESS & MODULATION SCHEMES Classes: 10

Signal Reflections in Flat And Hilly Terrain, Effect of Human Made Structures, Phase Difference Between Direct and Reflected Paths, Constant Standard Deviation, Straight Line Path Loss Slope, General Formula for Mobile Propagation Over V'later and Flat Open Area, Near and Long Distance Propagation, Path Loss From a Point to Point Prediction Model in Different Conditions, Merits of Lee Model. Cell Site and Mobile Antennas: Space Diversity Antennas, Umbrella Pattern Antennas, Minimum Separation of Cell Site Antennas, Mobile Antennas.

### UNIT-IV FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT Classes: 07

Numbering And Grouping, Setup Access And Paging Channels, Channel Assignments to Cell Sites and Mobile Units, Channel Sharing and Borrowing, Sectorization, Overlaid Cells, Non Fixed Channel Assignment.

### UNIT-V HANDOFFS AND DROPPED CALLS Classes: 08

Handoff Initiation, Types of Handoff, Delaying Handoff, Advantages of Handoff, Power Difference Handoff, Forced Handoff, Mobile Assisted and Soft Handoff, inter system Handoff, Introduction to Dropped Call Rates and their Evaluation.

**Text Books:**

1. Mobile Cellular Telecommunications — W.C.Y. Lee, Mc Graw Hill, 2nd Edn., 1989.
2. Wireless Communications – Theodore. S. Rapport, Pearson Education, 2nd Edn., 2002.
3. Mobile Cellular Communication – Gottapu sashibhushana Rao, Pearson, 2012.

**Reference Books:**

1. Principles of Mobile Communications — Gordon L. Stuber, Springer International, 2nd Edn., 2001
2. Modern Wireless Communications-Simon Haykin, Michael Moher, Pearson Education, 2005
3. Wireless Communications Theory and Techniques, Asrar U. H .Sheikh, Springer, 2004.

**Web References:**

1. [https://web.fe.up.pt/~mricardo/08\\_09/cmov-mieic/slides/introduction.pdf](https://web.fe.up.pt/~mricardo/08_09/cmov-mieic/slides/introduction.pdf)

**E-Text Books:**

1. <http://ee.sharif.edu/~pr.wireless.comm/references/Schwartz.pdf>

**MOOC Course:**

1. <http://nptel.ac.in/courses/117102062/>

## BASIC EMBEDDED SYSTEMS DESIGN

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EC70      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Explains about the Basic functions, Structure, Concepts and Applications of Embedded systems.
2. Build simple programs to solve well-defined problems on an Embedded platform
3. Understanding various communication interfaces available.
4. Identify different processors for various applications.
5. Explains about the tools used to develop in an embedded environment and Implement Real time applications on embedded platform

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explains about the Basic functions, Structure, Concepts and Applications of Embedded systems.
2. Build simple programs to solve well-defined problems on an Embedded platform
3. Understanding various communication interfaces available.
4. Identify different processors for various applications.
5. Explains about the tools used to develop in an Embedded environment and Implement Real time applications on Embedded platform

### UNIT-I INTRODUCTION TO EMBEDDED SYSTEMS Classes: 12

Definition and Classification – Overview of Processors and hardware units in an embedded system – Software embedded into the system – Exemplary Embedded Systems – Embedded Systems on a Chip (SoC) and the use of VLSI designed circuits.

### UNIT-II INTRODUCING THE 8051 MICROCONTROLLER FAMILY Classes: 10

Introduction, What's in a name, The external interface of the Standard 8051, Reset requirements, Clock frequency and performance, Memory issues, I/O pins, Timers, Interrupts, Serial interface, Power consumption, Conclusions

### UNIT-III DEVICES AND BUSES FOR DEVICES NETWORK Classes: 12

I/O Devices - Device I/O Types and Examples – Synchronous - Iso-synchronous and Asynchronous Communications from Serial Devices - Examples of Internal Serial-Communication Devices - UART and HDLC - Parallel Port Devices - Sophisticated interfacing features in Devices/Ports- Timer and Counting Devices - '12C', 'USB', 'CAN' and advanced I/O Serial high speed buses- ISA, PCI, PCI-X, cPCI and advanced buses.

### UNIT-IV PROGRAMMING CONCEPTS AND EMBEDDED PROGRAMMING IN C, C++ Classes: 14

Programming in assembly language (ALP) vs. High Level Language - C Program Elements, Macros and functions -Use of Pointers - NULL Pointers - Use of Function Calls – Multiple function calls in a Cyclic Order in the Main Function Pointers – Function Queues and Interrupt Service Routines Queues Pointers – Concepts of EMBEDDED PROGRAMMING in C++ - Objected Oriented Programming – Embedded Programming in C++, 'C' Program compilers – Cross compiler – Optimization of memory codes.

**UNIT-V****REAL TIME OPERATING SYSTEMS – PART - 1****Classes: 12**

Tasks and task states, tasks and data, semaphores, and shared data; message queues, mailboxes and pipes, timer functions, events, memory management, interrupt routines in an RTOS environment. Semaphores, queues, hard real-time scheduling considerations, saving memory and power, an example RTOS like uC-OS (open source).

**Text Books:**

1. Rajkamal, Embedded Systems Architecture, Programming and Design, TATA McGraw-Hill, First reprint Oct. 2003
2. Shibu K.V, Introduction to Embedded Systems, Mc Graw Hill.

**Reference Books:**

1. Steve Heath, Embedded Systems Design, Second Edition-2003, Newnes,
2. David E.Simon, An Embedded Software Primer, Pearson Education Asia, First Indian Reprint 2000.

**Web References:**

1. [https://www.bharathuniv.ac.in/colleges1/downloads/courseware\\_ece/notes/BEI605-%20Embedded-System.pdf](https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/notes/BEI605-%20Embedded-System.pdf)

**E-Text Books:**

1. <http://read.pudn.com/downloads158/ebook/707037/Embedded%20Systems%20Design%20-%202ed%20-%200750655461.pdf>

**MOOC Course:**

1. <https://nptel.ac.in/courses/108102045/>

## OPEN ELECTIVES OFFERED BY EEE DEPARTMENT

| OE1    |                                       | OE2    |                                     |
|--------|---------------------------------------|--------|-------------------------------------|
| A4EE52 | Electrical Wiring and Safety Measures | A4EE55 | Power Plant Engineering             |
| A4EE53 | Electrical Materials                  | A4EE56 | Analysis of Linear Systems          |
| A4EE54 | New trends Electrical Energy          | A4EE57 | Neural Networks and Fuzzy Logic     |
| OE3    |                                       | OE4    |                                     |
| A4EE58 | Electrical Engineering Materials      | A4EE61 | Instrumentation and Control         |
| A4EE59 | Non-Conventional Power Generation     | A4EE62 | Energy Audit and Management Systems |
| A4EE60 | Solar Energy and Applications         | A4EE63 | Energy Storage Systems              |



## ELECTRICAL WIRING AND SAFETY MEASURES

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE52      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring

### COURSE OUTCOMES:

The students should be able to

1. Know safety measures & state safety precautions.
2. Test single phase, three phase transformer, DC & AC machine as per IS.
3. Ascertain the condition of insulation & varnishing if necessary
4. Identify faults & measures to repair faults.

### UNIT-I

### BASICS OF ELECTRICAL INSTALLATIONS

Classes: 12

Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.

### UNIT-II

### EARTHING

Classes: 08

Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.

### UNIT-III

### SAFETY & PREVENTION OF ACCIDENTS

Classes: 08

Definition of terminology used in safety , I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation . Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.

### UNIT-IV

### RESIDENTIAL BUILDING ELECTRIFICATION

Classes: 10

General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub- circuits. Method of drawing single line diagram & wiring diagram.

### UNIT-V

### R ENDS& SUBSTATION AND METERS

Classes: 12

Rule 28 : Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises.

Rule 31: Cut-out on consumer's premises.

Rule 46: Periodical inspection & testing of consumer's installation.

Rule 47: Testing of consumer's installation.

Rule 54: Declared voltage of supply to consumer.

Rule 55: Declared frequency of supply to consumer.

Rule 56: Sealing of meters & cut-outs.

Rule 77: Clearances above ground of the lowest conductor.

Rule 79: Clearances between conductors & trolley wires.

Rule 87: Lines crossing or approaching each other.

Rule 88: Guarding.

### **Text Books:**

1. K.B. Raina, S.K. Bhattacharya Electrical Design; Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh.
2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B. Gupta
3. A course in Electrical Installation, Estimating & costing S.K. Kataria & sons, S.L. Uappal .
4. Electrical wiring Estimating and costing Khanna Publication. ,A.K. Sawhney

### **Reference Books:**

1. Electrical Machine Design Danpat Rai & co.
2. The Electricity Rule 2005 Universal Law Publishing Co. Pvt. Ltd. N. Alagapan S. Ekambaram
3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi , Surjit Singh
4. Tarlok Singh Installation, Commissioning & Maintenance of Electrical Equipment S.K. Kataria & Sons
5. B.V.S. Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd. Mumbai

### **Web References:**

1. <https://electrical-engineering-portal.com> > Technical Articles
2. <https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/>
3. <https://www.cpwd.gov.in/Publication/Internal2013.pdf>

### **E-Text Books:**

1. <https://books.google.co.in/books?isbn=0323170064>
2. <https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment>

### **MOOC Course:**

1. <https://nptel.ac.in/courses/103106071/5>
2. <https://nptel.ac.in/courses/108108099/28>
3. <https://nptel.ac.in/courses/124107001/>

## **ELECTRICAL MATERIALS**

**V Semester: OPEN ELECTIVE -I**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE53      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

To understand about various electrical engineering material

**COURSE OUTCOMES:**

The students should be able to

1. Analyze the various engineering materials.
2. Application of various engineering materials.

**UNIT-I****CONDUCTORS****Classes: 10**

Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.

**UNIT-II****SEMICONDUCTORS****Classes: 08**

General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semiconductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.

**UNIT-III****DIELECTRICS AND INSULATORS****Classes: 12**

Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.

**UNIT-IV****MAGNETIC MATERIALS****Classes: 10**

Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.

**UNIT-V****OPTICAL PROPERTIES OF SOLIDS****Classes: 10**

Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).

### Text Books:

1. "Electrical Engineering Materials", Dekker, PHI Pbs.
2. "Electrical Engineering Materials", Indulkar, S. Chand

### Reference Books:

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K. Rajput, Laxmi Pbs

### Web References:

1. <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
3. [web.mit.edu/course/6/6.732/www/6.732-pt2.pdf](http://web.mit.edu/course/6/6.732/www/6.732-pt2.pdf)

### E-Text Books:

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

### MOOC Course

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
3. [https://onlinecourses.nptel.ac.in/noc18\\_ee14/](https://onlinecourses.nptel.ac.in/noc18_ee14/)

## NEW TRENDS IN ELECTRICAL ENERGY

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE54      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. The fundamental principles of wireless power transfer (WPT) for cable-free transfer of power.
2. Theories for inductive power transfer (IPT) based on the coupled inductor model and low-order circuit compensation.
3. Specific converter topologies for lighting and battery charging applications.
4. Technology trends in the adoption of WPT for key consumer applications.

### COURSE OUTCOMES:

The students should be able to

1. Understand the characteristics of power transfer through coupled inductors and the significance of leakage inductance.
2. Analyze and design appropriate compensation circuits and efficient power converters for WPT applications.
3. Understand technical requirements for applications involving solid-state loads and battery loads using WPT technologies.
4. Appreciate the factors affecting adoption of WPT in consumer applications including lightings, charging of smart phones and electric vehicles.

### UNIT-I

### BASIC CIRCUIT THEORY

Classes: 10

Review of transformers. Leakage inductance. Circuit compensation principles. Low-order compensations; series and parallel compensations. Resonance and operating frequency. Efficiency equation.

### UNIT-II

### POWER CONVERTERS FUNDAMENTALS

Classes: 10

DC-DC converters. AC-DC converters and inverters. PWM and soft switching principles. Basic topologies with transformers. Input, output and transfer characteristics of power converters. Incorporation of leaky transformer. Control methods.

### UNIT-III

### COMPENSATION CONFIGURATIONS

Classes: 10

Types of compensation for inductor power transfer. Characteristics for various termination requirements. Design for load-independence output voltage and output current. Efficiency optimization.

### UNIT-IV

### APPLICATIONS

Classes: 10

Circuit requirements for various loading conditions. Characteristics of LED loads, resistors and battery loads. Appropriate compensation design. Lighting systems. Battery charging profiles. Electric vehicle charging. Energy efficiency metric for charging.

### UNIT-V

### TECHNOLOGY TRENDS

Classes: 10

Demand for safe power transfer and durable operation. Portable and smart devices. Mobile communication devices. IoT devices and systems. Sensors. Solid-state lighting development. Battery technologies. Electric vehicle development. Renewable source integration trends. Future trends and demand for wireless power transfer.

### Text Books:

1. C. T. Rim and C. Mi, Wireless Power Transfer for Electric Vehicles and Mobile Devices, New York: IEEE Press-Wiley, 2017.
2. J. I. Agbinya, Wireless Power Transfer, River Publishers, 2015

### Reference Books:

1. Z. Huang, S. C. Wong, and C. K. Tse, "Design of a single-stage inductive power-transfer converter for efficient EV battery charging," IEEE Transactions on Vehicular Technology, vol. 66, no. 7, pp. 5808-5821, July 2017.
2. L. Xu, Q. Chen, X. Ren, S. C. Wong, and C. K. Tse, "Self-oscillating resonant converter with contactless power transfer and integrated current sensing transformer," IEEE Transactions on Power Electronics, vol. 32, no. 6, pp. 4839-4851, June 2017.
3. W. Zhang, S. C. Wong, C. K. Tse, and Q. Chen, "Load-independent duality of current and voltage outputs of a series or parallel compensated inductive power transfer converter with optimized efficiency," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, pp. 137-146, March 2015.
4. J. Hou, Q. Chen, X. Ren, X. Ruan, S. C. Wong, and C. K. Tse, "Precise characteristics analysis of series/series-parallel compensated contactless resonant converter," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, 101-110, March 2015.

### Web References:

1. [https://www.researchgate.net/.../267512972\\_Wireless\\_Power\\_Transmission\\_Trends](https://www.researchgate.net/.../267512972_Wireless_Power_Transmission_Trends)
2. <https://ieeexplore.ieee.org/document/6403386>
3. <https://www.cse.wustl.edu/~jain/cse574-14/ftp/power/>

### E-Text Books:

1. [https://books.google.co.in/books/about/Wireless\\_Power\\_Transfer.html?id...](https://books.google.co.in/books/about/Wireless_Power_Transfer.html?id...)
2. <https://www.intechopen.com/books/wireless-power-transfer-principles-and-engineering-explorations>

### MOOC Course:

1. <https://ocw.mit.edu/.../6-452-principles-of-wireless-communications-spring-2006/>
2. [www.iitg.ac.in/e\\_mobility/WPT.html](http://www.iitg.ac.in/e_mobility/WPT.html)
3. <https://nptel.ac.in/courses/117102062/>
4. <https://nptel.ac.in/courses/117104099/2>
5. <https://nptel.ac.in/courses/117106108/150>

**VI Semester: OPEN ELECTIVE -II**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE55      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring

**COURSE OUTCOMES:**

The students should be able to

1. Know safety measures & state safety precautions.
2. Test single phase, three phase transformer, DC & AC machine as per IS.
3. Ascertain the condition of insulation & varnishing if necessary
4. Identify faults & measures to repair faults.

**UNIT-I INTRODUCTION TO THE SOURCES OF ENERGY Classes: 10**

Resources and Development of Power in India. Steam Power Plant: Plant Layout, Working of different Circuits, Fuel and handling equipment, types of coals, coal handling, choice of handling equipment, coal storage, Ash handling systems. Combustion Process: Properties of coal — overfeed and underfeed fuel beds, traveling grate stokers, spreader stokers, retort stokers, pulverized fuel burning system and its components, combustion needs and draught system, cyclone furnace, design and construction, Dust collectors, cooling towers and heat rejection. Corrosion and feed water treatment.

**UNIT-II INTERNAL COMBUSTION ENGINE PLANT Classes: 10**

DIESEL POWER PLANT: Introduction — IC Engines, types, construction— Plant layout with auxiliaries fuel supply system, air starting equipment,

**lubrication and cooling system-super charging. Gas Turbine Plant:**

Introduction — classification – construction — Layout with auxiliaries — Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison. Direct Energy Conversion: Solar energy, Fuel cells, Thermo electric and Thermo ionic, MHD generation.

**UNIT-III HYDRO ELECTRIC POWER PLANT Classes: 08**

Water power — Hydro logical cycle / flow measurement — drainage area characteristics — Hydro graphs — storage and Poundage — classification of dams and spill ways.

**Hydro Projects And Plant:** Classification — Typical layouts — plant auxiliaries — plant operation pumped storage plants. Power From Non-Conventional Sources: Utilization of Solar- Collectors- Principle of Working, Wind Energy — types — HAWT, VAWT Tidal Energy.

**UNIT-IV NUCLEAR POWER STATION Classes: 10**

Nuclear fuel — breeding and fertile materials — Nuclear reactor — reactor operation. Types of Reactors: Pressurized water reactor, Boiling water reactor, sodium-graphite reactor, fast Breeder Reactor, homogeneous Reactor, Gas cooled Reactor, Radiation hazards and shielding radioactive waste disposal.

**UNIT-V POWER PLANT ECONOMICS AND ENVIRONMENTAL CONSIDERATIONS Classes: 12**

Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration curve. Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor — related exercises. Effluents from power plants and Impact on environment — pollutants and pollution standards — Methods of Pollution control.

#### **Text Books:**

1. Power Plant Engineering! P.C.Sharma / S.K.Kataria Pub.
2. A Course in Power Plant Engineering: I Arora and S. Domkundwar.

#### **Reference Books:**

1. A Text Book of Power Plant Engineering I Rajput I Laxmi Publications.
2. Power Plant Engineering: P.K.Nag/ II Edition /TMH.
3. An Introduction to Power Plant Technology I G.D. Rai/Khanna Publishers.
4. Power plant Engg I Elanchezhian/ I.K. International Pub.
5. Power plant Engineering! Ramalingam / Scietech Publishers.

#### **Web References:**

1. [www.cbspd.co.in/power-plant-engineering-9788123919461-black-veatch.html](http://www.cbspd.co.in/power-plant-engineering-9788123919461-black-veatch.html)
2. <https://www.semcomaritime.com/en-en/references/power-generation>

#### **E-Text Books:**

1. <https://www.intechopen.com/books/subject/power-plant-engineering>
2. <https://books.google.co.in/books?isbn=1482221519>

#### **MOOC Course**

1. <https://nptel.ac.in/courses/108105058/8>
2. [www.nptelvideos.in/2012/11/energy-resources-and-technology.html](http://www.nptelvideos.in/2012/11/energy-resources-and-technology.html)
3. [www.ignou.ac.in/upload/Unit-2-58.pdf](http://www.ignou.ac.in/upload/Unit-2-58.pdf)

## **ANALYSIS OF LINEAR SYSTEMS**

### **VI Semester: OPEN ELECTIVE -II**



| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE56      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

1. To develop ability to analyze linear systems and signals
2. To develop critical understanding of mathematical methods to analyze linear systems and signals

**COURSE OUTCOMES:**

The students should be able to

1. Use mathematical modeling tools to represent linear systems
2. Use mathematical modeling tools to analyze linear systems

**UNIT-I****STATE VARIABLE ANALYSIS****Classes: 08**

Choice of state variables in Electrical networks-Formulation of state equations for Electrical networks Equivalent source method. Network topological method – Solution of state equations-Analysis of simple networks with state variable approach.

**UNIT-II****FOURIER SERIES AND FOURIER TRANSFORM REPRESENTATION****Classes: 12**

Introduction, Trigonometric form of Fourier series, Exponential form of Fourier series, Wave symmetry, Fourier integrals and transforms, Fourier transform of a periodic function, Properties of Fourier Transform, Parseval's theorem, Fourier transform of some common signals, Fourier transform relationship with Laplace Transform. Applications of Fourier series and Fourier Transform Representation: Introduction, Effective value, and average values of non sinusoidal periodic waves, currents, Power Factor, Effects of harmonics, Application in Circuit Analysis, Circuit Analysis using Fourier Series.

**UNIT-III****LAPLACE TRANSFORM APPLICATIONS****Classes: 10**

Application of Laplace transform Methods of Analysis – Response of RL, RC, RLC Networks to Step, Ramp, and impulse functions, Shifting Theorem – Convolution Integral – Applications Testing of Polynomials: Elements of realisability – Hurwitz polynomials-positive real functions-Properties-Testing-Sturm's Test, examples. Network Synthesis: Network synthesis: Synthesis of one port LC networks-Foster and Cauer methods-Synthesis of RL and RC one port networks-Foster and Cauer methods

**UNIT-IV****SAMPLING****Classes: 10**

Sampling theorem – Graphical and Analytical proof for Band Limited Signal impulse sampling, natural and Flat top Sampling, Reconstruction of signal from its samples, effect of under sampling – Aliasing, introduction to Band Pass sampling, Cross correlation and auto correlation of functions, properties of correlation function, Energy density spectrum, Power density spectrum, Relation between auto correlation function and Energy / Power spectral density function.

**UNIT-V****Z-TRANSFORMS****Classes: 10**

Fundamental difference between continuous and discrete time signals, discrete time complex, exponential and sinusoidal signals, periodicity of discrete time complex exponential, concept of Z Transform of a discrete sequence. Distinction between Laplace, Fourier, and Z-Transforms. Region of convergence in Z-Transforms, constraints on ROC for various classes of signals, Inverse Z-Transform properties of Z-Transforms.

**Text Books:**

1. B. P. Lathi, "Signals, Systems and Communications", BS Publications 2003.
2. "Umesh Sinha" "Network Analysis and Synthesis", Satya Prakashan Publications, 2013.

**Reference Books:**

1. "A. N. Tripathi", "Linear System Analysis", New Age International, 2nd Edition 1987.
2. "D. Roy Chowdhary", "Network and Systems", New Age International, 2005.
3. "Gopal G Bhise, Prem R. Chadha", Engineering Network Analysis and Filter Design, Umesh Publications 2009.
4. "A. Cheng", linear system analysis, Oxford publishers, 1999

**Web References:**

1. <https://archive.org/details/introductiontoli00brow>
2. <https://ieeexplore.ieee.org/jiel5/9/24171/01101971.pdf>

**E-Text Books:**

1. [www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete\\_04Mar10.pdf](http://www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_04Mar10.pdf)
2. <https://www.springer.com/gp/book/9780387975733>

**MOOC Course:**

1. [www.nptelvideos.in/2012/11/estimation-of-signals-and-systems.html](http://www.nptelvideos.in/2012/11/estimation-of-signals-and-systems.html)
2. <https://nptel.ac.in/courses/108104100/6>

## NEURAL NETWORKS AND FUZZY LOGIC

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE57      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

### COURSE OUTCOMES:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback network.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

### UNIT-I INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS Classes: 12

Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application

### UNIT-II SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS Classes: 10

Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training.

Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.

### UNIT-III ASSOCIATIVE MEMORIES-I Classes: 08

Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).

### UNIT-IV ASSOCIATIVE MEMORIES-II Classes: 10

Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hoptfield

Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications

**UNIT-V****FUZZY LOGIC****Classes: 10**

Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.

Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, De-fuzzification methods.

**Text Books:**

1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI.
2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publication-is.

**Reference Books:**

1. Artificial Neural Networks, B. Yegnanarayana, PHI.
2. Artificial Neural Networks, Zaruda, PHI.
3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI.
4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd.
5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education.
6. Neural networks by satish Kumar, TIVIH, 2004
7. Neural Networks, Simon Hakens, Pearson Education.
8. Neural Engineering, C.Eliasmith and CH.Anderson, PHI.

**Web References:**

1. [users.monash.edu/~app/CSE5301/Lnts/LaD.pdf](https://users.monash.edu/~app/CSE5301/Lnts/LaD.pdf)
2. <https://engineering.purdue.edu/~tsoukala/rational.html>
3. <https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf>

**E-Text Books:**

1. <https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book>
2. [www.crectirupati.com/sites/default/files/lecture\\_notes/NNFL.pdf](http://www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf)
3. [www.vssut.ac.in/lecture\\_notes/lecture1423723637.pdf](http://www.vssut.ac.in/lecture_notes/lecture1423723637.pdf)

**MOOC Course:**

1. [https://nptel.ac.in/noc/individual\\_course.php?id=noc19-ge07](https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07)
2. <https://nptel.ac.in/courses/108104049/16>
3. <https://nptel.ac.in/courses/117105084/>
4. <https://nptel.ac.in/syllabus/127105006/>

## ELECTRICAL ENGINEERING MATERIALS

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE58      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OUTCOMES:

The students should be able to

1. Analyze the various engineering materials.
2. Application of various engineering materials.

### UNIT-I

### CONDUCTORS

Classes: 10

Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.

### UNIT-II

### SEMICONDUCTORS

Classes: 08

General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semiconductors, extrinsic Semiconductors, hall effect, drift, mobility, diffusion in Semiconductors, Semiconductors and their applications, superconductors.

### UNIT-III

### DIELECTRICS AND INSULATORS

Classes: 12

Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.

### UNIT-IV

### MAGNETIC MATERIALS

Classes: 10

Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.

### UNIT-V

### OPTICAL PROPERTIES OF SOLIDS

Classes: 10

Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).

**Text Books:**

1. "Electrical Engineering Materials", Dekker, PHI Pbs.
2. "Electrical Engineering Materials", Indulkar, S.Chand

**Reference Books:**

1. "Electrical Engineering Materials", Tareev
2. "Electrical Engineering Materials", Yu. Koritsky.
3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs

**Web References:**

1. <https://physics.info/dielectrics/>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265>
3. [web.mit.edu/course/6/6.732/www/6.732-pt2.pdf](http://web.mit.edu/course/6/6.732/www/6.732-pt2.pdf)

**E-Text Books:**

1. <https://easyengineering.net/electrical-engineering-materials-by-dekker/>
2. <https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/>

**MOOC Course**

1. <https://nptel.ac.in/courses/108108076/>
2. <https://nptel.ac.in/courses/112104203/3>
3. [https://onlinecourses.nptel.ac.in/noc18\\_ee14/](https://onlinecourses.nptel.ac.in/noc18_ee14/)

## NON CONVENTIONAL POWER GENERATION

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE59      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. Demonstrate power generation systems including major subsystems.
2. Understand basic working principles of nuclear power generation systems.
3. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

#### COURSE OUTCOMES:

The students should be able to

1. Understand basic working principles of nuclear power generation systems.
2. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

#### UNIT-I

#### PRINCIPLES OF SOLAR RADIATION

**Classes: 08**

Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, the solar constant, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and Sun shine, solar radiation data.

#### UNIT-II

#### SOLAR ENERGY COLLECTORS

**Classes: 10**

Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

**STORAGE AND APPLICATIONS:** Different methods of solar energy storage, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating /cooling technique, solar distillation and drying.

#### UNIT-III

#### PHOTO VOLTAICS (PV)

**Classes: 12**

Fundamentals of solar cells, types of solar cells, semiconducting materials, band gap theory, absorption of photons, excitations and photoemission of electrons, band engineering.

**PV CELL PROPERTIES:** Solar cell properties and design, p-n junction photodiodes, depletion region, electrostatic field across the depletion layer, electron and holes transports, device physics, charge carrier generation, recombination and other losses, I-V characteristics, output power.

#### UNIT-IV

#### SOLAR CELL APPLICATIONS

**Classes: 10**

PV cell interconnection, module structure and module fabrication, Equivalent circuits, load matching, efficiency, fill factor and optimization for maximum power, Design of stand-alone PV systems, system sizing, device structures, device construction, DC to AC conversion, inverters, on-site storage and grid connections.

#### UNIT-V

#### COST ANALYSIS AND ENVIRONMENTAL ISSUES

**Classes: 10**

Cost analysis and pay back calculations for different types of solar panels and collectors, installation and operating costs, Environmental and safety issues, protection systems, performance monitoring.

**ALTERNATIVE ENERGY SOURCES:** Solar Energy: Types of devices for Solar Energy Collection, Thermal Storage System, Control Systems, Wind Energy, Availability, Wind Devices, Wind Characteristics, Performance of Turbines and systems.

### Text Books:

1. G. D. Rai (2009), *Non-Conventional Energy Sources*, 4th edition, Khanna Publishers, New Delhi.
2. Martin A. Green (2008), *Solar Cells: Operating Principles, Technology and system Applications*, 1st edition, Prentice Hall, New Delhi.

### Reference Books:

1. Sukatme (2008), *Solar Energy*, 3rd Edition, McGraw Hill Companies, New Delhi.
2. D. Yogi goswami, Frank Kreith, Jan F. Kreider (2000), *Principles of Solar Engineering*, 2nd edition, Taylor & Francis, USA.

### Web References:

1. [https://nptel.ac.in/courses/Webcourse-contents/IISc-BANG/notused/Non Conventional%20Energy%20Systems-/Learning%20Material%20-%20NCES.pdf](https://nptel.ac.in/courses/Webcourse-contents/IISc-BANG/notused/Non%20Conventional%20Energy%20Systems-/Learning%20Material%20-%20NCES.pdf)
2. <https://www.ukessays.com/essays/environmental-sciences/promoting-non-conventional-energy-resource>

### E-Text Books:

1. [bie.telangana.gov.in/files/Nonconventionalenergysources.pdf](http://bie.telangana.gov.in/files/Nonconventionalenergysources.pdf)
2. [https://www.researchgate.net/.../236857095\\_Non\\_Conventional\\_Energy\\_Resources](https://www.researchgate.net/.../236857095_Non_Conventional_Energy_Resources)

### MOOC Course

1. <https://nptel.ac.in/courses/108102047>
2. [https://onlinecourses.nptel.ac.in/noc19\\_ge11/cours](https://onlinecourses.nptel.ac.in/noc19_ge11/cours)
3. <https://nptel.ac.in/downloads/108108078/>



## SOLAR ENERGY AND APPLICATIONS

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE60      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To Study about solar modules and PV system design and their applications
2. To Deal with grid connected PV systems
3. To Discuss about different energy storage systems

### COURSE OUTCOMES:

The students should be able to

1. Understanding on solar energy storage systems
2. Basic knowledge on standalone PV system
3. Understand the issues in grid connected PV systems
4. Study about the modelling of different energy storage systems and their performances
5. Attain more on different applications of solar energy

### UNIT-I

### INTRODUCTION

**Classes: 10**

Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection

### UNIT-II

### STAND ALONE PV SYSTEM

**Classes: 10**

Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing

### UNIT-III

### GRID CONNECTED PV SYSTEMS

**Classes: 10**

PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs

### UNIT-IV

### ENERGY STORAGE SYSTEMS

**Classes: 10**

Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage

### UNIT-V

### APPLICATIONS

**Classes: 10**

Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.

### Text Books:

1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015.
2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994

### Reference Books:

1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011.
2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 S.P. Sukhatme, "Solar Energy", Tata McGraw Hill, 1987.

### Web References:

1. <https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html>
2. <https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798>
3. <https://link.springer.com/journal/11949>

### E-Text Books:

1. [https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar\\_energy\\_v1.1.pdf](https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf)
2. [bookstore.teri.res.in/books/9788179935736](http://bookstore.teri.res.in/books/9788179935736)

### MOOC Course:

1. <https://nptel.ac.in/courses/112105051/>
2. <https://nptel.ac.in/courses/121106014/18>
3. <https://nptel.ac.in/courses/112105050/>
4. <https://nptel.ac.in/syllabus/112105051/>

## INSTRUMENTATION AND CONTROL

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE61      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

#### COURSE OUTCOMES:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

#### UNIT-I INTRODUCTION TO MEASURING INSTRUMENTS Classes: 10

Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.

#### UNIT-II POTENTIOMETERS & INSTRUMENT TRANSFORMERS Classes: 08

Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications. CT and PT — Ratio and phase angle errors.

#### UNIT-III TRANSDUCERS & OSCILLOSCOPES Classes: 12

Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes.

**CRO:** Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.

#### UNIT-IV CLASSIFICATION OF CONTROL SYSTEMS Classes: 10

Concepts of Control Systems- Open Loop and closed loop control systems and their differences-Different examples of control systems- Classification of control systems

Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions .Block diagram algebra – Representation by Signal flow graph –

Reduction using mason's gain formula.

**UNIT-V****TIME RESPONSE ANALYSIS****Classes: 10**

Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.

**Text Books:**

1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd.
2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications.
3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009
4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.

**Reference Books:**

1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications.
2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd.
3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications.
4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998.
5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011.
6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.

**Web References:**

1. [home.mit.bme.hu/~virostek/docs/mt.../Principles\\_of\\_electrical\\_measurement.pdf](http://home.mit.bme.hu/~virostek/docs/mt.../Principles_of_electrical_measurement.pdf)
2. [https://www.mccdaq.com/handbook/chapt\\_4.aspx](https://www.mccdaq.com/handbook/chapt_4.aspx)

**E-Text Books:**

1. [www.vssut.ac.in/lecture\\_notes/lecture1423813026.pdf](http://www.vssut.ac.in/lecture_notes/lecture1423813026.pdf)
2. [www.vssut.ac.in/lecture\\_notes/lecture1423904331.pdf](http://www.vssut.ac.in/lecture_notes/lecture1423904331.pdf)
3. [www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf](http://www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf)

**MOOC Course:**

1. <https://nptel.ac.in/syllabus/108106070/>
2. <https://nptel.ac.in/courses/108105064/>
3. <https://nptel.ac.in/courses/108101037/>
4. <https://nptel.ac.in/courses/107106081/>

**VIII Semester: OPEN ELECTIVE -IV**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE62      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

1. To understand energy efficiency, scope, conservation and technologies.
2. To design energy efficient lighting systems.
3. To estimate/calculate power factor of systems and propose suitable compensation techniques.
4. To understand energy conservation in HVAC systems.
5. To calculate life cycle costing analysis and return on investment on energy efficient technologies.

**COURSE OUTCOMES:**

The students should be able to

1. Explain energy efficiency, conservation and various technologies.
2. Design energy efficient lighting systems.
3. Calculate power factor of systems and propose suitable compensation techniques.
4. Explain energy conservation in HVAC systems.
5. Calculate life cycle costing analysis and return on investment on energy efficient technologies.

**UNIT-I ENERGY AUDIT AND MANAGEMENT ENERGY AUDIT Classes: 12**

Basic Principles of Energy Audit and management Energy audit – Definitions – Concept – Types of audit – Energy index – Cost index – Pie charts – Sankey diagrams – Load profiles – Energy conservation schemes and energy saving potential – Numerical problems – Principles of energy management – Initiating, planning, controlling, promoting, monitoring, reporting – Energy manager – Qualities and functions – Language – Questionnaire – Check list for top management.

**UNIT-II LIGHTING Classes: 10**

Modification of existing systems – Replacement of existing systems – Priorities: Definition of terms and units – Luminous efficiency – Polar curve – Calculation of illumination level – Illumination of inclined surface to beam – Luminance or brightness – Types of lamps – Types of lighting – Electric lighting fittings (luminaries) – Flood lighting – White light LED and conducting Polymers – Energy conservation measures.

**UNIT-III POWER FACTOR AND ENERGY INSTRUMENTS Classes: 08**

Power factor – Methods of improvement – Location of capacitors – Power factor with non linear loads – Effect of harmonics on Power factor – Numerical problems. Energy Instruments – Watt-hour meter – Data loggers – Thermocouples – Pyrometers – Lux meters – Tong testers – Power analyzer.

**UNIT-IV SPACE HEATING AND VENTILATION Classes: 08**

Ventilation – Air-Conditioning (HVAC) and Water Heating: Introduction – Heating of buildings – Transfer of Heat-Space heating methods – Ventilation and air-conditioning – Insulation-Cooling load – Electric water heating systems – Energy conservation methods.

**UNIT-V ECONOMIC ASPECTS AND ANALYSIS Classes: 12**

Economics Analysis – Depreciation Methods – Time value of money – Rate of return – Present worth method – Replacement analysis – Life cycle costing analysis – Energy efficient motors (basic concepts) **Computation of Economic Aspects** : Calculation of simple payback method – Net present worth method – Power factor correction – Lighting – Applications of life cycle costing analysis – Return on investment.

### Text Books:

1. Energy management by W.R. Murphy & G. McKay Butterworth, Elsevier publications. 2012
2. Energy efficient electric motors by John .C. Andreas, Marcel Dekker Inc Ltd–2nd edition, 1995

### Reference Books:

1. Electric Energy Utilization and Conservation by S C Tripathy, Tata McGraw hill publishing company Ltd. New Delhi.
2. Energy management by Paul o' Callaghan, Mc-Graw Hill Book company–1st edition, 1998.
3. Energy management hand book by W.C.Turner, John wiley and sons.
4. Energy management and conservation –k v Sharma and pvenkatasessaiah-I K International Publishing House pvt.ltd,2011.

### Web References:

1. <https://beeindia.gov.in/sites/default/files/1Ch3.pdf>
2. [https://www.researchgate.net/publication/309463130\\_Energy\\_Audit\\_Management](https://www.researchgate.net/publication/309463130_Energy_Audit_Management)

### E-Text Books:

1. [lab.fs.uni-lj.si/kes/erasmus/Energy%20Management%20Handbook.pdf](http://lab.fs.uni-lj.si/kes/erasmus/Energy%20Management%20Handbook.pdf)
2. <https://fenix.tecnico.ulisboa.pt/downloadFile/.../Energy%20Audit%202013.pdf>
3. [enersupply.euke.sk/wp-content/uploads/Introduction-to-EM-and-EA\\_P1\\_Fotini.pdf](http://enersupply.euke.sk/wp-content/uploads/Introduction-to-EM-and-EA_P1_Fotini.pdf)

### MOOC Course:

1. <https://nptel.ac.in/courses/108106022>
2. <https://nptel.ac.in/courses/105107156/12>
3. [shodhganga.inflibnet.ac.in/bitstream/10603/46067/11/11\\_chapter%201.pdf](http://shodhganga.inflibnet.ac.in/bitstream/10603/46067/11/11_chapter%201.pdf)

## ENERGY STORAGE SYSTEMS

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4EE63      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To enable the student to understand the need for energy storage, devices and technologies available and their applications

#### COURSE OUTCOMES:

The students should be able to

1. Analyze the characteristics of energy from various sources and need for storage
2. Classify various types of energy storage and various devices used for the purpose
3. Identify various real time applications.

#### UNIT-I ELECTRICAL ENERGY STORAGE TECHNOLOGIES Classes: 12

Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.

#### UNIT-II NEEDS FOR ELECTRICAL ENERGY STORAGE Classes: 08

Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.

#### UNIT-III FEATURES OF ENERGY STORAGE SYSTEMS Classes: 08

Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES), Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H<sub>2</sub>), Synthetic natural gas (SNG).

#### UNIT-IV TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS Classes: 10

Electrical storage systems, Double-layer capacitors (DLC) , Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.

#### UNIT-V APPLICATIONS Classes: 12

Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications , Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems , Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA– aggregation of many dispersed batteries.

**Text Books:**

1. “James M. Eyer, Joseph J. Iannucci and Garth P. Corey “, “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004.
2. The Electrical Energy Storage by IEC Market Strategy Board.

**Reference Books:**

1. “Jim Eyer, Garth Corey”, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.

**Web References:**

1. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212>

**E-Text Books:**

1. [https://www.pewtrusts.org/~media/.../energy\\_storage-backs\\_up\\_power\\_supply.pdf](https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf)
2. <https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf>
3. <https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf>

**MOOC Course:**

1. [nptel.ac.in/courses/112105221/56](https://nptel.ac.in/courses/112105221/56)
2. [nptel.ac.in/courses/108108036/9](https://nptel.ac.in/courses/108108036/9)
3. <https://nptel.ac.in/courses/108102047/7>
4. <https://nptel.ac.in/courses/108105058/37>



## OPEN ELECTIVES OFFERED BY IT DEPARTMENT

| OE1    |                                             | OE2    |                                                |
|--------|---------------------------------------------|--------|------------------------------------------------|
| A4IT11 | Fundamentals of Data Structures             | A4IT14 | Basics of Python Programming                   |
| A4IT12 | Software Engineering Principles             | A4IT15 | Human Computer Interaction                     |
| A4IT13 | Computer Organization and Operating Systems | A4IT16 | Software Testing Fundamentals                  |
| OE3    |                                             | OE4    |                                                |
| A4IT17 | E- Commerce                                 | A4IT20 | Introduction to Mobile Application Development |
| A4IT18 | Cyber Forensics                             | A4IT21 | Distributed Databases                          |
| A4IT19 | Discrete Structures                         | A4IT22 | Big Data                                       |

## FUNDAMENTALS OF DATA STRUCTURES

### V Semester: OPEN ELECTIVE - I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT11      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Impart the basic concepts of data structures and algorithms.
2. Understand concepts linked lists and their applications.
3. Understand basic concepts about stacks, queues and their applications.
4. Understand basic concepts of trees, graphs and their applications.
5. Enable them to write algorithms for sorting and searching and hashing.
6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate algorithms in terms of time and memory complexity.
2. Formulate new solutions for problems or improve existing code using data structures and algorithms.
3. Implement basic data structures such as arrays, linked lists, stacks and queues.
4. Solve problem involving graphs, trees and heaps
5. Apply Algorithms for solving problems like sorting, searching, and hashing.
6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

### UNIT-I

### INTRODUCTION TO DATA STRUCTURES

**Classes: 12**

Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations

Introduction to Linear and Non Linear data structures-Singly Linked Lists- Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion.

### UNIT-II

### STACKS AND QUEUES

**Classes: 10**

Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation.

Queues-Queue ADT, definition and operations, array and linked Implementations in C, Circular queues- Insertion and deletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.

### UNIT-III

### TREES AND GRAPHS

**Classes: 14**

Trees – Terminology, Representation of Trees, binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals.

Max Priority Queue-ADT- implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from

a Max Heap. Graphs , Introduction, Definition, Terminology, Graph ADT, Graph Representations- Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.

**UNIT-IV****SEARCHING AND SORTING****Classes: 12**

**Searching-** Linear Search, Binary Search, Comparison of search techniques. **Sorting-** Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.

**UNIT-V****BINARY SEARCH TREES****Classes: 12**

**Search Trees-** Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees- Definition and Examples, Insertion into an AVL Tree, B-Trees, Definition, B-Tree of order m, operations- Insertion and Searching, Comparison of Search Trees.

**Pattern matching algorithm-** The Knuth-Morris-Pratt algorithm, Tries (examples only)

**Text Books:**

1. Fundamentals of Data Structures II, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and Susan Anderson-Freed, Universities Press

**Reference Books:**

1. Algorithms, Data Structures, and Problem Solving with C++, Illustrated Edition by Mark Allen Weiss, Addison- Wesley Publishing Company
2. How to Solve it by Computer II, 2nd Impression by R.G.Dromey, Pearson Education

## SOFTWARE ENGINEERING PRINCIPLES

### V Semester: OPEN ELECTIVE - I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT12      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Be familiar with basic Software engineering methods and practices, and its applications.
2. Apply the implementation of software engineering layered technology and Process framework.
3. Understand the concepts of software measurement and software risks.
4. Be familiar with software requirements and the SRS documents.
5. Be familiar with role of project management including planning, scheduling, risk management.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply basic knowledge and understanding of the analysis, synthesis and design of complex systems.
2. Apply software engineering principles and techniques.
3. Design and evaluate large-scale software systems.
4. Explain the principles, tools and practices of IT project management.
5. Apply the fundamental knowledge of science in emerging technologies.

### UNIT-I INTRODUCTION TO SOFTWARE ENGINEERING Classes: 12

The evolving role of software, Changing Nature of Software, Software myths.

**A Generic view of process:** Software engineering- A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models.

### UNIT-II PROCESS MODELS Classes: 12

The waterfall model, Incremental process models, Evolutionary process models, The Unified process.

**Software Requirements:** Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document.

**Requirements engineering process:** Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.

### UNIT-III SYSTEM MODELS Classes: 12

Context Models, Behavioral models, Data models, Object models, structured methods.

**Design Engineering:** Design process and Design quality, Design concepts, the design model.

**Creating an architectural design:** Software architecture, Data design, Architectural styles and patterns, Architectural Design.

### UNIT-IV OBJECT-ORIENTED DESIGN Classes: 12

Objects and object classes, An Object-Oriented design process, Design evolution.

**Performing User interface design:** Golden rules, User interface analysis and design, interface

analysis, interface design steps, Design evaluation.

**Testing Strategies:** A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging.

## UNIT-V

## RISK MANAGEMENT

**Classes: 12**

Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan.

**Quality Management:** Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

### Text Books:

1. Software Engineering a practitioner's Approach, Roger S Pressman, 6th edition. Mc. Graw Hill International Edition.
2. Software Engineering, Ian Somerville, 7th edition, Pearson education.

### Reference Books:

1. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010.
2. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw-Hill, 2008
3. Fundamentals of Software Engineering, Rajib Mall, PHI, 2005

## COMPUTER ORGANIZATION AND OPERATING SYSTEMS

### V Semester: OPEN ELECTIVE - I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT13      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To understand the basic structure and operation of a digital computer.
2. To discuss in detail the operation of the arithmetic unit including the algorithms & implementation of fixed-point and floating-point addition, subtraction, multiplication & division.
3. To study different ways of communicating with I/O devices and standard I/O interfaces.
4. To study hierarchical memory system including cache memories and virtual memory.
5. To demonstrate the knowledge of functions of operating system memory management scheduling, file system and interface, distributed systems, security and dead locks.
6. To implement a significant portion of an Operating System

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain different types of Addressing modes
2. Explain different synchronous and asynchronous data transfer techniques?
3. Explain different I/O data transfer techniques with performance comparison?
4. Outline various concepts and features of Operating systems.

### UNIT-I

### BASIC STRUCTURE OF COMPUTERS

Classes: 14

Computer Types, Functional UNIT, Basic Operational Concepts, Bus Structures, Software, Performance, Multiprocessors and Multi Computers, Data Representation, Fixed Point Representation, Floating - Point Representation.

**Register Transfer Language and Micro Operations:** Register Transfer Language, Register Transfer Bus and Memory Transfers, Arithmetic Micro Operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit, Instruction Codes, Computer Registers Computer Instructions - Instruction Cycle. Memory - Reference Instructions, Input - Output and Interrupt, STACK Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation.

### UNIT-II

### MICRO PROGRAMMED CONTROL

Classes: 13

Control Memory, Address Sequencing, Micro Program Examples, Design of Control Unit, Hard Wired Control, Micro Programmed Control.

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer Modes, Priority Interrupt, Direct Memory Access, Input-Output Processor (IOP).

### UNIT-III

### OPERATING SYSTEMS OVERVIEW

Classes: 09

Overview of Computer Operating Systems Functions, Protection and Security, Distributed Systems, Special Purpose Systems, Operating Systems Structures-Operating System Services and Systems Calls, System Programs, Operating System Generation.

Principles of Deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

**UNIT-IV****MEMORY MANAGEMENT****Classes: 12**

Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation, Virtual Memory, Demand Paging, Page-Replacement Algorithms, Allocation of Frames, Thrashing Case Studies -UNIX, Linux, Windows.

**UNIT-V****PRINCIPLES OF DEADLOCK****Classes: 12**

System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.

File System Interface: The Concept of a File, Access Methods, Directory Structure, File System Mounting, File Sharing, Protection. Allocation Methods, Free-Space Management.

**Text Books:**

1. Computer Organization - Carl Hamacher, Zvonks Vranesic, SafeaZaky, 5th Edition, McGraw Hill.
2. Computer System Architecture - M. moris mano, 3rd edition, Pearson
3. Operating System Concepts - Abreham Silberchatz, Peter B. Galvin, Greg Gagne, 8th Edition, John Wiley.

**Reference Books:**

1. Computer Organization and Architecture - William Stallings 6th Edition, Pearson
2. Structured Computer Organization - Andrew S. Tanenbaum, 4th Edition, PHI
3. Fundamentals of Computer Organization and Design - Sivaraama Dandamudi, Springer Int. Edition
4. Operating Systems - Internals and Design Principles, Stallings, 6th Edition - 2009, Pearson Education.
5. Modern Operating Systems, Andrew S Tanenbaum 2nd Edition, PHI
6. Principles of Operating System, B. L. Stuart, Cengage Learning, India Edition

## BASICS OF PYTHON PROGRAMMING

### VI Semester: OPEN ELECTIVE - II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT14      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To Describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To Describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methods through use of these in the context of a Python application.

### COURSE OUTCOMES:

Up on successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages. Specifically, the student will use data types, input, output, iterative, conditional, and functional components of the language in his or her programs.
2. Use object-oriented programming techniques to design and implement a clear, well-structured Python program. Specifically, the student will use and design classes and objects in his or her programs.
3. Outline the specific features of Python which made it more powerful programming language.

#### UNIT-I INTRODUCTION TO PYTHON Classes: 08

Overview, Basic. Python Installation, Comments in Python, Concept of Indentation in python.

#### UNIT-II DATA TYPES Classes: 08

Tuples, Lists More advanced data types (dictionary, string), Python operators, control flows, Loops, Functions.

#### UNIT-III MODULES AND PACKAGES Classes: 08

File handling in python, Module and Packages, Object oriented programming.

#### UNIT-IV ADVANCED FUNCTIONS Classes: 08

Exceptions, sorting, advanced function: map, filter, and reduce.

#### UNIT-V INTRODUCTION TO STANDARD LIBRARIES Classes: 08

Multi-Processing And Multi-Threading, Introduction To Standard Libraries (pandas,Turtle, numpy, os).



**Text Books:**

1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013.

**Reference Books:**

1. Python Programming for Beginners: Python Programming Languageby Joseph Joyner

## HUMAN COMPUTER INTERACTION

### VI Semester: OPEN ELECTIVE - II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT15      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain Computer components functions regarding interaction with human
3. Demonstrate Understanding of Interaction between the human and computer components.
4. Use Paradigms
5. Implement Interaction design basics
6. Use HCI in the software process

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Importance of user Interface – definition, importance of good design, Benefits of good design. A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics-Principles of user interface.

#### UNIT-II

#### DESIGN PROCESS

**Classes: 15**

Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.

#### UNIT-III

#### WINDOWS

**Classes: 12**

New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.

**UNIT-IV****SOFTWARE TOOLS****Classes: 08**

Specification methods, interface – Building Tools.

**UNIT-V****INTERACTION DEVICES****Classes: 08**

Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.

**Text Books:**

1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech.
2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.

**Reference Books:**

1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEAL, PEARSON.
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

## SOFTWARE TESTING FUNDAMENTALS

### VI Semester: OPEN ELECTIVE - II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT16      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

Software Testing has different goals and objectives.

1. Finding defects which may get created by the programmer while developing the software.
2. Gaining **confidence** in and providing information about the level of quality.
3. To prevent defects.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand software testing methods and apply various testing techniques
2. Design and conduct a software test process for a software testing project
3. Designing solutions for various software testing problems by selecting appropriate software test model
4. Implement various practice oriented software testing projects

#### UNIT-I

#### INTRODUCTION

**Classes: 08**

Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.

#### UNIT-II

#### TESTING METHODOLOGIES

**Classes: 12**

White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.

#### UNIT-III

#### INTEGRATION TESTING

**Classes: 10**

Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing

#### UNIT-IV

#### TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING

**Classes: 10**

Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.

#### UNIT-V

#### TEST MANAGEMENT AND AUTOMATION TEST PLANNING

**Classes: 10**

Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for

test tool framework, Test tool selection.

**Text Books:**

1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education.
2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.

**Reference Books:**

1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley
2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.

**E - COMMERCE****VII Semester: OPEN ELECTIVE - III**

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT17      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

**COURSE OBJECTIVES:**

To learn

1. To become familiar with state of the art Electronic Model, Payment Mechanisms.
2. To understand the basic principal of E-Government, Securities, Supply Chain Mechanisms.
3. To Evaluate and observe various online businesses Management.

**COURSE OUTCOMES:**

Upon successful completion of the course, the student is able to

1. Understand how the E-Commerce is done in the IT environment when the data is huge and in a variety of formats (e.g. multi media) like on the Internet and in digital libraries.
2. To enable them to use the search and retrieval techniques taught in any of the IT applications that they may develop in future.
3. Understand the concept of electronic data interchange.
4. To learn different types of payment modes.

**UNIT-I****ELECTRONIC COMMERCE****Classes: 08**

**Electronic Commerce** - Frame work, anatomy of E-Commerce applications, E-Commerce Consumer applications, and E-Commerce organization applications, Consumer Oriented Electronic commerce - Mercantile Process models.

**UNIT-II****ELECTRONIC PAYMENT SYSTEMS****Classes: 08**

**Electronic payment systems** – Digital Token-Based, Smart Cards, Credit Cards, Risks in Electronic Payment systems. Inter-Organizational Commerce – EDI, EDI Implementation, Value added networks.

**UNIT-III****INTRA ORGANIZATIONAL COMMERCE****Classes: 08**

**Intra Organizational Commerce** – work Flow, Automation Customization and internal Commerce, Supply chain Management.

**UNIT-IV****CORPORATE DIGITAL LIBRARY****Classes: 11**

**Corporate Digital Library** - Document Library, digital Document types, corporate Data Warehouses. Advertising and Marketing - Information based marketing, Advertising on Internet, on-line marketing process, market research.

**UNIT-V****CONSUMER SEARCH AND RESOURCE DISCOVERY & MULTIMEDIA****Classes: 12**

**Consumer Search And Resource Discovery:** Information search and Retrieval, Commerce Catalogues, Information Filtering

**Multimedia** - key multimedia concepts, Digital Video and electronic Commerce, Desktop video

processing, Desktop video conferencing.

**Text Books:**

1. Frontiers of electronic commerce – Kalakata, Whinston, Pearson.
2. E-Commerce fundamentals and applications Hendry Chan, Raymond Lee, Tharam Dillon, Elizabeth Chang, John Wiley.

**Reference Books:**

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turbon, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson.
4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

## CYBER FORENSICS

### VII Semester: OPEN ELECTIVE - III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT18      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. A brief explanation of the objective is to provide digital evidences which are obtained from digital media.
2. In order to understand the objectives of computer forensics, first of all, people have to recognize the different roles computer plays in a certain crime.
3. According to a snippet from the United States Security Service, the functions computer has in different kinds of crimes.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Students will understand the usage of computers in forensic, and how to use various forensic tools for a wide variety of investigations.
2. It gives an opportunity to students to continue their zeal in research in computer forensics.

### UNIT-I COMPUTER FORENSICS FUNDAMENTALS Classes: 15

What is Computer Forensics?, Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps taken by Computer Forensics Specialists Types of Computer Forensics Technology: Types of Military Computer Forensic Technology, Types of Law Enforcement — Computer Forensic Technology — Types of Business Computer Forensic Technology Computer Forensics Evidence and Capture: Data Recovery Defined — Data Back-up and Recovery — The Role of Back-up in Data Recovery — The Data-Recovery Solution.

### UNIT-II EVIDENCE COLLECTION AND DATA SEIZURE Classes: 15

Why Collect Evidence? Collection Options — Obstacles — Types of Evidence — The Rules of Evidence — Volatile Evidence — General Procedure — Collection and Archiving — Methods of Collection — Artifacts — Collection Steps — Controlling Contamination: The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene — Computer Evidence Processing Steps — Legal Aspects of Collecting and Preserving Computer Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication — Practical Consideration — Practical Implementation.

### UNIT-III COMPUTER FORENSICS ANALYSIS AND VALIDATION Classes: 12

Determining what data to collect and analyze, validating forensic data, addressing data-hiding techniques, performing remote acquisitions Network Forensics: Network forensics overview, performing live acquisitions, developing standard procedures for network forensics, using network tools, examining the honeynet project. Processing Crime and Incident Scenes: Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash, reviewing a case.

### UNIT-IV CURRENT COMPUTER FORENSIC TOOLS Classes: 14



Evaluating computer forensic tool needs, computer forensics software tools, computer forensics hardware tools, validating and testing forensics software E-Mail Investigations: Exploring the role of e-mail in investigation, exploring the roles of the client and server in e-mail, investigating e-mail crimes and violations, understanding e-mail servers, using specialized e-mail forensic tools. Cell phone and mobile device forensics: Understanding mobile device forensics, understanding acquisition procedures for cell phones and mobile devices.

**UNIT-V****WORKING WITH WINDOWS AND DOS SYSTEMS****Classes: 08**

Understanding file systems, exploring Microsoft File Structures, Examining NTFS disks, Understanding whole disk encryption, windows registry, Microsoft startup tasks, MS-DOS startup tasks, virtual machines.

**Text Books:**

1. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi.
2. Computer Forensics and Investigations by Nelson, Phillips Enfinger, Steuart, CENGAGE Learning

**Reference Books:**

1. Real Digital Forensics by Keith J. Jones, Richard Bejtich, Curtis W. Rose, Addison Wesley Pearson Education
2. Forensic Compiling, A Tractitioneris Guide by Tony Sammes and Brian Jenkinson, Springer International edition.
3. Computer Evidence Collection & Presentation by Christopher L.T. Brown, Firewall Media.
4. Homeland Security, Techniques & Technologies by Jesus Mena, Firewall Media.
5. Software Forensics Collecting Evidence from the Scene of a Digital Crime by Robert M. Slade, TMH 2005
6. Windows Forensics by Chad Steel, Wiley India Edition

## DISCRETE STRUCTURES

### VII Semester: OPEN ELECTIVE - III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT19      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Reason mathematically about basic data types and structures (such as numbers, sets, graphs, and trees) used in computer algorithms and systems;
2. Distinguish rigorous definitions and conclusions from merely plausible ones; synthesize elementary proofs, especially proofs by induction.
3. Model and analyze computational processes using analytic and combinatorial methods.
4. Apply principles of discrete probability to calculate probabilities and expectations of simple random processes.
5. Work in small teams to accomplish all the objectives above.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Use logical notation to define and reason about fundamental mathematical concepts such as sets, relations, functions, and integers.
2. Synthesize induction hypotheses and simple induction proofs.
3. Prove elementary properties of modular arithmetic and explain their applications in Computer Science, for example, in cryptography and hashing algorithms.
4. Apply graph theory models of data structures and state machines to solve problems of connectivity and constraint satisfaction, for example, scheduling.
5. Calculate numbers of possible outcomes of elementary combinatorial processes such as permutations and combinations.

#### UNIT-I

#### MATHEMATICAL LOGIC

**Classes: 15**

Statements and Notations, Connectives, Statement Formulas and Truth Tables, well formed formulas, Tautologies, Equivalence of Formulas, Normal Forms: Disjunctive Normal Forms (DNF), Conjunctive Normal Forms (CNF), Principle Disjunctive Normal Forms (PDNF), Principle Conjunctive Normal Forms (PCNF).

#### UNIT-II

#### PREDICATES

**Classes: 15**

The Predicate calculus, Free and Bound Variables, Rules of Inference, Consistency Of Premises and Indirect Method of Proof, Automatic Theorem Proving.

#### UNIT-III

#### RELATIONS AND ORDERING

**Classes: 15**

Relations, Properties of Binary Relations in a Set, Equivalence Relations, Compatibility Relations, and Partial Ordering, Partial Ordered Set – Representation and Associated Terminology.

**FUNCTIONS:** Definition and Introduction, Composition of Functions, Inverse Functions, Recursive Functions.

#### UNIT-IV

#### TREES

**Classes: 11**

Trees, Properties of trees, Representation of trees, Binary trees, Tree traversals, spanning trees, DFS, BFS, Minimal spanning trees, Prim's algorithm and Kruskal's algorithm.

**UNIT-V****GRAPHS****Classes: 10**

Representation of Graphs, Planar Graphs, Graph theory and applications. Basic concepts of Isomorphism and sub graphs, Multi graph and Euler circuits, Hamiltonian Graphs, chromatic number.

**Text Books:**

1. J. P. Trembly, R. Manohar (1997), Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, India.
2. Joe L. Mott, Abraham Kandel, Theodore P. Baker (2011), Discrete Mathematics for Computer Scientists and Mathematicians, 2nd edition, Prentice Hall of India Learning Private Limited, New Delhi, India.

**Reference Books:**

1. Kenneth H. Rosen (2007), Discrete Mathematics and its Applications, 6th edition, Tata McGraw Hill, India.
2. C.L. Liu, D.P. Mohapatra (2008), Elements of Discrete Mathematics, 3rd edition, McGraw Hill, India.
3. Ralph P. Grimaldi, B.V. Ramana (2006), Discrete and Combinatorial Mathematics - An Applied Introduction, 5th Edition, Pearson Education, India.

## INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT20      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Produce apps for Android platform devices
2. Gain a basic understanding of computer architecture and object-oriented programming
3. Develop a working knowledge of Apple's Xcode app development tool
4. Understand mobile design principles
5. Identify need and opportunity in app markets

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe those aspects of mobile programming that make it unique from programming for other platforms,
2. Critique mobile applications on their design pros and cons,
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces,
4. Program mobile applications for the Android operating system that use basic and advanced phone features, and
5. Deploy applications to the Android marketplace for distribution.

#### UNIT-I INTRODUCTION TO ANDROID Classes: 10

The Android Platform, Android SDK, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file.

#### UNIT-II ANDROID APPLICATION DESIGN ESSENTIALS Classes: 08

Android terminologies, Application Context, Activities.

#### UNIT-III ANDROID USER INTERFACE DESIGN ESSENTIALS Classes: 08

User Interface Screen elements, Designing User Interfaces with Layouts.

#### UNIT-IV TESTING ANDROID APPLICATIONS Classes: 08

Testing Android applications, Publishing Android application, Using Android preferences, managing Application resources in a hierarchy, working with different types of resources.

#### UNIT-V USING COMMON ANDROID APIS Classes: 10

Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers.

**Text Books:**

1. Lauren Darcey and Shane Conder, —Android Wireless Application Developmentll, Pearson Education, 2<sup>nd</sup> ed. (2011)

**Reference Books:**

1. R1. Reto Meier, —Professional Android 2 Application Developmentll, Wiley India Pvt Ltd
2. R2. Mark L Murphy, —Beginning Androidll, Wiley India Pvt Ltd.
3. R3. Android Application Development All in one for Dummies by Barry Burd, Edition.

## DISTRIBUTED DATABASES

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT21      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Introducing Distributed Database Management System and its Design issues
2. Exploring several algorithms for processing queries and be able to use them
3. Describe the methods to translate complex conceptual data models into logical and Physical database designs
4. Demonstrating query optimization and its algorithms
5. Enumerating the concepts behind distributed transaction processing

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify the introductory distributed database concepts and its structures.
2. Describe terms related to distributed object database design and management.
3. Produce the transaction management and query processing techniques in DDBMS.
4. Relate the importance and application of emerging database technology.

**Pre-requisites:** Basic concept of DBMS and RDBMS.

#### UNIT-I INTRODUCTORY CONCEPTS AND DESIGN OF (DDBMS) Classes: 12

Data Fragmentation; Replication, and allocation techniques for DDBMS; Methods for designing and implementing DDBMS, designing a distributed relational database; Architectures for DDBMS: cluster federated, parallel databases and client server architecture.

#### UNIT-II QUERY PROCESSING & TRANSACTION MANAGEMENT Classes: 12

Overview Of Query Processing: Query processing problem; Objectives of Query Processing; Complexity of Relational Algebra operations; characterization of Query processors; Layers of Query Processing Introduction To Transaction Management: Definition of Transaction, Properties of Transaction, types of transaction ; Distributed Concurrency Control: Serializability theory; Taxonomy of concurrency control mechanisms; locking bases concurrency control algorithms.

#### UNIT-III DISTRIBUTED OBJECT DATABASE MANAGEMENT SYSTEMS Classes: 08

Fundamental Object concepts and Object models, Object distribution design, Architectural issues, Object management, Distributed object storage, object query processing.

#### UNIT-IV CURRENT TRENDS & DEVELOPMENTS RELATED TO DISTRIBUTED DATABASE APPLICATIONS TECHNOLOGIES Classes: 10

Distributed Object/component-based DBMS; Database Interoperability including CORBA; DCOM and Java RMI; Distributed document-based systems; XML and Workflow management.

#### UNIT-V EMERGING RELATED DATABASE TECHNOLOGIES Classes: 08

Parallel Database Mobile database; Multimedia Database; Spatial Database and Web Databases.

**Text Books:**

1. 1 Distributed Databases - Principles and Systems; Stefano Ceri; Guiseppe Pelagatti; Tata McGraw Hill; 1985.
2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia.
3. Database System Concepts; Korth & Sudarshan; TMH.
4. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall.

**Reference Books:**

1. Data Base Management System; Leon & Leon; Vikas Publications
2. Introduction to Database Systems; Bipin C Desai; Galgotia.

## BIG DATA

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4IT22      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. **Compare** various file systems and **use** an appropriate file system for storing different types of data.
2. **Demonstrate** the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. **Apply** the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. **Connect** to web data sources for data gathering, **Integrate** data sources with hadoop components to process streaming data.
5. **Tabulate** and **examine** the results generated using hadoop components.

#### UNIT-I

#### INTRODUCTION TO BIG DATA

Classes: 12

Data and its importance, Big Data- Definition, V's of Big Data, Hadoop Ecosystem

##### HADOOP ARCHITECTURE

**Hadoop Storage** : HDFS, Hadoop

**Processing** : Map Reduce Framework

**Hadoop Server Roles** : Name Node, Secondary Name Node and Data Node, Job Tracker, Task Tracker

##### HDFS-HADOOP DISTRIBUTED FILE SYSTEM

Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE- File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

#### UNIT-II

#### MAPREDUCE PROGRAMMING MODEL

Classes: 12

Introduction to Map Reduce Programming model to process BigData, key features of MapReduce, MapReduce Job skeleton, Introduction to MapReduce API, Hadoop Data Types, Develop MapReduce Job using Eclipse, build a MapReduce Job export it as a java archive(.jar file).

**MAPREDUCE JOB LIFE CYCLE:** How Mapreduce Works? Understanding Mapper ,Combiner, Partitioner ,Shuffle & Sort and Reduce phases of MapReduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

**MAPREDUCE API:** Understanding new MapReduce API from org.apache.hadoop.mapreduce and its sub packages to develop MapReduce applications ,key difference between old MapReduce API and the new MapReduce API.



**UNIT-III****INTRODUCTION TO PIG****Classes: 12**

Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage.

**PIG LATIN LANGUAGE–DATA TYPES IN PIG**

Pig Latin Basics, Key words ,Pig Data types ,Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell ,expressions, Data processing operators, using Built in functions.

**WRITING PIG SCRIPTS USING PIG LATIN:** Writing pig scripts and saving them in text editor, running pig scripts from command line.

**UNIT-IV****INTRODUCTION TO HIVE****Classes: 12**

Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write.

**HIVE QL DATA TYPES, SEMANTICS:** Introduction to Hive QL (Query Language), Language semantics, Hive Data Types.

**HIVE DDL, DML AND HIVE SCRIPTS:** Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and commands line.

**UNIT-V****SQOOP, FLUME, OOZIE****Classes: 12**

**SQOOP:** Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS.

**FLUME:** Introduction to Flume agent, understanding Flume components Source, Channel and Sink, Writing flume configuration file, running flume configuration file to ingest the data into HDFS.

**OOZIE:** Introduction to Oozie, Understanding work flow and how to create Work flow using Work Flow definition language in XML, running a basic Oozie workflow to run a Map Reduce job.

**Text Books:**

1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media
2. Big Data and Hadoop- Learn by Example, Mayank Bhushan

**Reference Books:**

1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.
2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.

## OPEN ELECTIVES OFFERED BY MECH DEPARTMENT

| OE1    |                                                 | OE2    |                                   |
|--------|-------------------------------------------------|--------|-----------------------------------|
| A4ME60 | Elements of Mechanical Engineering              | A4ME63 | Fundamentals of Mechatronics      |
| A4ME61 | Fundamentals of Engineering Materials           | A4ME64 | Basics of Thermodynamics          |
| A4ME62 | Fundamentals of Theory of Machines              | A4ME65 | Fundamentals of Smart Materials   |
| OE3    |                                                 | OE4    |                                   |
| A4ME66 | Fundamentals of Operation Research              | A4ME69 | Introduction to Material Handling |
| A4ME67 | Fundamentals of Energy Sources and Applications | A4ME70 | Non Conventional Energy Sources   |
| A4ME68 | Basics of Robotics                              | A4ME71 | Aspects of Heat Transfer          |

## ELEMENTS OF MECHANICAL ENGINEERING

### V Semester: OPEN ELECTIVE-I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME60      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course aims to enable the students learn basic concepts of mechanics, materials, thermodynamics, manufacturing and CAD/CAM.

### COURSE OUTCOMES:

After completing this course the student must demonstrate the knowledge and ability:

1. To demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. To explain fundamentals of metallurgy and material science.
3. To discuss basic concepts and principles of thermo dynamics.
4. To demonstrate the elements of fabrication processes and equipment.
5. To explain basic concepts of CAD/CAM.

### UNIT-I

### MECHANICS

**Classes: 14**

MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law-derivation and problems, polygon law, resolution of forces in coplanar-concurrent force system-problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, definition of centroid and centre of gravity, pappus and guldinus theorems, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia.

MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strain, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.

### UNIT-II

### METALLURGY & MATERIAL SCIENCE

**Classes: 12**

Bonds in Solids, crystallization of metals, Classification of steels, , Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide diagram, Heat treatment- Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.

**UNIT-III****THERMO DYNAMICS****Classes: 12**

System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry, First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.

**UNIT-IV****MANUFACTURING****Classes: 11**

Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies; Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.

**UNIT-V****CAD/CAM****Classes: 11**

Fundamentals of CAD/CAM- Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, dis-advantages, wire frame entities-analytic entities and synthetic entities; Surface modeling: Definition, advantages, dis-advantages, surface entities-analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.

**Text Books:**

1. Singer's Engineering Mechanics by K.Vijay Kumar Reddy and J.Suresh Kumar, B.S Publications
2. Strength of Materials By Dr.Sadhu Singh, Khanna Publishers.
3. Introduction to physical metallurgy by Avner, TATA McGrawHill Edition.
4. Thermodynamics by P.K Nag, McGrawHill Education.
5. Production Technology by R.K Jain, Khanna Publishers.
6. CAD/CAM Theory & Practice by Ibrahim Zeid & R.Sivasubramanian, McGrawHill Education.

**Reference Books:**

1. Engineering Mechanics By R.K.Bansal, Laxmi Publications
2. Mechanics of Solids By GVV Satyanarayana & Pakirappa, Durga Publishing House
3. Material science and metallurgy by OP Khanna, Dhanpat Ray Publications
4. Engineering Thermodynamics By Pakirappa, V.Naveen Kumar, V.Naresh, Durga Publishing House
5. Production Technology by Dr. P.C Sharma, S.CHAND
6. CAD/CAM By P.Radha Krishnan, S.Subramanyan and V.Raju, New Age International Publishers.

**Web References:**

1. <https://www.iitg.ac.in/rkbc/me101/Presentation/L01-03.pdf>
2. <https://nptel.ac.in/courses/113106032/>
3. <https://nptel.ac.in/downloads/112108148/>
4. <https://lecturenotes.in/subject/195/cad-and-cam-cac>

**E-Text Books:**

1. [https://books.google.co.in/books?id=8kA8BAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=8kA8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=XYorDAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=XYorDAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course:**

1. <https://ocw.mit.edu/courses/mechanical-engineering/>

## FUNDAMENTALS OF ENGINEERING MATERIALS

### V Semester: OPEN ELECTIVE-I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME61      | PCC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

### COURSE OUTCOMES:

At the end of course students are able to

- 1.Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
- 2.Demonstrate the concept of alloying and formation of different types of phases in alloys.
- 3.Differentiate ferrous and non ferrous alloys.
- 4.Explain various heat treatment processes.
- 5.Classify and explain polymers, ceramics and composites.

#### UNIT-I

#### CRYSTAL STRUCTURE

**Classes: 10**

Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.

#### UNIT-II

#### ALLOYS & PHASE DIAGRAMS

**Classes: 10**

Alloys- substitutional and interstitial solid solutions.

Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.

#### UNIT-III

#### FERROUS AND NON FERROUS ALLOYS

**Classes: 12**

Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons; copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.

#### UNIT-IV

#### HEAT TREATMENT OF STEEL

**Classes: 12**

Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.

#### UNIT-V

#### POYMERS, CERAMICS AND COMPOSITES

**Classes: 8**

Classification, properties and applications of polymers, ceramics, composites and nano materials.

**Text Books:**

1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999.
2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011.
3. Sidney H. Avener (2007,) *Introduction to Physical Metallurgy*, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.

**Reference Books:**

1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India.
2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002.
3. V. D. Kodgire (2006), Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India.

**Web References:**

1. <https://nptel.ac.in/courses/113106032/>
2. <http://web.pdx.edu/~pmoeck/phy381/lecture1.pdf>

**E-Text Books:**

1. [https://books.google.co.in/books?id=XXE8BAAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=XXE8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=kBM8BAAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=kBM8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course:**

1. <https://ocw.mit.edu/courses/materials-science-and-engineering/>
2. <https://www.mooc-list.com/tags/materials-science>

## FUNDAMENTALS OF THEORY OF MACHINES

### V Semester: OPEN ELECTIVE-I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME62      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

This course aims to enable students to learn basic concepts of kinematics and dynamics of machinery.

### COURSE OUTCOMES:

At the end of the course the students are able to:

1. Demonstrate various kinematic links, pairs, chains, mechanisms and their inversions.
2. Illustrate and analyze exact and approximate straight line motion mechanisms.
3. Explain concepts of higher pairs.
4. Demonstrate and analyze the effect of gyroscopic couples on stability of motor cycle, motor car and aeroplane.
5. Illustrate and analyze various governors.

#### UNIT-I

#### MECHANISMS

**Classes: 14**

Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains.

#### UNIT-II

#### STRAIGHT LINE MOTION MECHANISMS

**Classes: 12**

Exact, approximately copied and generated types – Peaucellier - Hart and Scott Russel – Grasshopper – Watt – Tchebichef's and Robert Mechanisms - Pantograph.

#### UNIT-III

#### GEARS & GEAR TRAINS

**Classes: 12**

Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion, Form of teeth-cycloid and involute profiles, phenomena of interference.

Gear trains: Introduction – Types – Simple, compound and reverted gear trains-simple problems.

#### UNIT-IV

#### GYROSCOPES

**Classes: 11**

Gyroscopes – effect of precession – motion on the stability of moving vehicles such as motorcycle motorcar and aeroplane.

#### UNIT-V

#### GOVERNORS

**Classes: 11**

Governors: Types of governors - Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. (No derivations)



**Text Books:**

1. Theory of Machines by S.S.Rattan.
2. Theory of Machines by R.S.Khurmi

**Reference Books:**

1. Theory of Machines by Shigley, Mc Graw Hill Publishers
2. Theory of Machines by Thomas Bevan, CBS Publishers
3. Theory of Machines by R.K.Bansal (Lakshmi publications)
4. Mechanism and Machine Theory by JS Rao and RV Duggipati, New Age International

**Web References:**

1. <http://nptel.ac.in/courses/112104121/>
2. <https://freevideolectures.com/course/2359/kinematics-of-machines>
3. <https://nptel.ac.in/courses/112104114/>
4. <https://freevideolectures.com/course/2364/dynamics-of-machines>

**E-Text Books:**

1. [https://books.google.co.in/books?id=T\\_WlquM4EI4C&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=T_WlquM4EI4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course:**

1. <https://cosmolearning.org/courses/kinematics-of-machines>
2. <https://nptel.ac.in/courses/112101096/>
3. <https://www.mooc-list.com/course/mechanics-kinematics-and-dynamics-edx>

## FUNDAMENTALS OF MECHATRONICS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME63      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

- 1) To understand the basic principles of mechatronics, Differentiating between conventional and mechatronics design process and product.
- 2) Differentiate various sensors, actuators, electromechanical systems.
- 3) To understand and apply interfacing of electromechanical products to microcontrollers and microprocessor.
- 4) To understand the working of various programmable logic controllers (PLCs), Architecture of PLCs.
- 5) Evaluate different mechatronic systems.

#### COURSE OUTCOMES:

At the end of the course students are able to

- 1) Demonstrate various elements underlying mechatronic systems, electronics, control systems and able to differentiate their purpose in the system.
- 2) Analyze and select sensors, actuators, electromechanical components needed for an application.
- 3) To evaluate microprocessor and microcontroller interfacing to mechanical application.
- 4) To choose PLC to a mechanical application.
- 5) To design mechatronic systems for mechanical applications.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Introduction to Mechatronics —Mechatronics systems - Mechatronics design process - Mechatronics in Manufacturing —Adaptive and distributed control systems – Modelling and simulation of mechatronics systems.

#### UNIT-II

#### SENSORS AND ACTUATORS

**Classes: 12**

**Sensors and actuators:** Overview of sensors and transducers – Microsensors - Signal conditioning – Operational amplifiers – Protection – Filtering - Analog and Digital converters. Electro – pneumatics and Electro – hydraulics - Solenoids – Direct Current motors – Servomotors – Stepper motors - Micro actuators; Drives selection and application.

**UNIT-III****INTERFACING****Classes: 12**

**Interfacing:** Microprocessor based Controllers Architecture of microprocessor and microcontroller – System interfacing for a sensor, keyboard, display and motors - Application cases for temperature control, warning and process control systems

**UNIT-IV****PLCs****Classes: 12**

**PLCs:** Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/Output modules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting.

**UNIT-V****ARTIFICIAL INTELLIGENCE****Classes: 12**

**AI:** Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer – based instrumentation - Real-time Data Acquisition and Control – Software integration – Man Machine interface -Vision system – Mechatronics system case studies.

**Text Books:**

1. Introduction to Mechatronics and Measurement Systems, Tata McGraw Hill

**Reference Books:**

1. Designing Intelligent Machines, Michel B. Hsiao and David G. Alciatore, Open University London
2. Control Sensors and Actuators, ICW. Desha, Prentice Hall

**Web References:**

1. <https://lecturenotes.in/subject/137/mechatronics-mech>
2. <https://sites.google.com/site/profnarendralakal/Home/presentations/mtrx>

**E-Text Books:**

1. [https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course:**

1. <https://ocw.mit.edu/courses/mechanical-engineering/2-737-mechatronics-fall-2014/>
2. <https://nptel.ac.in/courses/112103174/>
3. <https://www.igi-global.com/chapter/massive-online-open-course-assisted-mechatronics-learning/137327>

## BASICS OF THERMODYNAMICS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME64      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To understand the treatment of classical Thermodynamics and to apply the First and Second laws of Thermodynamics for the analysis of thermal equipment

### COURSE OUTCOMES:

At the end of the course, the student should be able to

1. Demonstrate basic concepts of thermodynamics.
2. Explain laws of thermodynamics.
3. Explain pure substances and power cycles.
4. Explain perfect gas laws and concepts of mixture of perfect gases.
5. Discuss fundamental concepts of psychrometry.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Introduction: System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Exact & Inexact Differentials, Cycle – Reversibility – Quasi – static Process, Irreversible Process, Causes of Irreversibility – Energy in State and in Transition, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry

#### UNIT-II

#### LAWS OF THERMO DYNAMICS

**Classes: 12**

First law of Thermodynamics – Corollaries – First law applied to a Closed System – applied to a flow system – Steady Flow Energy Equation. Limitations of the First Law  
Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements and their Equivalence / Corollaries, Entropy, Principle of Entropy Increase – Energy Equation, Availability and Irreversibility – Elementary Treatment of the Third Law of Thermodynamics

#### UNIT-III

#### PURE SUBSTANCES & POWER CYCLES

**Classes: 12**

Pure Substances, p-V-T- surfaces, T-S and h-s diagrams, Mollier Charts, Phase Transformations – Triple point at critical state properties during change of phase, Dryness Fraction – Clausius – Clapeyron Equation Property tables.

Power Cycles: Otto, Diesel, Dual Combustion cycles, Sterling Cycle – Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis – comparison of Cycles.

#### UNIT-IV

#### PERFECT GAS LAWS & MIXTURE OF PERFECT GASES

**Classes: 12**

Perfect Gas Laws – Equation of State, specific and Universal Gas constants – various Non-flow processes, properties, end states

Mixtures of perfect Gases – Mole Fraction, Mass fraction Gravimetric and volumetric Analysis – Dalton's Law of partial pressure, Avogadro's Laws of additive volumes – Mole fraction, Volume fraction and partial pressure.

#### UNIT-V

#### PSYCHROMETRY

**Classes: 12**

Psychrometric Properties – Dry bulb Temperature, Wet Bulb Temperature, Dew point Temperature, Thermodynamic Wet Bulb Temperature, Specific Humidity, Relative Humidity, saturated Air, Vapour pressure, Degree of saturation – Adiabatic Saturation , Carrier's Equation – Psychrometric chart.

#### **Text Books:**

1. Engineering Thermodynamics, P.K. Nag, 6<sup>th</sup> Edition, Mc Graw Hill Education.
2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8<sup>th</sup> Edition, Mc Graw Hill Companies.
3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7<sup>th</sup> Edition, Wiley.

#### **Reference Books:**

1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2<sup>nd</sup> Edition, Prentice hall of India Pvt Ltd., 2006.
2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi.
3. Applied Thermodynamics, Onkar Singh, 3<sup>rd</sup> Edition, New Age, India

#### **Web References:**

1. <https://www.askiitians.com/revision-notes/physics/thermodynamics/>
2. <http://www.ncert.nic.in/ncerts//kech106.pdf>

#### **E-Text Books:**

1. [https://books.google.co.in/books?id=91MZ2ZOb3n8C&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=91MZ2ZOb3n8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

#### **MOOC Course**

1. <https://www.coursera.org/learn/thermodynamics-intro>
2. <https://www.edx.org/course/thermodynamics-2>

## FUNDAMENTALS OF SMART MATERIALS

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME65      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The objective of this course is to enable the students to learn principles of smart materials such as electronic materials, ionic polymer matrix composites etc.

### COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various smart materials and their principles.
2. Explain different types of electro-active materials.
3. Explain principles of piezoelectricity.
4. Discuss principles of Magnetostriction.
5. Explain about various shape memory alloys.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Overview of Smart Materials: Introduction and definition of Smart Materials, principles of smart materials, input – output decision ability, devices based on conductivity changes, devices based on changes in optical response, biological systems smart materials. Devices based on magnetization, artificial structures, surfaces, hetero structures, polycrystalline, amorphous, liquid crystalline materials

#### UNIT-II

#### ELECTRO ACTIVE MATERIALS & PIEZO ELECTRICITY

**Classes: 12**

Electro-active Materials: Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composites (IPMC)

#### UNIT-III

#### PIEZOELECTRICITY

**Classes: 12**

Piezoelectricity: Principles of Piezoelectricity, Perovskite Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers

#### UNIT-IV

#### MAGNETOSTRICTION

**Classes: 12**

Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magnetoresistance Effect, magnetorheological fluids, electrorheological materials and their applications

**UNIT-V****SHAPE MEMORY ALLOYS****CLASSES: 12**

Shape Memory Alloys: Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Self- healing materials.

**Text Books:**

1. B. Culshaw, Smart Structures and Materials, Artech House, 2000
2. P. Gauenzi, Smart Structures: Physical Behaviour, Mathematical Modelling and Applications, Wiley Publishers, 2009
3. W. G. Cady, Piezoelectricity, Dover Publication, 1964
4. M. Addington, Schodek, L. Daniel.: Smart materials and new technologies, Architectural Press, 2005
5. W.D Callister, Materials Science and Engineering an Introduction, Willey 1999

**Reference Books:**

1. M. Schwartz, Encyclopedia of Smart Materials, Volumes 1-2 ,Willey, 2002
2. Smith, C.: Smart material systems, Ralph, SIAM, 2005
3. K Vijay, K.Varadan, J. Vinoy, S.Gopalakrisham, Smart Material Systems and MEMS: Design and Development Methodologies , Willey 2006

**Web References:**

1. <https://www.sciencedirect.com/topics/chemistry/smart-material>
2. <https://www.mide.com/collections/smart-materials>

**E-Text Books:**

1. [https://books.google.co.in/books?id=KdBr38J-bxoC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=KdBr38J-bxoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=CQxwDQAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=CQxwDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course**

1. [https://nptel.ac.in/noc/individual\\_course.php?id=noc18-me60](https://nptel.ac.in/noc/individual_course.php?id=noc18-me60)
2. <https://nptel.ac.in/courses/112104251/>

## FUNDAMENTALS OF OPERATIONS RESEARCH

### VII Semester: OPEN ELECTIVE-III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME66      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The objectives of this course are to learn quantitative methods and techniques for effective decisions-making; model formulation and applications that are used in solving business decision problems.

#### COURSE OUTCOMES:

At the end of course students will be able to

1. Describe types of models and solve linear programming problem.
2. Solve transportation and assignment problems.
3. Analyze sequencing and replacement models and apply them for optimization.
4. Apply gaming theory for optimal decision making.
5. Analyze inventory models to optimize the cost

#### UNIT-I INTRODUCTION & ALLOCATION Classes: 12

Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications.

ALLOCATION: Linear Programming Problem - Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two-phase method, Big-M method; Duality Principle.

#### UNIT-II TRANSPORTATION & ASSIGNMENT PROBLEMS Classes: 12

TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy.

ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.

#### UNIT-III SEQUENCING & REPLACEMENT Classes: 12

SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines

REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.

#### UNIT-IV THEORY OF GAMES Classes: 12

THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games –m x 2 & 2 x n games - graphical method – m x n games - dominance principle.

#### UNIT-V INVENTORY Classes: 12



INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.

**Text Books:**

1. Operation Research by J.K.Sharma, MacMilan.
2. Operations Research by ACS Kumar, Yesdee

**Reference Books:**

1. Operations Research: Methods and Problems by Maurice Saseini, Arhur Yaspan and Lawrence Friedman
2. Operations Research by A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Pearson Education.
3. Operations Research by Wagner, PHI Publications.
4. Introduction to O.R by Hillier & Libermann, TMH.

**Web References:**

1. <http://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf>
2. [http://www.mathcity.org/msc/notes/operation\\_research](http://www.mathcity.org/msc/notes/operation_research)
3. [http://biobharati.com/admin/img/gall/1467763196\\_BCA-504.pdf](http://biobharati.com/admin/img/gall/1467763196_BCA-504.pdf)

**E-Text Books:**

1. [https://books.google.co.in/books?id=6khDDAAQBAJ&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=6khDDAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
3. [https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

**MOOC Course:**

1. <http://www.orcomplete.com/internet/enesbilgin/open-courses-on-operations-research>
2. <https://www.coursera.org/learn/wharton-operations>

## FUNDAMENTALS OF ENERGY SOURCES AND APPLICATIONS

### VII Semester: OPEN ELECTIVE-III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME67      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The objectives of this course are to study classifications and comparison of various renewable energy sources, flat plate and concentrating collectors, wind energy, principles of bio mass, geothermal and ocean energy sources and principles of direct energy conversion.

### COURSE OUTCOMES:

At the end of the course, the student shall be able to

1. Classify renewable energy sources and explain environmental impact of solar energy, solar radiation, their measurement and instruments.
2. Classify concentrating collectors and explain in detail about various solar gadgets.
3. Explain about wind mill, types of wind mills, its construction, wind energy conversion systems and principles of bio mass.
4. Explain in detail geothermal and ocean energies.
5. Demonstrate the concepts of direct energy conversion.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Introduction – Renewable energy sources – Limitations – Classifications – Comparison – Global and Indian energy scenario. Solar Energy: Introduction – Environmental impact of solar power – extraterrestrial and terrestrial solar radiation – solar radiation on tilted surface – Solar Radiation Measurement and Instruments

#### UNIT-II

#### SOLAR ENERGY

**Classes: 12**

Flat plate and concentrating collectors – classification of concentrating collectors – orientation and thermal analysis – advanced collectors. Solar Gadgets: Solar furnace, Solar air heater, Solar water heater, Solar cooker, solar still, solar pond, solar pump, solar dryer, solar cell.

#### UNIT-III

#### WIND ENERGY & BIO MASS

**Classes: 12**

Wind energy utilization: Introduction – Origin and nature of winds – Types of wind mills and its construction – Wind energy conversion systems – performance characteristics.

Bio-Mass: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, combustion characteristics of bio-gas, applications and economic aspects.

#### UNIT-IV

#### GEO THERMAL ENERGY & OCEAN ENERGY

**Classes: 12**

Geothermal Energy: Resources, types of wells, methods of harnessing the energy, potential in India.

Ocean Energy: OTEC, Principles utilization, setting of OTEC plants. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.

#### UNIT-V

#### DIRECT ENERGY CONVERSION

**Classes: 12**

Direct Energy Conversion: Need, limitations, principles of DEC. Thermo-electric generators, seebeck, peltier and Joule-Thomson effects – applications, MHD generator principle, Fuel cells: Classification of fuel cells; Operating principles. Bio-fuels, Urban waste to energy conversion

**Text Books:**

1. Rai G.D, Non-Conventional Energy Sources, Khanna Publishers, New Delhi, 2011.
2. Twidell, J.W. & Weir, A., Renewable Energy Sources, EFN Spon Ltd., UK, 2006

**Reference Books:**

1. Ashok V Desai, *Non-Conventional Energy*, Wiley Eastern Ltd, New Delhi, 2003.
2. Mittal K M, *Non-Conventional Energy Systems*, Wheeler Publishing Co. Ltd, New Delhi, 2003.
3. Khan B H, *Non-Conventional Energy Resources*, Tata McGraw Hill, 2<sup>nd</sup> Edn. New Delhi, 2009.
4. Ramesh R & Kumar K U, *Renewable Energy Technologies*, Narosa Publishing House, New Delhi, 2004
5. Tiwari. G.N., *Solar Energy – Fundamentals Design, Modelling & Applications*, Narosa Publishing House, New Delhi, 2002.

**Web References:**

1. [http://www.afdc.energy.gov/fuels/natural\\_gas.html](http://www.afdc.energy.gov/fuels/natural_gas.html)
2. <https://cleancities.energy.gov/>
3. <http://www.firmgreen.com/>
4. <https://www.renewableresourcescoalition.org/alternative-energy-sources/>
5. <https://www.edx.org/learn/renewable-energy>

**E-Text Books:**

1. <https://www.sciencedirect.com/book/9780126561531/renewable-energy>
2. <https://www.pdfdrive.com/renewable-energy-resources-e14705840.html>
3. <https://www.pdfdrive.com/search?q=renewable&pagecount=&pubyear=&searchin=>

**MOOC Course:**

1. <https://www.coursera.org/learn/renewable-energy-entrepreneurship>
2. <https://www.coursera.org/learn/wind-for-renewable-energies>
3. <https://online-learning.tudelft.nl/courses/sustainable-energy-design-a-renewable-future/>

## BASICS OF ROBOTICS

### VII Semester: OPEN ELECTIVE-III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME68      | OEC      | 3            |   | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The objectives of this course are to

1. Study about basic components, architecture, anatomy and specifications of robot.
2. Study about various end effectors and actuators.
3. Study homogeneous transformation, D-H notation and direct kinematics
4. Study Inverse kinematics and Trajectory planning techniques
5. Study feedback components and applications of robots.

### COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots.
2. Explain various types of end effectors
3. Evaluate rotation matrices, forward kinematics of RR, RP and 3R Manipulators.
4. Explain inverse kinematics of RR manipulator, RP manipulator and trajectory planning techniques.
5. Explain feedback components used in robots and industrial applications.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Introduction: Automation and Robotics, Asimov's laws, Robot Architecture, Components, , Anatomy of robot, Factors to be considered in the selection of robot, present and future applications, Specifications- Degree of freedom, Pay load, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems

#### UNIT-II

#### END EFFECTORS & ACTUATORS

**Classes: 12**

End effectors: Mechanical and Non-mechanical grippers, requirements for the design of grippers, considerations for the selection of grippers, Types of actuation mechanisms.

Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators

#### UNIT-III

#### MOTION ANALYSIS & DIRECT KINEMATICS

**Classes: 12**

Motion Analysis: Basic Rotation Matrices, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems.

Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP and 3R manipulators

#### UNIT-IV

#### INVERSE KINEMATICS & TRAJECTORY PLANNING

**Classes: 12**

Inverse kinematics : Inverse kinematics of 2R and RP manipulators.

Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic polynomial trajectory

#### UNIT-V

#### FEEDBACK COMPONENTS & APPLICATIONS

**Classes: 12**

Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor.

Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.

#### Text Books:

1. Industrial Robotics by Groover M P, Pearson Edu.
2. Robotics by Fu K S, McGraw Hill.
3. Theory of Applied Robotics (kinematics, Dynamics and Control-Jazar, Springer.

#### Reference Books:

1. Robotics and Control by Mittal R K & Nagrath I J, TMH.
2. Robot Dynamics and Controls by Spony and Vidyasagar, John Wiley
3. Robot Analysis and control by Asada and Slotine, Wiley Inter-Science
4. Introduction to Robotics by John J Craig, Pearson Education

#### Web References:

1. [https://www.sciencedaily.com/terms/industrial\\_robot.htm](https://www.sciencedaily.com/terms/industrial_robot.htm)
2. <https://www.robotics.org/robotics/industrial-robot-industry-and-all-it-entails>

#### E-Text Books:

1. [https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

#### MOOC Course:

1. <https://www.classcentral.com/tag/robotics>
2. <https://www.mooc-list.com/tags/robotics>

## INTRODUCTION TO MATERIAL HANDLING

### VIII Semester: OPEN ELECTIVE-IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME69      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

The aim of this course is to enable the students to learn basic concepts of material handling, consideration in material handling system, materials transport and storage system, various inventory models and forecasting techniques.

#### COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts and principles of material handling.
2. Discuss about various material transport systems.
3. Explain conventional and automated storage systems.
4. Explain various inventory models.
5. Demonstrate different forecasting techniques.

#### UNIT-I

#### INTRODUCTION

**Classes: 12**

Introduction: Overview of material handling equipment, considerations in material handling system design, The 10 principles of material handling.

#### UNIT-II

#### MATERIAL TRANSPORT SYSTEMS

**Classes: 12**

Material Transport Systems: Industrial trucks, Automated guided vehicle systems, Monorails and other rail guided vehicles, conveyor systems, cranes and hoists, Analysis of material transport systems.

#### UNIT-III

#### STORAGE SYSTEMS

**Classes: 12**

Storage Systems: Storage system performance, storage location strategies, conventional storage methods and equipment, automated storage systems and analysis of automated storage systems.

#### UNIT-IV

#### INVENTORY

**Classes: 12**

Inventory: Introduction, Inventory management, Static Inventory models, ABC Analysis, VED analysis, Just in Time, Aggregate inventory analysis, Inventory models with quantity discounts.

#### UNIT-V

#### FORECASTING

**Classes: 12**

Forecasting: Forecasting techniques, Make or Buy decision, Acceptance sampling, Materials requirement planning, objectives of master schedule.

#### Text Books:

1. Mikell P. Groover, Automation Production Systems and Computer Integrated Manufacturing, Pearson International Edition.
2. A.K Singh, Materials Management, Laxmi Publications (P) Ltd.

#### Reference Books:

1. Dr.K.C Arora, Vikas V.Shinde, Aspects of Material Handling, Laxmi Publications (P) Ltd.
2. Charles Reese, Material Handling Systems, Taylor & Francis Network

### Web References:

1. <https://www.scribd.com/doc/222647028/Material-Handling-Full-Notes>
2. [http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20\(MH\).pdf](http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20(MH).pdf)

### E-Text Books:

1. [https://books.google.co.in/books?id=AN8dzzzlp1wC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=AN8dzzzlp1wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)
2. [https://books.google.co.in/books?id=d0mhPIDEm0gC&printsec=frontcover&source=gbs\\_ge\\_summary\\_r&cad=0#v=onepage&q&f=false](https://books.google.co.in/books?id=d0mhPIDEm0gC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false)

### MOOC Course

1. <https://www.mooc-list.com/course/materials-and-materials-processing-saylororg>
2. <https://www.mooc-list.com/tags/material-processing>

## NON CONVENTIONAL ENERGY SOURCES

### VIII Semester: OPEN ELECTIVE-IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME70      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To impart the knowledge of basics of different non conventional types of power generation & power plants in detail so that it helps them in understanding the need and role of Non-Conventional Energy sources particularly when the conventional sources are scarce in nature.

### COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various energy sources and their availability.
2. Explain concepts and applications of solar radiation
3. Discuss elaboratively the principles of wind energy conversion
4. Explain Biomass conversion techniques and Geothermal Sources and resources
5. Explain about tidal energy and Ocean Thermal Energy.

### UNIT- I ENERGY SOURCES & THEIR AVAILABILITY

Classes: 12

**Energy Sources & their Availability** - Importance of Non Conventional Energy Resources - Classification of NCES - Solar, Wind, Geothermal, Bio-mass, Nuclear Energy & Fuel Cells, Ocean Energy Sources, Comparison of these Energy Sources, Prospects of Renewable Energy Sources - Criteria for Assessing the Potential of NCES, Statistics on Conventional Energy Sources and supply in Developing Countries

### UNIT- II INTRODUCTION TO SOLAR RADIATION & APPLICATIONS OF SOLAR ENERGY

Classes: 15

**Introduction to Solar radiation** - beam and diffuse radiation, solar constant, earth sun angles, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, collector efficiency, Focusing and non-focusing type

**Applications of Solar Energy** - solar water heater- Solar Cooker-Box type- Solar dryer - solar greenhouse— Summer and winter greenhouse-solar electric power generation- Solar Photo- Voltaic, Solar Cell Principle, Conversion efficiency and power output, Basic Photo Voltaic System for Power Generation.

### UNIT- III PRINCIPLE OF WIND ENERGY CONVERSION

Classes: 12

**Principle of Wind Energy Conversion** - Basic components of Wind Energy Conversion Systems : Wind data and energy estimation, Site Selection Considerations - Wind mill Components, Various Types and their Constructional Features - Effect of Density, Frequency Variances, Angle of attack, and Wind Speed - Design Considerations of Horizontal and Vertical Axis Wind Machines - Analysis of Aerodynamic Forces Acting on Wind Mill Blades and Estimation of Power Output.

### UNIT- IV BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES

Classes: 15

**Biomass conversion techniques** - Biogas Generation - Factors affecting biogas Generation-Types of biogas plants - Advantages and disadvantages of biogas plants - site selection, digester design consideration, filling a digester for starting, maintaining biogas production, Fuel properties of bio gas, utilization of biogas - urban



waste to energy conversion.

**Geothermal Sources and resources** like hydrothermal, geo-pressured hot dry rock, magma. advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India - exhaust types of conventional steam turbines.

## **UNIT- V                      TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION                      Classes: 12**

**Tidal Energy** - Principle of working, performance and limitations. **Wave Energy** - Principle of working, performance and limitations.

**Ocean Thermal Energy Conversion** - Availability, theory and working principle, performance and limitations. OTEC power plants, Operational of small cycle experimental facility, Economics of OTEC, Environmental impacts of OTEC.

### **Text Books:**

1. G. D. Rai, "Non-Conventional Energy Sources", 4th Edition, Khanna Publishers, 2000
2. B H Khan, " Non-Conventional Energy Resources", 2nd Edition, Tata Mc Graw Hill Education Pvt Ltd, 2011

### **Reference Books:**

1. S.Hasan Saeed and D.K.Sharma , "Non-Conventional Energy Resources", 3rd Edition, S.K.Kataria & Sons, 2012
2. Renewable energy sources and conversion technology by N.K. Bansal, M. Kleemann, M. Heliss, Tata McGraw Hill 1990.
3. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003
4. K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003
5. S. P. Sukhatme", "Solar Energy Principles and Application", TMH, 2009

### **Web References:**

1. <https://lecturenotes.in/subject/57/non-conventional-energy-systems-nces>
2. <https://www.pdfdrive.com/non-conventional-energy-sources-e10086374.html>
3. [http://www.vssut.ac.in/lecture\\_notes/lecture1428910296.pdf](http://www.vssut.ac.in/lecture_notes/lecture1428910296.pdf)

### **E-Text Books:**

1. <http://nptel.ac.in/courses/121106014>
2. <http://nptel.ac.in/courses/121106014/13>
3. <http://nptel.ac.in/courses/12110601427>

### **MOOC Course:**

1. <https://nptel.ac.in/courses/121106014/>
2. <https://nptel.ac.in/syllabus/121106014/>
3. <https://nptel.ac.in/downloads/121106014/>

## ASPECTS OF HEAT TRANSFER

### VIII Semester: OPEN ELECTIVE-IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIE           | SEE | Total |
| A4ME71      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To understand the mechanisms of heat transfer under steady and transient conditions.
2. To understand the concepts of heat transfer through free and forced convection.
3. To learn the thermal analysis and sizing of heat exchangers.
4. To understand the principles of radiation heat transfer.

#### COURSE OUTCOMES:

1. Demonstrate basic modes of heat transfer and compute temperature distribution in steady state and unsteady state heat conduction.
2. Analyze the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
3. Interpret and analyze the free and forced convection heat transfer.
4. Comprehend the concepts of free convection and heat exchangers.
5. Explain the principles of heat transfer with phase change.

#### UNIT-I

#### INTRODUCTION

**Classes: 09**

Introduction: Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier rate equation, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.

#### UNIT-II

#### ONE DIMENSIONAL STEADY STATE HEAT CONDUCTION

**Classes: 15**

One dimensional steady state heat conduction: Homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy, Critical radius of insulation. Variable thermal conductivity and Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips.

#### UNIT-III

#### CONVECTIVE HEAT TRANSFER

**Classes: 12**

Convective Heat Transfer: Classification of systems based on causation of flow, condition of flow, configuration of flow and medium of flow – Dimensional analysis as a tool for experimental investigation – Buckingham Theorem and method, application for developing semi – empirical non-dimensional correlation for convection heat transfer – Significance of non-dimensional numbers – Concepts of Continuity, Momentum and Energy Equations.

#### UNIT-IV

#### FREE CONVECTION & HEAT EXCHANGERS

**Classes: 12**

Free Convection: Development of Hydrodynamic and thermal boundary layer along a vertical plate - Use of empirical relations for Vertical plates and pipes.

Heat Exchangers: Classification of heat exchangers – overall heat transfer Coefficient and fouling factor .

**UNIT-V****RADIATION HEAT TRANSFER****Classes: 12**

Radiation Heat Transfer : Emission characteristics and laws of black-body radiation – Irradiation – total and monochromatic quantities – laws of Planck, Wien, Kirchoff, Lambert, Stefan and Boltzmann– heat exchange between two black bodies – concepts of shape factor – Emissivity – heat exchange between grey bodies – radiation shields – electrical analogy for radiation networks.

**Text Books:**

1. Yunus A. Cengel (2012), Heat Transfer a Practical Approach, 4th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. R. C. Sachdeva (2012), Fundamentals of Engineering, Heat and Mass Transfer, 3rd edition, New Age, New Delhi, India.

**Reference Books:**

1. Holman (2012), Heat Transfer (SI Units), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. P. S. Ghoshdastidar (2012), Heat Transfer, 2nd edition, Oxford University Press, New Delhi, India.
3. Incropera, Dewitt (2012), Fundamentals of Heat Transfer, 6th edition, John Wiley, UK.

**Web References:**

1. <http://nptel.ac.in/courses/112104113/>
2. <http://nptel.ac.in/courses/112105123/>
3. <https://www.udemy.com/fundamentals-of-engineering-heat transfer>

**E-Text Books:**

1. <https://www.pdfdrive.net/heat-transfer-e18770201.html>
2. <https://www.pdfdrive.net/fundamentals-of-heat-transfer-e18762062.html>
3. <https://www.pdfdrive.net/heat-transfer-e15057935.html>

**MOOC Course:**

1. <https://onlinecourses.nptel.ac.in/>
2. <https://www.coursera.org/learn/heat-transfer-intro>
3. [https://swayam.gov.in/courses/4642-july-2018-conduction and convection](https://swayam.gov.in/courses/4642-july-2018-conduction-and-convection)

## OPEN ELECTIVES OFFERED BY HS DEPARTMENT

| OE1    |                                           | OE2    |                                      |
|--------|-------------------------------------------|--------|--------------------------------------|
| A4HS10 | Business Economics and Financial Analysis | A4HS13 | Advanced Entrepreneurship            |
| A4HS11 | Basics of Entrepreneurship                |        |                                      |
| A4HS12 | Human Values and Professional Ethics      |        |                                      |
| OE3    |                                           | OE4    |                                      |
| A4HS14 | Indian Ethos & Business Ethics            | A4HS15 | Management Science                   |
|        |                                           | A4HS16 | Intellectual Property Rights         |
|        |                                           | A4HS17 | Number Theory                        |
|        |                                           | A4HS18 | Physics and Technology of Thin films |
|        |                                           | A4HS19 | Polymer chemistry                    |

## BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS10      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

### COURSE OUTCOMES:

The students should be able to

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Analyse how capital budgeting decisions are carried out.
4. Understanding the framework for both manual and computerised accounting process
5. Know how to analyse and interpret the financial statements through ratio analysis.

### UNIT-I

### INTRODUCTION & DEMAND ANALYSIS

**Classes: 10**

**Introduction & Demand Analysis:** Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

### UNIT-II

### PRODUCTION & COST ANALYSIS

**Classes: 10**

**Production & Cost Analysis:** Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.

### UNIT-III

### MARKETS & NEW ECONOMIC ENVIRONMENT

**Classes: 10**

**Markets & New Economic Environment:** Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership,

Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.

**UNIT-IV****CAPITAL BUDGETING****Classes: 10**

**Capital Budgeting:** Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).

**UNIT-V****INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS****Classes: 10**

**Introduction to Financial Accounting & Financial Analysis:** Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart

**Text Books:**

1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009.
2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013.
3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.

**Reference Books:**

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012.
3. Lipsey & Chrystel, Economics, Oxford University Press, 2012.
4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012.
5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012.
6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012.
7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012.
8. Dwivedi: Managerial Economics, Vikas, 2012.
9. Shailaja & Usha: MEFA, University Press, 2012.
10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012.
11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011.
12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.

## BASIC OF ENTREPRENEURSHIP

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS11      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

### UNIT-I DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING Classes: 10

Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style, Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples,

Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"

### UNIT-II CUSTOMER, BUSINESS MODEL, VALIDATION Classes: 10

**CUSTOMER** :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and “Jobs-To- Be-Done” (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch .

**BUSINESS MODEL**: Get Started with Lean Canvas.

**VALIDATION**: Develop the Solution, Sizing the Opportunity, Building an MVP.

### UNIT-III MONEY AND TEAM Classes: 12

**MONEY**: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs -

Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each, What investors expect from you,

Practice Pitching to Investors and Corporates.

**TEAM:** Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

#### UNIT-IV     **MARKETING & SALES**

**Classes: 10**

**MARKETING & SALES:** Positioning - Understand the difference between product and brand and the link between them, • Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

**Channels & Strategy :** • Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

#### UNIT-V     **PLANNING & TRACKING**

**Classes: 08**

**Sales Planning :** Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, • Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale

**Planning & Tracking :** Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

**Business Regulation :** Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

#### **Text Books:**

1. Ries, E., 2011. The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses. The Crown Publishing Group
2. Maurya, A., 2012. Running Lean: Iterate from Plan A to a Plan That Works. O'Reilly Media
3. Blank, S., 2005. The Four Steps to the Epiphany: Successful Strategies for Startups That Win. K&S Ranch
4. Roy, R., 2012. Entrepreneurship. Oxford University Press
5. Gupta, T. S., 2011. Intellectual Property Law in India. Kluwer Law International
6. Hisrich, R., 2012. Entrepreneurship. McGraw-Hill Education



## HUMAN VALUES AND PROFESSIONAL ETHICS

### V Semester: OPEN ELECTIVE -I

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS12      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behaviour and mutually enriching interaction with Nature.

### COURSE OUTCOMES:

The students should be able to

1. It ensures students sustained happiness through identifying the essentials of human values and skills.
2. It facilitates a correct understanding between profession and happiness
3. It helps students understand practically the importance of trust, mutually satisfying human behaviour and enriching interaction with nature.
4. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

### UNIT-I

### COURSE INTRODUCTION

**Classes: 10**

**Course Introduction** - Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfilment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfil the above human aspirations: understanding and living in harmony at various levels.

### UNIT-II

### UNDERSTANDING HARMONY IN THE HUMAN BEING

**Classes: 10**

**Understanding Harmony in the Human Being** - Harmony in Myself! : Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' ( I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.

### **UNIT-III      UNDERSTANDING HARMONY IN THE FAMILY AND SOCIETY      Classes: 10** **- HARMONY IN HUMAN - HUMAN RELATIONSHIP**

**Understanding Harmony in the Family and Society - Harmony in Human - Human Relationship:** Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; **Trust (Vishwas) and Respect ( Samman) as the foundational values of relationship.** Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society ( society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (Akhand Samaj), Universal Order ( Sarvabhaum Vyawastha) - from family to world family!

### **UNIT-IV      UNDERSTANDING HARMONY IN THE NATURE AND      Classes: 10** **EXISTENCE - WHOLE EXISTENCE AS CO-EXISTENCE**

**Understanding Harmony in the nature and Existence - Whole existence as Co-existence:** Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.

### **UNIT-V      IMPLICATIONS OF THE ABOVE HOLISTIC UNDERSTANDING      Classes: 10** **OF HARMONY ON PROFESSIONAL ETHICS**

**Implications of the above Holistic Understanding of Harmony on Professional Ethics:** Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics:

- a. Ability to utilize the professional competence for augmenting universal human order,
- b. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
- c. Ability to identify and develop appropriate technologies and management patterns for above production systems.

Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order.

- a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
- b. At the level of society: as mutually enriching institutions and organizations.

#### **Text Books:**

1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.

**Reference Books:**

1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
2. E. F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
3. A Nagraj, 1998 Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
4. Susan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991.
5. P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers.
6. A. N. Tripathy, 2003, Human Values, New Age International Publishers.
7. Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
8. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books.
9. E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press.
10. M Govindrajan, S Natrajan & V. S Senthil kumar, Engineering Ethics ( including Human Values), Eastern Economy Edition, Prentice Hall of India Ltd.

**Relevant CDs, Movies, Documentaries & Other Literature:**

1. value Education website, <http://www.uptu.ac.in>
2. Story of Stuff, <http://www.storyofstuff.com>
3. Al Gore, An Inconvenient Truth, Paramount Classics, USA
4. Charle Chaplin, Modern Times, United Artists, USA
5. IIT Delhi, Modern Technology - the Untold Story

## ADVANCED ENTREPRENEURSHIP

### VI Semester: OPEN ELECTIVE -II

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS13      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

### UNIT-I ORIENTATION TO GROWTH Classes: 10

**ORIENTATION TO GROWTH:** Getting Ready for Growth-Why growth stage is different compared to start up phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.

### UNIT-II CUSTOMER, BUSINESS MODEL, VALIDATION Classes: 10

**CUSTOMERS :** Expanding Customer Base-Revisit your business model and develop few variants (more business model types), Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize.

**BUSINESS MODEL:** Get Started with Lean Canvas.

**VALIDATION:** Develop the Solution, Sizing the Opportunity, Building an MVP.

### UNIT-III TRACTION AND TEAM Classes: 12

**TRACTION :** Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value(LTV), Identifying waste in your operations and focusing your team on what is important for traction.

Channels and Strategy - The Bullseye framework, Identify Channels using Bulls Eye Framework, Measuring the effectiveness of selected channels, Budgeting and planning.

### UNIT-IV MONEY & SALES Classes: 10

**MONEY:** Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams(licensing, franchising), Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening

skills, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers' needs, not on product features, Follow-up and closing a sale; Asking for the sale.

Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets.

Improving Margins - Testing price elasticity, Optimizing costs and operational expenses,

Advanced concepts of unit costing. Financial Modelling - Financial modelling of your venture's growth, Analyzing competitor and peer's financial models.

## UNIT-V

## SUPPORT

Classes: 08

**SUPPORT:** Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks.

Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan

### Text Books:

1. Clancy, Ann L. & Binkert, Jacqueline, 2017, Pivoting- A coach's guide to igniting substantial change. Palgrave Macmillan US
2. Porter, Michael, E., 2004, Competitive Advantage: Creating and Sustaining Superior Performance, Simon & Schuster
3. Schwetje, Gerald & Vaseghi Sam. The Business Plan, Springer
4. LeMay, Matt, Product Management in Practice, O'Reilly Media Inc.
5. Smart, Geoff & Randy, Street. 2008, Who: The A method of hiring, Ballantine books
6. Blokdyk, Gerardus. 2018, Customer Lifecycle Management a complete guide, 5starcooks
7. Kanell, Leo, 2017. The Business Funding Formula: How Entrepreneurs Are Jump Starting Their Businesses With Powerful Funding Strategies, Amazon Digital Services LLC
8. Lerman, Scott, 2013. Building better Brands: A competitive guide to brand strategy and identity development, HOW books, COM edition

## INDIAN ETHOS AND BUSINESS ETHICS

### VII Semester: OPEN ELECTIVE -III

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS14      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. To understand the importance of ethics in business
2. To acquire knowledge and capability to develop ethical practices for effective management
3. To understand the Business Ethics and to provide best practices of business ethics.
4. To learn the values and implement in their careers to become a good managers.
5. To develop various ethical Responsibilities and practice in their professional life
6. To imbibe the ethical issues and to adhere to the ethical codes.

### COURSE OUTCOMES:

1. Understand the dynamics of morality
2. Identify the constant in morality
3. Recognize the variable values in morality
4. Students will be able to understand the business ethics.
5. The student will be able to analyze various ethical codes in corporate governance
6. Student will be able to analyze the Employees conditions and Business Ethics

### UNIT-I

#### INTRODUCTION TO INDIAN ETHOS

**Classes: 10**

**Introduction to Indian Ethos:** History & Relevance, Principles Practiced by Indian Companies, Role of Indian Ethos in Managerial Practices, Management Lessons from Vedas, Mahabharata, Bible and Quran.

### UNIT-II

#### UNDERSTANDING VALUES IN BUSINESS

**Classes: 10**

**Understanding Values in Business:** Kautilya's Arthashastra, Indian Heritage in Business, Management-Production and Consumption. Ethics v/s Ethos, Indian v/s Western Management, Work Ethos and Values for Indian Managers- Relevance of Value Based Management in Global Change- Impact of Values on Stakeholders, Trans-Cultural Human Values, Secular v/s Spiritual Values, Value System in Work Culture, Stress Management-Meditation for mental health, Yoga.

### UNIT-III

#### CONTEMPORARY APPROACHES TO INDIAN ETHOS

**Classes: 12**

**Contemporary Approaches to Indian Ethos:** Contemporary Approaches to Leadership- Joint Hindu Family Business-Leadership Qualities of Karta, Indian Systems of Learning-Gurukul System of Learning, Advantages- Disadvantages of Karma, importance of Karma to Managers-Nishkama Karma-Laws of Karma, Law of Creation- Law of Humility- Law of Growth- Law of Responsibility- Law

of Connection-Corporate Karma Leadership.

**UNIT-IV****UNDERSTANDING THE BUSINESS ETHICS****Classes: 10**

**Understanding the Business Ethics :** Understanding the need for ethics, Ethical values, myths and ambiguity, ethical codes, Ethical Principles in Business; Theories of Ethics, Absolutism verses Relativism, Teleological approach, the Deontological approach, Kohlberg's six stages of moral development (CMD).

**UNIT-V****ETHICAL CULTURE IN ORGANIZATION****Classes: 08**

**Ethical Culture in Organization:** Ethical Culture in Organization, Developing codes of Ethics and conduct, Ethical and value based leadership. Role of scriptures in understanding ethics, Indian wisdom & Indian approaches towards business ethics.

**Text Books:**

1. M.G. Velasquez, Business Ethics, Prentice Hall India Limited, New Delhi,
2. R.C. Sekhar, Ethical Choices in Business, Response Books, New Delhi, 2007

**Reference Books:**

1. Chakraborty S.K., —Management Transformation by ValuesII, New Delhi, Sage Publication, 1990.
2. Chakraborty, S.K., Ethics in Management-Vedantic Approach, New Delhi, Oxford India Ltd. 1995.
3. Fernando A.C., Business Ethics: An Indian Perspective, Pearson, 2009.
4. Kautilya's Arthashastra, King, Governance, and Law in Ancient India, Oxford University Press, 2016.
5. Murthy, C.S.R. Business Ethics, Himalaya Publishing House, Mumbai, 2009.
6. Narayana G., —The Responsible Leader: A Journey through Gitall, Ahmedabad, AMA 2000

**E Resources**

1. <http://lsib.co.uk/lms/wp-content/uploads/2015/02/Indian-Ethos-and-Management.pdf>
2. [www.vikaspublishing.com/books/business-economics/management/ethics](http://www.vikaspublishing.com/books/business-economics/management/ethics)

## MANAGEMENT SCIENCE

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS15      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

1. Familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices.
2. Understand the structure & Designing of an Organization.
3. Knowledge on the aspects of Production.
4. Analyze the market and the strategies involved in Marketing.
5. Knowledge on concepts related to Human Resources.
6. Understand the techniques used in Project Management.
7. Familiarize with strategies used for analysis of an Organization.
8. Understand the Contemporary Management Issues.
9. Familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.

### COURSE OUTCOMES:

1. Knowledge on management theories and practices.
2. Understanding designing organizational structure.
3. Understanding on the methods & charts used in operations management.
4. Ability to understand the market and its environment.
5. Understand the processes, functions etc in Human Resources Management.
6. Ability to solve problems in managing the Project.
7. Knowledge on Strategic alternatives.
8. Familiar with the practices implemented in management.
9. Understand the social responsibilities of Management.
10. Understand the basic concepts of Management.

### UNIT-I INTRODUCTION TO MANAGEMENT AND ORGANIZATION Classes: 10

**INTRODUCTION TO MANAGEMENT AND ORGANIZATION:** Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management- Taylor's Scientific Management Theory- Fayal's Principles of Management- Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Herzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.

### UNIT-II OPERATIONS AND MARKETING MANAGEMENT Classes: 10

**OPERATIONS AND MARKETING MANAGEMENT:** Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study



and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.

### **UNIT-III                      HUMAN RESOURCES MANAGEMENT(HRM)                      Classes: 12**

**HUMAN RESOURCES MANAGEMENT(HRM):** Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.

### **UNIT-IV                      PROJECT MANAGEMENT (PERT/ CPM)                      Classes: 10**

**PROJECT MANAGEMENT (PERT/ CPM):** Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems).

### **UNIT-V                      STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES                      Classes: 08**

**STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES:** Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.

#### **Text Books:**

1. A.R.Aryasri : Management Science, TMH, (Latest edition)
2. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.

#### **Reference Books:**

1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012.
2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012.
3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012.
4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012.
5. Samuel C. Certo: Modern Management, 2012.
6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012.

7. Parnell: Strategic Management, Cengage, 2012.
8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.

### E Resources

1. <https://www.smartzworld.com/notes/management-science-pdf-notes-ms-notes-pdf>
2. <https://www.alljntuworld.in/download/management-science-notes/>

## INTELLECTUAL PROPERTY RIGHTS

### VIII Semester: OPEN ELECTIVE -IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS16      | OEC      | 3            | - | - | 3       | 30            | 70  | 100   |

#### COURSE OBJECTIVES:

1. To introduce fundamental aspects of Intellectual property Rights to students who are going to play a major role in development and management of innovative projects in industries.
2. To disseminate knowledge on patents, patent regime in India and abroad and registration aspects
3. To disseminate knowledge on copyrights and its related rights and registration aspects
4. To aware about current trends in IPR and Govt. steps in fostering IPR
5. To familiarize them with the kind of rights, remedies and licensing regime associated with each kind of intellectual property so that students can have a basic understanding of Intellectual Property laws.

#### COURSE OUTCOMES:

1. Skill to understand the concept of intellectual property rights.
2. Develops procedural knowledge to Legal System and solving the problem relating to intellectual property rights.
3. Skill to pursue the professional programs in Company Secretary ship, Law, Business(MBA), International Affairs, Public Administration and Other fields.
4. Employability as the Compliance Officer, Public Relation Officer and Liaison Officer.
5. Establishment of Legal Consultancy and service provider.

#### UNIT-I INTRODUCTION TO INTELLECTUAL PROPERTY Classes: 10

**Introduction to Intellectual property:** Introduction, types of intellectual property, international organizations, agencies and treaties, importance of intellectual property rights.

#### UNIT-II TRADE MARKS Classes: 10

**Trade Marks:** Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.

#### UNIT-III LAW OF COPY RIGHTS Classes: 12

**Law of copy rights:** Fundamental of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law.

**Law of patents:** Foundation of patent law, patent searching process, ownership rights and transfer.

#### **UNIT-IV                      TRADE SECRETS & UNFAIR COMPETITION                      Classes: 10**

**Trade Secrets:** Trade secret law, determination of trade secret status, liability for misappropriations of trade secrets, protection for submission, trade secret litigation.

**Unfair competition:** Misappropriation right of publicity, False advertising.

#### **UNIT-V                      NEW DEVELOPMENT OF INTELLECTUAL PROPERTY                      Classes: 08**

**New development of intellectual property:** new developments in trade mark law; copy right law, patent law, intellectual property audits. International overview on intellectual property, international - trade mark law, copy right law, international patent law, international development in trade secrets law.

#### **Text Books:**

1. Intellectual property right, Deborah, E. Bouchoux, cengage learning.
2. Intellectual property right - Unleashing the knowledge economy, prabuddha ganguli, Tata McGraw Hill Publishing Company Ltd.

#### **Reference Books:**

1. Deborah. E. Bouchoux (2009), Intellectual property, Cengage learning, India.
2. Deborah. E. Bouchoux (2001), Protecting your companies intellectual property, AMACOM, USA.
3. Prabuddaganguli (2003), Intellectual property right, Tata McGraw Hill Publishing company Ltd., India.
4. Robert Hisrich, Michael P. Peter, Dean A. Shepherd (201), Entrepreneurship, Tata McGraw Hill., India

#### **E Resources**

1. Subramanian, N., & Sundararaman, M. (2018). Intellectual Property Rights – An Overview. Retrieved from <http://www.bdu.ac.in/cells/ipr/docs/ipr-eng-ebook.pdf>
2. World Intellectual Property Organisation. (2004). WIPO Intellectual property Handbook. Retrieved from [https://www.wipo.int/edocs/pubdocs/en/intproperty/489/wipo\\_pub\\_489.pdf](https://www.wipo.int/edocs/pubdocs/en/intproperty/489/wipo_pub_489.pdf)      Reference Journal: 1. Journal of Intellectual Property Rights (JIPR): NISCAIR

## NUMBER THEORY

### VIII Semester: OPEN ELECTIVE - IV

| Course Code: | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|--------------|----------|--------------|---|---|---------|---------------|-----|-------|
|              |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS17       | HSMC     | 3            | 0 | - | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

To learn

1. Divisibility test and Euclidian Algorithms.
2. The Arithmetic function and Dirichlet multiplication.
3. The properties of congruences.
4. Application of Lagrange's theorem.
5. Quadratic Residues and Reciprocity Law.

### COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply the Fundamental theorem of Arithmetic..
2. Identify the Liouville's function and divisor function.
3. Apply the Euler Fermat theorem in polynomial congruence modulo  $P$
4. Apply Chinese remainder theorem.
5. Find Quadratic residues.

### UNIT-I

Classes: 10

Divisibility, GCD, Prime Numbers, Fundamental theorem of Arithmetic, the series of reciprocal of the Primes, The Euclidean Algorithm.

### UNIT-II

Classes: 08

A Arithmetic function and Dirichlet Multiplication, The functions  $\phi(n)$ ,  $\mu(n)$  and a relation connecting them, Product formulae for  $\phi(n)$ , Dirichlet Product, Dirichlet inverse and Mobius inversion formula and Mangoldt function  $\Lambda(n)$ , multiplication function, multiplication function and Dirichlet multiplication, Inverse of a completely multiplication function, Liouville's function  $\lambda(n)$ , the divisor function is  $\sigma_a(n)$

### UNIT-III

Classes: 08

Congruences, Properties of congruences, Residue Classes and complete residue system, linear congruences conversion, reduced residue system and Euler Fermat theorem, polynomial congruence modulo  $P$

### UNIT-IV

Classes: 08

Lagrange's theorem, Application of Lagrange's theorem, Chinese remainder theorem and its application, polynomial congruences with prime power module.

**UNIT-V****Classes: 08**

Quadratic residue and quadratic reciprocity law, Quadratic residues, Legendre's symbol and its properties, evaluation of  $(-1/p)$  and  $(2/p)$ , Gauss Lemma, the quadratic reciprocity law and its applications

**Text Books:**

1. "Introduction to analytic Number Theory " by Tom M. Apostol. Chapters 1, 2, 5, 9.

**Reference Books:**

1. Number Theory by Joseph H. Silverman.
2. Theory of Numbers by K.Ramchandra.
3. Elementary Number Theory by James K Strayer.
4. Elementary Number Theory by James Tattusall.

## PHYSICS AND TECHNOLOGY OF THIN FILMS

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS18      | HSMC     | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the bonding, nucleation process in materials.
2. Gain the knowledge on fabrication of thin films.
3. Learn the kinetics of different thin films.
4. Understand the properties of thin films by using characterization methods.
5. Learn the applications of thin films in different fields.

### COURSE OUTCOMES:

The student will able to:

1. Analyze the defects in solids and kinetics.
2. Evaluate the epitaxy of thin films and growth process.
3. Justify how the growth kinetics is more efficient than nucleation kinetics in thin film technology.
4. Recommend appropriate synthesis method and explain the characterization techniques.
5. Analyze which thin films are used in different applications.

### UNIT-I

#### A Review of Materials Science

Classes: 06

Introduction, Structure Defects in Solids Bonding of Materials, Thermodynamics of Materials, Kinetics, Nucleation and Conclusion.

### UNIT-II

#### Preparation of Thin-films:

Classes: 10

**Classifications of vacuum ranges** – Vacuum pumps - Rotary, Diffusion, Turbo molecular and Ion Pumps –Thin film (epitaxy) – definition & advantages – Types of epitaxy.

**Different Growth Techniques:** Liquid Phase Epitaxy, Molecular Beam Epitaxy, Metal Organic Vapour Phase Epitaxy, Sputtering (RF & DC), Pulsed Laser Deposition. Thickness Measurement: Microbalance technique, Photometry, Interferometry (MBI, FECO)

### UNIT-III

#### Kinetics of Thin films

Classes: 08

**Nucleation Kinetics:** types of nucleation – kinetic theory of nucleation – energy formation of a nucleus – critical nucleation parameters; spherical and non-spherical nucleus (cap, disc and cubic shaped) on the substrates.

**Growth Kinetics:** Kinetics of binary (GaAs, InP, etc.), ternary ( $\text{Al}_{1-x}\text{Ga}_x\text{As}$ ,  $\text{Ga}_{1-x}\text{In}_x\text{P}$ , etc.) and quaternary ( $\text{Ga}_{1-x}\text{In}_x\text{As}_{1-y}\text{P}_y$ , etc.) semiconductors – derivation of growth rate and composition expressions.

### UNIT-IV

#### Properties & Characterization of Thin Films

Classes: 10

**Properties of Thin Films:** Dielectric properties – Important parameters, Measurement of dielectric

properties- Effect of annealing and film thickness. Optical properties – Optical constants, determination of optical constants by Brewster angle method, Normal incidence method and graphical method. Mechanical properties – Concept and origin of stress and strain, Lattice misfit, Thermal misfit, Hardness test and Bulge test.

### **Characterization**

X-ray diffraction –Photoluminescence –UV-Vis-IR spectrophotometer – Atomic Force Microscope – Scanning Electron Microscope – Hall Effect – Vibrational Sample Magnetometer – Secondary Ion MassSpectrometry.

## **UNIT-V Applications of Thin Films**

**Classes: 08**

Optoelectronic devices: LED, LASER and Solar cell – Micro Electromechanical Systems (MEMS) – Fabrication of thin film capacitor – application of ferromagnetic thin films; Data storage, Giant Magneto resistance (GMR).

### **Text Books:**

1. A. Goswami, Thin Film Fundamentals, New Age international (P) Ltd. Publishers, New Delhi(1996).

### **Reference Books:**

1. K. L. Chopra, Thin Film Phenomena, McGraw- Hill book company New York,(1969).
2. L. Eckertova, Physics of Thin Films, Plenum press, New York(1977).
3. Milton Ohring, Material Science of Thin films, 2<sup>nd</sup> Edition, Academic Press(2002).

### **E-Text Books:**

1. <https://www.safaribooksonline.com/library/view/materials-science-of/9780125249751/?ar&orpg>
2. <http://doi-ds.org/doi/10.1016/j.jmr.2018.07.001>
3. <https://ieeexplore.ieee.org/document/5488896>

### **MOOC Course:**

1. <https://nptel.ac.in/courses/113106034/44>
2. [https://swayam.gov.in/nd1\\_noc19\\_mm13](https://swayam.gov.in/nd1_noc19_mm13)
3. [www.facweb.iitkgp.ac.in/~jdas/mfm/Teaching.html](http://www.facweb.iitkgp.ac.in/~jdas/mfm/Teaching.html)



## POLYMER CHEMISTRY

### VIII Semester: OPEN ELECTIVE - IV

| Course Code | Category | Hours / Week |   |   | Credits | Maximum Marks |     |       |
|-------------|----------|--------------|---|---|---------|---------------|-----|-------|
|             |          | L            | T | P |         | CIA           | SEE | Total |
| A4HS19      | OEC      | 3            | 0 | 0 | 3       | 30            | 70  | 100   |

### COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the structure of the polymers, identify and explain types of polymerization.
2. Gain the knowledge to use techniques of polymerization.
3. Understand the properties, processing and applications of natural rubber.
4. Gain the knowledge to design polymers for industrial applications.
5. Learn the structure properties of polymers, inorganic and its characterization techniques

### COURSE OUTCOMES:

The student will be able to:

1. Describe the structures of the polymers, identify and explain types of polymerization and account for the concept of the molecular weight.
2. Explain the various properties of polymers and use techniques of polymerization.
3. Account the properties, processing and applications of natural rubber.
4. Design polymers for industrial application using suitable additives.
5. Correlate structure-properties of polymer composite, inorganic, conducting polymers and its characterization techniques.

### UNIT-I

#### BASICS OF POLYMERS

Classes: 10

**Introduction:** Polymer Definition, Classifications of Polymers: Natural & Synthetic, Biopolymers, Thermoplastic, Thermosets, Elastomers, Fibers etc. Types of Polymerization: Addition & Condensation Polymerization. Molecular Weight : No. Average and Weight Average Molecular Weight of Polymers, Distribution & Poly dispersity Index, Structure of Polymers: Amorphous, Semi crystalline and Crystalline states

### UNIT-II

#### PROPERTIES & POLYMERIZATION TECHNIQUES

Classes: 10

**Polymerization Techniques:** Mass Polymerization, Bulk Polymerization, Solution Polymerization, Emulsion Polymerization, Suspension Polymerization, Mechanisms with explanation. Structure Property Relationship in Polymers: glass transition, melting and crystallization temperature. Effect of structure on the chemical, mechanical, electrical and optical properties of polymers.

### UNIT-III

#### NATURAL RUBBER: PROPERTIES, PROCESSING AND APPLICATIONS

Classes: 10

**Natural Rubber (NR):** Origin – Natural Rubber Latex, Tapping, Processing- Vulcanization, Properties and Applications – Conversion of Latex into Dry Rubber – Properties of Dry Rubber. Synthetic Polyisoprene (IR) Rubbers: Preparation, Properties and Applications, Thermoplastic Elastomers Based on Plastics.

**UNIT-IV****NON RUBBER ADDITIVES****Classes: 10**

**Non Rubber Additives:** Part A: Vulcanizing ingredients & other additives: Vulcanizing ingredients & their sequence of mixing: Activators and Accelerators: mechanisms of action Part B: Fillers Carbon black-Its preparation, structure, properties and their effect on rubber properties Silica fillers & coupling agents, Other fillers: Clay, Calcium carbonate, titania etc. Nano-fillers: Reinforcement by filler: Reinforcement, Part C: Processing aids & other additives: Processing aids, plasticizers, process additives, release agents, Prevention of Ozone Attack with the use of waxes & saturated polymer for Ozone Protection.

**UNIT-V****INORGANIC, COMPOSITE AND CONDUCTING POLYMER,  
CHARACTERIZATION TECHNIQUES OF POLYMERS****Classes: 08**

**Inorganic Polymers:** Synthesis, Structure- property Relationship and Applications of Polyphosphates, Polysiloxanes, Polymer Composite- Properties and its Applications. Conducting Polymers- Poly Aniline (PaNi)- Mechanism of Conduction and Applications.

**Characterization Techniques of Polymers:** Thermo gravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC).

**Reference Books:**

1. F. W. Billmeyer Jr., Text Book of Polymer Science, Ed. Wiley-Interscience, 1984.
2. V. T. Gowariker, N. V. Viswanathan, and J. Sreedar, Polymer Science, 1988.
3. M. Morton, Rubber Technology, Chapman Hall, 1995.
4. Robert J. Young and Peter A. Lovell, Introduction to Polymers, 3<sup>rd</sup> Edition, 2011.